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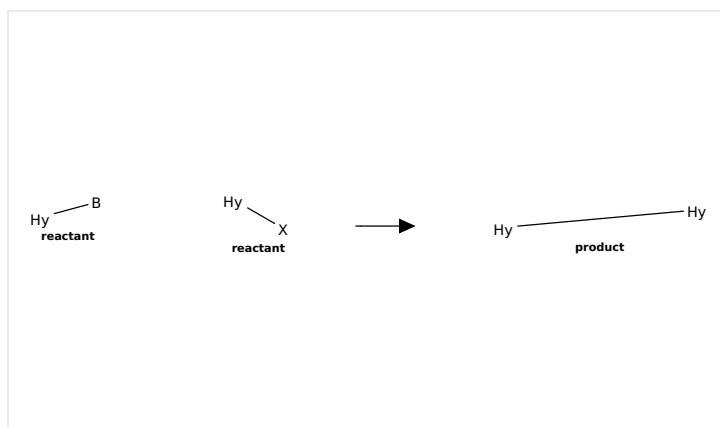
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Publication 2010 to 2025

Year:

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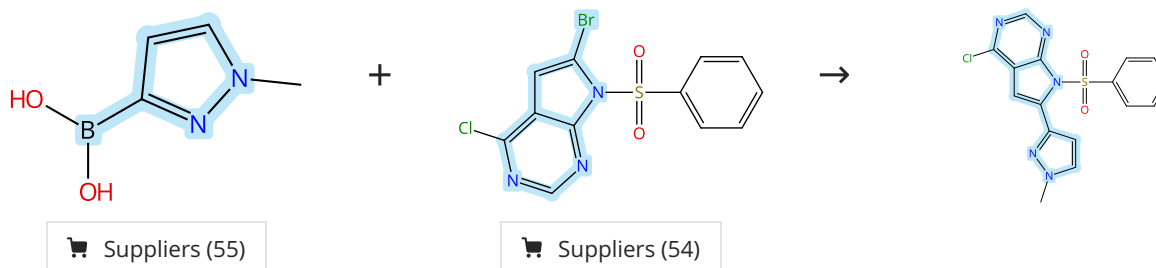
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Reactions (500)

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Scheme 1 (1 Reaction)

Steps: 1 Yield: 63%



31-179-CAS-14387274

Steps: 1 Yield: 63%

Discovery of novel Bruton's tyrosine kinase (BTK) inhibitors bearing a pyrrolo[2,3-d]pyrimidine scaffold

By: Zhao, Xinge; et al

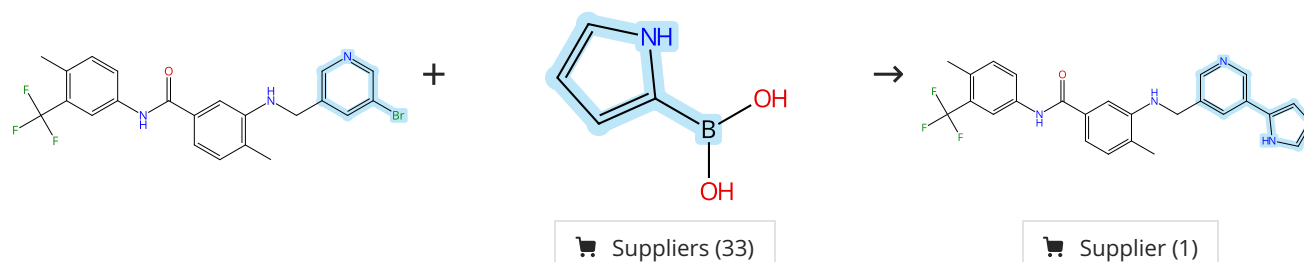
Bioorganic & Medicinal Chemistry (2015), 23(4), 891-901.

1.1 **Reagents:** Sodium carbonate
Catalysts: Dichloro[1,1'-bis(diphenylphosphino)ferrocene] palladium(II) dichloromethane adduct
Solvents: 1,4-Dioxane, Water; 5 h, 80 °C

[Experimental Protocols](#)

Scheme 2 (1 Reaction)

Steps: 1 Yield: 63%



31-179-CAS-22561065

Steps: 1 Yield: 63%

Discovery of VU6015929: A Selective Discoidin Domain Receptor 1/2 (DDR1/2) Inhibitor to Explore the Role of DDR1 in Antifibrotic Therapy

By: Jeffries, Daniel E.; et al

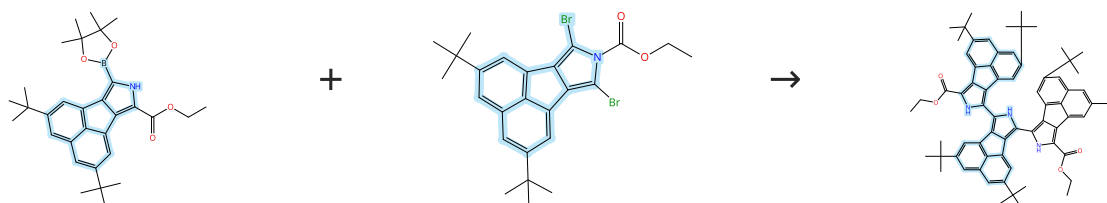
ACS Medicinal Chemistry Letters (2020), 11(1), 29-33.

1.1 **Reagents:** Cesium carbonate
Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium
Solvents: 1,4-Dioxane, Water; 3 h, 90 °C

[Experimental Protocols](#)

Scheme 3 (1 Reaction)

Steps: 1 Yield: 63%



31-179-CAS-23538664

Steps: 1 Yield: 63%

Cyclo[9]pyrrole: Selective Synthesis of [34] Nonaphyrin (0.0.0.0.0.0.0.0)

By: Matsumoto, Hiroki; et al

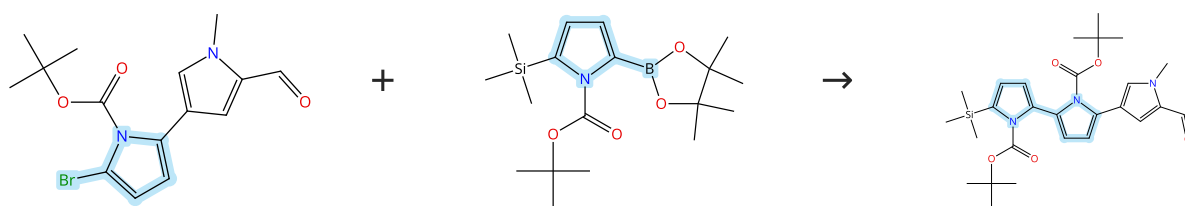
Organic Letters (2021), 23(9), 3442-3446.

1.1 **Reagents:** Potassium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: 1,4-Dioxane, Water; 14 h, reflux

[Experimental Protocols](#)

Scheme 4 (1 Reaction)

Steps: 1 Yield: 63%



Suppliers (2)

31-179-CAS-22693161

Steps: 1 Yield: 63%

1.1 **Reagents:** Potassium carbonate
Catalysts: Dichloro[1,1'-bis(diphenylphosphino)ferrocene] palladium(II) dichloromethane adduct
Solvents: 1,4-Dioxane, Water; 22 °C; 1 h, 50 °C

Experimental Protocols

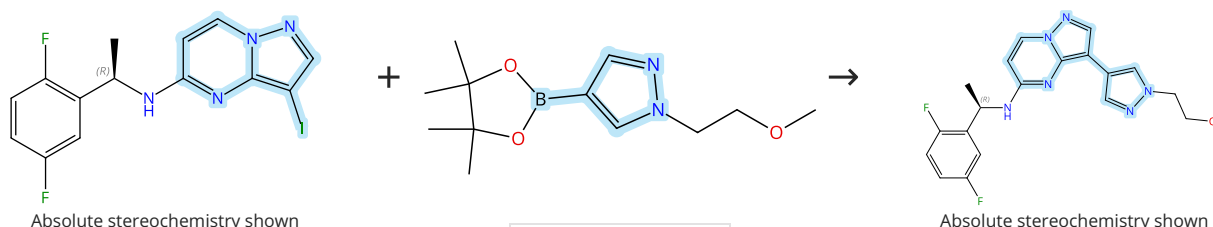
Iterative Suzuki-Miyaura Cross-coupling/Bromo-desilylation Reaction Sequences for the Assembly of Chemically Well-defined, Acyclic Oligopyrrole/Benzenoid Hybrids Embodying Mixed Modes of Connectivity

By: Lee, BoRa; et al

Chemistry - An Asian Journal (2020), 15(19), 3059-3081.

Scheme 5 (1 Reaction)

Steps: 1 Yield: 63%



Suppliers (83)

31-614-CAS-34551890

Steps: 1 Yield: 63%

1.1 **Reagents:** Potassium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Dimethylformamide, Water; 2 h, 100 °C

Experimental Protocols

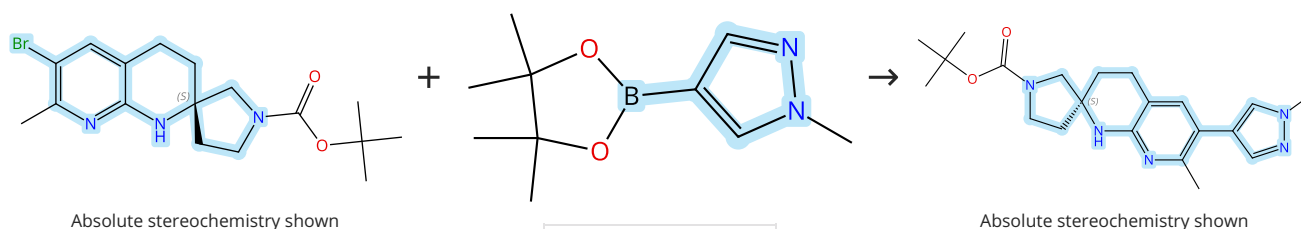
Discovery of 3-pyrazolyl-substituted pyrazolo[1,5-a]pyrimidine derivatives as potent TRK inhibitors to overcome clinically acquired resistance

By: Gong, Yi; et al

European Journal of Medicinal Chemistry (2022), 241, 114654.

Scheme 6 (1 Reaction)

Steps: 1 Yield: 63%



Suppliers (5)

Suppliers (126)

31-614-CAS-38847502

Steps: 1 Yield: 63%

1.1 **Reagents:** Sodium carbonate
Catalysts: Dichloro[1,1'-bis(diphenylphosphino)ferrocene] palladium(II) dichloromethane adduct
Solvents: 1,4-Dioxane; overnight, rt → 100 °C

Experimental Protocols

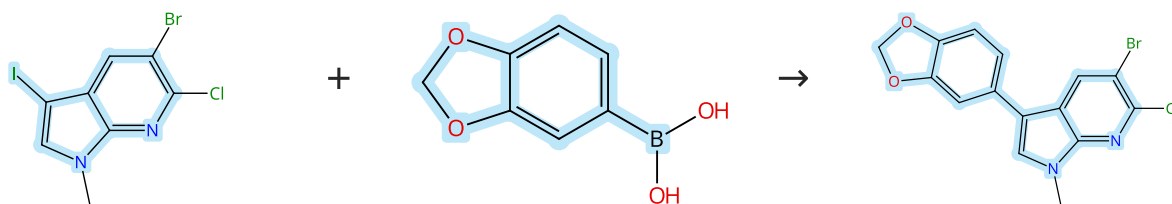
Discovery of the Potent and Selective MC4R Antagonist PF-07258669 for the Potential Treatment of Appetite Loss

By: Garnsey, Michelle R.; et al

Journal of Medicinal Chemistry (2023), 66(5), 3195-3211.

Scheme 7 (1 Reaction)

Steps: 1 Yield: 63%



Suppliers (99)

31-179-CAS-20704851

Steps: 1 Yield: 63%

Pd-Catalyzed Sequential Arylation of 7-Azaindoles: Aggregate-Induced Emission of Tetra-Aryl 7-Azaindoles

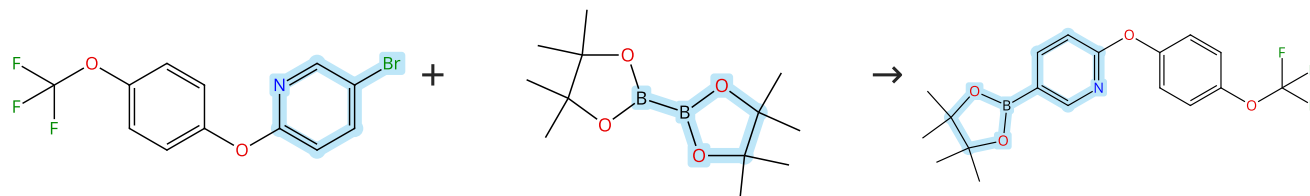
By: Cardoza, Savio; et al

Journal of Organic Chemistry (2019), 84(21), 14015-14029.

1.1 **Reagents:** Cesium carbonate
Catalysts: 1,1-Bis(diphenylphosphino)ferrocene, Tris (dibenzylideneacetone)dipalladium
Solvents: Ethanol, Toluene; 0.25 h, 60 °C

Scheme 8 (1 Reaction)

Steps: 1 Yield: 63%



Suppliers (11)

Suppliers (131)

Suppliers (3)

31-142-CAS-689042

Steps: 1 Yield: 63%

Discovery, Synthesis, and Optimization of Antimalarial 4(1H)-Quinolone-3-Diarylethers

By: Nilsen, Aaron; et al

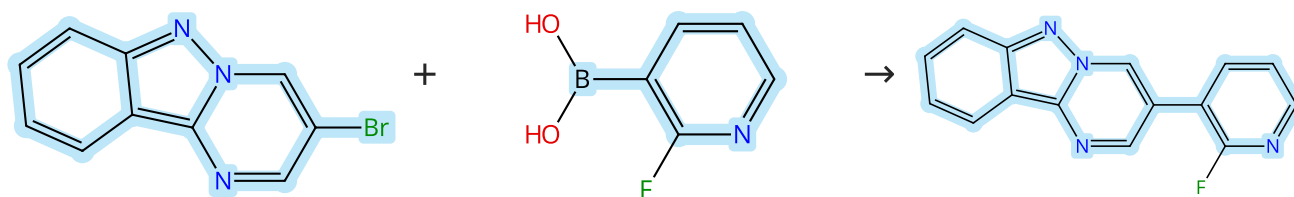
Journal of Medicinal Chemistry (2014), 57(9), 3818-3834.

1.1 **Reagents:** Potassium acetate
Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium
Solvents: Dimethylformamide; rt → 80 °C; 18 h, 80 °C

Experimental Protocols

Scheme 9 (1 Reaction)

Steps: 1 Yield: 63%



Suppliers (14)

Suppliers (101)

Supplier (1)

31-179-CAS-22871714

Steps: 1 Yield: 63%

Pyrimido[1,2-b]indazole derivatives: Selective inhibitors of human monoamine oxidase B with neuroprotective activity

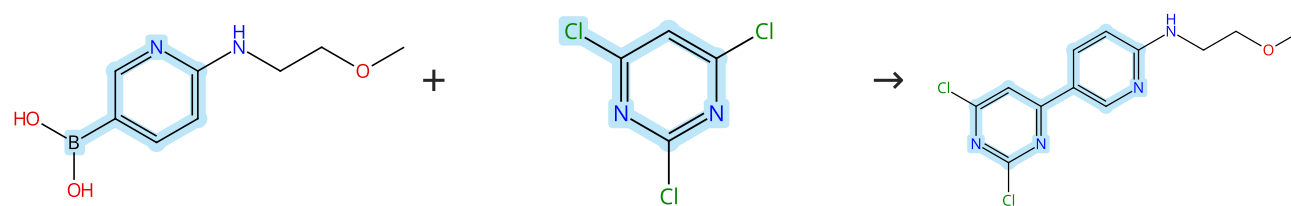
By: Jismy, Badr; et al

European Journal of Medicinal Chemistry (2021), 209, 112911.

1.1 **Reagents:** Sodium carbonate
Catalysts: Dichlorobis(triphenylphosphine)palladium
Solvents: 1,4-Dioxane, Water; 3 h, rt → 110 °C

Experimental Protocols

Scheme 10 (1 Reaction)



Suppliers (4)

Suppliers (90)

31-614-CAS-34480598

Steps: 1 Yield: 63%

1.1 Reagents: Sodium carbonate
Catalysts: Palladium diacetate
Solvents: Acetonitrile, Water; 4 h, 50 °C

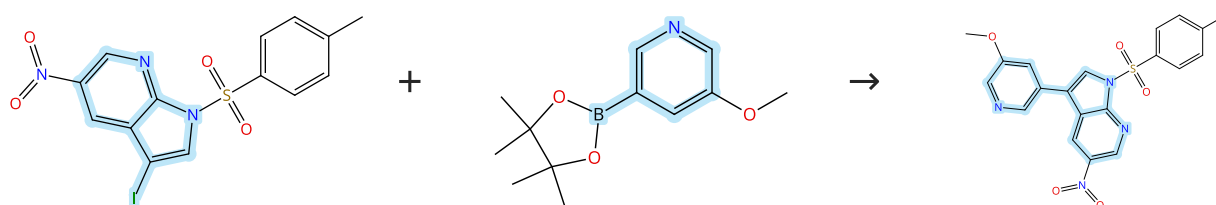
Experimental Protocols

Structure-activity relationships of 2-pyrimidinecarboxydrizides as utrophin modulators for the potential treatment of Duchenne muscular dystrophy

By: Chatzopoulou, Maria; et al

Bioorganic & Medicinal Chemistry (2022), 69, 116812.

Scheme 11 (1 Reaction)



Suppliers (90)

31-614-CAS-36218767

Steps: 1 Yield: 63%

1.1 Reagents: Potassium carbonate
Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium
Solvents: Ethanol, 1,4-Dioxane, Water; 3 h, 90 °C

1.2 Reagents: Water

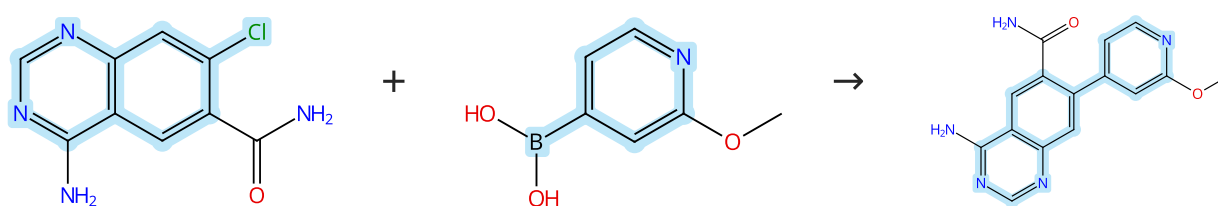
Experimental Protocols

Rational design and appraisal of selective Cdc2- Like kinase 1 (Clk1) inhibitors as novel autophagy inducers for the treatment of acute liver injury (ALI)

By: Yang, Tao; et al

European Journal of Medicinal Chemistry (2023), 250, 115168.

Scheme 12 (1 Reaction)



Suppliers (99)

31-614-CAS-34609651

Steps: 1 Yield: 63%

1.1 Reagents: Tripotassium phosphate
Catalysts: Tris(dibenzylideneacetone)dipalladium, [2'-(Amino-κM)[1,1'-biphenyl]-2-yl-κC][dicyclohexyl(2',6'-dimethoxy[1,1'-biphenyl]-2-yl)phosphine-κP](methanesulfonato-κO)palladium
Solvents: 1,4-Dioxane, Water; 24 h, 95 °C

Experimental Protocols

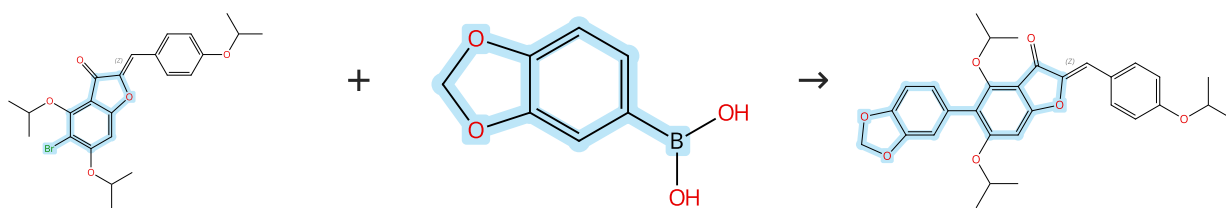
Structure-based design and modular synthesis of novel PI4K class II inhibitors bearing a 4- aminoquinazoline scaffold

By: Misehe, Mbilo; et al

Bioorganic & Medicinal Chemistry Letters (2022), 76, 129010.

Scheme 13 (1 Reaction)

Steps: 1 Yield: 63%



Double bond geometry shown

Suppliers (99)

Double bond geometry shown

31-179-CAS-17587098

Steps: 1 Yield: 63%

Synthesis and anti-oxidant activity evaluation of (±)-Anastatins A, B and their analogs

By: Pan, Guojun; et al

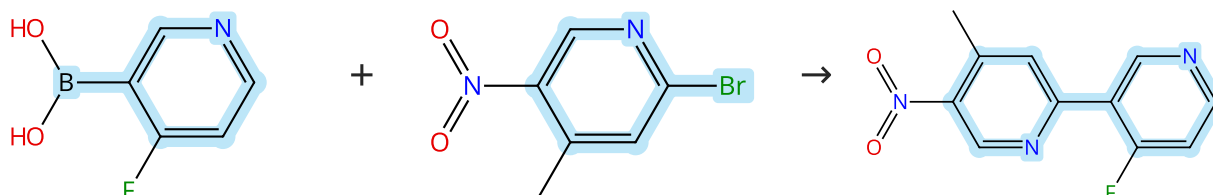
European Journal of Medicinal Chemistry (2017), 138, 577-589.

1.1 **Reagents:** Cesium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Dimethylformamide; rt → 100 °C; 12 h, 100 °C

Experimental Protocols

Scheme 14 (1 Reaction)

Steps: 1 Yield: 63%



Suppliers (45)

Suppliers (80)

31-179-CAS-14456253

Steps: 1 Yield: 63%

Discovery of 3,5-substituted 6-azaindazoles as potent pan-Pim inhibitors

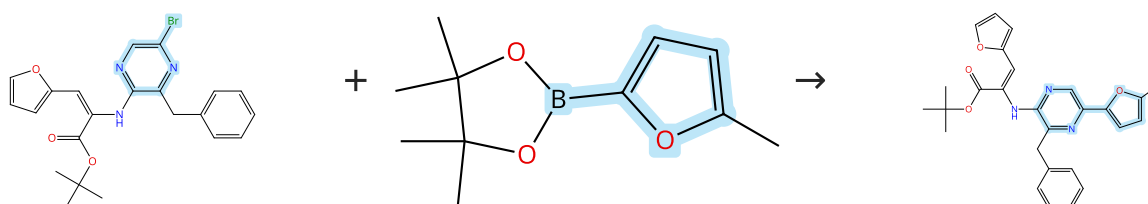
By: Hu, Huiyong; et al

Bioorganic & Medicinal Chemistry Letters (2015), 25(22), 5258-5264.

1.1 **Reagents:** Potassium acetate, Sodium carbonate
Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium
Solvents: Acetonitrile, Dichloromethane; 30 min, 130 °C

Scheme 15 (1 Reaction)

Steps: 1 Yield: 63%



Suppliers (87)

31-179-CAS-22258161

Steps: 1 Yield: 63%

Novel NanoLuc-type substrates with various C-6 substitutions

By: Yan, Chongzheng; et al

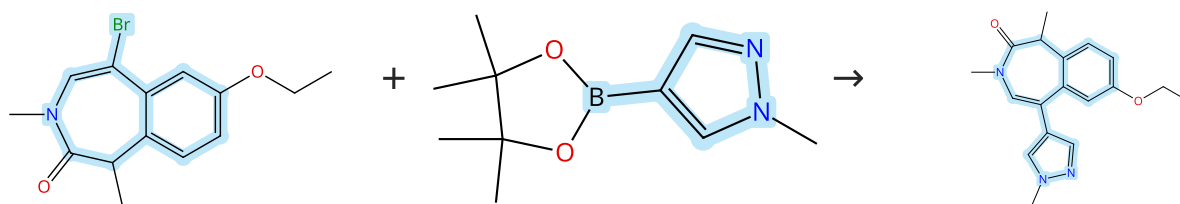
Bioorganic & Medicinal Chemistry Letters (2020), 30(9), 127085.

1.1 **Reagents:** Cesium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: 1,4-Dioxane, Water; 3 h, rt → 85 °C

Experimental Protocols

Scheme 16 (1 Reaction)

Steps: 1 Yield: 63%



Suppliers (126)

31-614-CAS-40832781

Steps: 1 Yield: 63%

Identification and Optimization of a Ligand-Efficient Benzoazepinone Bromodomain and Extra Terminal (BET) Family Acetyl-Lysine Mimetic into the Oral Candidate Quality Molecule I-BET432

By: Humphreys, Philip G.; et al

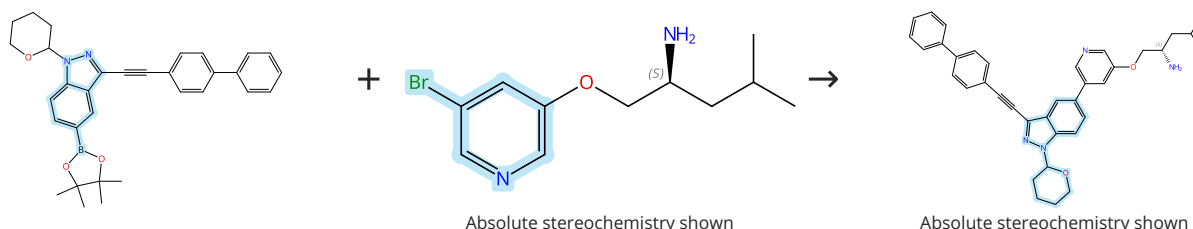
Journal of Medicinal Chemistry (2022), 65(22), 15174-15207.

1.1 **Reagents:** Potassium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Isopropanol, Water; 1 h, 100 °C

Experimental Protocols

Scheme 17 (1 Reaction)

Steps: 1 Yield: 63%



Suppliers (2)

31-614-CAS-39864156

Steps: 1 Yield: 63%

Structure-Guided Discovery of Potent and Selective CLK2 Inhibitors for the Treatment of Knee Osteoarthritis

By: Sun, Yongqiang; et al

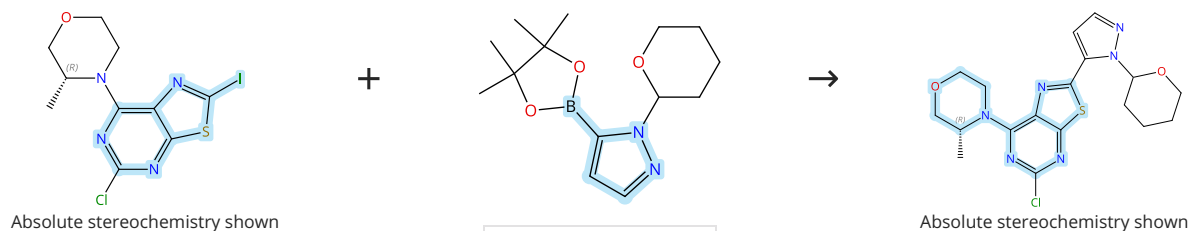
Journal of Medicinal Chemistry (2024), 67(6), 4603-4623.

1.1 **Reagents:** Sodium carbonate
Catalysts: Dichlorobis(triphenylphosphine)palladium
Solvents: 1,4-Dioxane, Water; 6 h, rt → reflux

Experimental Protocols

Scheme 18 (1 Reaction)

Steps: 1 Yield: 63%



Suppliers (102)

31-614-CAS-24031068

Steps: 1 Yield: 63%

Discovery of a Brain-Penetrant ATP-Competitive Inhibitor of the Mechanistic Target of Rapamycin (mTOR) for CNS Disorders

By: Bonazzi, Simone; et al

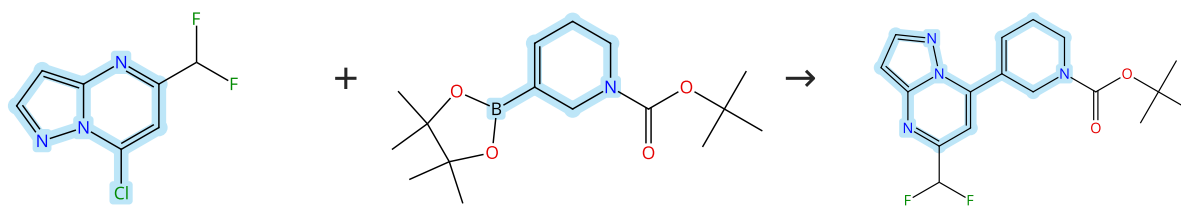
Journal of Medicinal Chemistry (2020), 63(3), 1068-1083.

1.1 **Reagents:** Potassium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Acetonitrile, Water; 45 min, 150 °C

Experimental Protocols

Scheme 19 (1 Reaction)

Steps: 1 Yield: 63%



Suppliers (6)

Suppliers (81)

31-177-CAS-23686374

Steps: 1 Yield: 63%

[1,2,4]Triazolo[1,5-a]pyrimidine Phosphodiesterase 2A Inhibitors: Structure and Free-Energy Perturbation-Guided Exploration

By: Tresadern, Gary; et al

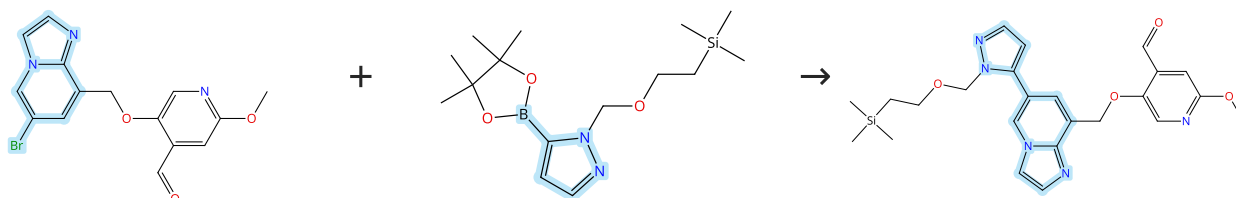
Journal of Medicinal Chemistry (2020), 63(21), 12887-12910.

1.1 **Reagents:** Sodium carbonate
Catalysts: [1,3-Bis[2,6-bis(1-methylethyl)phenyl]-2-imidazolide inylidene]chloro(η^3 -2-propen-1-yl)palladium
Solvents: Ethanol, 1,4-Dioxane, Water; 2 h, 120 °C

Experimental Protocols

Scheme 20 (1 Reaction)

Steps: 1 Yield: 63%



Supplier (1)

Suppliers (56)

31-179-CAS-17004553

Steps: 1 Yield: 63%

Discovery of GBT440, an Orally Bioavailable R-State Stabilizer of Sick Cell Hemoglobin

By: Metcalf, Brian; et al

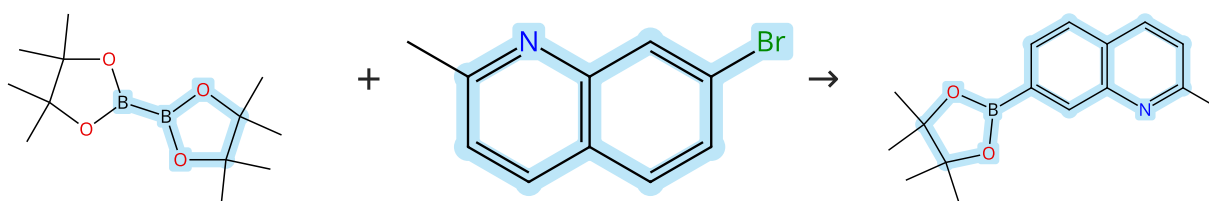
ACS Medicinal Chemistry Letters (2017), 8(3), 321-326.

1.1 **Reagents:** Potassium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Dimethylformamide; 30 min, 125 °C

Experimental Protocols

Scheme 21 (1 Reaction)

Steps: 1 Yield: 63%



Suppliers (131)

Suppliers (67)

Suppliers (33)

31-614-CAS-43159433

Steps: 1 Yield: 63%

Ruthenium-Catalyzed Carbocycle-Selective Hydrogenation of Fused Heteroarenes

By: Luo, Chenguang; et al

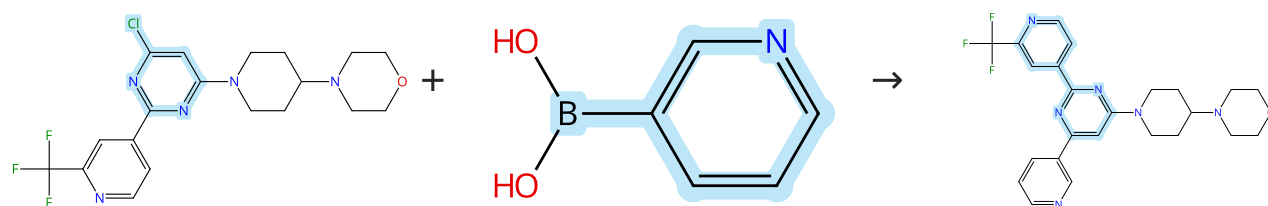
Journal of the American Chemical Society (2024), 146(51), 35043-35056.

1.1 **Reagents:** Potassium acetate
Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium
Solvents: 1,4-Dioxane; rt → 90 °C; overnight, 90 °C

Experimental Protocols

Scheme 22 (1 Reaction)

Steps: 1 Yield: 63%



Suppliers (115)

Supplier (1)

31-179-CAS-15736964

Steps: 1 Yield: 63%

Trisubstituted Pyrimidines as Efficacious and Fast-Acting Antimalarials

1.1 **Reagents:** Tripotassium phosphate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: 1,2-Dimethoxyethane, Water; 20 min, 120 °C

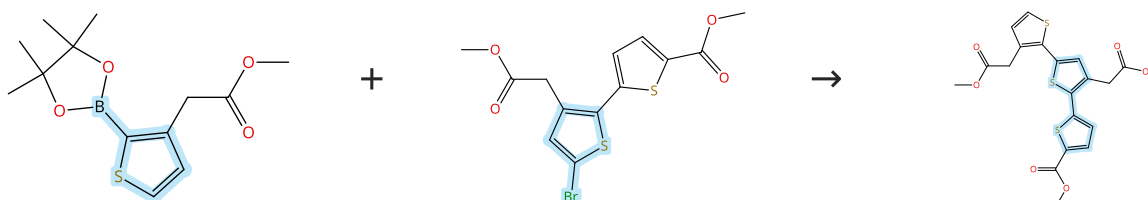
By: Norcross, Neil R.; et al

Journal of Medicinal Chemistry (2016), 59(13), 6101-6120.

Experimental Protocols

Scheme 23 (1 Reaction)

Steps: 1 Yield: 63%



Suppliers (13)

31-179-CAS-8277270

Steps: 1 Yield: 63%

Distinct Spacing Between Anionic Groups: An Essential Chemical Determinant for Achieving Thiophene-Based Ligands to Distinguish β -Amyloid or Tau Polymorphic Aggregates

1.1 **Reagents:** Potassium carbonate
Catalysts: (*SP*-4-1)-[1,3-Bis[2,6-bis(1-methylethyl)phenyl]-1,3-dihydro-2*H*-imidazol-2-ylidene]dichloro(3-chloropyridine- κ N) palladium
Solvents: Methanol, 1,4-Dioxane; 20 min, rt $\rightarrow$ 70 °C; 70 °C $\rightarrow$ rt

By: Klingstedt, Therese; et al

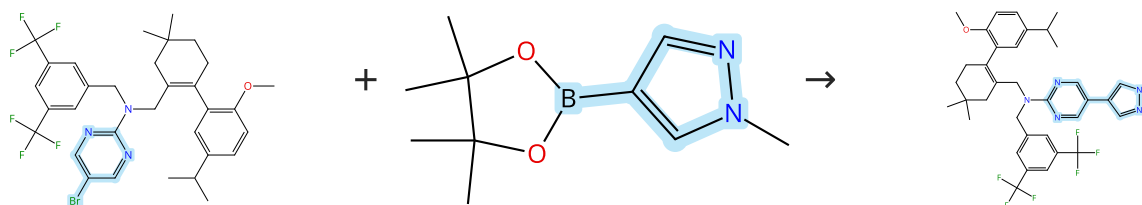
Chemistry - A European Journal (2015), 21(25), 9072-9082.

1.2 **Reagents:** Hydrochloric acid
Solvents: Water; pH 4, rt

Experimental Protocols

Scheme 24 (1 Reaction)

Steps: 1 Yield: 63%



Suppliers (126)

31-179-CAS-16113769

Steps: 1 Yield: 63%

Design, synthesis and biological evaluation of novel cholesterol ester transfer protein inhibitors bearing a cycloalkene scaffold

1.1 **Reagents:** Sodium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: 1,2-Dimethoxyethane, Water; 12 h, 90 °C

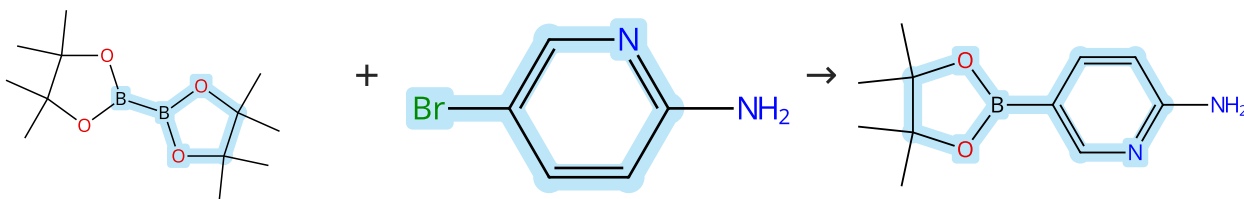
By: Liu, Chunchi; et al

European Journal of Medicinal Chemistry (2016), 123, 419-430.

Experimental Protocols

Scheme 25 (1 Reaction)

Steps: 1 Yield: 62%



Suppliers (131)

Suppliers (102)

Suppliers (93)

31-614-CAS-30485496

Steps: 1 Yield: 62%

Visible Light-Induced Borylation of C-O, C-N, and C-X Bonds

1.1 **Reagents:** Cesium carbonate
Catalysts: Phenothiazine
Solvents: Acetonitrile; 48 h

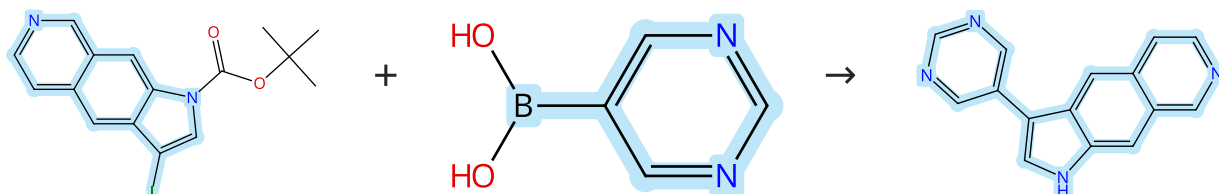
By: Jin, Shengfei; et al

Journal of the American Chemical Society (2020), 142(3), 1603-1613.

Experimental Protocols

Scheme 26 (1 Reaction)

Steps: 1 Yield: 62%



Suppliers (113)

31-614-CAS-39485400

Steps: 1 Yield: 62%

Synthesis and biological evaluation of 1 H-pyrrolo[3,2-g]isoquinolines

1.1 **Reagents:** Sodium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Dimethylformamide, Water; 30 min, 120 °C

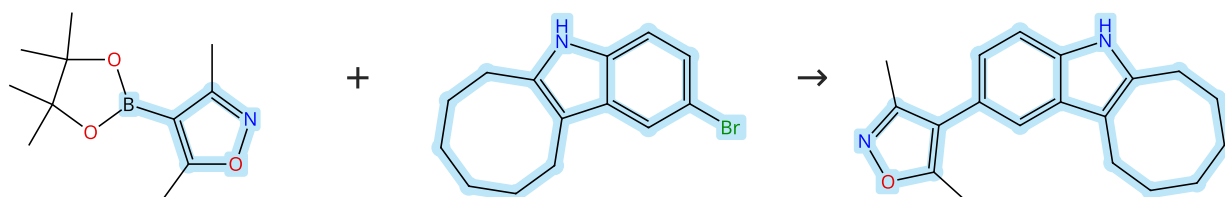
By: Defois, Mathilde; et al

Bioorganic & Medicinal Chemistry (2024), 100, 117619.

Experimental Protocols

Scheme 27 (1 Reaction)

Steps: 1 Yield: 62%



Suppliers (93)

Suppliers (26)

31-179-CAS-13141704

Steps: 1 Yield: 62%

Discovery and Optimization of Small-Molecule Ligands for the CBP/p300 Bromodomains

1.1 **Reagents:** Sodium bicarbonate
Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium
Solvents: 1,2-Dimethoxyethane, Water; 24 h, 100 °C

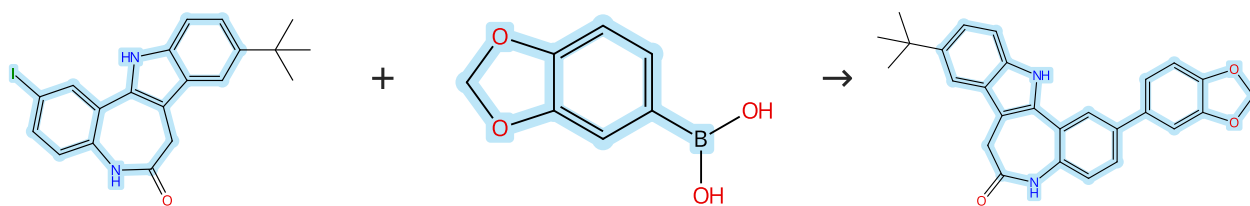
By: Hay, Duncan A.; et al

Journal of the American Chemical Society (2014), 136(26), 9308-9319.

Experimental Protocols

Scheme 28 (1 Reaction)

Steps: 1 Yield: 62%



Supplier (1)

Suppliers (99)

Supplier (1)

31-179-CAS-8588199

Steps: 1 Yield: 62%

2-Arylpallones are selective antitrypanosomal agents

1.1 **Reagents:** Sodium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: 1-Propanol, Water; 5 min, rt → 150 °C; 20 min, 250 psi, 150 °C

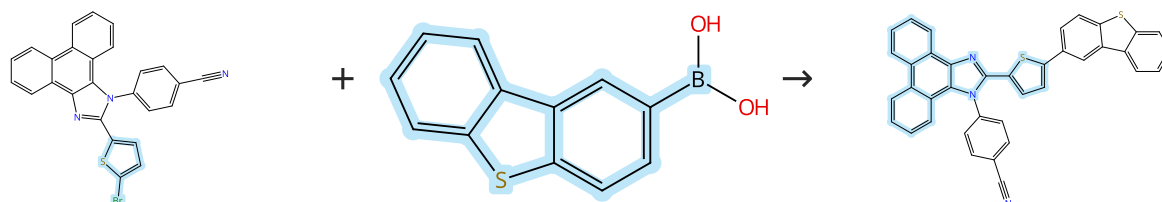
By: Ryczak, Jasmin; et al

European Journal of Medicinal Chemistry (2013), 64, 396-400.

Experimental Protocols

Scheme 29 (1 Reaction)

Steps: 1 Yield: 62%



Suppliers (92)

31-614-CAS-44094743

Steps: 1 Yield: 62%

Exploring Single Atom Substitution In Phenanthro[9,10-d] Imidazole -Based D-π-A Architectures with Fluorene and its Heteroanalogues for Non-Volatile Resistive WORM Memory Device Applications

1.1 **Reagents:** Sodium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Tetrahydrofuran, Water; 12 h, 70 °C

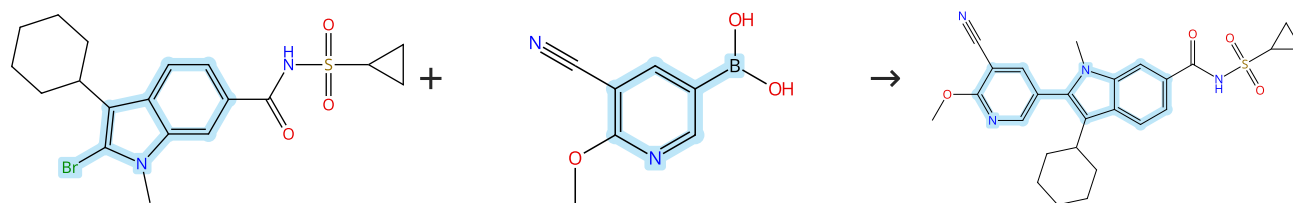
By: Akshaya, Madanan; et al

Chemistry - A European Journal (2025), 31(12), e202404337.

Experimental Protocols

Scheme 30 (1 Reaction)

Steps: 1 Yield: 62%



Suppliers (11)

Supplier (1)

31-179-CAS-20265164

Steps: 1 Yield: 62%

Discovery of new indole-based acylsulfonamide Na_v1.7 inhibitors

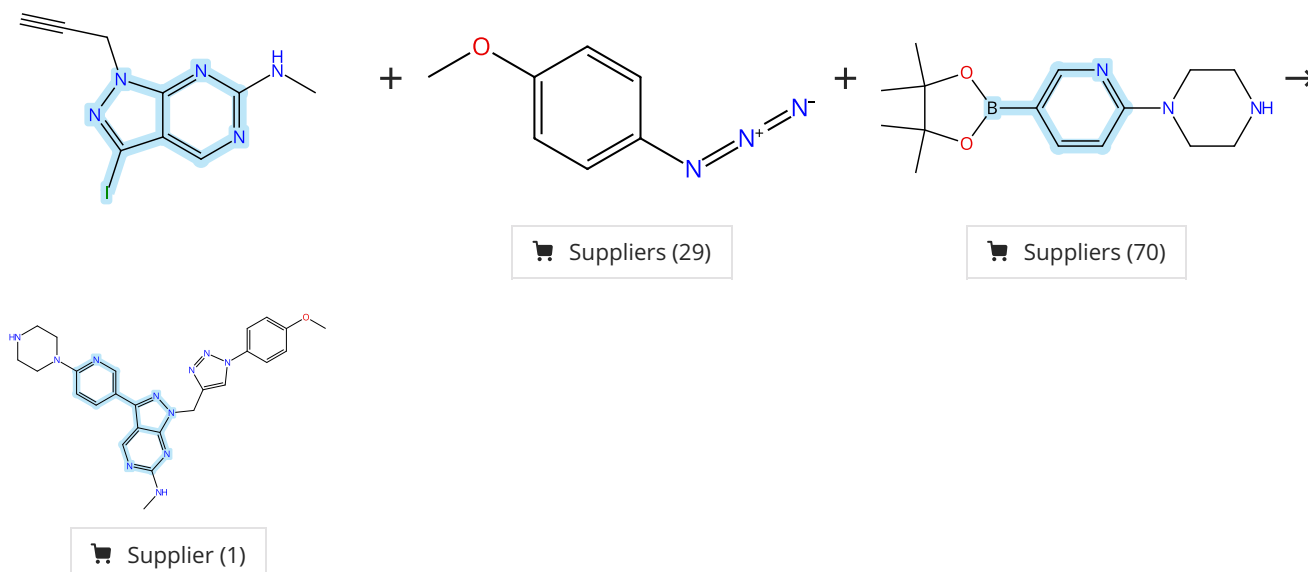
1.1 **Reagents:** Sodium carbonate
Catalysts: Dichlorobis(triphenylphosphine)palladium
Solvents: 1,4-Dioxane, Water; 30 - 75 min, 150 °C

By: Wu, Yong-Jin; et al

Bioorganic & Medicinal Chemistry Letters (2019), 29(4), 659-663.

Scheme 31 (1 Reaction)

Steps: 1 Yield: 62%



31-179-CAS-18785422

Steps: 1 Yield: 62%

Development of Potent Inhibitors of Receptor Tyrosine Kinases by Ligand-Based Drug Design and Target-Biased Phenotypic Screening

By: Myers, Samuel H.; et al

Journal of Medicinal Chemistry (2018), 61(5), 2104-2110.

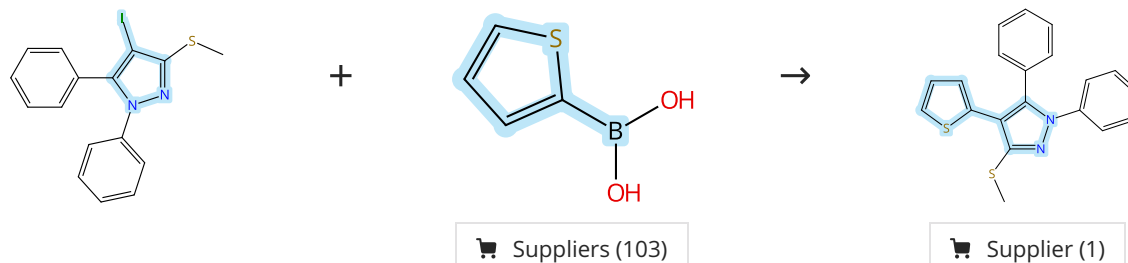
1.1 **Catalysts:** Triethylamine, Sodium ascorbate, Cuprous iodide
Solvents: Dichloromethane, 1,4-Dioxane, Water; 24 h, rt

1.2 **Reagents:** Potassium carbonate
Catalysts: Triphenylphosphine, Palladium diacetate
Solvents: 1,4-Dioxane, Water; 1 h, 120 °C

Experimental Protocols

Scheme 32 (1 Reaction)

Steps: 1 Yield: 62%



31-179-CAS-5028296

Steps: 1 Yield: 62%

Regioselective rapid analog synthesis of 1,3-(or 1,5-)diphenyl-4-aryl/heteroaryl-5-(or 3-)(methylthio)pyrazoles via Suzuki cross-coupling

By: Khan, Taukeer A.; et al

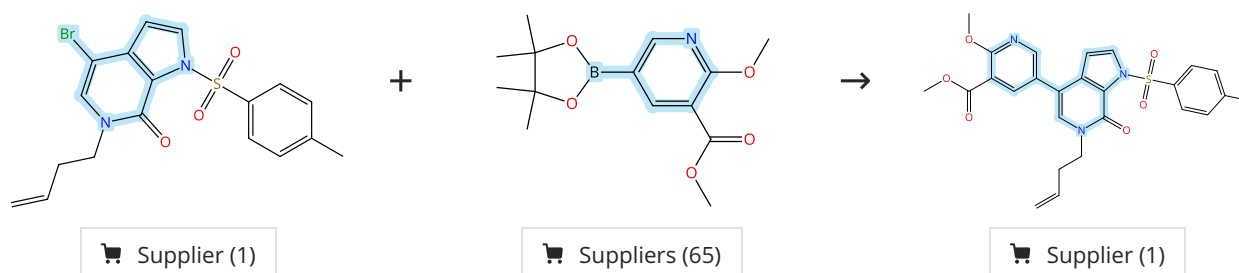
Tetrahedron (2011), 67(16), 2961-2968.

1.1 **Catalysts:** Tetrakis(triphenylphosphine)palladium
Solvents: Ethanol, Toluene, Water; 10 - 12 h, 90 - 120 °C

Experimental Protocols

Scheme 33 (1 Reaction)

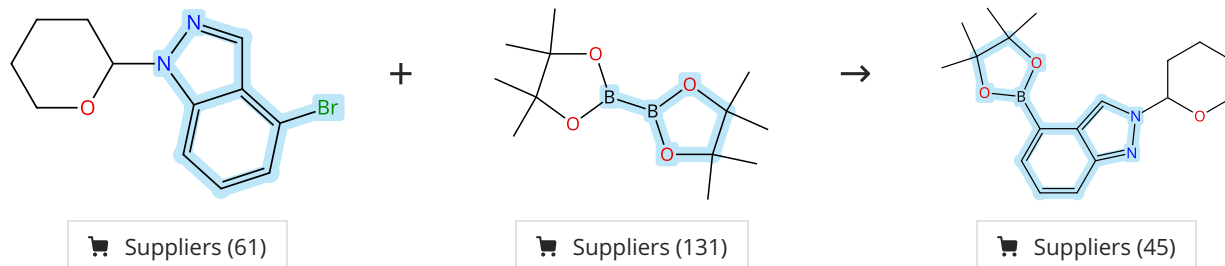
Steps: 1 Yield: 62%



31-179-CAS-19577492 Steps: 1 Yield: 62% 1.1 Reagents: Cesium carbonate Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: 1,4-Dioxane, Water; 2 h, 110 °C; cooled Experimental Protocols	GNE-371, a Potent and Selective Chemical Probe for the Second Bromodomains of Human Transcription-Initiation-Factor TFIID Subunit 1 and Transcription-Initiation-Factor TFIID Subunit 1-like By: Wang, Shumei; et al Journal of Medicinal Chemistry (2018), 61(20), 9301-9315.
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Scheme 34 (1 Reaction)

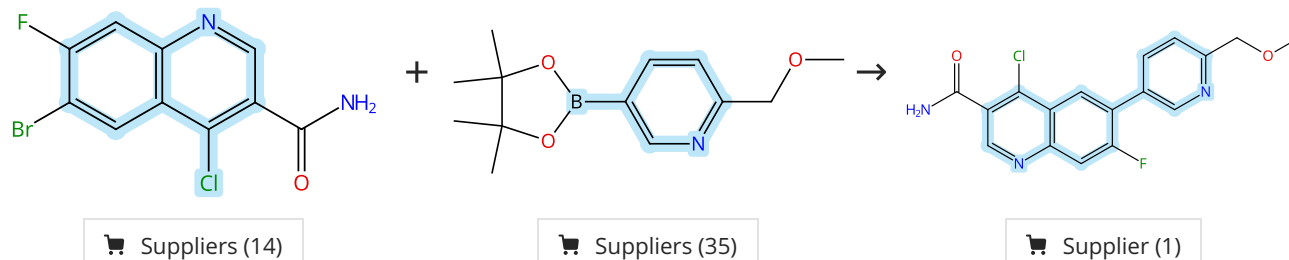
Steps: 1 Yield: 62%



31-614-CAS-26778898 Steps: 1 Yield: 62% 1.1 Reagents: Potassium acetate Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: 1,4-Dioxane; 16 h, 100 °C Experimental Protocols	Discovery of Novel and Orally Bioavailable Inhibitors of PI3 Kinase Based on Indazole Substituted Morpholino-Triazines By: Dugar, Sundeep; et al ACS Medicinal Chemistry Letters (2015), 6(12), 1190-1194.
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Scheme 35 (1 Reaction)

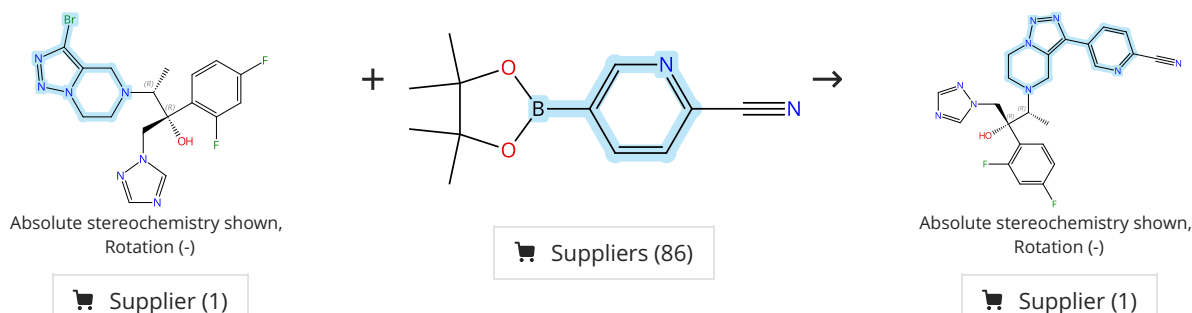
Steps: 1 Yield: 62%



31-179-CAS-15928086 Steps: 1 Yield: 62% 1.1 Reagents: Cesium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: 1,4-Dioxane, Water; 20 °C; 5 h, 100 °C Experimental Protocols	Discovery of Novel 3-Quinoline Carboxamides as Potent, Selective, and Orally Bioavailable Inhibitors of Ataxia Telangiectasia Mutated (ATM) Kinase By: Degorce, Sebastien L.; et al Journal of Medicinal Chemistry (2016), 59(13), 6281-6292.
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Scheme 36 (1 Reaction)

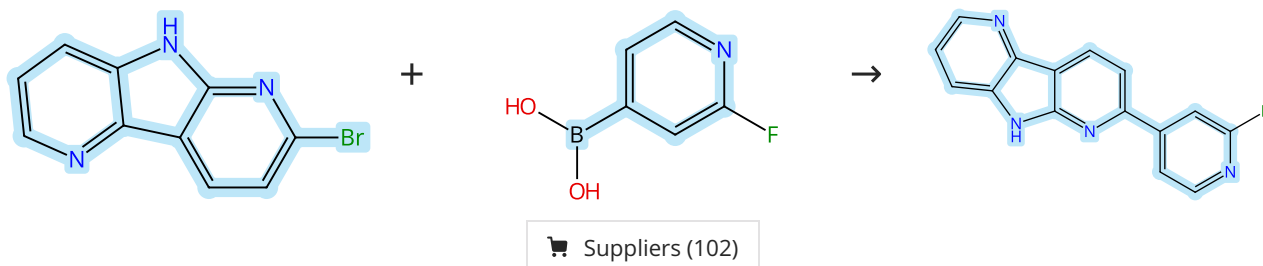
Steps: 1 Yield: 62%



31-179-CAS-5627534	Steps: 1 Yield: 62%	Design, Synthesis, and Structure-Activity Relationship Studies of Novel Fused Heterocycles-Linked Triazoles with Good Activity and Water Solubility
1.1 Reagents: Cesium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: 1,4-Dioxane, Water; 12 h, 80 °C		By: Cao, Xufeng; et al
Experimental Protocols		Journal of Medicinal Chemistry (2014), 57(9), 3687-3706.

Scheme 37 (1 Reaction)

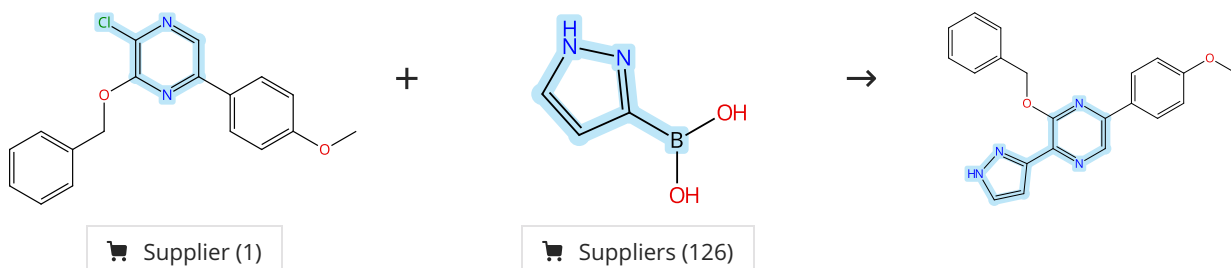
Steps: 1 Yield: 62%



31-614-CAS-32826614	Steps: 1 Yield: 62%	Structure-activity relationship around PI-2620 highlights the importance of the nitrogen atom position in the tricyclic core
1.1 Reagents: Sodium carbonate Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: Dichloromethane, Dimethylacetamide, Water; 1 h, 150 °C		By: Kroth, Heiko; et al
Experimental Protocols		Bioorganic & Medicinal Chemistry (2021), 52, 116528.

Scheme 38 (1 Reaction)

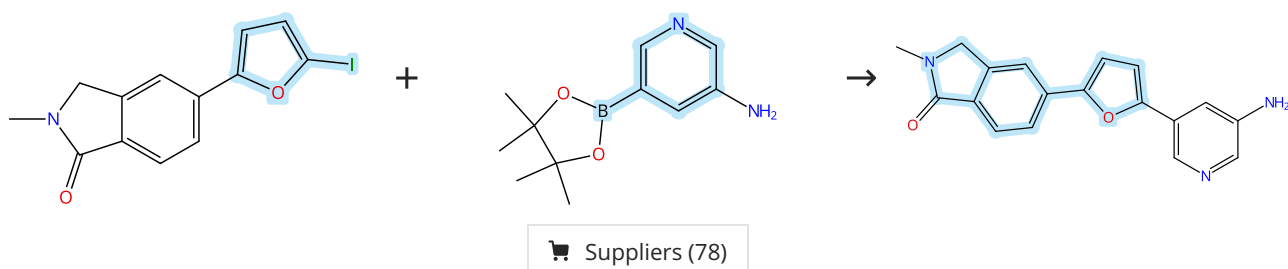
Steps: 1 Yield: 62%



31-179-CAS-1695676	Steps: 1 Yield: 62%	Design and synthesis of 2(1H)-pyrazinones as inhibitors of protein kinases
1.1 Reagents: Sodium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Acetonitrile, Water; 20 min, 150 °C		By: Caldwell, John J.; et al
Experimental Protocols		Tetrahedron (2012), 68(47), 9713-9728.

Scheme 39 (1 Reaction)

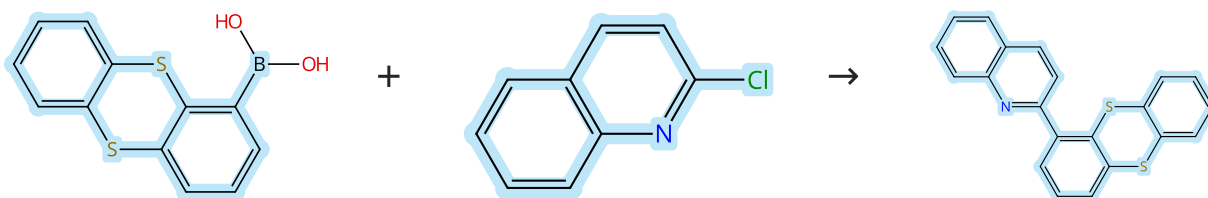
Steps: 1 Yield: 62%



31-179-CAS-17456605 Steps: 1 Yield: 62% 1.1 Reagents: Sodium carbonate Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: Ethanol, Toluene, Water; 2 h, reflux; overnight, reflux Experimental Protocols	Benzenesulphonamide inhibitors of the cytolytic protein perforin By: Spicer, Julie A.; et al Bioorganic & Medicinal Chemistry Letters (2017), 27(4), 1050-1054.
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Scheme 40 (1 Reaction)

Steps: 1 Yield: 62%



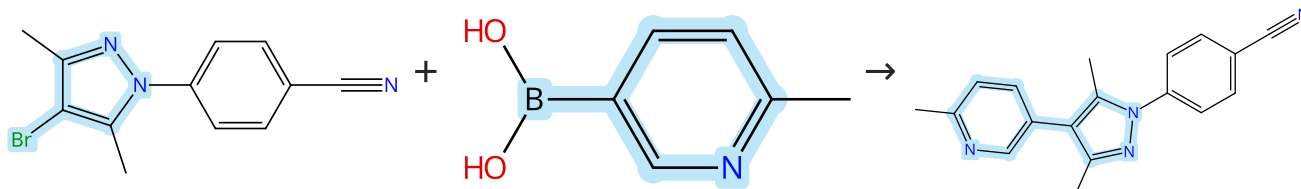
Suppliers (77)

Suppliers (95)

31-179-CAS-18375381 Steps: 1 Yield: 62% 1.1 Reagents: Potassium carbonate, Oxygen Catalysts: X-Phos, 2103932-14-3 Solvents: Water; 18 h, 100 °C Experimental Protocols	A Zwitterionic Palladium(II) Complex as a Precatalyst for Neat-Water-Mediated Cross-Coupling Reactions of Heteroaryl, Benzyl, and Aryl Acid Chlorides with Organoboron Reagents By: Ramakrishna, Visannagari; et al European Journal of Organic Chemistry (2017), 2017(48), 7238-7255.
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Scheme 41 (1 Reaction)

Steps: 1 Yield: 62%



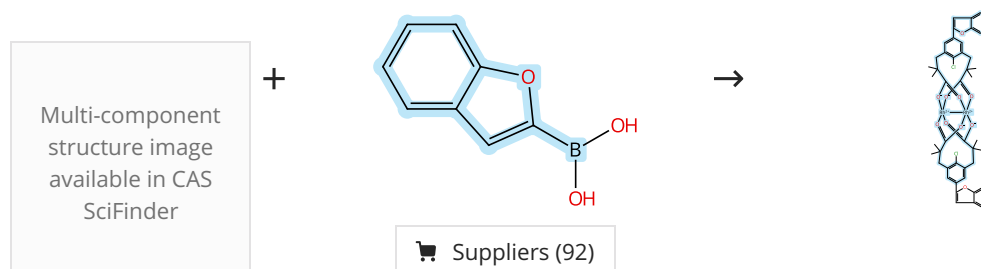
Suppliers (3)

Suppliers (81)

31-179-CAS-21844997 Steps: 1 Yield: 62% 1.1 Reagents: Sodium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Dimethylformamide, Water; 15 - 20 min, 140 °C	Identification of Isoform 2 Acid-Sensing Ion Channel Inhibitors as Tool Compounds for Target Validation Studies in CNS By: Bencheva, Leda Ivanova; et al ACS Medicinal Chemistry Letters (2019), 10(4), 627-632.
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Scheme 42 (1 Reaction)

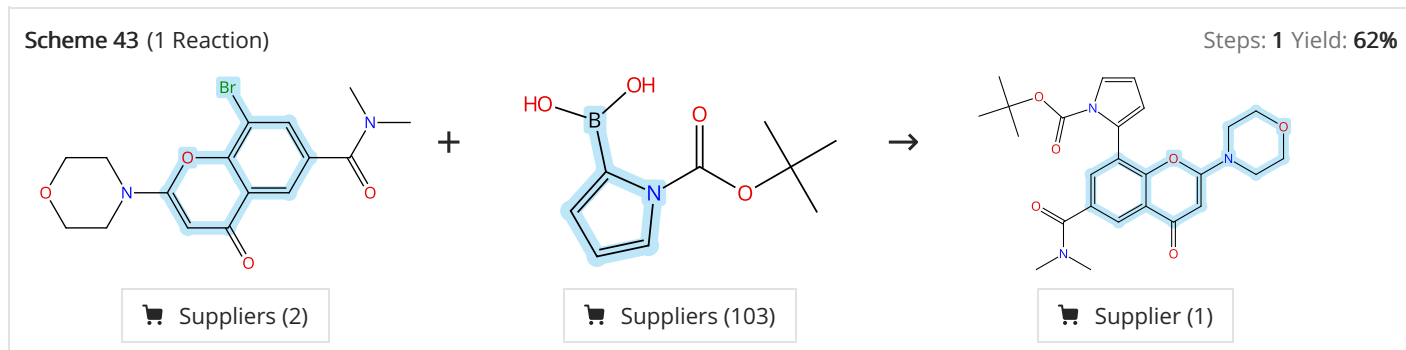
Steps: 1 Yield: 62%



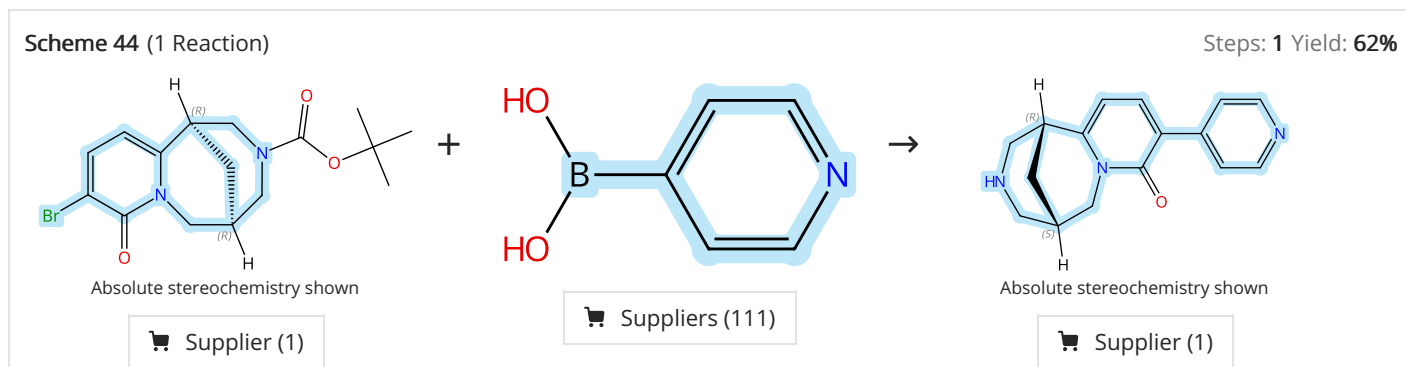
Multi-component
structure image
available in CAS
SciFinder

Suppliers (92)

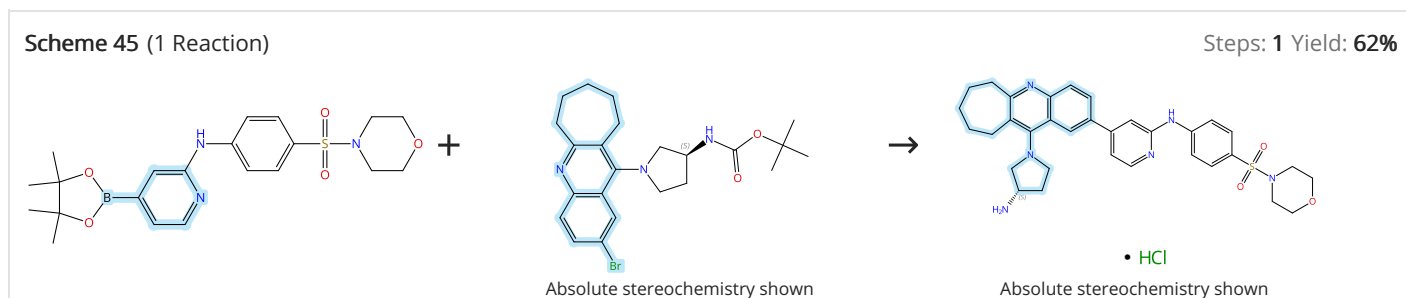
31-614-CAS-40725850 Steps: 1 Yield: 62% 1.1 Reagents: Tripotassium phosphate Catalysts: [2'-(Amino-κM)[1,1'-biphenyl]-2-yl-κC][dicyclohexyl (2',6'-dimethoxy[1,1'-biphenyl]-2-yl)phosphine-κP](methanesulfonato-κO)palladium Solvents: Toluene; rt → 110 °C; 12 h, 110 °C Experimental Protocols	Locking the Conformation of a Paddlewheel Rhodium Complex: Design, Synthesis, and Applications in Catalytic Nitrene Transfers By: Tang, Xinxin; et al Angewandte Chemie, International Edition (2023), 62(51), e202311027.
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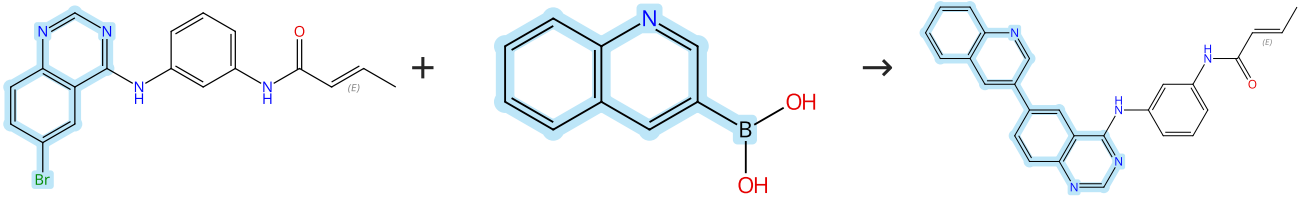
31-179-CAS-17228519 Steps: 1 Yield: 62% 1.1 Reagents: Sodium carbonate Catalysts: Dichlorobis(triphenylphosphine)palladium Solvents: 1,2-Dimethoxyethane, Water; 8 h, 80 °C	Discovery of a series of 8-(1-phenylpyrrolidin-2-yl)-6-carboxamide-2-morpholino-4H-chromen-4-one as PI3Kβ/δ inhibitors for the treatment of PTEN-deficient tumors By: Barlaam, Bernard; et al Bioorganic & Medicinal Chemistry Letters (2017), 27(9), 1949-1954.
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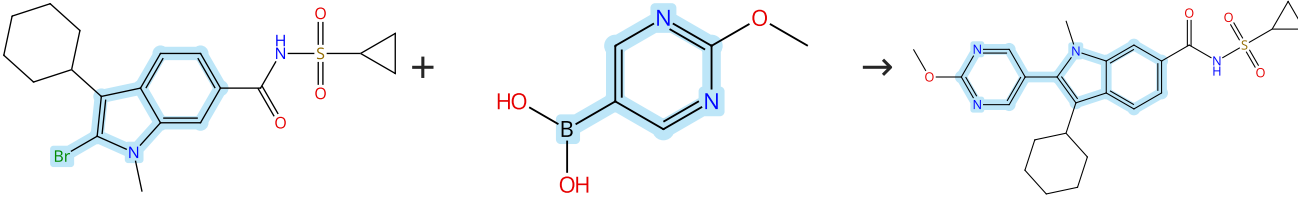
31-179-CAS-2111294 Steps: 1 Yield: 62% 1.1 Reagents: Tripotassium phosphate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: 1,2-Dimethoxyethane, Water; rt → 80 °C; 90 min, 80 °C 1.2 Solvents: Water; rt → 150 °C; 30 min, 150 °C Experimental Protocols	The 3,7-diazabicyclo[3.3.1]nonane scaffold for subtype selective nicotinic acetylcholine receptor ligands. Part 2: Carboxamide derivatives with different spacer motifs By: Eibl, Christoph; et al Bioorganic & Medicinal Chemistry (2013), 21(23), 7309-7329.
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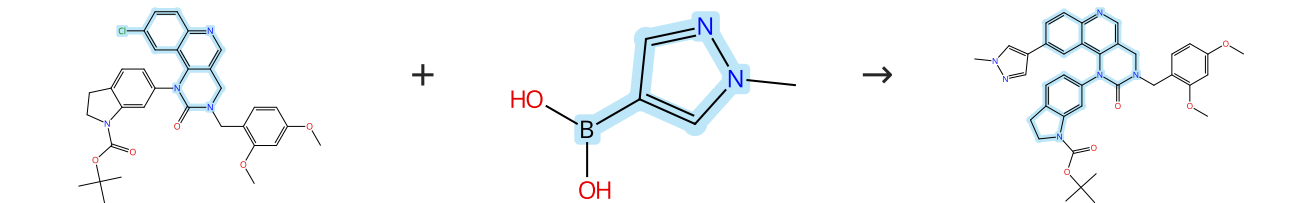
31-614-CAS-34602981 Steps: 1 Yield: 62% 1.1 Reagents: Cesium carbonate Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: 1,4-Dioxane, Water; 12 h, 95 °C 1.2 Reagents: Hydrochloric acid Solvents: Ethyl acetate; 1 h, rt Experimental Protocols	Development and structure-activity relationship of tacrine derivatives as highly potent CDK2/9 inhibitors for the treatment of cancer By: Wu, Limeng; et al European Journal of Medicinal Chemistry (2022), 242, 114701.
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Scheme 46 (1 Reaction)  Double bond geometry shown  Supplier (1)  Suppliers (107) Double bond geometry shown  Supplier (1)	Steps: 1 Yield: 62%
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31-179-CAS-22765040 Steps: 1 Yield: 62% 1.1 Reagents: Potassium carbonate Catalysts: Dichlorobis(triphenylphosphine)palladium Solvents: Dimethylformamide; 8 - 12 h, 100 °C Experimental Protocols	Discovery of novel quinazoline derivatives as potent PI3Kδ inhibitors with high selectivity By: Teng, Yu; et al European Journal of Medicinal Chemistry (2020), 208, 112865.
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Scheme 47 (1 Reaction)   Suppliers (104)  Supplier (1)	Steps: 1 Yield: 62%
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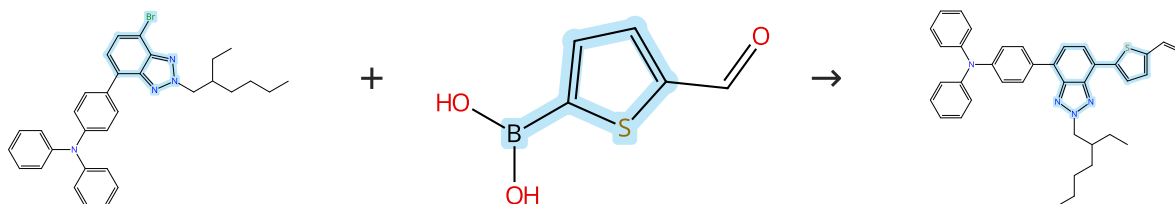
31-179-CAS-20265165 Steps: 1 Yield: 62% 1.1 Reagents: Sodium carbonate Catalysts: Dichlorobis(triphenylphosphine)palladium Solvents: 1,4-Dioxane, Water; 30 - 75 min, 150 °C	Discovery of new indole-based acylsulfonamide Na_v1.7 inhibitors By: Wu, Yong-Jin; et al Bioorganic & Medicinal Chemistry Letters (2019), 29(4), 659-663.
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Scheme 48 (1 Reaction)   Suppliers (77)	Steps: 1 Yield: 62%
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31-179-CAS-17828813 Steps: 1 Yield: 62% 1.1 Reagents: Sodium carbonate Catalysts: Dichlorobis(triphenylphosphine)palladium, Bis(1,1-dimethylethyl)[2',4',6'-tris(1-methylethyl)[1,1'-biphenyl]-2-yl]phosphine Solvents: 1,4-Dioxane, Water; overnight, 80 °C Experimental Protocols	Structure-Activity Relationship study of QL47-a Broad-spectrum Antiviral agent By: Liang, Yanke; et al ACS Medicinal Chemistry Letters (2017), 8(3), 344-349.
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Scheme 49 (1 Reaction)

Steps: 1 Yield: 62%

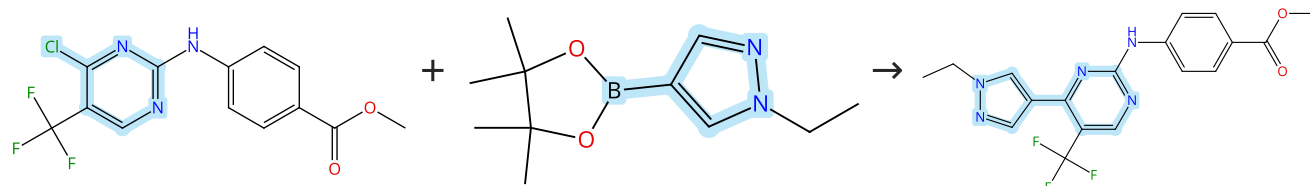


Suppliers (88)

31-179-CAS-2910862 Steps: 1 Yield: 62% 1.1 Reagents: Potassium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Tetrahydrofuran, Water; 30 min, rt → 55 °C 1.2 Solvents: Tetrahydrofuran; 12 h, reflux	Benzotriazole-Bridged Sensitizers Containing a Furan Moiety for Dye-Sensitized Solar Cells with High Open-Circuit Voltage Performance By: Mao, Jiangyi; et al Chemistry - An Asian Journal (2012), 7(5), 982-991.
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Scheme 50 (1 Reaction)

Steps: 1 Yield: 62%

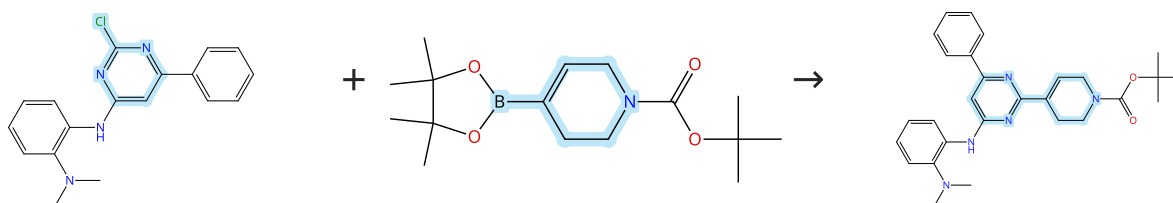


Suppliers (91)

31-179-CAS-1096282 Steps: 1 Yield: 62% 1.1 Reagents: Sodium carbonate Catalysts: Dichlorobis(triphenylphosphine)palladium Solvents: 1,4-Dioxane, Water; 14 h, reflux Experimental Protocols	Synthesis and pharmacological evaluation of trifluoromethyl containing 4-(2-pyrimidinylamino)benzamides as Hedgehog signaling pathway inhibitors By: Xin, Minhang; et al Bioorganic & Medicinal Chemistry (2016), 24(5), 1079-1088.
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Scheme 51 (1 Reaction)

Steps: 1 Yield: 62%

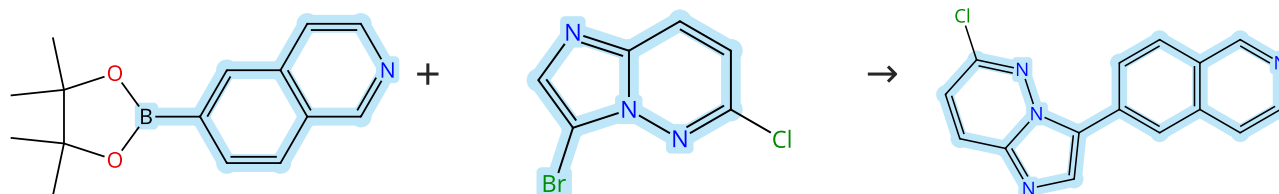


Suppliers (127)

31-614-CAS-40725253 Steps: 1 Yield: 62% 1.1 Reagents: Potassium carbonate Catalysts: Dichloro[1,1'-bis(diphenylphosphino)ferrocene]palladium(II) dichloromethane adduct Solvents: 1,4-Dioxane, Water; 2 h, 120 °C Experimental Protocols	Design, Synthesis, In Vitro and In Vivo Characterization of CD C42 GTPase Interaction Inhibitors for the Treatment of Cancer By: Brindani, Nicoletta; et al Journal of Medicinal Chemistry (2023), 66(8), 5981-6001.
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Scheme 52 (1 Reaction)

Steps: 1 Yield: 62%



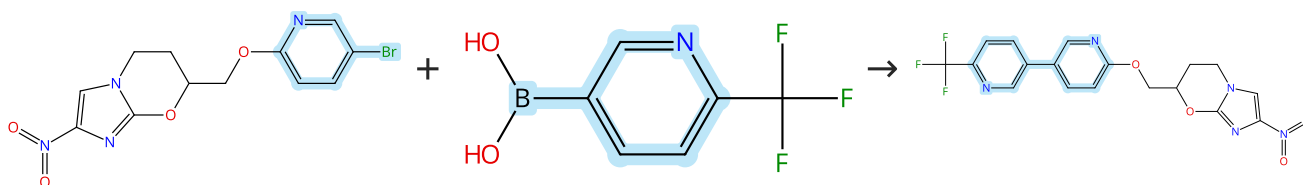
Suppliers (64)

Suppliers (81)

31-179-CAS-18882491 Steps: 1 Yield: 62% 1.1 Reagents: Potassium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: 1,2-Dimethoxyethane, Water; rt; overnight, 85 °C	Structure-based Design of Pyridone-Aminal eFT508 Targeting Dysregulated Translation by Selective Mitogen-activated Protein Kinase Interacting Kinases 1 and 2 (MNK1/2) Inhibition By: Reich, Siegfried H.; et al Journal of Medicinal Chemistry (2018), 61(8), 3516-3540.
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Scheme 53 (1 Reaction)

Steps: 1 Yield: 62%



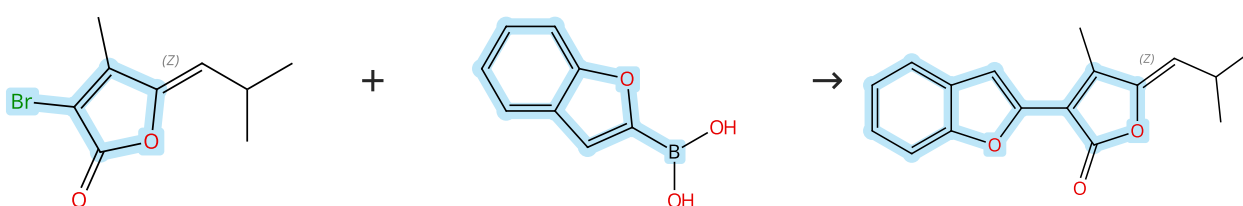
Suppliers (90)

Supplier (1)

31-179-CAS-17228834 Steps: 1 Yield: 62% 1.1 Reagents: Potassium bicarbonate Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: Dimethylformamide, Water; 4.5 h, 70 °C 1.2 Reagents: Sodium chloride Solvents: Water Experimental Protocols	7-Substituted 2-Nitro-5,6-dihydroimidazo[2,1-b][1,3]oxazines: Novel Antitubercular Agents Lead to a New Preclinical Candidate for Visceral Leishmaniasis By: Thompson, Andrew M.; et al Journal of Medicinal Chemistry (2017), 60(10), 4212-4233.
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Scheme 54 (1 Reaction)

Steps: 1 Yield: 62%



Double bond geometry shown

Suppliers (92)

Double bond geometry shown

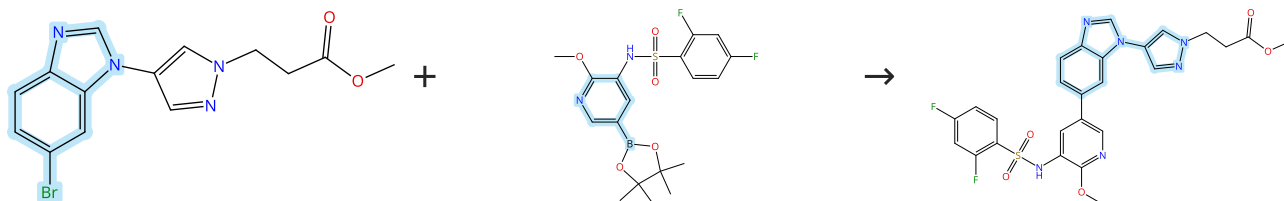
Supplier (1)

Supplier (1)

31-179-CAS-22035771	Steps: 1 Yield: 62%	Synthesis and evaluation of heterocycle structures as potential inhibitors of Mycobacterium tuberculosis UGM
1.1 Reagents: Sodium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Ethanol, Toluene, Water; 8 h, 80 °C		By: Maaliki, Carine; et al
Experimental Protocols		Bioorganic & Medicinal Chemistry (2020), 28(13), 115579.

Scheme 55 (1 Reaction)

Steps: 1 Yield: 62%

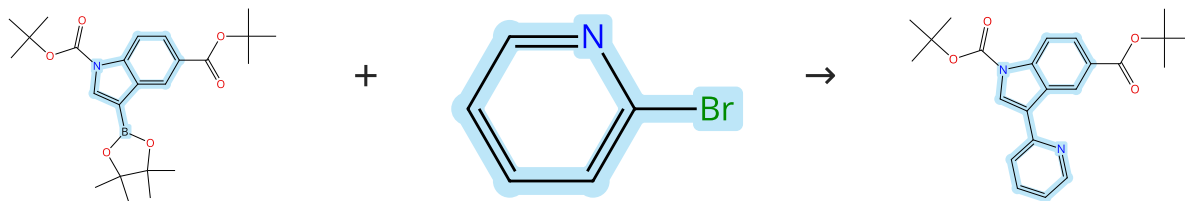


Suppliers (54)

31-179-CAS-20866905	Steps: 1 Yield: 62%	Design, synthesis and evaluation of some 1,6-disubstituted-1 H-benzo[d]imidazoles derivatives targeted PI3K as anticancer agents
1.1 Reagents: Cesium carbonate Catalysts: Dichlorobis(triphenylphosphine)palladium Solvents: Dimethylformamide, Water; 4 h, 95 °C		By: Ding, Huai-Wei; et al
Experimental Protocols		Bioorganic Chemistry (2019), 93, 103283pp..

Scheme 56 (1 Reaction)

Steps: 1 Yield: 62%



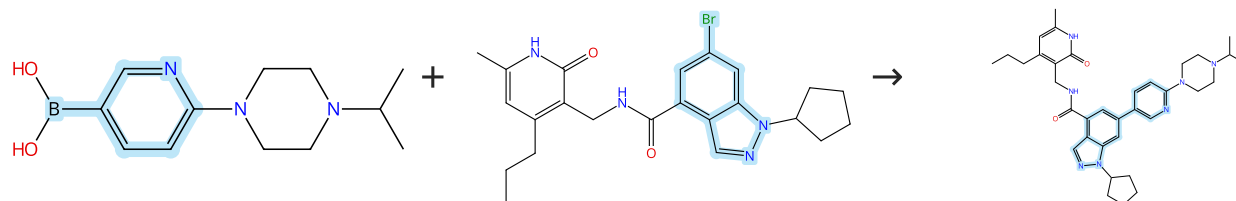
Supplier (1)

Suppliers (89)

31-179-CAS-7755717	Steps: 1 Yield: 62%	Copper-Catalyzed N-Alkynylation of N-tert-Butoxycarbonyl (B OC)-Protected Indoles
1.1 Reagents: Cesium carbonate Catalysts: Palladium diacetate, 1,1-Bis(diphenylphosphino) ferrocene Solvents: Methanol, Toluene; 8 h, 75 °C		By: Song, Jiahui; et al
Experimental Protocols		Asian Journal of Organic Chemistry (2013), 2(10), 877-881.

Scheme 57 (1 Reaction)

Steps: 1 Yield: 62%



Suppliers (4)

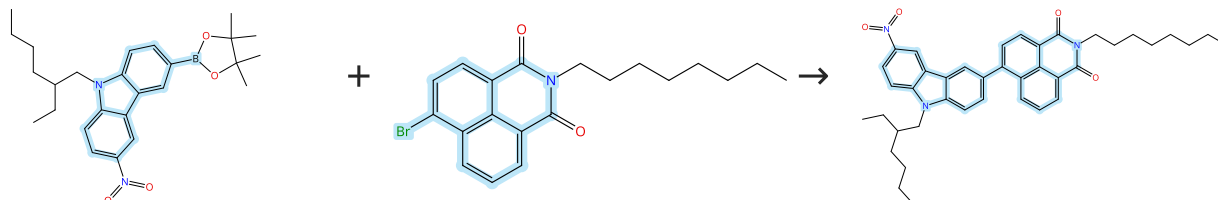
Supplier (1)

Supplier (1)

31-179-CAS-16055744	Steps: 1 Yield: 62%	Structure-Activity Relationship Studies for Enhancer of Zeste Homologue 2 (EZH2) and Enhancer of Zeste Homologue 1 (EZ H1) Inhibitors By: Yang, Xiaobao; et al Journal of Medicinal Chemistry (2016), 59(16), 7617-7633.
1.1 Reagents: Potassium acetate Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: 1,4-Dioxane, Water; overnight, rt → 80 °C		
Experimental Protocols		

Scheme 58 (1 Reaction)

Steps: 1 Yield: 62%

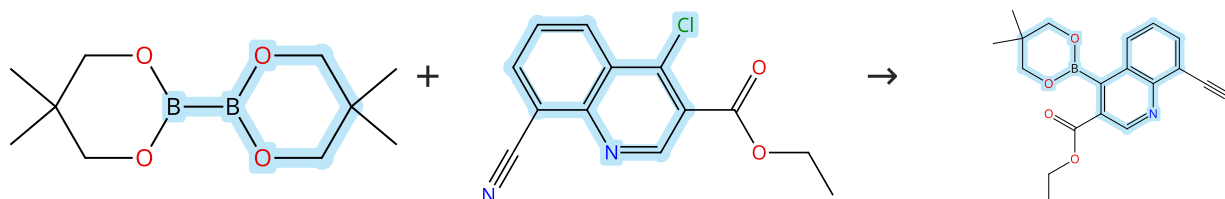


Suppliers (7)

31-179-CAS-15865213	Steps: 1 Yield: 62%	Inserting Thienyl Linkers into Conjugated Molecules for Efficient Multilevel Electronic Memory: A New Understanding of Charge-Trapping in Organic Materials By: Li, Yang; et al Chemistry - An Asian Journal (2016), 11(6), 906-914.
1.1 Reagents: Potassium carbonate Catalysts: Tri- <i>o</i> -tolylphosphine, Tris(dibenzylideneacetone) dipalladium Solvents: Toluene, Water; 24 h, 90 °C		
Experimental Protocols		

Scheme 59 (1 Reaction)

Steps: 1 Yield: 62%



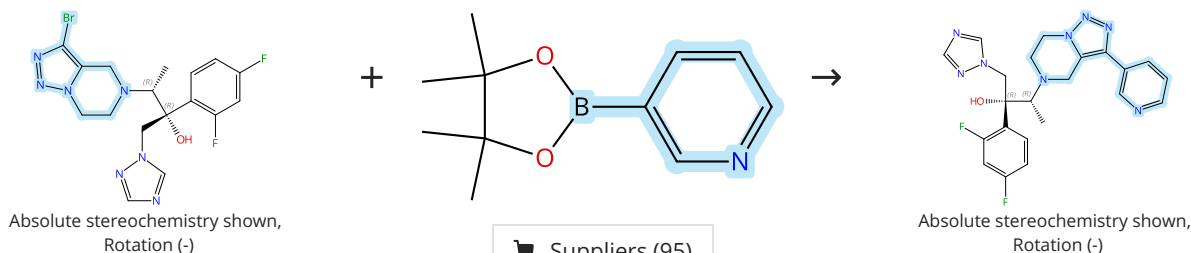
Suppliers (94)

Suppliers (51)

31-614-CAS-38098365	Steps: 1 Yield: 62%	One-pot borylation/arylation of 4-chloroquinolines and its application for the synthesis of various quinoline-based pharmacophores By: Yadav, Pratik; et al Tetrahedron Letters (2023), 131, 154780.
1.1 Reagents: Potassium acetate Catalysts: Dichlorobis(triphenylphosphine)palladium Solvents: 1,4-Dioxane; 16 h, 80 °C		
Experimental Protocols		

Scheme 60 (1 Reaction)

Steps: 1 Yield: 62%


 Absolute stereochemistry shown,
 Rotation (-)

Suppliers (95)

 Absolute stereochemistry shown,
 Rotation (-)

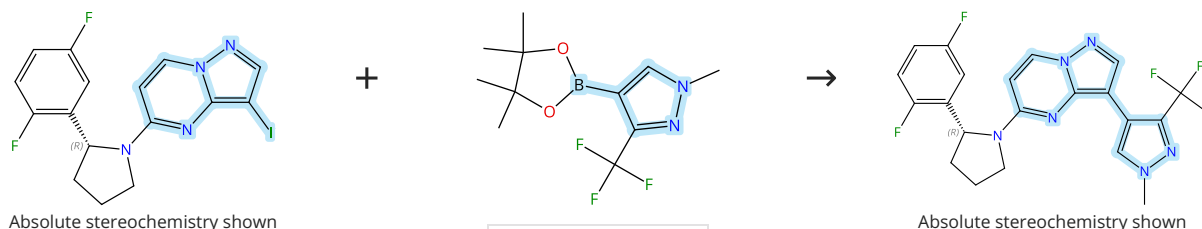
Supplier (1)

Supplier (1)

31-179-CAS-15085674	Design, Synthesis, and Structure-Activity Relationship Studies of Novel Fused Heterocycles-Linked Triazoles with Good Activity and Water Solubility
1.1 Reagents: Cesium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: 1,4-Dioxane, Water; 12 h, 80 °C	By: Cao, Xufeng; et al
Experimental Protocols	Journal of Medicinal Chemistry (2014), 57(9), 3687-3706.

Scheme 61 (1 Reaction)

Steps: 1 Yield: 62%

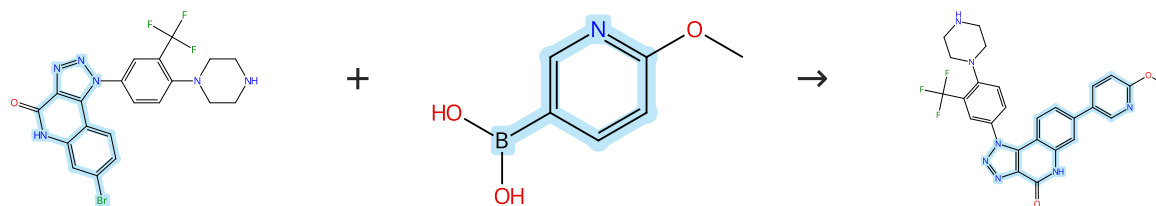


Suppliers (61)

31-614-CAS-34713696	Conformational adjustment overcomes multiple drug-resistance mutants of tropomyosin receptor kinase
1.1 Reagents: Potassium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Dimethylformamide, Water; rt; overnight, rt → 110 °C	By: Mei, Long-Can; et al
Experimental Protocols	European Journal of Medicinal Chemistry (2022), 237, 114406.

Scheme 62 (1 Reaction)

Steps: 1 Yield: 62%

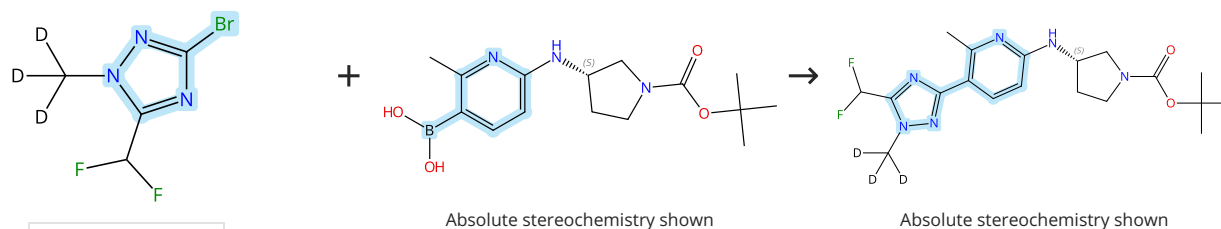


Suppliers (100)

31-614-CAS-32517464	Discovery of 8-(6-Methoxypyridin-3-yl)-1-(4-(piperazin-1-yl)-3-(trifluoromethyl)phenyl)-1,5-dihydro-4H-[1,2,3]triazolo[4,5-c]quinolin-4-one (CQ211) as a Highly Potent and Selective R1O K2 Inhibitor
1.1 Reagents: Hydrochloric acid Solvents: Ethanol, Water; overnight, 120 °C	By: Ouyang, Yifan; et al
1.2 Reagents: Cesium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Dimethylformamide, Water; overnight, 80 °C	Journal of Medicinal Chemistry (2022), 65(11), 7833-7842.
1.3 Reagents: Water	
Experimental Protocols	

Scheme 63 (1 Reaction)

Steps: 1 Yield: 62%



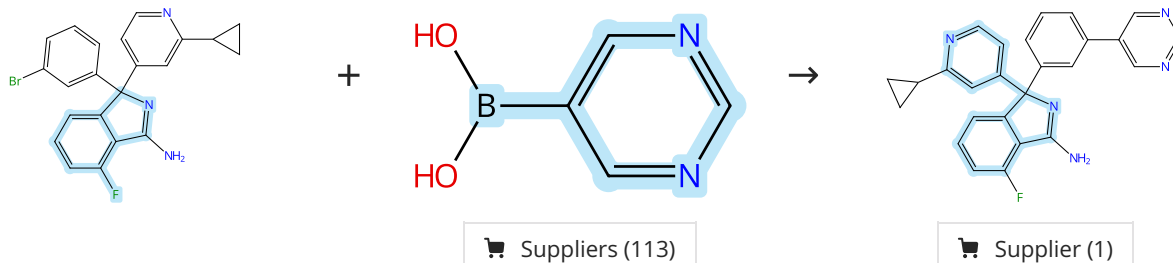
Supplier (1)

Suppliers (2)

31-614-CAS-38847478 Steps: 1 Yield: 62% 1.1 Reagents: Tripotassium phosphate Catalysts: Dichloro[1,1'-bis(diphenylphosphino)ferrocene] palladium(II) dichloromethane adduct Solvents: Tetrahydrofuran, Water; 18 h, rt → 70 °C Experimental Protocols	Discovery of the Potent and Selective MC4R Antagonist PF-07258669 for the Potential Treatment of Appetite Loss By: Garnsey, Michelle R.; et al Journal of Medicinal Chemistry (2023), 66(5), 3195-3211.
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Scheme 64 (1 Reaction)

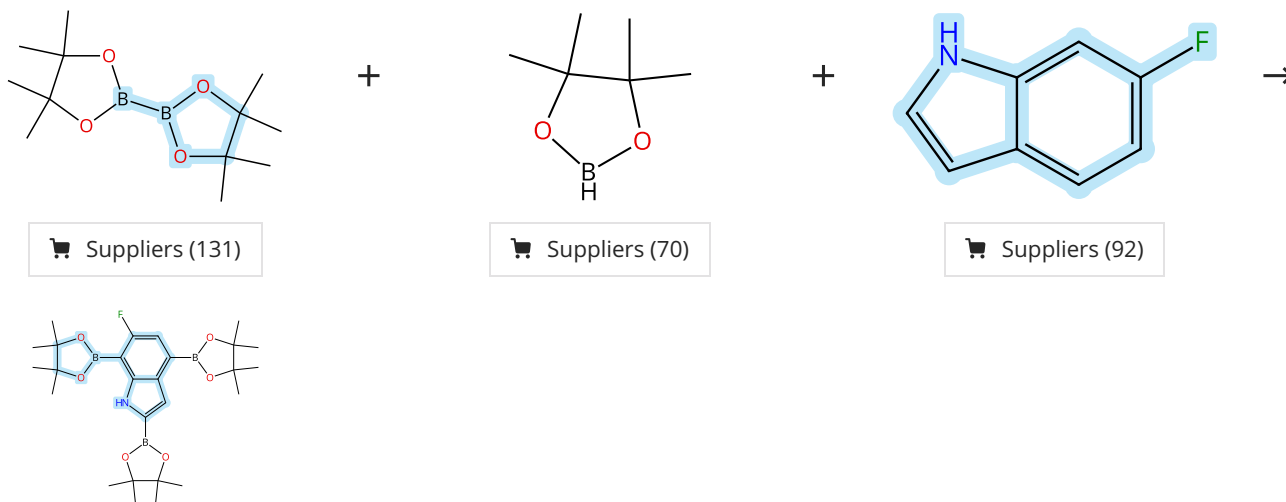
Steps: 1 Yield: 62%



31-179-CAS-6591094 Steps: 1 Yield: 62% 1.1 Reagents: Cesium carbonate Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: Ethanol, 1,2-Dimethoxyethane, Water; 20 min, 100 °C Experimental Protocols	Design and Synthesis of β-Site Amyloid Precursor Protein Cleaving Enzyme (BACE1) Inhibitors with in Vivo Brain Reduction of β-Amyloid Peptides By: Swahn, Britt-Marie; et al Journal of Medicinal Chemistry (2012), 55(21), 9346-9361.
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Scheme 65 (1 Reaction)

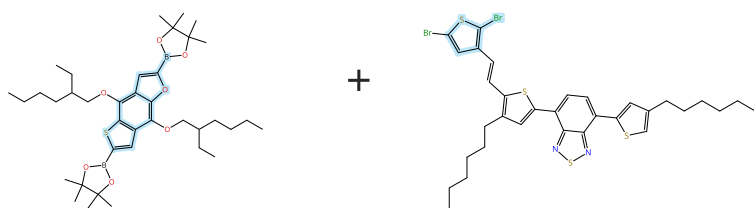
Steps: 1 Yield: 62%



31-142-CAS-11948396 Steps: 1 Yield: 62% 1.1 Catalysts: Bis[(1,2,5,6-η)-1,5-cyclooctadiene]di-μ-methoxydiiridium, 4,4'-Bis(1,1-dimethylethyl)-2,2'-bipyridine Solvents: Tetrahydrofuran; 1 min, rt; 24 h, 70 °C Experimental Protocols	Bismuth Acetate as a Catalyst for the Sequential Protodeborylation of Di- and Triborylated Indoles By: Shen, Fangyi; et al Organic Letters (2016), 18(7), 1554-1557.
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Scheme 66 (1 Reaction)

Steps: 1 Yield: 62%



Multi-component
structure image
available in CAS
SciFinder

31-614-CAS-25823015

Steps: 1 Yield: 62%

Directed Metalation Cascade To Access Highly Functionalized Thieno[2,3f]benzofuran and Exploration as Building Blocks for Organic Electronics

By: Aeschi, Yves; et al

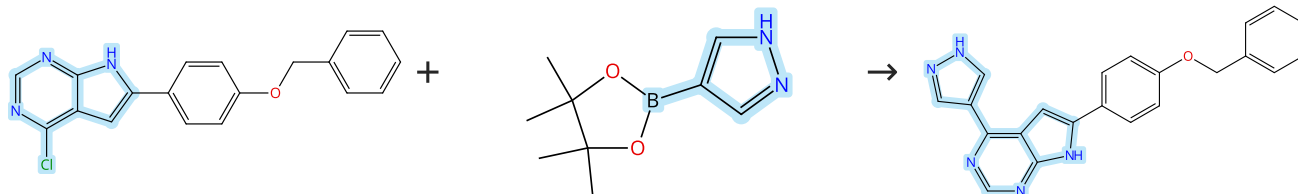
Organic Letters (2013), 15(21), 5586-5589.

- 1.1 **Reagents:** Potassium carbonate
Catalysts: Tetrabutylammonium bromide, Tetrakis(triphenyl phosphine)palladium
Solvents: Toluene; 72 h, reflux
- 1.2 **Reagents:** Phenylboronic acid; 10 h, reflux
- 1.3 **Reagents:** Iodobenzene; 6 h, heated

Experimental Protocols

Scheme 67 (1 Reaction)

Steps: 1 Yield: 62%



Suppliers (2)

Suppliers (108)

31-614-CAS-32861989

Steps: 1 Yield: 62%

Structure-Aided Design, Synthesis, and Biological Evaluation of Potent and Selective Non-Nucleoside Inhibitors Targeting Protein Arginine Methyltransferase 5

By: Rong, Deqin; et al

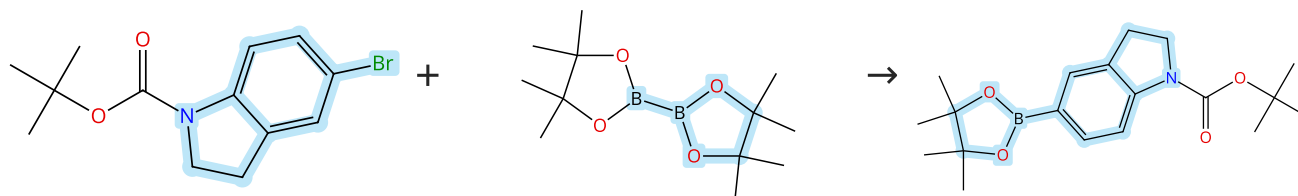
Journal of Medicinal Chemistry (2022), 65(11), 7854-7875.

- 1.1 **Reagents:** Potassium carbonate
Catalysts: Dichloro[1,1'-bis(diphenylphosphino)ferrocene] palladium(II) dichloromethane adduct
Solvents: 1,4-Dioxane, Water; 4 h, 110 °C

Experimental Protocols

Scheme 68 (1 Reaction)

Steps: 1 Yield: 61%



Suppliers (79)

Suppliers (131)

Suppliers (71)

31-614-CAS-37770725

Steps: 1 Yield: 61%

An electrophilic fragment screening for the development of small molecules targeting caspase-2

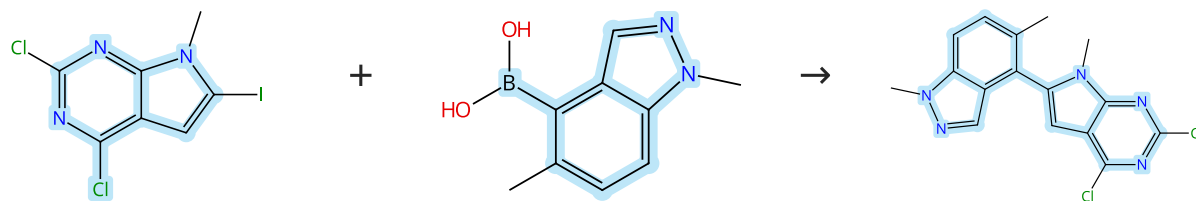
By: Cuellar, Matthew E.; et al

European Journal of Medicinal Chemistry (2023), 259, 115632.

- 1.1 **Reagents:** Potassium acetate
Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium
Solvents: Dimethyl sulfoxide; overnight, 80 °C

Scheme 69 (1 Reaction)

Steps: 1 Yield: 61%



Suppliers (42)

Suppliers (55)

31-614-CAS-34862394

Steps: 1 Yield: 61%

Identification of novel Pyrrolo[2,3-d]Pyrimidine-based KRAS G12C inhibitors with anticancer effects

1.1 Reagents: Potassium carbonate

Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium

Solvents: 1,4-Dioxane, Water; 12 h, 95 °C

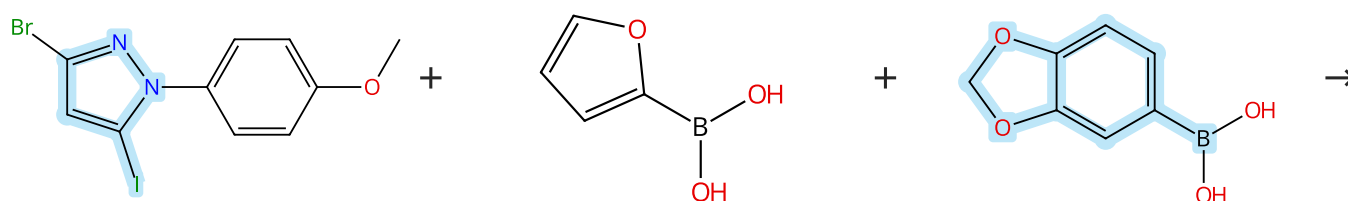
By: Song, Zhendong; et al

European Journal of Medicinal Chemistry (2023), 245(Part_1), 114907.

Experimental Protocols

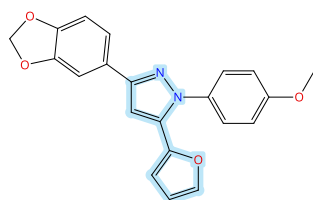
Scheme 70 (1 Reaction)

Steps: 1 Yield: 61%



Suppliers (93)

Suppliers (99)



31-179-CAS-10473009

Steps: 1 Yield: 61%

A Modular Sydnone Cycloaddition/Suzuki-Miyaura Cross-Coupling Strategy to Unsymmetrical 3,5-Bis(hetero)aromatic Pyrazoles

1.1 Reagents: Tripotassium phosphate

Catalysts: Tetrakis(triphenylphosphine)palladium

Solvents: Dimethylformamide, Water; 3 h, 50 °C

By: Delaunay, Thierry; et al

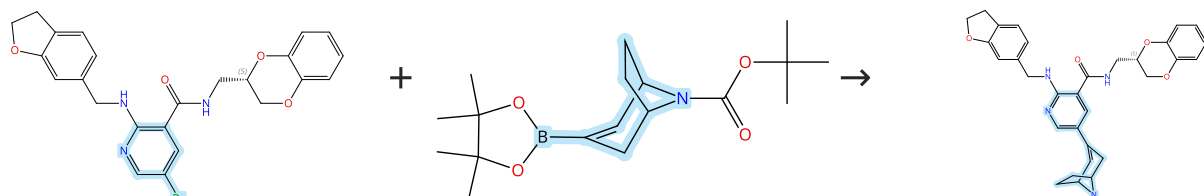
1.2 overnight, 80 °C

Organic Letters (2010), 12(15), 3328-3331.

Experimental Protocols

Scheme 71 (1 Reaction)

Steps: 1 Yield: 61%



Absolute stereochemistry shown

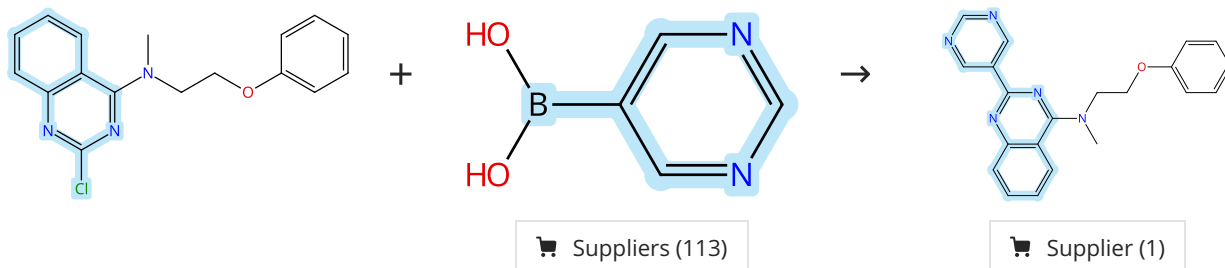
Absolute stereochemistry shown

Suppliers (70)

31-177-CAS-6499655 Steps: 1 Yield: 61% 1.1 Reagents: Tripotassium phosphate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: 1,4-Dioxane, Water; 7 h, 100 °C; 100 °C → 0 °C 1.2 Solvents: Water 1.3 Reagents: Trifluoroacetic acid Solvents: Dichloromethane; 1 h, rt	Orally active ghrelin receptor inverse agonists and their actions on a rat obesity model By: Takahashi, Bitoku; et al Bioorganic & Medicinal Chemistry (2015), 23(15), 4792-4803.
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Scheme 72 (1 Reaction)

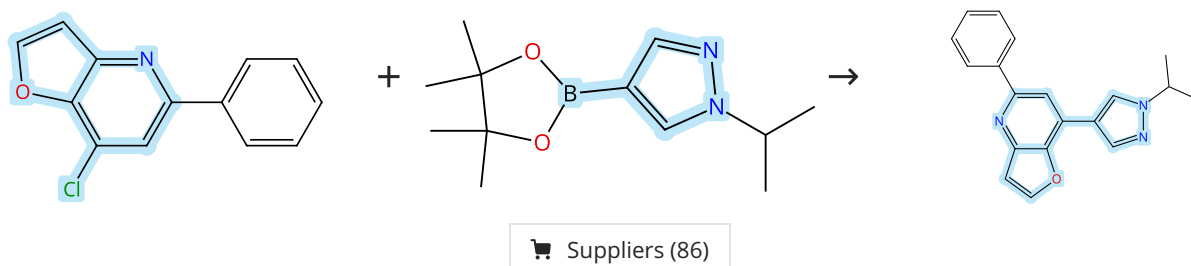
Steps: 1 Yield: 61%



31-179-CAS-20076408 Steps: 1 Yield: 61% 1.1 Reagents: Potassium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: 1,4-Dioxane, Water; 16 h, 100 °C	Conversion of Quinazoline Modulators from Inhibitors to Activators of β-Glucocerebrosidase By: Zheng, Jianbin; et al Journal of Medicinal Chemistry (2019), 62(3), 1218-1230.
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Scheme 73 (1 Reaction)

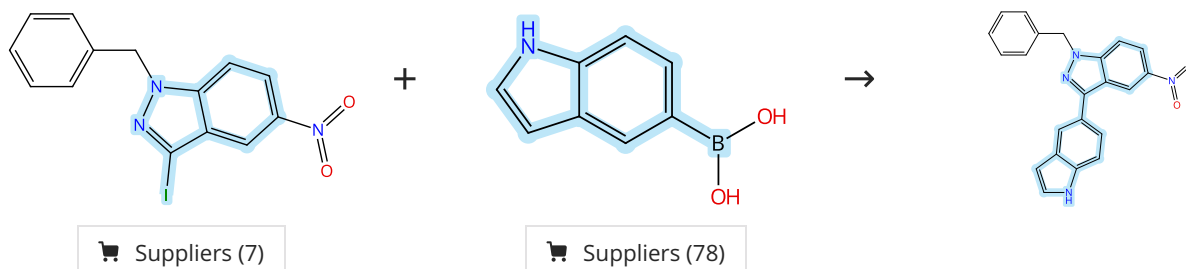
Steps: 1 Yield: 61%



31-179-CAS-19761215 Steps: 1 Yield: 61% 1.1 Reagents: Triethylamine Catalysts: Palladium diacetate, 2-Dicyclohexylphosphino-2',6'-dimethoxybiphenyl Solvents: 1-Butanol, Water; 30 min, 100 °C Experimental Protocols	Furo[3,2-b]pyridine: A Privileged Scaffold for Highly Selective Kinase Inhibitors and Effective Modulators of the Hedgehog Pathway By: Nemec, Vaclav; et al Angewandte Chemie, International Edition (2019), 58(4), 1062-1066.
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Scheme 74 (1 Reaction)

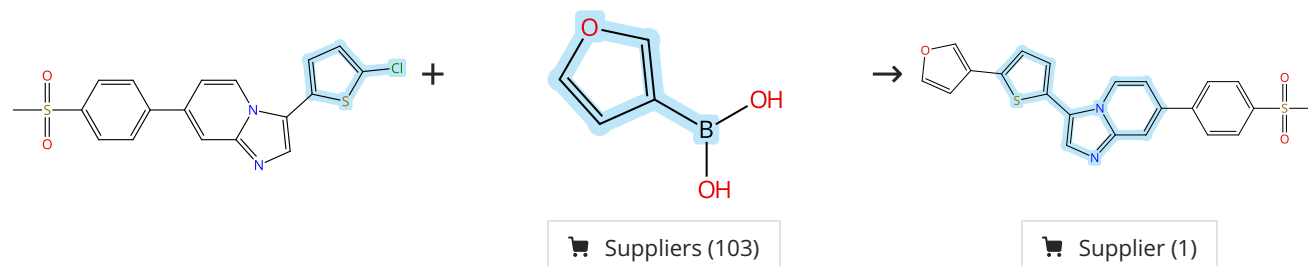
Steps: 1 Yield: 61%



31-179-CAS-20129451 Steps: 1 Yield: 61% 1.1 Reagents: Sodium carbonate Catalysts: Dichloro[1,1'-bis(diphenylphosphino)ferrocene] palladium(II) dichloromethane adduct Solvents: Tetrahydrofuran; rt → reflux; 5 h, reflux; reflux → rt	Discovery of 3-(Indol-5-yl)-indazole Derivatives as Novel Myeloid Differentiation Protein 2/Toll-like Receptor 4 Antagonists for Treatment of Acute Lung Injury By: Liu, Zhiguo; et al Journal of Medicinal Chemistry (2019), 62(11), 5453-5469.
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Scheme 75 (1 Reaction)

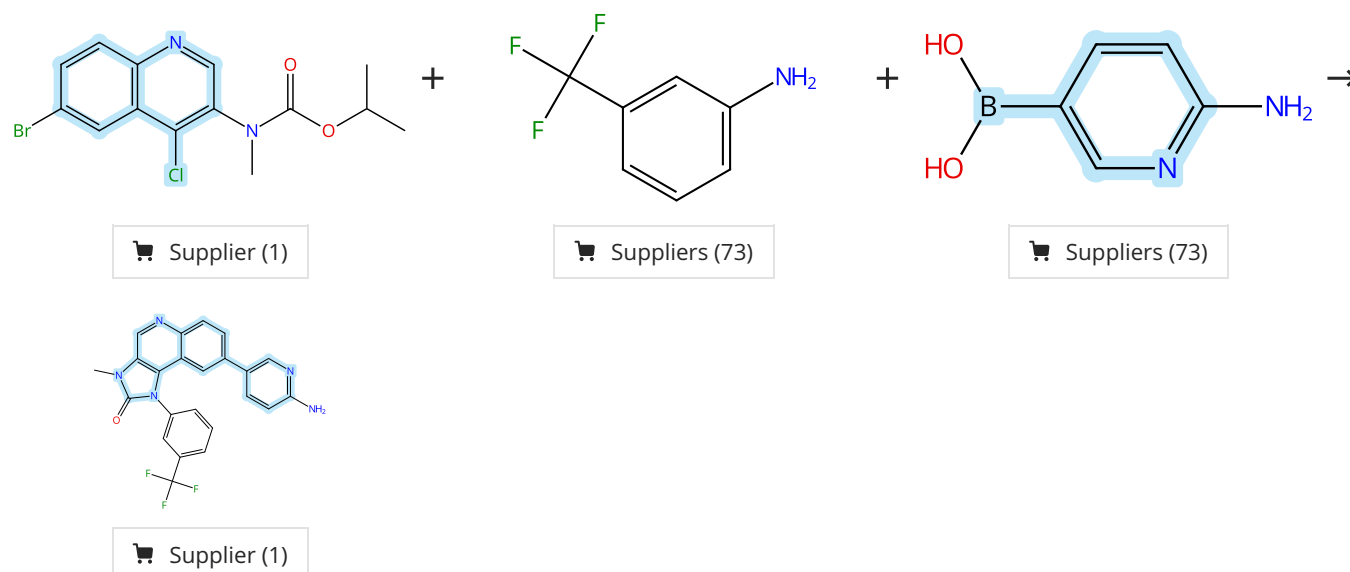
Steps: 1 Yield: 61%



31-614-CAS-32142872 Steps: 1 Yield: 61% 1.1 Reagents: Sodium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: 1,4-Dioxane, Water; 1.5 h, 130 °C Experimental Protocols	Discovery of imidazo[1,2-a]pyridine-thiophene derivatives as FLT3 and FLT3 mutants inhibitors for acute myeloid leukemia through structure-based optimization of an NEK2 inhibitor By: Zhang, Lingtian; et al European Journal of Medicinal Chemistry (2021), 225, 113776.
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Scheme 76 (1 Reaction)

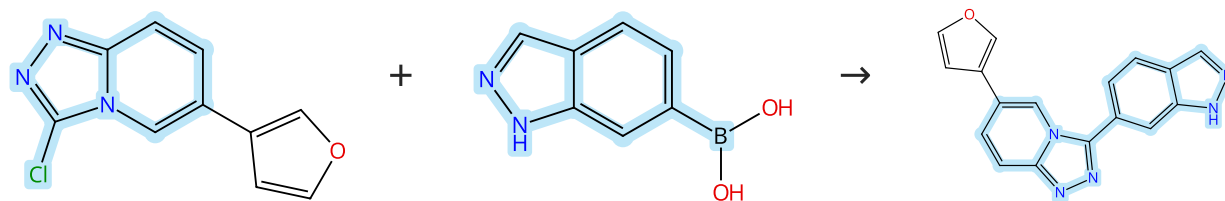
Steps: 1 Yield: 61%



31-172-CAS-18801173 Steps: 1 Yield: 61% 1.1 Reagents: Hydrochloric acid Solvents: 1,4-Dioxane, 3-Pentanol; 20 min, 230 °C; 230 °C → rt 1.2 Reagents: Tripotassium phosphate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Dimethylformamide, Water; 10 min, 150 °C; 150 °C → rt 1.3 Reagents: Sodium bicarbonate Solvents: Water; rt Experimental Protocols	Acid-Promoted Cascade Reaction of N-(4-Chloroquinolin-3-yl) carbamates with Amines: One-Pot Assembly of Imidazo[4,5-c]quinolin-2-ones By: Lu, Xiao; et al European Journal of Organic Chemistry (2018), 2018(13), 1572-1580.
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Scheme 77 (1 Reaction)

Steps: 1 Yield: 61%



Suppliers (5)

Suppliers (72)

Supplier (1)

31-614-CAS-24297835

Steps: 1 Yield: 61%

Discovery of Potent and Selective Non-Nucleotide Small Molecule Inhibitors of CD73

1.1 Reagents: Sodium carbonate
 Catalysts: Tetrakis(triphenylphosphine)palladium
 Solvents: 1,4-Dioxane, Water; 20 h, rt → 100 °C

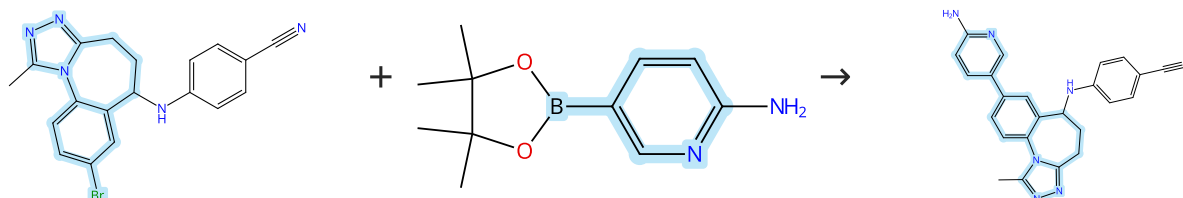
By: Beatty, Joel W.; et al

Journal of Medicinal Chemistry (2020), 63(8), 3935-3955.

Experimental Protocols

Scheme 78 (1 Reaction)

Steps: 1 Yield: 61%



Suppliers (93)

Supplier (1)

31-179-CAS-21471156

Steps: 1 Yield: 61%

Discovery, structural insight, and bioactivities of BY27 as a selective inhibitor of the second bromodomains of BET proteins

1.1 Reagents: Sodium carbonate
 Catalysts: Dichloro[1,1'-bis(diphenylphosphino)ferrocene]palladium(II) dichloromethane adduct
 Solvents: 1,2-Dimethoxyethane, Water; 12 h, 95 °C

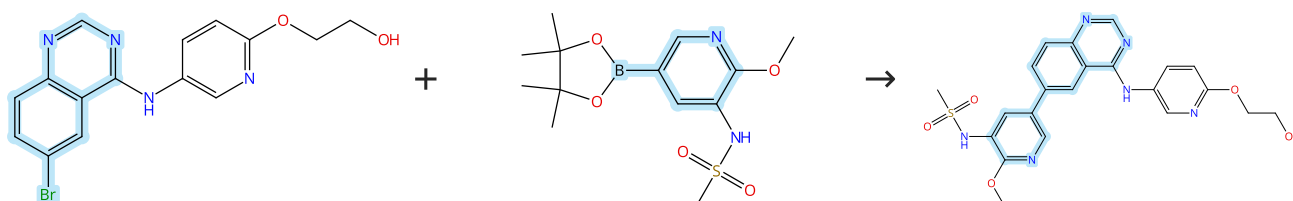
By: Chen, Deheng; et al

European Journal of Medicinal Chemistry (2019), 182, 111633.

Experimental Protocols

Scheme 79 (1 Reaction)

Steps: 1 Yield: 61%



Suppliers (60)

Supplier (1)

31-179-CAS-20385968

Steps: 1 Yield: 61%

Design, synthesis, and biological evaluation of some novel 4-aminoquinazolines as Pan-PI3K inhibitors

1.1 Reagents: Cesium carbonate
 Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium
 Solvents: Dimethylformamide; 4 h, 90 °C

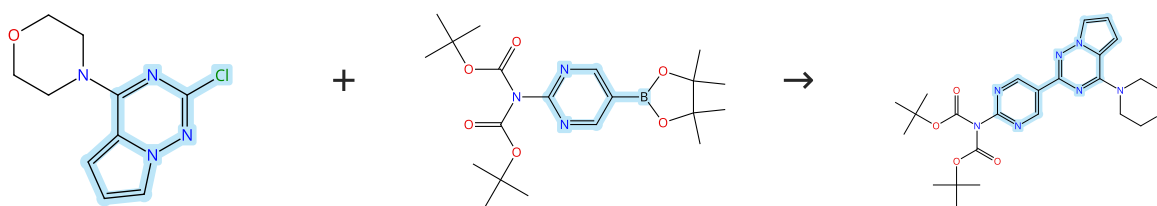
By: Ding, Huai-Wei; et al

Bioorganic & Medicinal Chemistry (2019), 27(13), 2729-2740.

Experimental Protocols

Scheme 80 (1 Reaction)

Steps: 1 Yield: 61%



Suppliers (43)

Suppliers (53)

31-179-CAS-3919650

Steps: 1 Yield: 61%

Synthesis and evaluation of pyrrolotriazine based molecules as PI3 kinase inhibitors

By: Dugar, Sundeep; et al

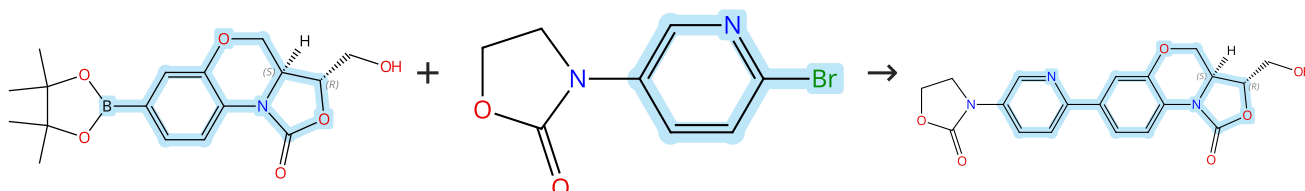
Bioorganic & Medicinal Chemistry Letters (2015), 25(16), 3142-3146.

1.1 **Reagents:** Sodium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Ethanol, Toluene, Water; 1 h, 150 °C

Experimental Protocols

Scheme 81 (1 Reaction)

Steps: 1 Yield: 61%



Absolute stereochemistry shown

Absolute stereochemistry shown

Suppliers (13)

31-179-CAS-13217324

Steps: 1 Yield: 61%

Solubility-Driven Optimization of (Pyridin-3-yl) Benzoxazinyl-oxazolidinones Leading to a Promising Antibacterial Agent

By: Guo, Bin; et al

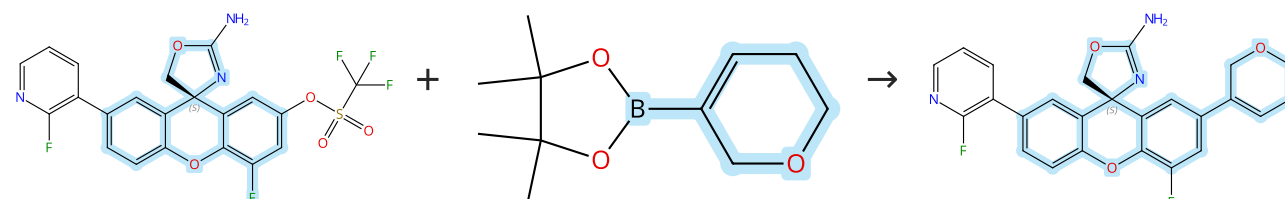
Journal of Medicinal Chemistry (2013), 56(6), 2642-2650.

1.1 **Reagents:** Cesium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: 1,4-Dioxane, Water; overnight, 80 °C

Experimental Protocols

Scheme 82 (1 Reaction)

Steps: 1 Yield: 61%



Absolute stereochemistry shown

Absolute stereochemistry shown

Suppliers (61)

Supplier (1)

31-177-CAS-15377216

Steps: 1 Yield: 61%

Lead Optimization and Modulation of hERG Activity in a Series of Aminooxazoline Xanthene β-Site Amyloid Precursor Protein Cleaving Enzyme (BACE1) Inhibitors

By: Epstein, Oleg; et al

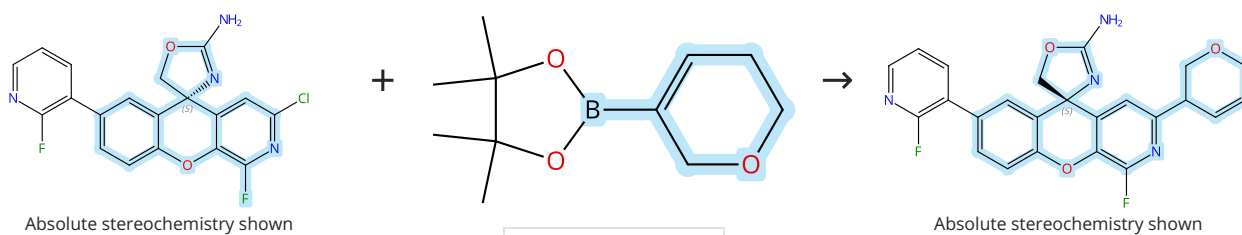
Journal of Medicinal Chemistry (2014), 57(23), 9796-9810.

1.1 **Reagents:** Sodium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Dimethylformamide, Water; rt

Experimental Protocols

Scheme 83 (1 Reaction)

Steps: 1 Yield: 61%



Suppliers (61)

31-117-CAS-5573842

Steps: 1 Yield: 61%

An Orally Available BACE1 Inhibitor That Affords Robust CNS Aβ Reduction without Cardiovascular Liabilities

By: Cheng, Yuan; et al

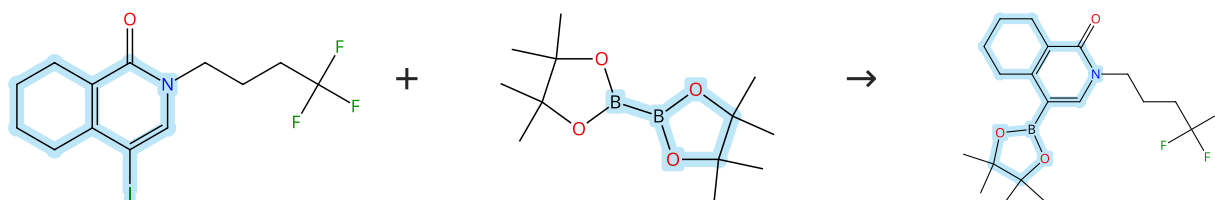
ACS Medicinal Chemistry Letters (2015), 6(2), 210-215.

1.1 **Reagents:** Tripotassium phosphate
Catalysts: (*SP*-4-1)-Bis[4-[bis(1,1-dimethylethyl)phosphino-κ^P]-*N,N*-dimethylbenzenamine]dichloropalladium
Solvents: 1,4-Dioxane, Water; 25 min, 135 °C

Experimental Protocols

Scheme 84 (1 Reaction)

Steps: 1 Yield: 61%



Supplier (1)

Suppliers (131)

Supplier (1)

31-142-CAS-21327558

Steps: 1 Yield: 61%

Discovery of Isoquinolinone Indole Acetic Acids as Antagonists of Chemoattractant Receptor Homologous Molecule Expressed on Th2 Cells (CRTH2) for the Treatment of Allergic Inflammatory Diseases

By: Kaila, Neelu; et al

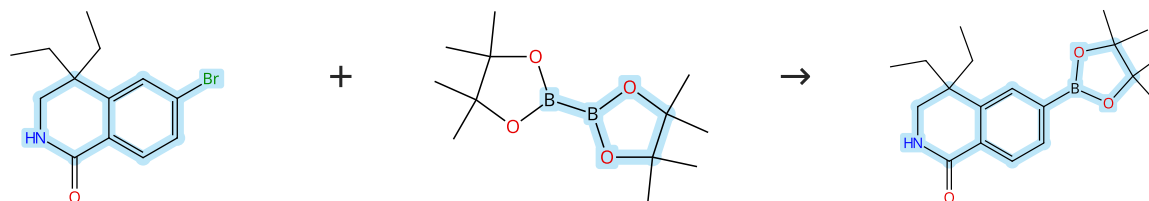
Journal of Medicinal Chemistry (2014), 57(4), 1299-1322.

1.1 **Reagents:** Potassium acetate
Catalysts: Dichloro[1,1'-bis(diphenylphosphino)ferrocene] palladium(II) dichloromethane adduct
Solvents: Dimethyl sulfoxide; rt → 80 °C; 16 h, 80 °C

Experimental Protocols

Scheme 85 (1 Reaction)

Steps: 1 Yield: 61%



Suppliers (131)

31-614-CAS-45123267

Steps: 1 Yield: 61%

Discovery of a Novel Dihydroisoquinolinone Derivative as a Potent CDK9 Inhibitor Capable of Overcoming L156F Mutant for the Treatment of Hematologic Malignancies

By: Shi, Chenliang; et al

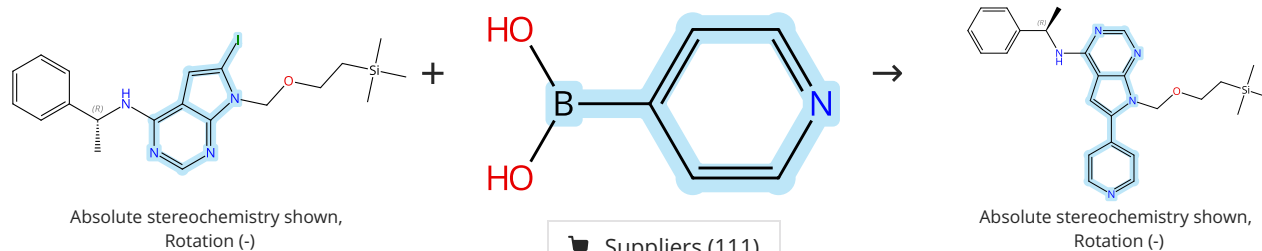
Journal of Medicinal Chemistry (2025), 68(8), 8106-8123.

1.1 **Reagents:** Potassium acetate
Catalysts: Dichloro[1,1'-bis(diphenylphosphino)ferrocene] palladium(II) dichloromethane adduct
Solvents: 1,4-Dioxane; 4 h, 100 °C

Experimental Protocols

Scheme 86 (1 Reaction)

Steps: 1 Yield: 61%



Suppliers (111)

31-179-CAS-16134931

Steps: 1 Yield: 61%

Balancing potency, metabolic stability and permeability in pyrrolopyrimidine-based EGFR inhibitors

1.1 **Reagents:** Potassium carbonate
Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium
Solvents: 1,4-Dioxane, Water; 2 h, 100 °C

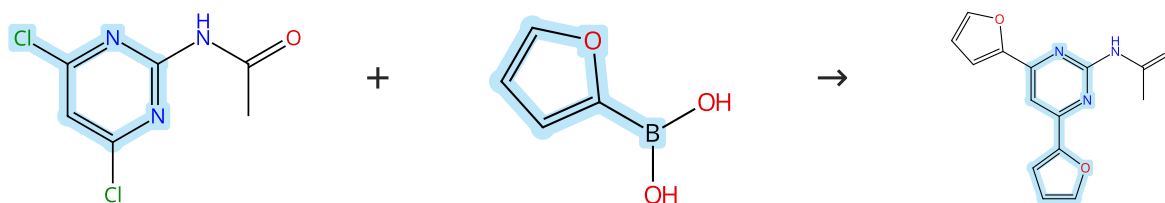
By: Han, Jin; et al

European Journal of Medicinal Chemistry (2016), 124, 583-607.

Experimental Protocols

Scheme 87 (1 Reaction)

Steps: 1 Yield: 61%



Suppliers (9)

Suppliers (93)

Suppliers (2)

31-179-CAS-10464561

Steps: 1 Yield: 61%

Pyrimidine derivatives as potent and selective A₃ adenosine receptor antagonists

1.1 **Reagents:** Sodium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: 1,2-Dimethoxyethane, Water; 12 h, 110 °C

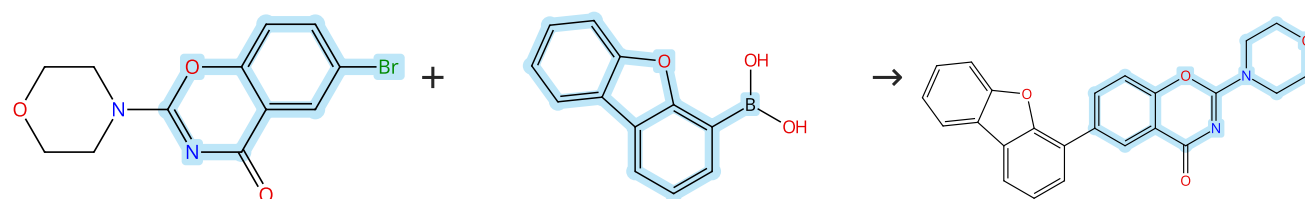
By: Yaziji, Vicente; et al

Journal of Medicinal Chemistry (2011), 54(2), 457-471.

Experimental Protocols

Scheme 88 (1 Reaction)

Steps: 1 Yield: 61%



Suppliers (2)

Suppliers (94)

Suppliers (2)

31-179-CAS-9187208

Steps: 1 Yield: 61%

Synthesis, structure elucidation, DNA-PK and PI3K and anti-cancer activity of 8- and 6- aryl-substituted-1-3-benzoxazines

1.1 **Reagents:** Potassium carbonate
Catalysts: Dichlorobis(triphenylphosphine)palladium
Solvents: 1,4-Dioxane, Water; 16 h, 50 °C

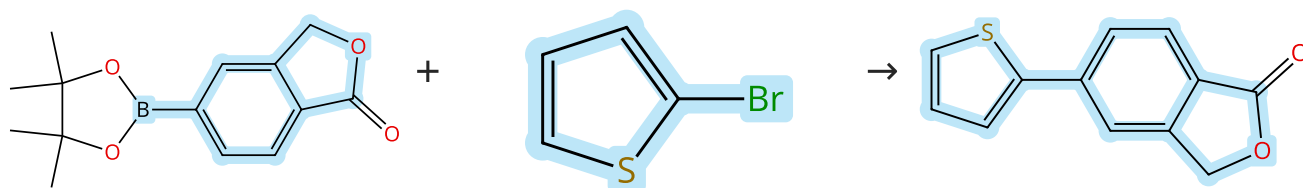
By: Morrison, Rick; et al

European Journal of Medicinal Chemistry (2016), 110, 326-339.

Experimental Protocols

Scheme 89 (1 Reaction)

Steps: 1 Yield: 61%



Suppliers (59)

Suppliers (76)

Suppliers (3)

31-179-CAS-3799942

Steps: 1 Yield: 61%

Inhibition of the pore-forming protein perforin by a series of aryl-substituted isobenzofuran-1(3H)-ones

By: Spicer, Julie A.; et al

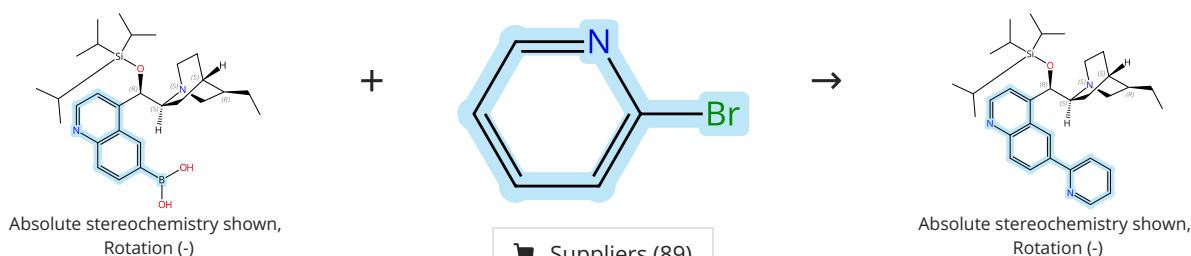
Bioorganic & Medicinal Chemistry (2012), 20(3), 1319-1336.

- 1.1 Reagents: Sodium carbonate
 Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium
 Solvents: Ethanol, Toluene, Water; 2 h, reflux

Experimental Protocols

Scheme 90 (1 Reaction)

Steps: 1 Yield: 61%



Suppliers (89)

31-614-CAS-37170680

Steps: 1 Yield: 61%

A Palladium-Catalyzed Borylation/Silica Gel Promoted Hydrolysis Sequence for the Synthesis of Hydroquinine-6'-Boric Acid and Its Applications

By: Liang, Wei; et al

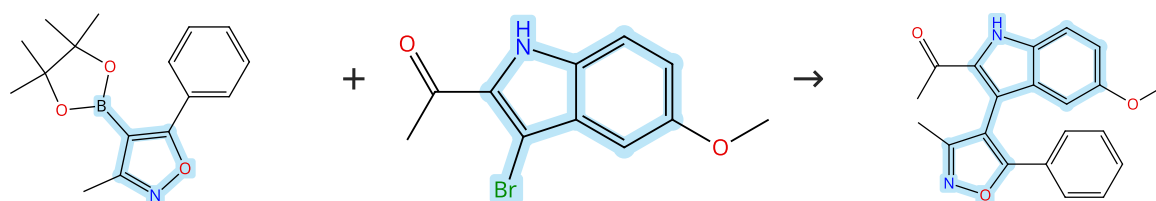
Journal of Organic Chemistry (2023), 88(15), 11284-11298.

- 1.1 Reagents: Potassium carbonate
 Catalysts: Palladium diacetate, CataCXium A
 Solvents: Butyl ether; 5 h, rt → 120 °C

Experimental Protocols

Scheme 91 (1 Reaction)

Steps: 1 Yield: 61%



Suppliers (3)

31-179-CAS-21954708

Steps: 1 Yield: 61%

A Novel Approach to Highly Substituted β-Carbolines via Reductive Ring Transformation of 2-Acyl-3-isoxazolyindoles

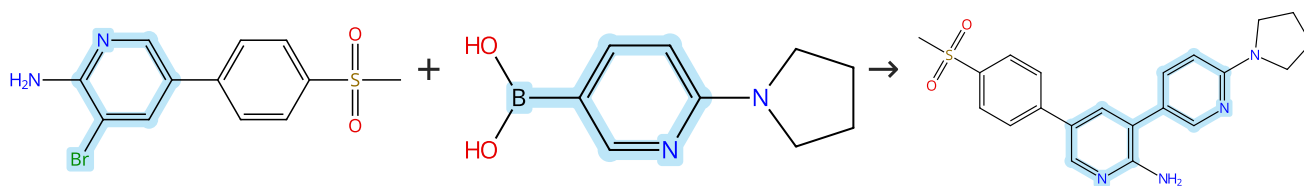
By: Kamlah, Alexandra; et al

European Journal of Organic Chemistry (2020), 2020(18), 2708-2719.

- 1.1 Reagents: Cesium carbonate
 Catalysts: Tetrakis(triphenylphosphine)palladium
 Solvents: 1,4-Dioxane, Water; rt; 19 h, 75 °C; 75 °C → rt
- 1.2 Reagents: Ammonium chloride
 Solvents: Water; rt

Scheme 92 (1 Reaction)

Steps: 1 Yield: 61%



Suppliers (8)

Suppliers (71)

Supplier (1)

31-179-CAS-12685908

Steps: 1 Yield: 61%

1.1 **Reagents:** Potassium carbonate
Catalysts: Dichlorobis(triphenylphosphine)palladium
Solvents: 1,4-Dioxane, Water; 16 h, 110 °C

Experimental Protocols

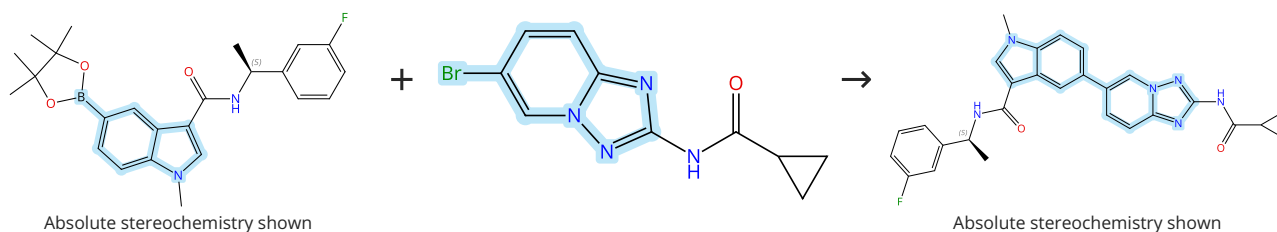
3,5-Diaryl-2-aminopyridines as a Novel Class of Orally Active Antimalarials Demonstrating Single Dose Cure in Mice and Clinical Candidate Potential

By: Younis, Yassir; et al

Journal of Medicinal Chemistry (2012), 55(7), 3479-3487.

Scheme 93 (1 Reaction)

Steps: 1 Yield: 61%



31-614-CAS-39000361

Steps: 1 Yield: 61%

1.1 **Reagents:** Cesium carbonate
Catalysts: Tricyclohexylphosphine, Dichloro[1,1'-bis(diphenylphosphino)ferrocene]palladium(II) dichloromethane adduct
Solvents: 1,4-Dioxane, Water; 8 h, 100 °C

Experimental Protocols

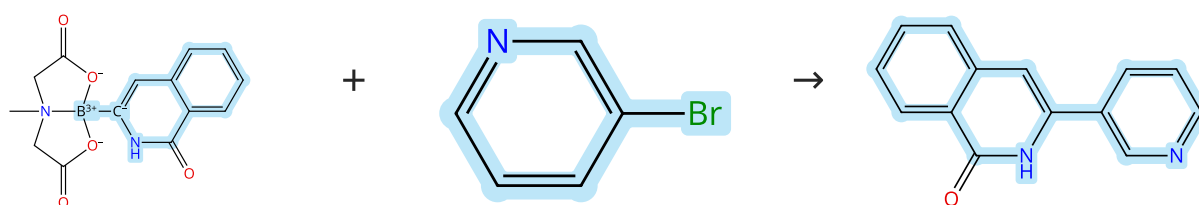
From Hit to Lead: Structure-Based Optimization of Novel Selective Inhibitors of Receptor-Interacting Protein Kinase 1 (RIPK1) for the Treatment of Inflammatory Diseases

By: Zhang, Liting; et al

Journal of Medicinal Chemistry (2024), 67(1), 754-773.

Scheme 94 (1 Reaction)

Steps: 1 Yield: 61%



Supplier (1)

Suppliers (92)

Suppliers (4)

31-177-CAS-4528089

Steps: 1 Yield: 61%

1.1 **Reagents:** Tripotassium phosphate
Catalysts: Palladium diacetate, 2-Dicyclohexylphosphino-2',6'-dimethoxybiphenyl
Solvents: Tetrahydrofuran, Water; 4 h, 60 °C

Experimental Protocols

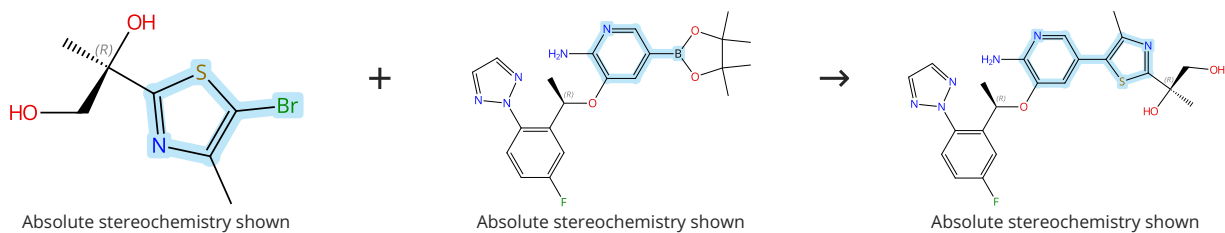
Mild Rh(III)-catalyzed C-H activation and annulation with alkyne MIDA boronates: short, efficient synthesis of heterocyclic boronic acid derivatives

By: Wang, Honggen; et al

Journal of the American Chemical Society (2012), 134(48), 19592-19595.

Scheme 95 (1 Reaction)

Steps: 1 Yield: 61%



Suppliers (10)

Supplier (1)

Suppliers (8)

31-179-CAS-7152207

Steps: 1 Yield: 61%

Design of Potent and Selective Inhibitors to Overcome Clinical Anaplastic Lymphoma Kinase Mutations Resistant to Crizotinib

By: Huang, Qinhua; et al

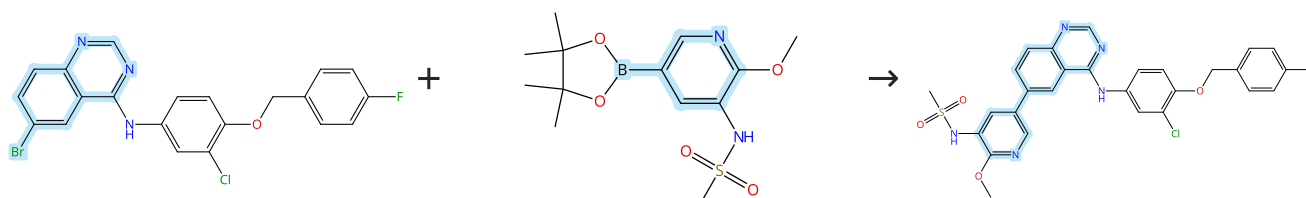
Journal of Medicinal Chemistry (2014), 57(4), 1170-1187.

1.1 **Reagents:** Cesium fluoride
Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium
Solvents: Methanol; 1 h, 120 °C

Experimental Protocols

Scheme 96 (1 Reaction)

Steps: 1 Yield: 61%



Suppliers (60)

31-179-CAS-19093711

Steps: 1 Yield: 61%

Design, synthesis and biological evaluation of novel 4-aminoquinazolines as dual target inhibitors of EGFR-PI3Ka

By: Ding, Huai-Wei; et al

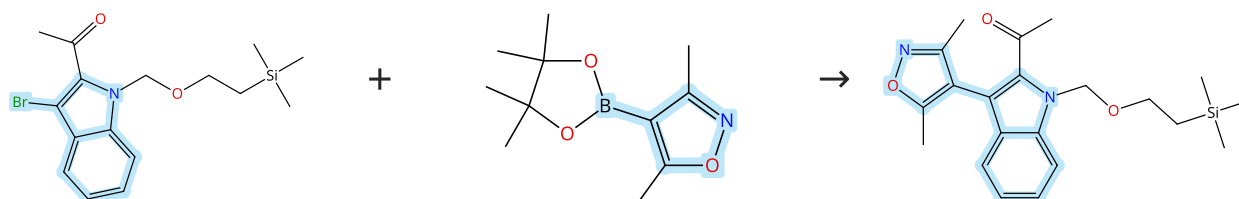
European Journal of Medicinal Chemistry (2018), 146, 460-470.

1.1 **Reagents:** Cesium carbonate
Catalysts: Dichloro[1,1'-bis(diphenylphosphino)ferrocene]palladium(II) dichloromethane adduct
Solvents: Dimethylformamide; 4 h, 90 °C

Experimental Protocols

Scheme 97 (1 Reaction)

Steps: 1 Yield: 61%



Suppliers (93)

31-179-CAS-21954695

Steps: 1 Yield: 61%

A Novel Approach to Highly Substituted β -Carbolines via Reductive Ring Transformation of 2-Acyl-3-isoxazolyindoles

By: Kamlah, Alexandra; et al

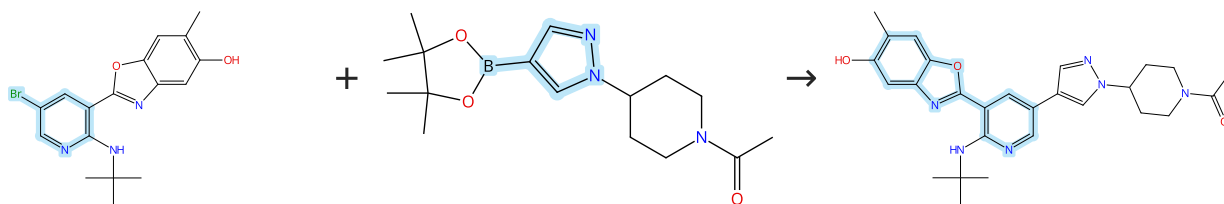
European Journal of Organic Chemistry (2020), 2020(18), 2708-2719.

1.1 **Reagents:** Cesium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: 1,4-Dioxane, Water; rt; 19 h, 75 °C; 75 °C $\rightarrow$ rt

1.2 **Reagents:** Ammonium chloride
Solvents: Water; rt

Scheme 98 (1 Reaction)

Steps: 1 Yield: 61%



Suppliers (51)

31-179-CAS-8248537

Steps: 1 Yield: 61%

Synthesis and structure-activity relationship of aminopyridines with substituted benzoxazoles as c-Met kinase inhibitors

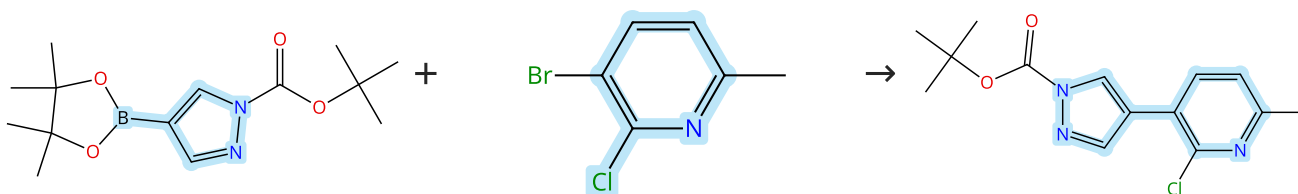
By: Lee, Jongkook; et al

Bioorganic & Medicinal Chemistry Letters (2012), 22(12), 4044-4048.

1.1 **Reagents:** Sodium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: 1,4-Dioxane

Scheme 99 (1 Reaction)

Steps: 1 Yield: 61%



Suppliers (96)

Suppliers (69)

Supplier (1)

31-614-CAS-41449003

Steps: 1 Yield: 61%

Discovery of 2-Methyl-5-(1H-pyrazol-4-yl)pyridines and Related Heterocycles as Promising M₄ mAChR Positive Allosteric Modulators for the Treatment of Neurocognitive Disorders

By: Liu, Boqun; et al

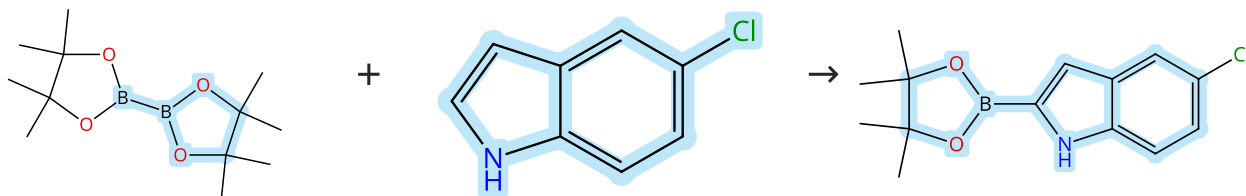
Journal of Medicinal Chemistry (2024), 67(15), 13286-13304.

1.1 **Reagents:** Cesium carbonate
Catalysts: Dichlorobis(triphenylphosphine)palladium
Solvents: Tetrahydrofuran; 80 °C

Experimental Protocols

Scheme 100 (1 Reaction)

Steps: 1 Yield: 61%



Suppliers (131)

Suppliers (91)

Suppliers (56)

31-142-CAS-20156496

Steps: 1 Yield: 61%

Iridium(I)-Catalyzed C-H Borylation in Air by Using Mechanochemistry

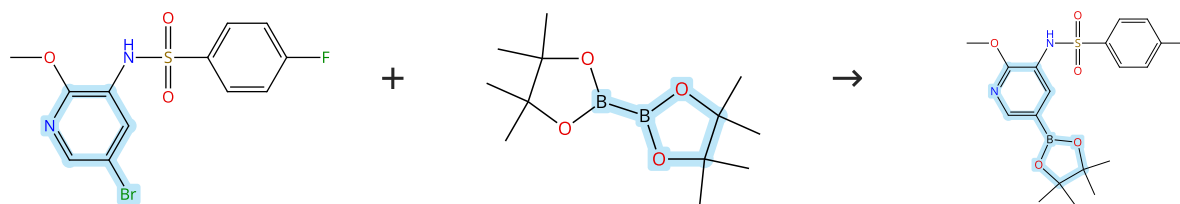
By: Pang, Yadong; et al

Chemistry - A European Journal (2019), 25(18), 4654-4659.

1.1 **Reagents:** Tetrahydrofuran, Oxygen
Catalysts: Bis[(1,2,5,6-η)-1,5-cyclooctadiene]di-μ-methoxydiiridium, 4,4'-Bis(1,1-dimethylethyl)-2,2'-bipyridine; 99 min

Scheme 101 (1 Reaction)

Steps: 1 Yield: 60%



Suppliers (4)

Suppliers (131)

Suppliers (6)

31-614-CAS-25907928

Steps: 1 Yield: 60%

Discovery and Optimization of 2-Amino-4-methylquinazoline Derivatives as Highly Potent Phosphatidylinositol 3-Kinase Inhibitors for Cancer Treatment

By: Lin, Songwen; et al

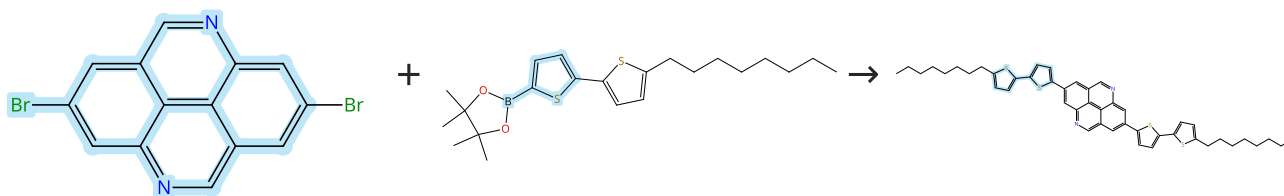
Journal of Medicinal Chemistry (2018), 61(14), 6087-6109.

1.1 **Reagents:** Potassium acetate
Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium
Solvents: 1,4-Dioxane; 4 h, 100 °C

Experimental Protocols

Scheme 102 (1 Reaction)

Steps: 1 Yield: 60%



Supplier (1)

31-614-CAS-42380619

Steps: 1 Yield: 60%

Isomeric diazapyrene-thiophene conjugated systems: synthesis, characterization, and transport properties

By: Li, Honglei; et al

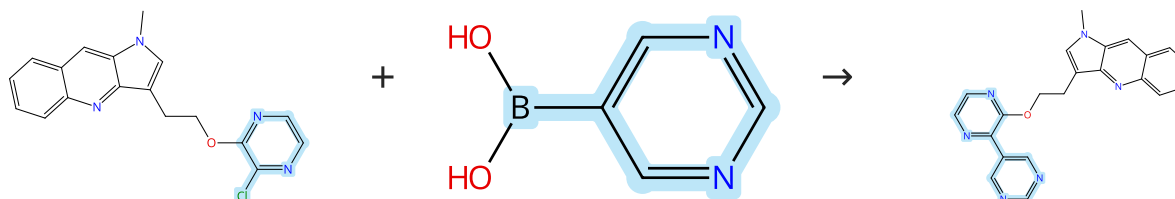
Organic Chemistry Frontiers (2024), 11(24), 7205-7213.

1.1 **Reagents:** Potassium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Toluene, Water; rt → 100 °C; 24 h, 100 °C

Experimental Protocols

Scheme 103 (1 Reaction)

Steps: 1 Yield: 60%



Suppliers (113)

Supplier (1)

31-179-CAS-2858420

Steps: 1 Yield: 60%

Design, synthesis and pharmacological evaluation of novel polycyclic heteroarene ethers as PDE10A inhibitors

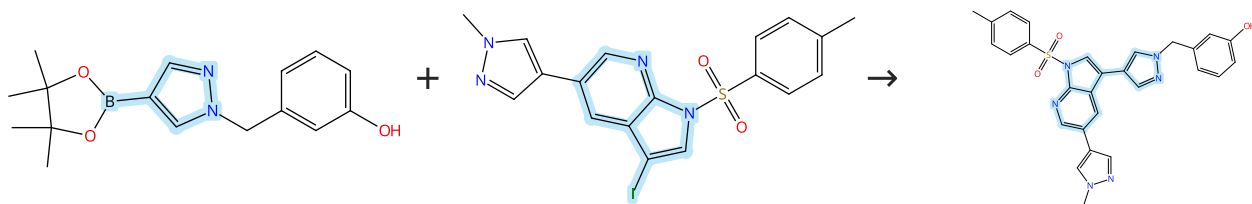
By: Das, Sanjib; et al

Bioorganic & Medicinal Chemistry Letters (2014), 24(9), 2073-2078.

1.1 **Reagents:** Sodium carbonate
Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium
Solvents: Dimethyl sulfoxide, Water; rt → 80 °C; 16 h, 80 °C

Scheme 104 (1 Reaction)

Steps: 1 Yield: 60%



Suppliers (2)

Suppliers (2)

31-179-CAS-15474783

Steps: 1 Yield: 60%

Discovery of 7-azaindole based anaplastic lymphoma kinase (ALK) inhibitors: Wild type and mutant (L1196M) active compounds with unique binding mode

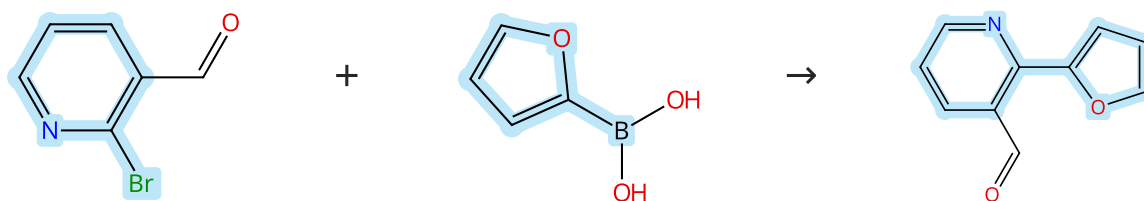
By: Gummadi, Venkateshwar Rao; et al

Bioorganic & Medicinal Chemistry Letters (2013), 23(17), 4911-4918.

1.1 **Reagents:** Sodium carbonate
Catalysts: Dichlorobis(triphenylphosphine)palladium
Solvents: Ethanol, Toluene, Water; 15 h, 110 °C

Scheme 105 (1 Reaction)

Steps: 1 Yield: 60%



Suppliers (83)

Suppliers (93)

Suppliers (20)

31-179-CAS-2740419

Steps: 1 Yield: 60%

Photoassisted Diversity-Oriented Synthesis: Accessing 2,6-Epoxyazocane (Oxamorphane) Cores

By: Mukhina, Olga A.; et al

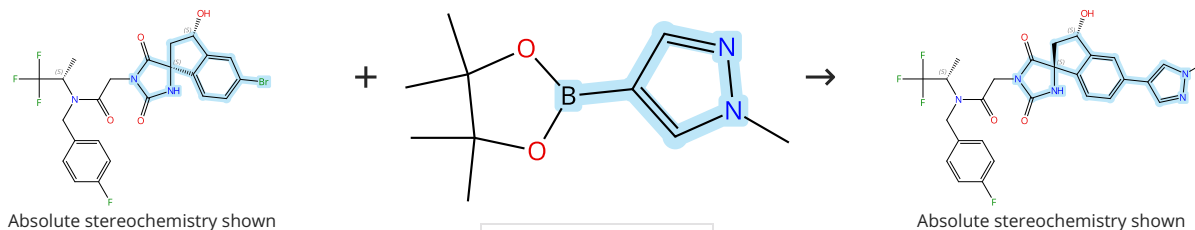
Journal of Organic Chemistry (2014), 79(22), 10956-10971.

1.1 **Reagents:** Potassium carbonate
Catalysts: Dichlorobis(triphenylphosphine)palladium
Solvents: Dimethylformamide, Water; 110 °C; 110 °C → rt

Experimental Protocols

Scheme 106 (1 Reaction)

Steps: 1 Yield: 60%



Absolute stereochemistry shown

Suppliers (126)

Absolute stereochemistry shown

Supplier (1)

31-614-CAS-25109200

Steps: 1 Yield: 60%

Discovery of spirohydantoin as selective, orally bioavailable inhibitors of p300/CBP histone acetyltransferases

By: Ji, Zhiqin; et al

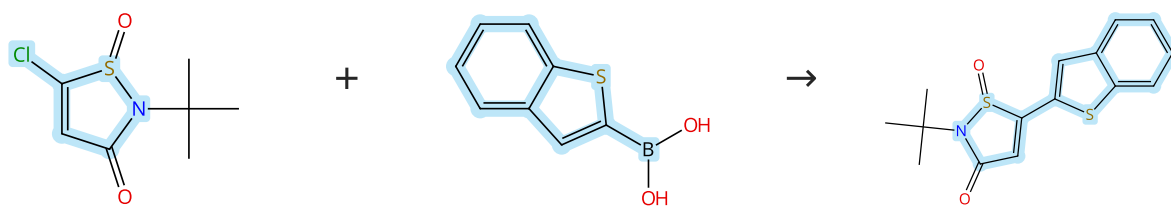
Bioorganic & Medicinal Chemistry Letters (2021), 39, 127854.

1.1 **Reagents:** Potassium carbonate
Catalysts: Dichloro[1,1'-bis(diphenylphosphino)ferrocene] palladium(II) dichloromethane adduct
Solvents: 1,4-Dioxane, Water; 3 h, 85 °C

Experimental Protocols

Scheme 107 (1 Reaction)

Steps: 1 Yield: 60%



Suppliers (6)

Suppliers (91)

Supplier (1)

31-179-CAS-20307227

Steps: 1 Yield: 60%

5-Arylisothiazol-3(2H)-one-1,1(1)-(di)oxides: A new class of selective tumor-associated carbonic anhydrases (hCA IX and XII) inhibitors

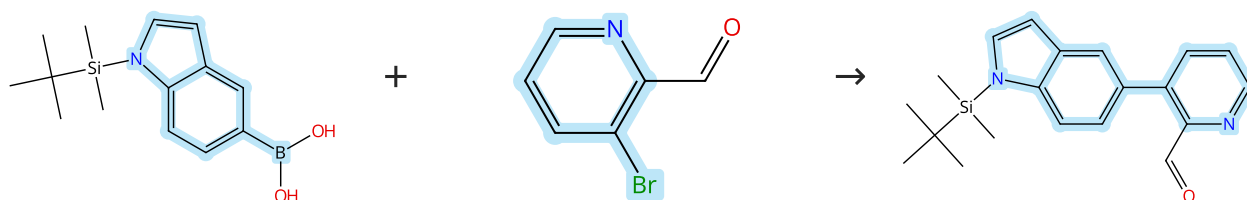
By: Cornelio, Benedetta; et al

European Journal of Medicinal Chemistry (2019), 175, 40-48.

1.1 **Reagents:** Potassium carbonate
Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium
Solvents: 1,2-Dimethoxyethane; 5.75 h, 80 °C

Scheme 108 (1 Reaction)

Steps: 1 Yield: 60%



Suppliers (49)

Suppliers (77)

31-614-CAS-32086062

Steps: 1 Yield: 60%

A data-driven approach to the development and understanding of chiroptical sensors for alcohols with remote γ -stereocenters

By: Dotson, Jordan J.; et al

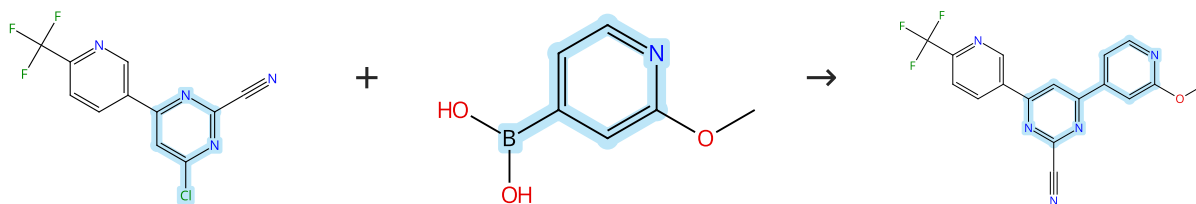
Journal of the American Chemical Society (2021), 143(45), 19187-19198.

1.1 **Reagents:** Cesium carbonate
Catalysts: [2'-(Amino- κ M)[1,1'-biphenyl]-2-yl- κ C][dicyclohexyl (2',6'-dimethoxy[1,1'-biphenyl]-2-yl)phosphine- κ P](methanesulfonato- κ O)palladium
Solvents: 1,4-Dioxane, Water; 2 h, 100 °C

Experimental Protocols

Scheme 109 (1 Reaction)

Steps: 1 Yield: 60%



Suppliers (99)

31-614-CAS-34480410

Steps: 1 Yield: 60%

Structure-activity relationships of 2-pyrimidinecarboxamides as utrophin modulators for the potential treatment of Duchenne muscular dystrophy

By: Chatzopoulou, Maria; et al

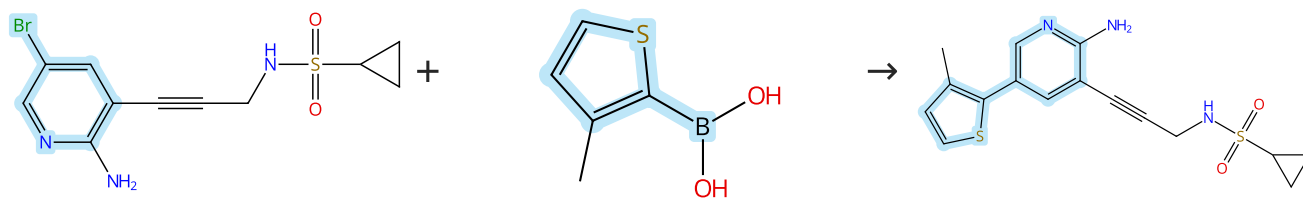
Bioorganic & Medicinal Chemistry (2022), 69, 116812.

1.1 **Reagents:** Potassium carbonate
Catalysts: Dichloro[1,1'-bis(diphenylphosphino)ferrocene]palladium(II) dichloromethane adduct
Solvents: 1,4-Dioxane, Water; 3 - 12 h, 90 °C

Experimental Protocols

Scheme 110 (1 Reaction)

Steps: 1 Yield: 60%



Suppliers (72)

31-179-CAS-6810492

Steps: 1 Yield: 60%

1.1 **Reagents:** Potassium carbonate
Catalysts: Dichloro[1,1'-bis(diphenylphosphino)ferrocene]
 palladium(II) dichloromethane adduct
Solvents: 1,2-Dimethoxyethane, Water; 18 h, reflux

Experimental Protocols

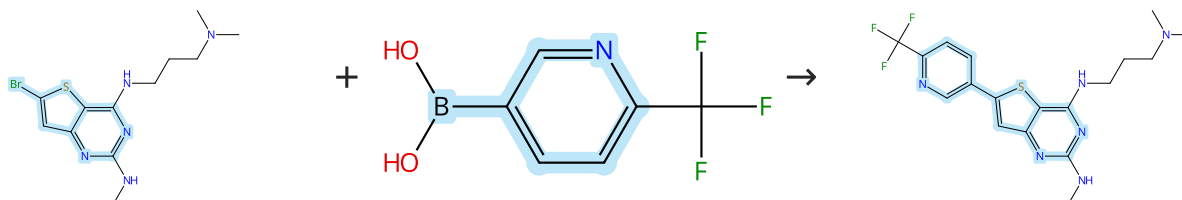
Exploring the Ribose Sub-Pocket of the Substrate-Binding Site in Escherichia coli IspE: Structure-Based Design, Synthesis, and Biological Evaluation of Cytosines and Cytosine Analogues

By: Schuetz, Andri P.; et al

European Journal of Organic Chemistry (2012), 2012(17), 3278-3287, S3278/1-S3278/78.

Scheme 111 (1 Reaction)

Steps: 1 Yield: 60%



Supplier (1)

Suppliers (90)

Supplier (1)

31-179-CAS-5301520

Steps: 1 Yield: 60%

1.1 **Reagents:** Potassium carbonate
Catalysts: Dichlorobis(triphenylphosphine)palladium
Solvents: 1,4-Dioxane, Water; 14 h, 90 °C

Experimental Protocols

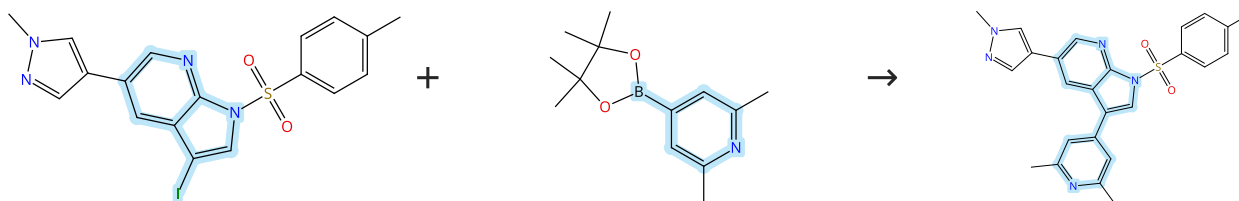
2,4-Diaminothienopyrimidines as Orally Active Antimalarial Agents

By: Gonzalez Cabrera, Diego; et al

Journal of Medicinal Chemistry (2014), 57(3), 1014-1022.

Scheme 112 (1 Reaction)

Steps: 1 Yield: 60%



Suppliers (2)

Suppliers (68)

31-614-CAS-25136701

Steps: 1 Yield: 60%

1.1 **Reagents:** Potassium carbonate
Catalysts: Dichloro[1,1'-bis(diphenylphosphino)ferrocene]
 palladium(II) dichloromethane adduct
Solvents: 1,4-Dioxane, Water; 30 min, 120 °C

Experimental Protocols

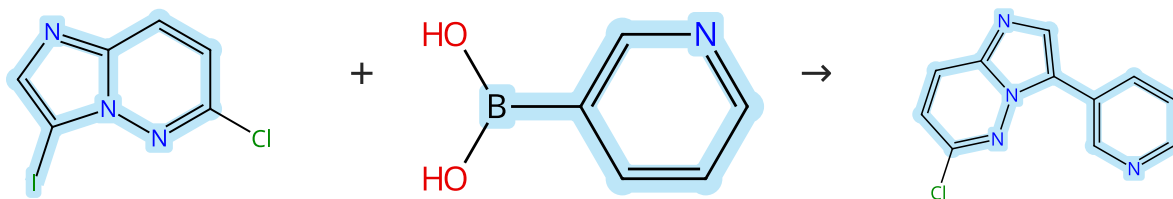
Lead Optimization of 3,5-Disubstituted-7-Azaindoles for the Treatment of Human African Trypanosomiasis

By: Klug, Dana M.; et al

Journal of Medicinal Chemistry (2021), 64(13), 9404-9430.

Scheme 113 (1 Reaction)

Steps: 1 Yield: 60%



Suppliers (68)

Suppliers (115)

Suppliers (4)

31-179-CAS-7609079

Steps: 1 Yield: 60%

Medicinal Chemistry Optimization of Antiplasmodial Imidazopyridazine Hits from High Throughput Screening of a SoftFocus Kinase Library: Part 1

By: Le Manach, Claire; et al

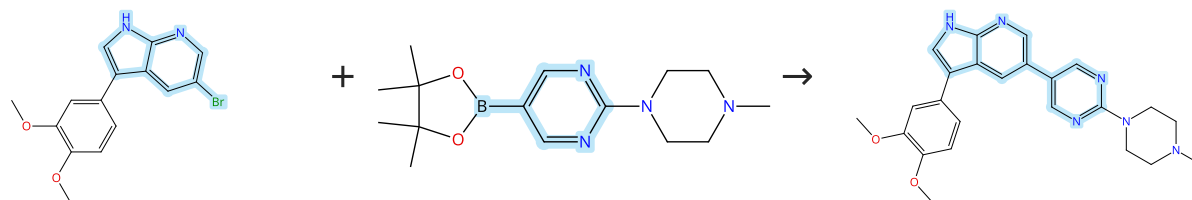
Journal of Medicinal Chemistry (2014), 57(6), 2789-2798.

1.1 **Reagents:** Potassium carbonate
Catalysts: Dichlorobis(triphenylphosphine)palladium
Solvents: Dimethylformamide, Water; rt → 80 °C; 12 h, 80 °C

Experimental Protocols

Scheme 114 (1 Reaction)

Steps: 1 Yield: 60%



Suppliers (2)

Suppliers (60)

31-614-CAS-39559475

Steps: 1 Yield: 60%

Discovery of novel, potent, selective and orally bioavailable H PK1 inhibitor for enhancing the efficacy of anti-PD-L1 antibody

By: Zeng, Shenxin; et al

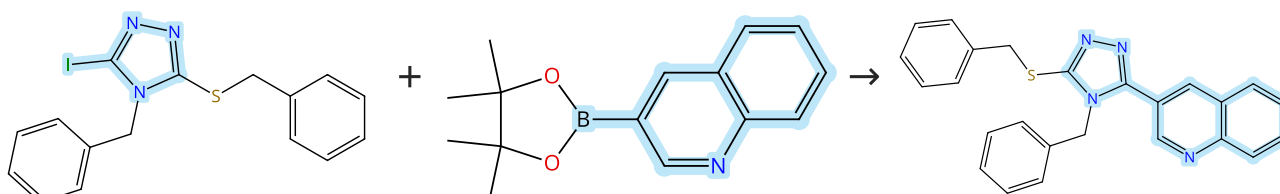
European Journal of Medicinal Chemistry (2024), 267, 116206.

1.1 **Reagents:** Potassium carbonate
Catalysts: Dichloro[1,1'-bis(diphenylphosphino)ferrocene] palladium(II) dichloromethane adduct
Solvents: 1,4-Dioxane, Water; 8 h, 90 °C

Experimental Protocols

Scheme 115 (1 Reaction)

Steps: 1 Yield: 60%



Suppliers (85)

31-179-CAS-5566195

Steps: 1 Yield: 60%

Synthesis of 5-substituted 3-mercapto-1,2,4-triazoles via Suzuki-Miyaura reaction

By: Katkevica, Sarmite; et al

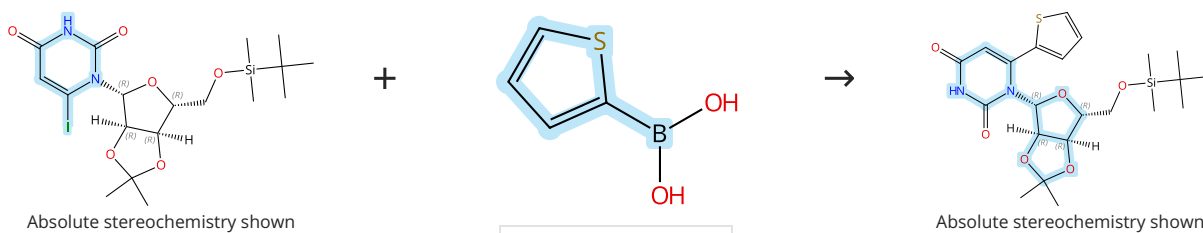
Tetrahedron Letters (2013), 54(34), 4524-4525.

1.1 **Reagents:** Potassium carbonate, Tetrakis(triphenylphosphine) palladium
Solvents: Tetrahydrofuran, Water; 24 - 48 h, 80 °C

Experimental Protocols

Scheme 116 (1 Reaction)

Steps: 1 Yield: 60%



Absolute stereochemistry shown

Absolute stereochemistry shown

Suppliers (3)

Suppliers (103)

31-179-CAS-11053690

Steps: 1 Yield: 60%

Practical synthesis of 6-aryluridines via palladium(II) acetate catalyzed Suzuki-Miyaura cross-coupling reaction

1.1 **Reagents:** Sodium carbonate
Catalysts: Palladium diacetate, 1,1-Bis(diphenylphosphino)ferrocene
Solvents: 1,2-Dimethoxyethane, Water; 16 h, reflux; reflux → rt

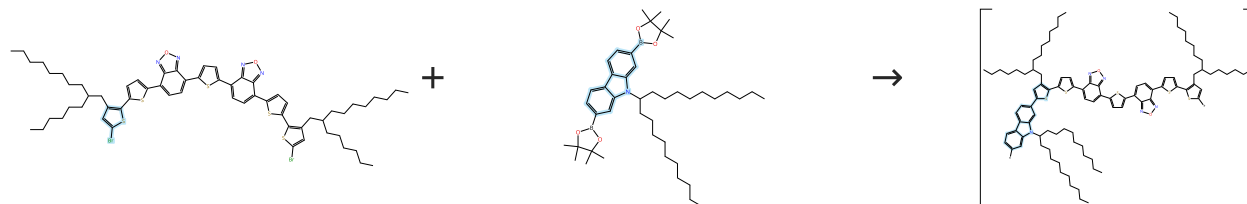
By: Shih, Yu-Chiao; et al

Tetrahedron (2011), 67(21), 3915-3923.

Experimental Protocols

Scheme 117 (1 Reaction)

Steps: 1 Yield: 60%



Suppliers (33)

31-614-CAS-25177593

Steps: 1 Yield: 60%

Synthesis of the (X-DADAD)_n-type conjugated polymers with 2,1,3-benzoxadiazole acceptor blocks and their application in organic solar cells

1.1 **Reagents:** Potassium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Toluene, Water; 3 - 6 h, reflux

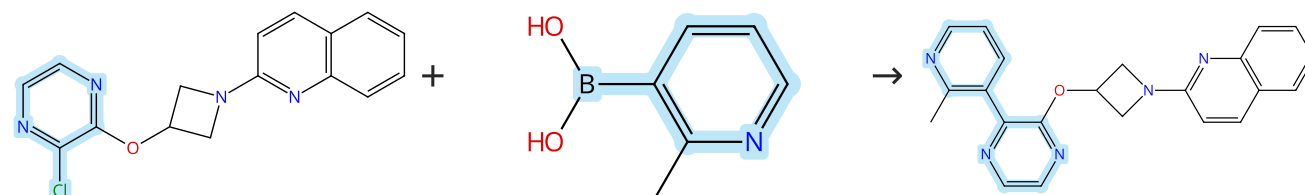
By: Akkuratov, Alexander V.; et al

Tetrahedron Letters (2017), 58(1), 97-100.

Experimental Protocols

Scheme 118 (1 Reaction)

Steps: 1 Yield: 60%



Suppliers (4)

Suppliers (71)

Supplier (1)

31-179-CAS-1742762

Steps: 1 Yield: 60%

Synthesis and preliminary biological evaluation of potent and selective 2-(3-alkoxy-1-azetidyl) quinolines as novel PDE10A inhibitors with improved solubility

1.1 **Reagents:** Sodium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: 1,4-Dioxane, Water; 64 h, 120 °C

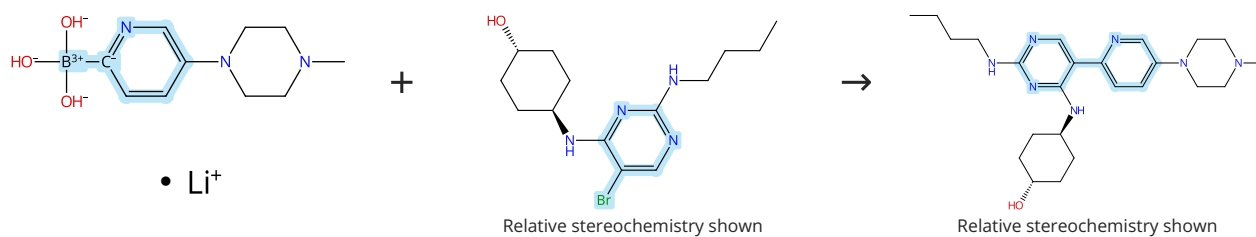
By: Rzasa, Robert M.; et al

Bioorganic & Medicinal Chemistry (2014), 22(23), 6570-6585.

Experimental Protocols

Scheme 119 (1 Reaction)

Steps: 1 Yield: 60%



Supplier (1)

Supplier (1)

31-326-CAS-1450632

Steps: 1 Yield: 60%

Pseudo-Cyclization through Intramolecular Hydrogen Bond Enables Discovery of Pyridine Substituted Pyrimidines as New Mer Kinase Inhibitors

By: Zhang, Weihe; et al

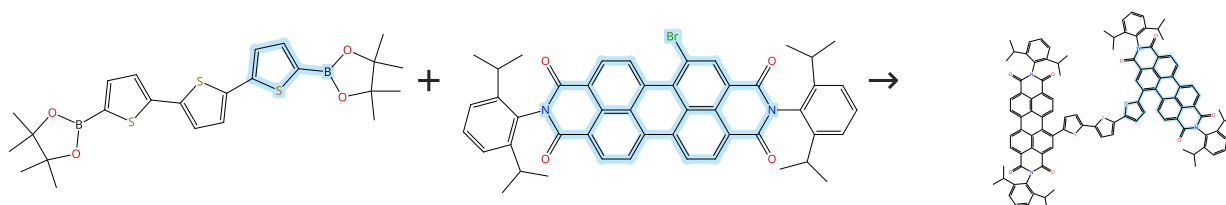
Journal of Medicinal Chemistry (2013), 56(23), 9683-9692.

1.1 **Reagents:** Potassium carbonate
Catalysts: Copper bromide (CuBr₂), [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium
Solvents: Dimethylformamide, Water; 30 min, 120 °C

Experimental Protocols

Scheme 120 (1 Reaction)

Steps: 1 Yield: 60%



Supplier (1)

31-614-CAS-30395201

Steps: 1 Yield: 60%

Controllable Supramolecular Architectures for Modulating Optical Properties on the Molecular Aggregation Level

By: Jiang, Runsheng; et al

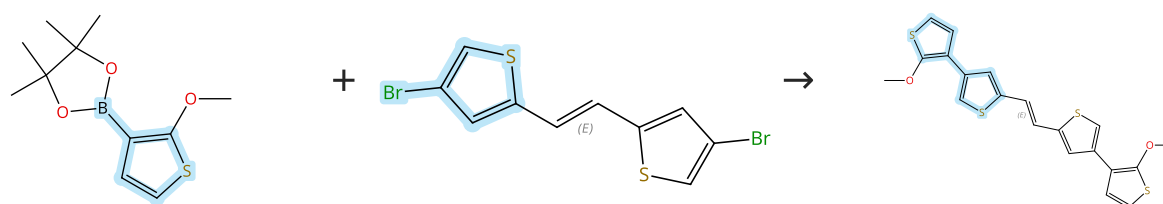
European Journal of Organic Chemistry (2014), 2014(23), 5004-5009.

1.1 **Reagents:** Potassium carbonate
Solvents: Tetrahydrofuran, Water; 10 min, rt
 1.2 **Catalysts:** Tetrakis(triphenylphosphine)palladium; overnight, 75 °C

Experimental Protocols

Scheme 121 (1 Reaction)

Steps: 1 Yield: 60%



Suppliers (18)

Double bond geometry shown

Double bond geometry shown

31-614-CAS-43157661

Steps: 1 Yield: 60%

(BO)₂-doped tetrathia[7]helicenes: synthesis and property-change induced by "BO bond inversion"

By: Menduti, Luigi; et al

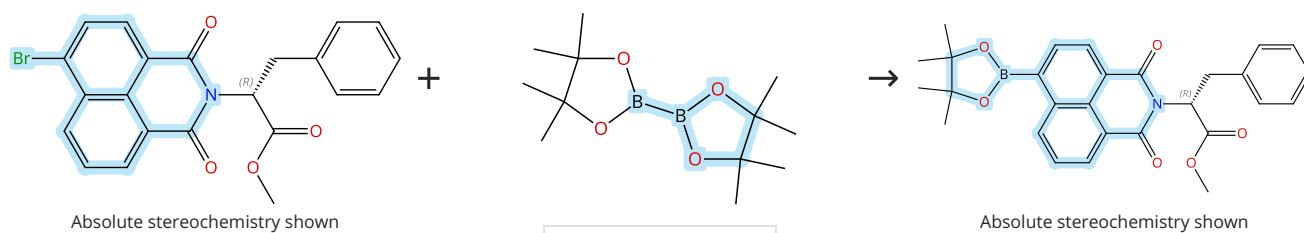
Organic Chemistry Frontiers (2025), 12(3), 725-735.

1.1 **Reagents:** Sodium hydroxide
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Tetrahydrofuran, Water; 15 h, rt → 80 °C

Experimental Protocols

Scheme 122 (1 Reaction)

Steps: 1 Yield: 60%



Suppliers (131)

31-614-CAS-27810958

Steps: 1 Yield: 60%

Enabling Light Work in Helical Self-Assembly for Dynamic Amplification of Chirality with Photoreversibility

By: Cai, Yunsong; et al

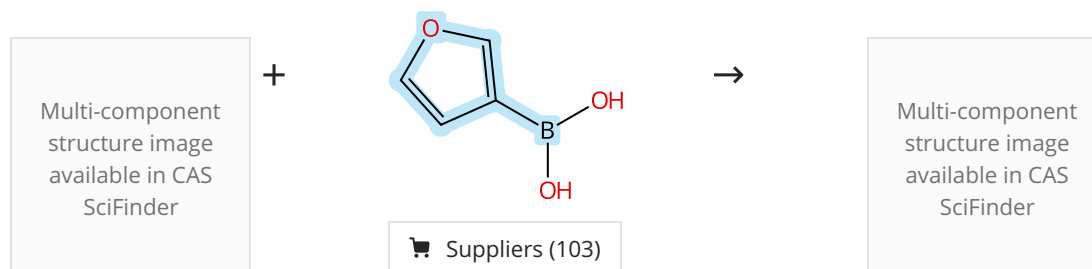
Journal of the American Chemical Society (2016), 138(7), 2219-2224.

1.1 **Reagents:** Potassium acetate
Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium
Solvents: 1,4-Dioxane; 12 h, 85 °C

Experimental Protocols

Scheme 123 (1 Reaction)

Steps: 1 Yield: 60%



Suppliers (103)

31-614-CAS-40118651

Steps: 1 Yield: 60%

Functionalization of Dodecaborates by Mild and Efficient Pd-Catalyzed Formation of B-C Bonds with Boronic Acids**

By: Al-Joumhawry, Mahmoud K.; et al

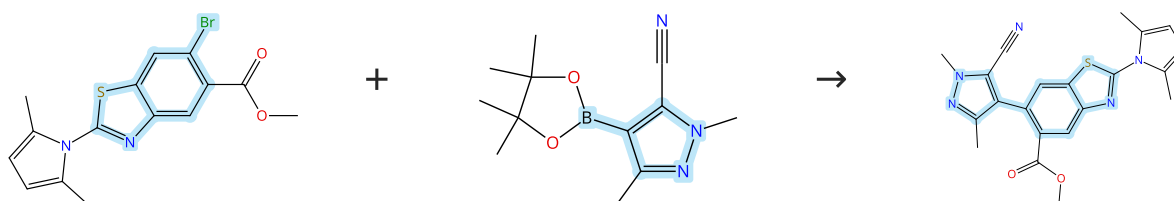
Chemistry - A European Journal (2023), 29(67), e202302466.

1.1 **Reagents:** Potassium hydroxide
Catalysts: Tris(dibenzylideneacetone)dipalladium, 2'-(Dicyclohexylphosphino)-N,N-dimethyl[1,1'-biphenyl]-2-amine
Solvents: Acetonitrile-*d*₃; 3 h, 90 °C

Experimental Protocols

Scheme 124 (1 Reaction)

Steps: 1 Yield: 60%



Suppliers (20)

31-614-CAS-46057519

Steps: 1 Yield: 60%

The synthesis and evaluation of a series of novel fused rings based benzothiazole derivatives as potent Polθ inhibitors

By: Kan, Chao; et al

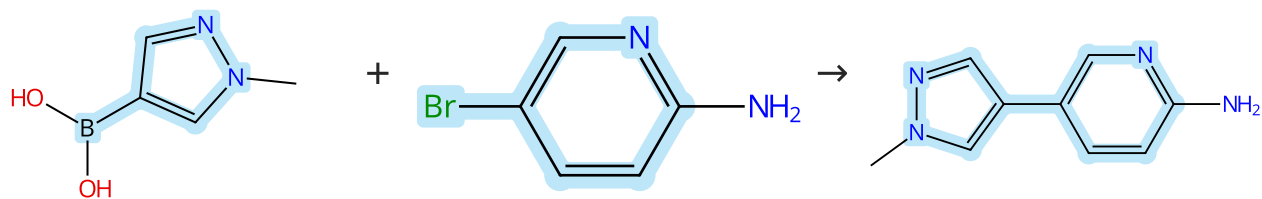
Tetrahedron (2025), 184, 134734.

1.1 **Reagents:** Potassium carbonate
Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium
Solvents: 1,4-Dioxane, Water; 2 h, 95 °C

Experimental Protocols

Scheme 125 (1 Reaction)

Steps: 1 Yield: 60%



Suppliers (77)

Suppliers (102)

Suppliers (46)

31-614-CAS-35492146

Steps: 1 Yield: 60%

Discovery and In Vivo Anti-inflammatory Activity Evaluation of a Novel Non-peptidyl Non-covalent Cathepsin C Inhibitor

By: Chen, Xing; et al

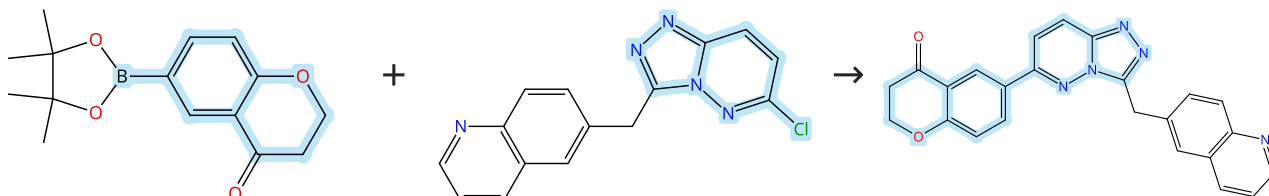
Journal of Medicinal Chemistry (2021), 64(16), 11857-11885.

1.1 **Reagents:** Sodium carbonate
Catalysts: Dichlorobis(triphenylphosphine)palladium
Solvents: 1,4-Dioxane, Water; 10 h, 90 °C

Experimental Protocols

Scheme 126 (1 Reaction)

Steps: 1 Yield: 60%



Suppliers (35)

Suppliers (4)

31-614-CAS-38905965

Steps: 1 Yield: 60%

Discovery of potent and selective c-Met inhibitors for MET-amplified hepatocellular carcinoma treatment

By: Min, Wenjian; et al

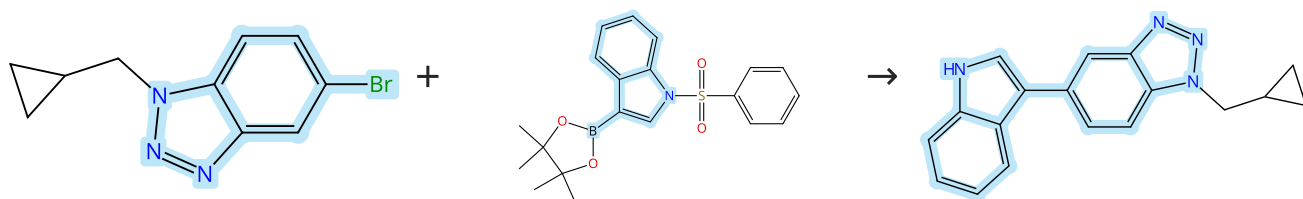
European Journal of Medicinal Chemistry (2024), 264, 116025.

1.1 **Reagents:** Potassium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: 1,4-Dioxane, Water; 8 h, 90 °C

Experimental Protocols

Scheme 127 (1 Reaction)

Steps: 1 Yield: 60%



Suppliers (4)

Suppliers (64)

Supplier (1)

31-614-CAS-34289547

Steps: 1 Yield: 60%

Rational Design of Original Fused-Cycle Selective Inhibitors of Tryptophan 2,3-Dioxygenase

By: Kozlova, Arina; et al

Journal of Medicinal Chemistry (2021), 64(15), 10967-10980.

1.1 **Reagents:** Tripotassium phosphate
Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium, 2-Dicyclohexylphosphino-2',6'-dimethoxybiphenyl
Solvents: 1,4-Dioxane, Water; 3 - 16 h, 70 °C; 70 °C → rt

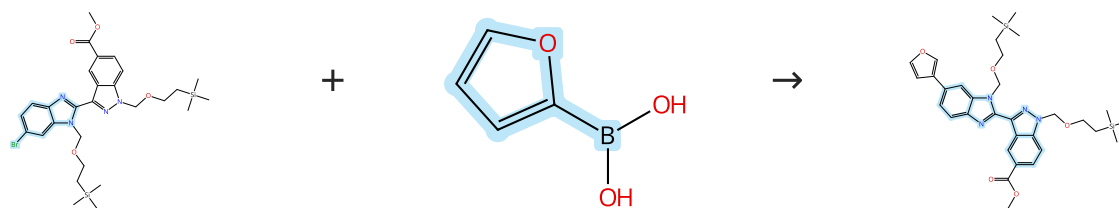
1.2 **Reagents:** Sodium hydroxide
Solvents: Water; rt

1.3 **Reagents:** Potassium hydroxide
Solvents: Methanol, Water; 0.5 - 48 h, reflux

Experimental Protocols

Scheme 128 (1 Reaction)

Steps: 1 Yield: 60%



Suppliers (93)

31-614-CAS-38257743

Steps: 1 Yield: 60%

Discovery of benzimidazole-indazole derivatives as potent FL T3-tyrosine kinase domain mutant kinase inhibitors for acute myeloid leukemia

By: Ko, Bongki; et al

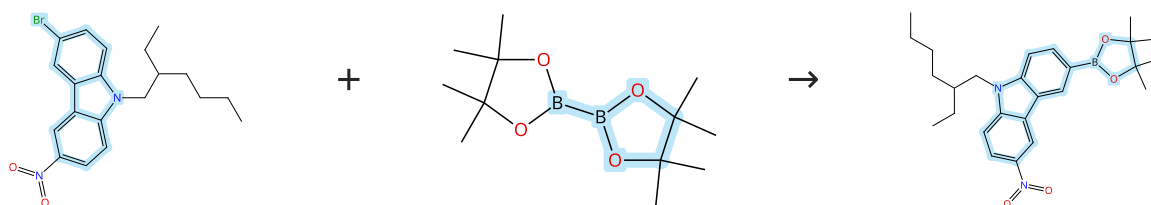
European Journal of Medicinal Chemistry (2023), 262, 115860.

1.1 **Reagents:** Potassium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Toluene, Water; 3 h, 60 °C

Experimental Protocols

Scheme 129 (1 Reaction)

Steps: 1 Yield: 60%



Suppliers (131)

31-614-CAS-28101894

Steps: 1 Yield: 60%

Inserting Thienyl Linkers into Conjugated Molecules for Efficient Multilevel Electronic Memory: A New Understanding of Charge-Trapping in Organic Materials

By: Li, Yang; et al

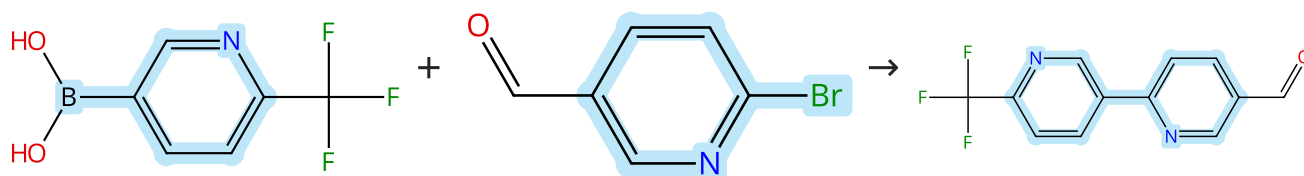
Chemistry - An Asian Journal (2016), 11(6), 906-914.

1.1 **Reagents:** Potassium acetate
Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium
Solvents: Toluene; 16 h, 70 °C

Experimental Protocols

Scheme 130 (1 Reaction)

Steps: 1 Yield: 60%



Suppliers (90)

Suppliers (105)

Suppliers (2)

31-179-CAS-9574883

Steps: 1 Yield: 60%

Identification, Design and Biological Evaluation of Heterocyclic Quinolones Targeting Plasmodium falciparum Type II NADH:Quinone Oxidoreductase (PfNDH2)

By: Leung, Suet C.; et al

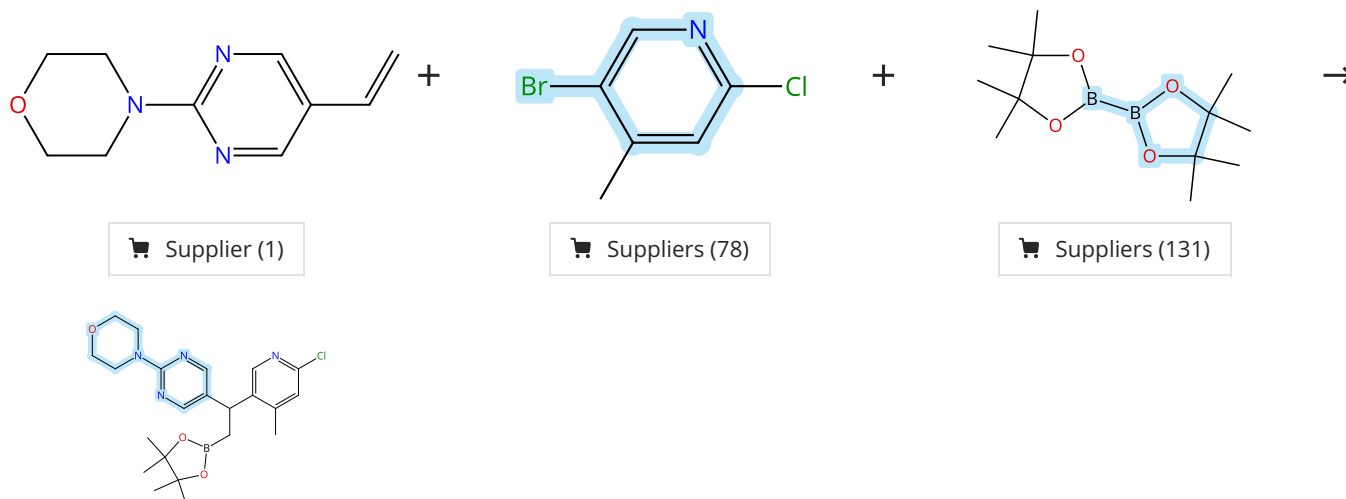
Journal of Medicinal Chemistry (2012), 55(5), 1844-1857.

1.1 **Reagents:** Sodium carbonate
Catalysts: Tri-*tert*-butylphosphine, Dichlorobis(triphenylphosphine)palladium
Solvents: 1,4-Dioxane; 16 h, rt → 100 °C

Experimental Protocols

Scheme 131 (1 Reaction)

Steps: 1 Yield: 60%



31-177-CAS-19953880

Steps: 1 Yield: 60%

Synthesis of Bisheteroarylalkanes by Heteroarylboration: Development and Application of a Pyridylidene-Copper Complex

By: Huang, Yuan; et al

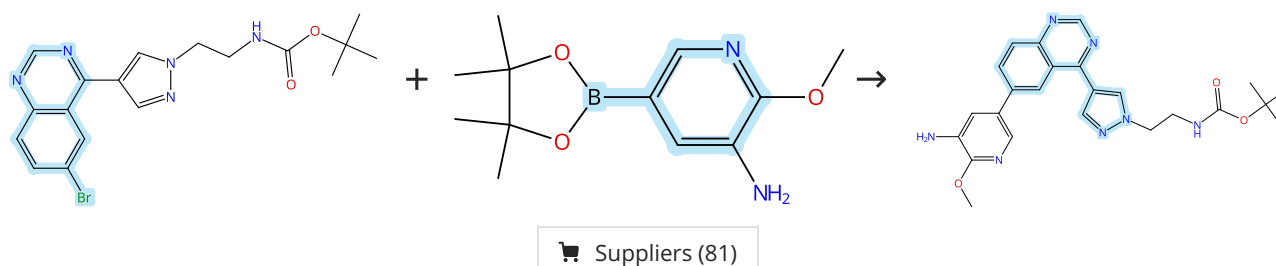
Angewandte Chemie, International Edition (2019), 58(18), 6048-6052.

- 1.1 **Reagents:** Sodium *tert*-butoxide
Catalysts: 2268801-39-2, 2304898-62-0
Solvents: Toluene; 12 h, rt
- 1.2 **Reagents:** Potassium hydroxide
Solvents: Water; rt

Experimental Protocols

Scheme 132 (1 Reaction)

Steps: 1 Yield: 60%



31-614-CAS-24893738

Steps: 1 Yield: 60%

Omipalisib inspired macrocycles as dual PI3K/mTOR inhibitors

By: Alvarez, Rosa M.; et al

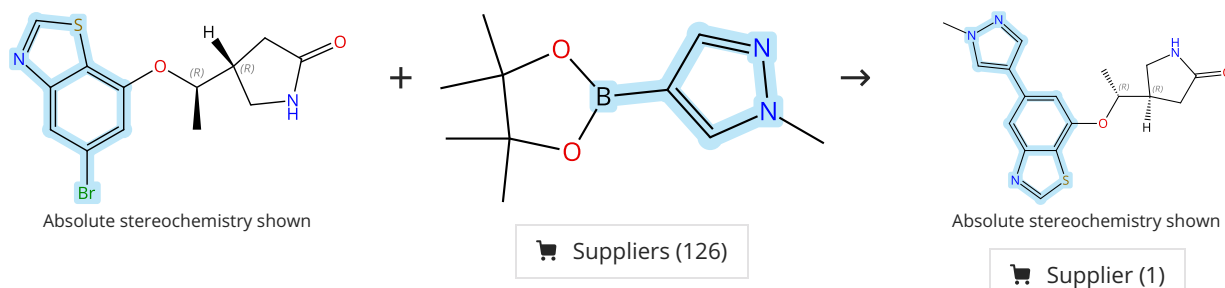
European Journal of Medicinal Chemistry (2021), 211, 113109.

- 1.1 **Reagents:** Potassium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: 1,4-Dioxane, Water; 2 h, reflux

Experimental Protocols

Scheme 133 (1 Reaction)

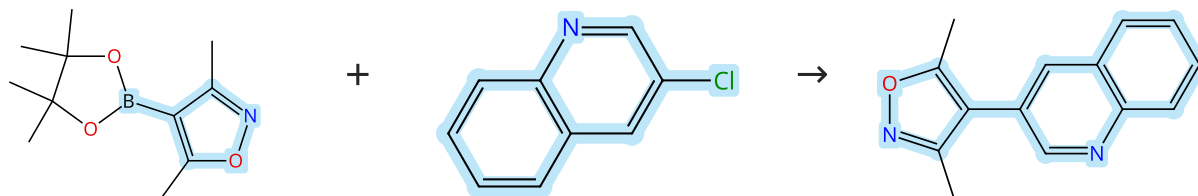
Steps: 1 Yield: 60%



31-179-CAS-6438319 Steps: 1 Yield: 60% 1.1 Reagents: Sodium carbonate Catalysts: Dichlorobis(triphenylphosphine)palladium Solvents: 1,4-Dioxane; 16 h, 100 °C	Orally bioavailable Syk inhibitors with activity in a rat P K/PD model By: Thoma, Gebhard; et al Bioorganic & Medicinal Chemistry Letters (2015), 25(20), 4642-4647.
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Scheme 134 (1 Reaction)

Steps: 1 Yield: 60%



Suppliers (93)

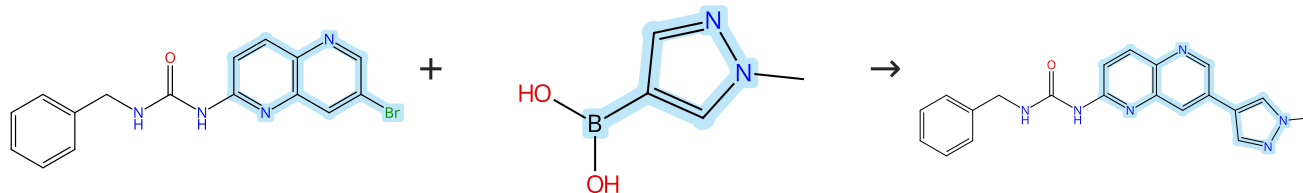
Suppliers (64)

Supplier (1)

31-179-CAS-22683833 Steps: 1 Yield: 60% 1.1 Reagents: 1,8-Diazabicyclo[5.4.0]undec-7-ene Catalysts: [2'-(Amino-κM)[1,1'-biphenyl]-2-yl-κC][dicyclohexyl [2',4',6'-tris(1-methylethyl)[1,1'-biphenyl]-2-yl]phosphine] (methanesulfonato-κO)palladium Solvents: Tetrahydrofuran, Water; 10 min, 110 °C Experimental Protocols	Optimizing chemical reaction conditions using deep learning: a case study for the Suzuki-Miyaura cross-coupling reaction By: Fu, Zunyun; et al Organic Chemistry Frontiers (2020), 7(16), 2269-2277.
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Scheme 135 (1 Reaction)

Steps: 1 Yield: 60%



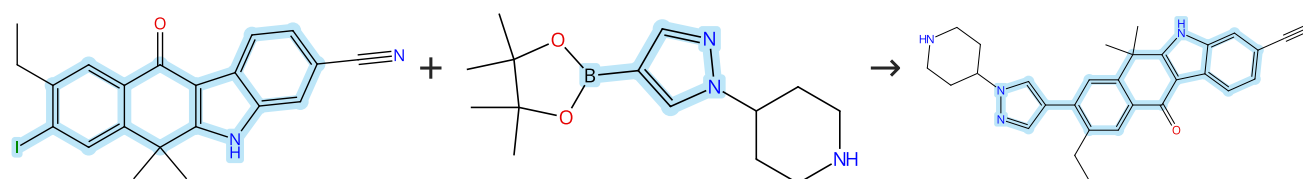
Suppliers (77)

Supplier (1)

31-179-CAS-1955616 Steps: 1 Yield: 60% 1.1 Reagents: Sodium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Dimethylformamide, Water; 2 - 16 h, 80 °C	Discovery of (7-aryl-1,5-naphthyridin-2-yl)ureas as dual inhibitors of ERK2 and Aurora B kinases with antiproliferative activity against cancer cells By: Defaux, Julien; et al Bioorganic & Medicinal Chemistry Letters (2014), 24(16), 3748-3752.
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Scheme 136 (1 Reaction)

Steps: 1 Yield: 60%



Suppliers (56)

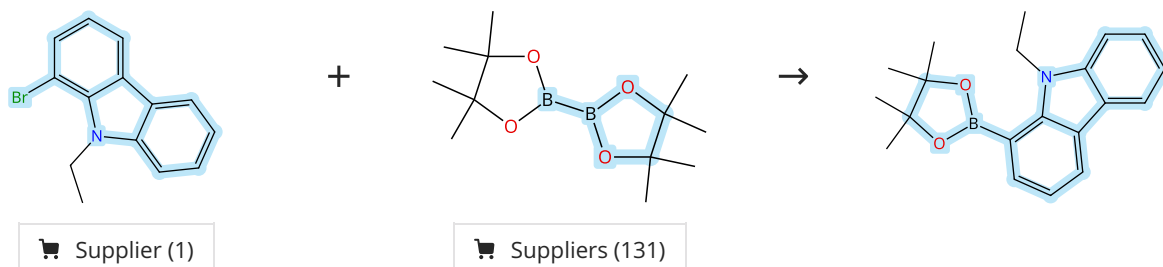
Suppliers (44)

Supplier (1)

31-179-CAS-7278262 Steps: 1 Yield: 60% 1.1 Reagents: Sodium carbonate Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium, Dicyclohexyl[2',4',6'-tris(1,1-dimethylethyl)[1,1'-biphenyl]-2-yl]phosphine Solvents: 1,4-Dioxane, Water; rt → 100 °C; 1 h, 100 °C Experimental Protocols	Discovery of Inhibitors That Overcome the G1202R Anaplastic Lymphoma Kinase Resistance Mutation By: Hatcher, John M.; et al Journal of Medicinal Chemistry (2015), 58(23), 9296-9308.
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Scheme 137 (1 Reaction)

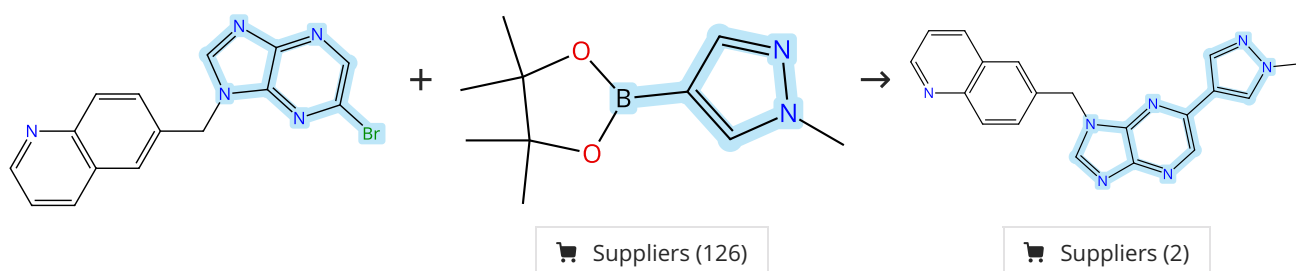
Steps: 1 Yield: 60%



31-614-CAS-40655084 Steps: 1 Yield: 60% 1.1 Reagents: Potassium acetate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: 1,4-Dioxane; 4 h, 85 °C Experimental Protocols	Dibenzoheterole-Fused s-Indacenes By: Saha, Hemonta Kumar; et al Journal of Organic Chemistry (2023), 88(23), 16248-16258.
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Scheme 138 (1 Reaction)

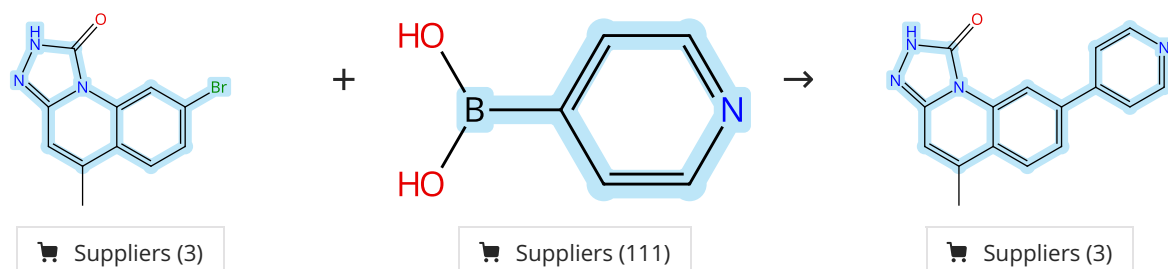
Steps: 1 Yield: 60%



31-179-CAS-16073535 Steps: 1 Yield: 60% 1.1 Reagents: Potassium carbonate Catalysts: Dichloro[1,1'-bis(diphenylphosphino)ferrocene]palladium(II) dichloromethane adduct Solvents: 1,4-Dioxane, Water; 4 h, 110 °C Experimental Protocols	Discovery and optimization of a series of imidazo[4,5-b]pyrazine derivatives as highly potent and exquisitely selective inhibitors of the mesenchymal-epithelial transition factor (c-Met) protein kinase By: Zhao, Fei; et al Bioorganic & Medicinal Chemistry (2016), 24(18), 4281-4290.
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Scheme 139 (1 Reaction)

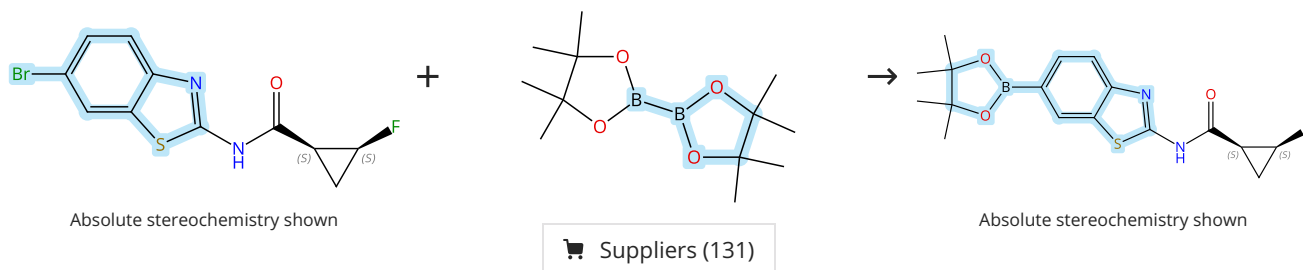
Steps: 1 Yield: 60%



31-179-CAS-1095565	Steps: 1 Yield: 60%	Discovery of a novel class of triazolones as Checkpoint Kinase inhibitors-Hit to lead exploration
1.1 Catalysts: Tetrakis(triphenylphosphine)palladium; 20 min, 165 °C		By: Oza, Vibha; et al Bioorganic & Medicinal Chemistry Letters (2010), 20(17), 5133-5138.

Scheme 140 (1 Reaction)

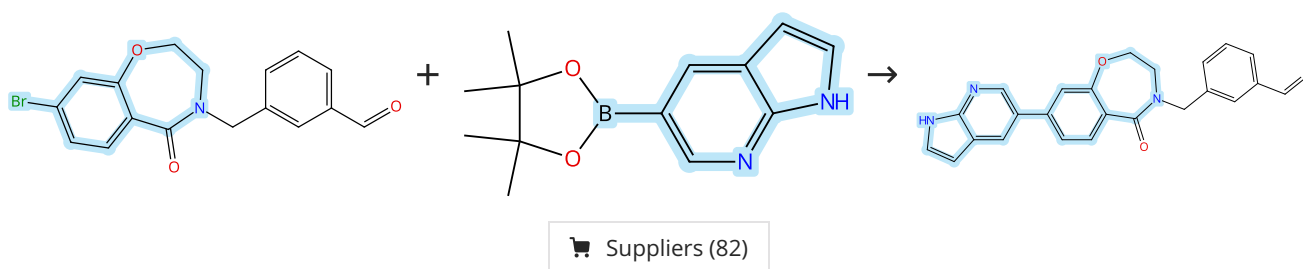
Steps: 1 Yield: 60%



31-614-CAS-40234158	Steps: 1 Yield: 60%	Development of ¹⁸F-Labeled PET Tracer Candidates for Imaging of the Abelson Non-receptor Tyrosine Kinase in Parkinson's Disease
1.1 Reagents: Potassium acetate Catalysts: Dichloro[1,1'-bis(diphenylphosphino)ferrocene] palladium(II) dichloromethane adduct Solvents: 1,4-Dioxane; 12 h, 110 °C		By: Steen, E. Johanna L.; et al Journal of Medicinal Chemistry (2023), 66(18), 12990-13006.
Experimental Protocols		

Scheme 141 (1 Reaction)

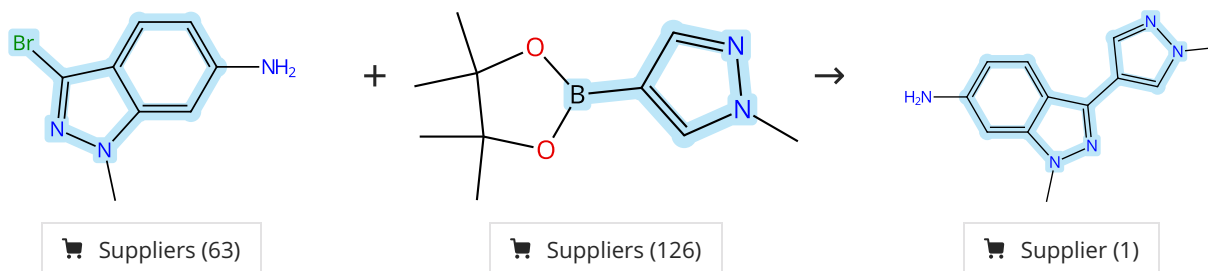
Steps: 1 Yield: 60%



31-614-CAS-38846959	Steps: 1 Yield: 60%	Discovery of 3,4-Dihydrobenzo[f][1,4]oxazepin-5(2H)-one Derivatives as a New Class of Selective TNIK Inhibitors and Evaluation of Their Anti-Colorectal Cancer Effects
1.1 Reagents: Potassium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: 1,4-Dioxane, Water; rt; 12 h, 100 °C		By: Li, Yueshan; et al Journal of Medicinal Chemistry (2022), 65(3), 1786-1807.
Experimental Protocols		

Scheme 142 (1 Reaction)

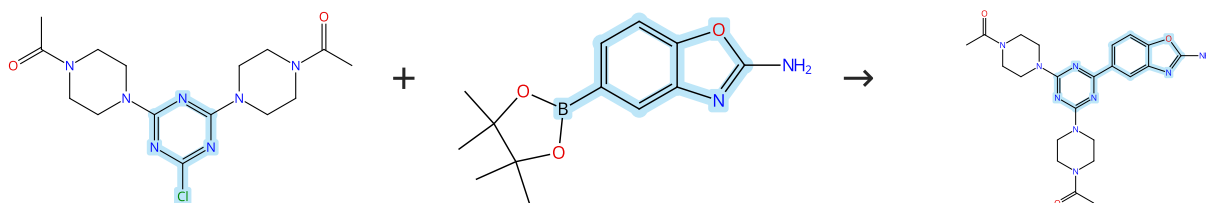
Steps: 1 Yield: 60%



31-614-CAS-32637792	Steps: 1 Yield: 60%	Fragment-based lead discovery of indazole-based compounds as AXL kinase inhibitors
1.1 Reagents: Tripotassium phosphate Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: 1,4-Dioxane, Water; 16 h, 100 °C		By: Ng, Pearly Shuyi; et al Bioorganic & Medicinal Chemistry (2021), 49, 116437.
Experimental Protocols		

Scheme 143 (1 Reaction)

Steps: 1 Yield: 60%

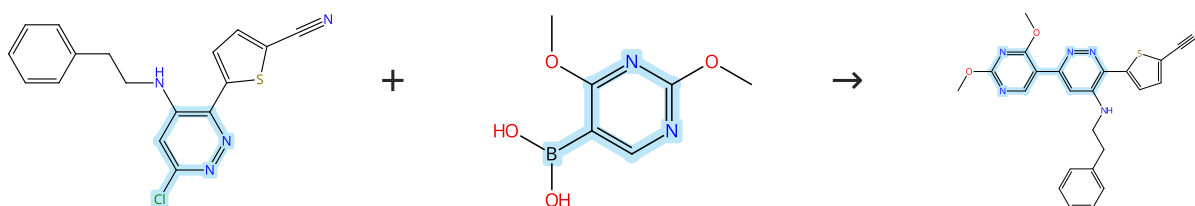


Suppliers (82)

31-614-CAS-40064344	Steps: 1 Yield: 60%	Discovery of Novel Phosphoinositide-3-Kinase α Inhibitors with High Selectivity, Excellent Bioavailability, and Long-Acting Efficacy for Gastric Cancer
1.1 Reagents: Sodium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: 1,2-Dimethoxyethane, Water; 24 h, rt $\rightarrow$ 92 °C		By: Hou, Yi; et al Journal of Medicinal Chemistry (2022), 65(14), 9873-9892.
Experimental Protocols		

Scheme 144 (1 Reaction)

Steps: 1 Yield: 60%

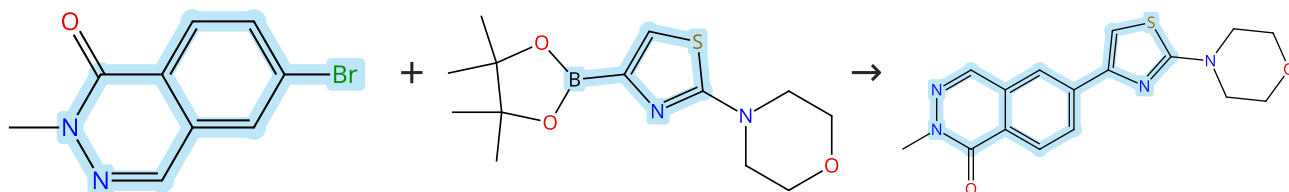


Suppliers (85)

31-614-CAS-42060182	Steps: 1 Yield: 60%	Discovery of Novel 5-(pyridazin-3-yl)pyrimidine-2,4(1H,3H)-dione Derivatives as Potent and Orally Bioavailable Inhibitors Targeting Ecto-5'-Nucleotidase
1.1 Reagents: Potassium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: 1,4-Dioxane, Water; overnight, 90 °C		By: Xu, Yu; et al Journal of Medicinal Chemistry (2024), 67(20), 18491-18511.
Experimental Protocols		

Scheme 145 (1 Reaction)

Steps: 1 Yield: 60%

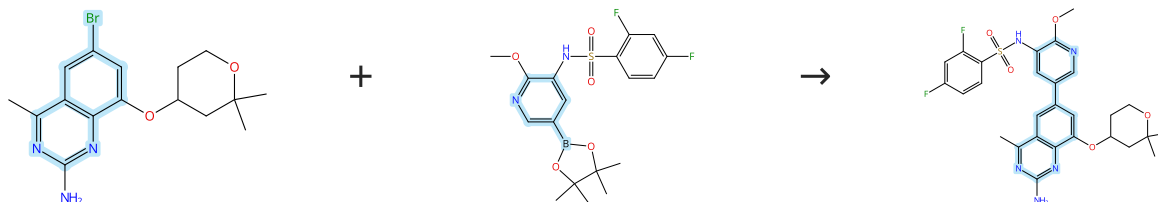


Suppliers (34)

Suppliers (40)

31-179-CAS-16412315 Steps: 1 Yield: 60% 1.1 Reagents: Potassium carbonate Catalysts: X-Phos, (<i>SP</i>-4-4)-[2'-(Amino-κM)[1,1'-biphenyl]-2-yl-κC]chloro[dicyclohexyl[2',4',6'-tris(1-methylethyl)[1,1'-biphenyl]-2-yl]phosphine]palladium Solvents: Tetrahydrofuran, Water; 18 h, 60 °C; 60 °C → rt Experimental Protocols	Rhodium(III)-Catalyzed C-H Functionalization of 1-(2H)-Phthalazinones at C8 By: Huestis, Malcolm P. Journal of Organic Chemistry (2016), 81(24), 12545-12552.
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Scheme 146 (1 Reaction)

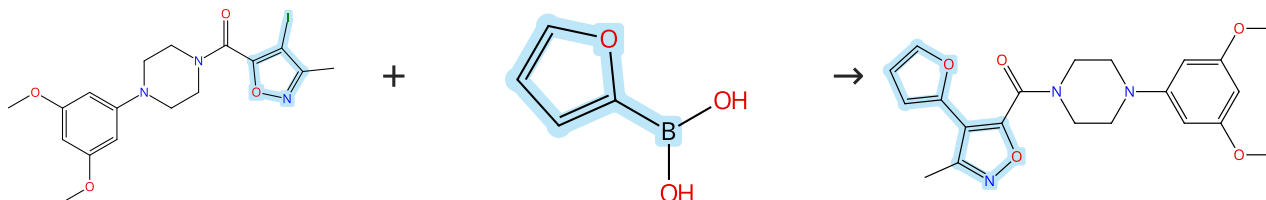
Steps: **1** Yield: **60%**

Suppliers (54)

Supplier (1)

31-179-CAS-19092963 Steps: 1 Yield: 60% 1.1 Reagents: Potassium carbonate Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: 1,4-Dioxane, Water; 5 h, 100 °C; 100 °C → rt 1.2 Reagents: Hydrochloric acid Solvents: Ethyl acetate, Water; pH 5 - 6, rt Experimental Protocols	Discovery and Optimization of 2-Amino-4-methylquinazoline Derivatives as Highly Potent Phosphatidylinositol 3-Kinase Inhibitors for Cancer Treatment By: Lin, Songwen; et al Journal of Medicinal Chemistry (2018), 61(14), 6087-6109.
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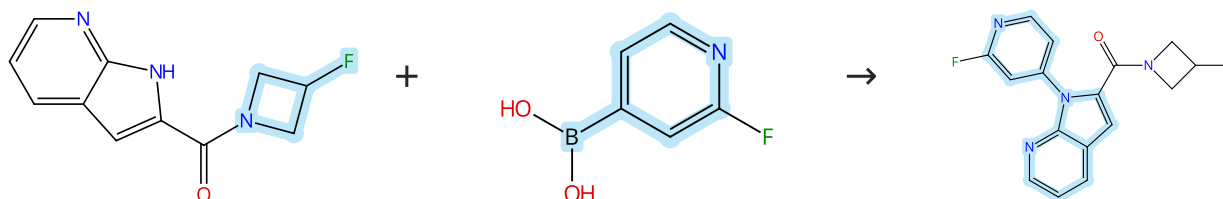
Scheme 147 (1 Reaction)

Steps: **1** Yield: **60%**

Suppliers (93)

31-179-CAS-2077628 Steps: 1 Yield: 60% 1.1 Reagents: Sodium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: 1,4-Dioxane, Water; rt; 10 h, 100 °C Experimental Protocols	Synthesis and Biological Evaluation of Aryloxazole Derivatives as Antimitotic and Vascular-Disrupting Agents for Cancer Therapy By: Choi, Min Jeong; et al Journal of Medicinal Chemistry (2013), 56(22), 9008-9018.
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Scheme 148 (1 Reaction)

Steps: **1** Yield: **60%**

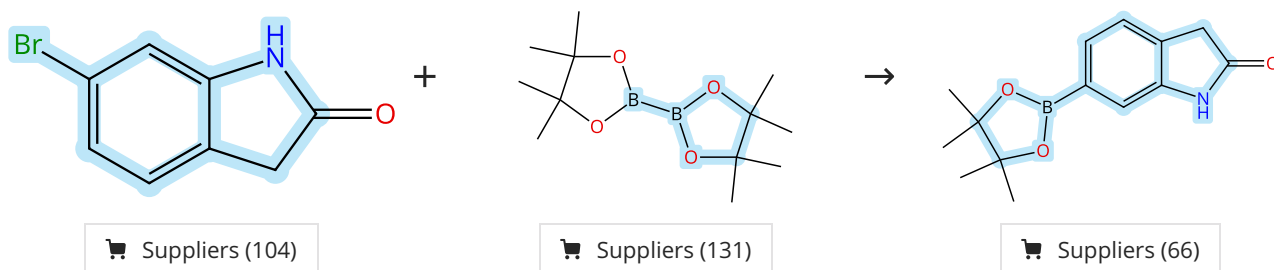
Suppliers (102)

Supplier (1)

31-614-CAS-27877216	Steps: 1 Yield: 60%	Synthesis and SAR Studies of 1 H-Pyrrolo[2,3-b]pyridine-2-carboxamides as Phosphodiesterase 4B (PDE4B) Inhibitors By: Vadukoot, Anish K.; et al ACS Medicinal Chemistry Letters (2020), 11(10), 1848-1854.
1.1 Reagents: Pyridine Catalysts: Cupric acetate Solvents: Dichloromethane; 12 h, rt		

Scheme 149 (1 Reaction)

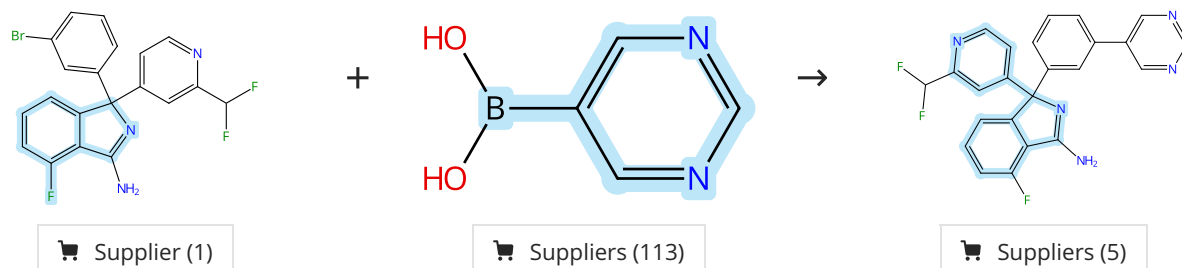
Steps: 1 Yield: 60%



31-614-CAS-38905976	Steps: 1 Yield: 60%	Discovery of potent and selective c-Met inhibitors for MET-amplified hepatocellular carcinoma treatment By: Min, Wenjian; et al European Journal of Medicinal Chemistry (2024), 264, 116025.
1.1 Reagents: Potassium acetate Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: 1,4-Dioxane; overnight, 100 °C		
Experimental Protocols		

Scheme 150 (1 Reaction)

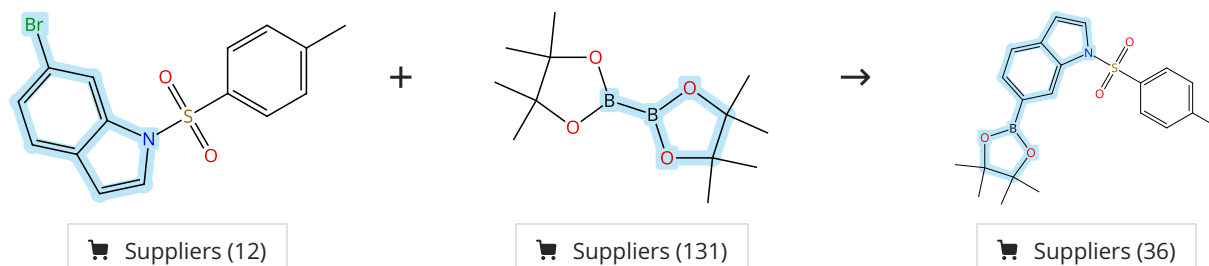
Steps: 1 Yield: 60%



31-179-CAS-4135419	Steps: 1 Yield: 60%	Design and Synthesis of β-Site Amyloid Precursor Protein Cleaving Enzyme (BACE1) Inhibitors with in Vivo Brain Reduction of β-Amyloid Peptides By: Swahn, Britt-Marie; et al Journal of Medicinal Chemistry (2012), 55(21), 9346-9361.
1.1 Reagents: Cesium carbonate Catalysts: Dichloro[1,1'-bis(diphenylphosphino)ferrocene]palladium(II) dichloromethane adduct Solvents: Ethanol, 1,2-Dimethoxyethane, Water; 40 min, 150 °C		
Experimental Protocols		

Scheme 151 (1 Reaction)

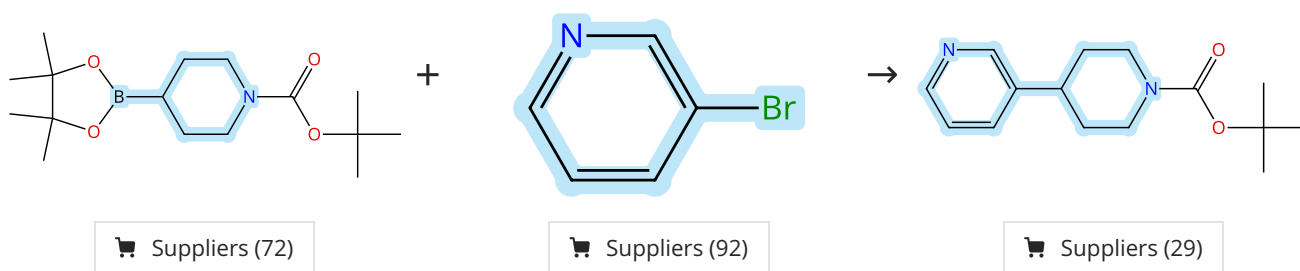
Steps: 1 Yield: 60%



31-614-CAS-26922081 Steps: 1 Yield: 60% 1.1 Reagents: Potassium acetate Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: Dimethylformamide; rt → 100 °C; 3.5 h, 100 °C; 100 °C → rt 1.2 Reagents: Diethyl ether, Ethyl acetate Experimental Protocols	Preparation of no-carrier-added 6-[¹⁸F]fluoro-L-tryptophan via Cu-mediated radiofluorination By: Schaefer, Dominique; et al European Journal of Organic Chemistry (2016), 2016(27), 4621-4628.
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Scheme 152 (1 Reaction)

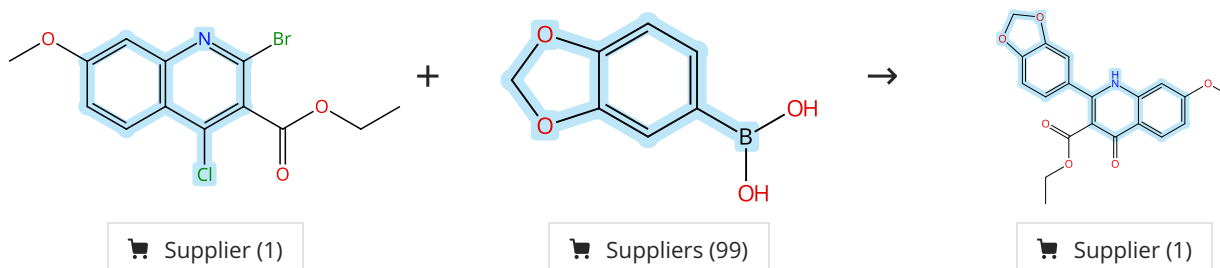
Steps: 1 Yield: 59%



31-614-CAS-33630972 Steps: 1 Yield: 59% 1.1 Reagents: 2-<i>tert</i>-Butyl-1,1,3,3-tetramethylguanidine Catalysts: Quinoline, Tris[2-[4-(1,1-dimethylethyl)-2-pyridinyl-κM]-3,5-difluorophenyl-κC]iridium, Nickel, [4,4'-bis(1,1-dimethylethyl)-2,2'-bipyridine-κN¹,κN¹]dibromo-, (7-4)- Solvents: Tetrahydrofuran; 24 h, rt Experimental Protocols	Ni/Photoredox-Catalyzed C(sp²)-C(sp³) Cross-Coupling of Alkyl Pinacolboronates and (Hetero)Aryl Bromides By: Cauley, Anthony N.; et al Organic Letters (2022), 24(31), 5663-5668.
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Scheme 153 (1 Reaction)

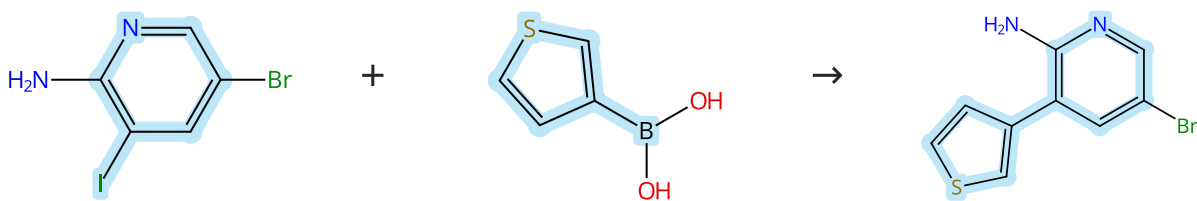
Steps: 1 Yield: 59%



31-179-CAS-13058361 Steps: 1 Yield: 59% 1.1 Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: 1,4-Dioxane; rt 1.2 Reagents: Cesium carbonate Solvents: Water; rt; 3 h, 75 °C; 75 °C → rt 1.3 Reagents: Acetic acid Solvents: Water; 1 h, 120 °C 1.4 Reagents: Ammonium hydroxide Solvents: Water; neutralized, rt Experimental Protocols	Synthesis and structure-activity relationships of antimalarial 4-oxo-3-carboxyl quinolones By: Zhang, Yiqun; et al Bioorganic & Medicinal Chemistry (2010), 18(7), 2756-2766.
--	---

Scheme 154 (1 Reaction)

Steps: 1 Yield: 59%



Suppliers (98)

Suppliers (101)

Suppliers (4)

31-179-CAS-9444767

Steps: 1 Yield: 59%

Identification and characterisation of 2-aminopyridine inhibitors of checkpoint kinase 2

1.1 Reagents: Sodium carbonate

Catalysts: Tetrakis(triphenylphosphine)palladium

Solvents: 1,2-Dimethoxyethane, Water; 35 min, 135 °C

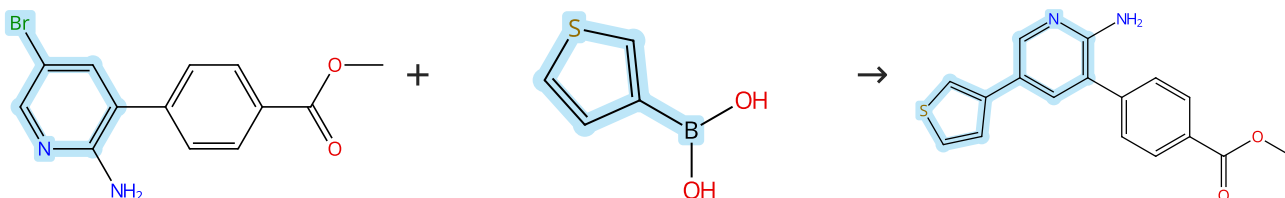
By: Hilton, Stephen; et al

Bioorganic & Medicinal Chemistry (2010), 18(2), 707-718.

Experimental Protocols

Scheme 155 (1 Reaction)

Steps: 1 Yield: 59%



Supplier (1)

Suppliers (101)

Suppliers (2)

31-179-CAS-1919027

Steps: 1 Yield: 59%

Identification and characterisation of 2-aminopyridine inhibitors of checkpoint kinase 2

1.1 Reagents: Sodium carbonate

Catalysts: Tetrakis(triphenylphosphine)palladium

Solvents: 1,2-Dimethoxyethane, Water; 30 min, 135 °C

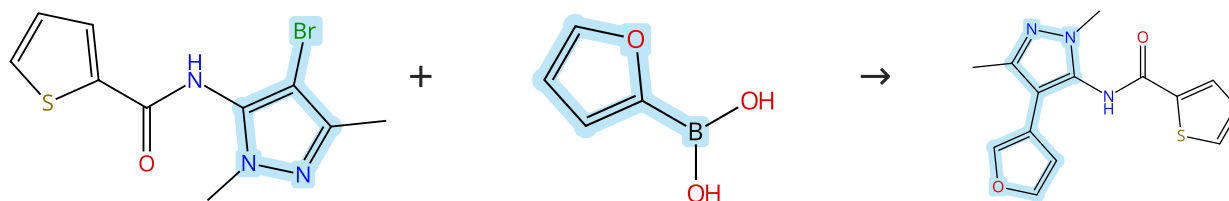
By: Hilton, Stephen; et al

Bioorganic & Medicinal Chemistry (2010), 18(2), 707-718.

Experimental Protocols

Scheme 156 (1 Reaction)

Steps: 1 Yield: 59%



Supplier (1)

Suppliers (93)

31-614-CAS-29501475

Steps: 1 Yield: 59%

The Suzuki-Miyaura Cross-Coupling Reaction of Halogenated Aminopyrazoles: Method Development, Scope, and Mechanism of Dehalogenation Side Reaction

1.1 Reagents: Potassium carbonate

Catalysts: X-Phos, (SP-4-4)-[2'-(Amino-κM)[1,1'-biphenyl]-2-yl-κC]chloro[dicyclohexyl[2',4',6'-tris(1-methylethyl)[1,1'-biphenyl]-2-yl]phosphine]palladium

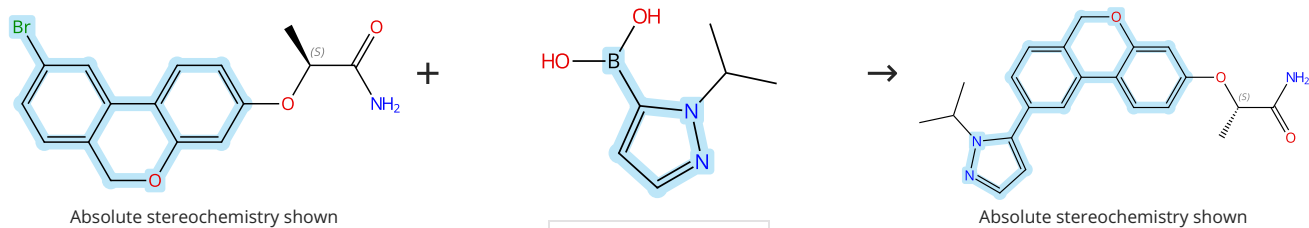
Solvents: Ethanol, Water; 20 min, 110 °C

By: Jedinak, Lukas; et al

Journal of Organic Chemistry (2017), 82(1), 157-169.

Scheme 157 (1 Reaction)

Steps: 1 Yield: 59%



Suppliers (55)

31-614-CAS-42202572

Steps: 1 Yield: 59%

Rational Design of a Novel 6H-Benzo[c]chromen Series as Selective PI3Kα Inhibitors

By: Shi, Xue; et al

Journal of Medicinal Chemistry (2024), 67(17), 15387-15410.

1.1 Reagents: Potassium carbonate

Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium

Solvents: 1,4-Dioxane, Water; 12 h, rt → 100 °C

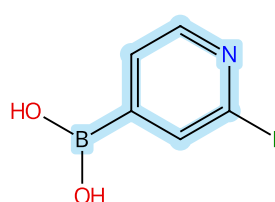
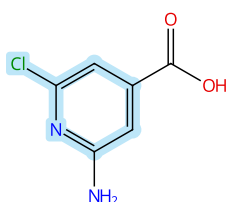
Experimental Protocols

Scheme 158 (1 Reaction)

Steps: 1 Yield: 59%

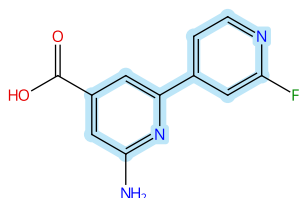
3053724-30-1

Image Not Available



Suppliers (62)

Suppliers (102)



DNA-encoded library conjugates

31-614-CAS-41796237

Steps: 1 Yield: 59%

Groebke-Blackburn-Bienayme Reaction for DNA-Encoded Library Technology

By: An, Yujin; et al

Organic Letters (2023), 25(24), 4445-4450.

1.1 Reagents: Diisopropylethylamine, *O*-(7-Azabenzotriazol-1-yl)-*N,N,N,N*-tetramethyluronium hexafluorophosphate

Solvents: Dimethylacetamide, Water; overnight, pH 9.5, 40 °C

1.2 Reagents: Cesium hydroxide

Catalysts: Palladate(1-), [2'-(amino-κM)[1,1'-biphenyl]-2-yl-κC]chloro[2'-(dicyclohexylphosphino-κP)-2,6-dimethoxy[1,1'-biphenyl]-3-sulfonato]-, sodium (1:1)

Solvents: 1,4-Dioxane, Dimethylacetamide, Water; 15 min, 80 °C; 10 min, rt

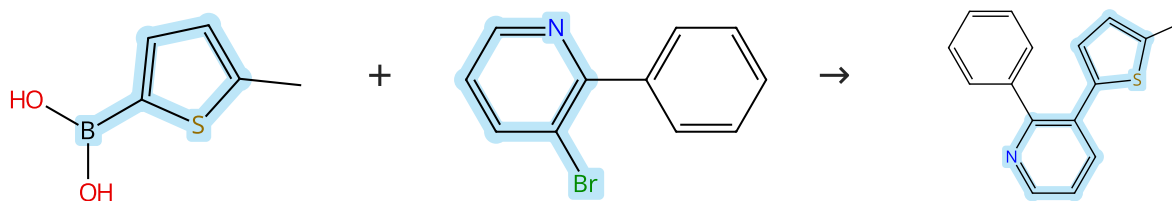
1.3 Reagents: Sodium diethyldithiocarbamate

Solvents: Water; 30 min, 60 °C

Experimental Protocols

Scheme 159 (1 Reaction)

Steps: 1 Yield: 59%



Suppliers (84)

Suppliers (67)

31-179-CAS-20726197

Steps: 1 Yield: 59%

Syntheses of Benzofuranoquinolines and Analogues via Photoinduced Acceptorless Dehydrogenative Annulation of o-Phenylfuranylpriidines

By: Fan, Jinming; et al

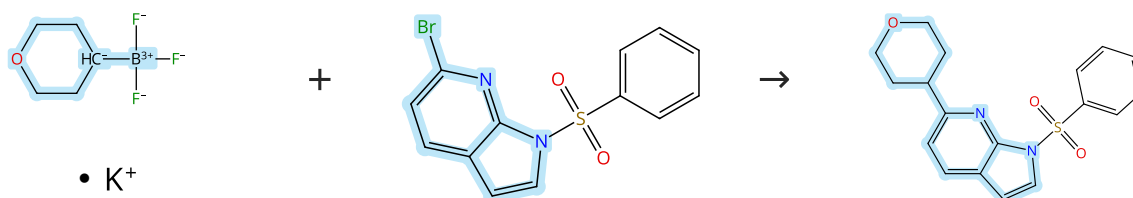
Organic Letters (2019), 21(22), 9183-9187.

1.1 **Reagents:** Cesium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: 1,4-Dioxane, Water; 6 - 12 h, 85 °C

Experimental Protocols

Scheme 160 (1 Reaction)

Steps: 1 Yield: 59%



Suppliers (50)

Suppliers (41)

31-179-CAS-21154887

Steps: 1 Yield: 59%

Synthesis of Cycloalkyl Substituted 7-Azaindoles via Photoredox Nickel Dual Catalytic Cross-Coupling in Batch and Continuous Flow

By: Palaychuk, Natalie; et al

Organic Letters (2016), 18(23), 6180-6183.

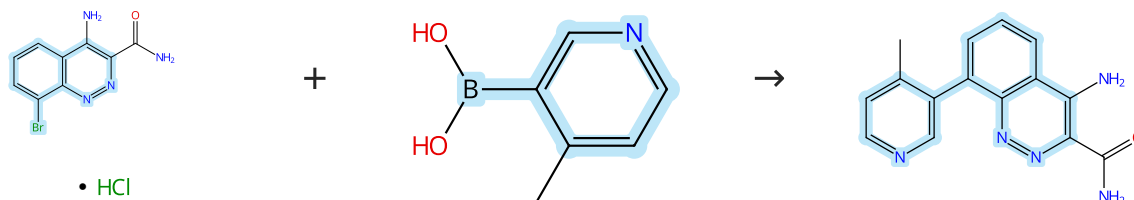
1.1 **Catalysts:** Dichloro[1,2-di(methoxy-κO)ethane]nickel, 4,4'-Bis(1,1-dimethylethyl)-2,2'-bipyridine
Solvents: Dimethylacetamide; 5 min, rt

1.2 **Reagents:** 2,6-Lutidine
Catalysts: Iridium(1+), [4,4'-bis(1,1-dimethylethyl)-2,2'-bipyridine-κN¹,κN¹']bis[3,5-difluoro-2-[5-(trifluoromethyl)-2-pyridinyl-κM]phenyl-κC]-, (OC-6-33)-, hexafluorophosphate(1-)(1:1)
Solvents: 1,4-Dioxane; 30 °C

Experimental Protocols

Scheme 161 (1 Reaction)

Steps: 1 Yield: 59%



Suppliers (7)

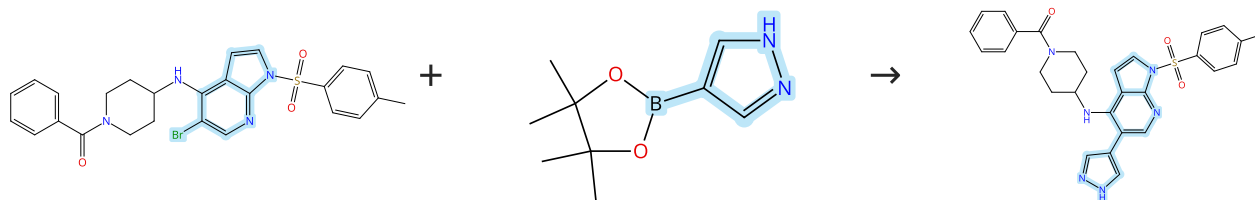
Suppliers (84)

Supplier (1)

31-179-CAS-11207177	Steps: 1 Yield: 59%	Fragment-Based Discovery of a Small Molecule Inhibitor of Bruton's Tyrosine Kinase
1.1 Reagents: Sodium carbonate Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: 1,4-Dioxane, Water; 1 h, 140 °C		By: Smith, Christopher R.; et al Journal of Medicinal Chemistry (2015), 58(14), 5437-5444.
Experimental Protocols		

Scheme 162 (1 Reaction)

Steps: 1 Yield: 59%

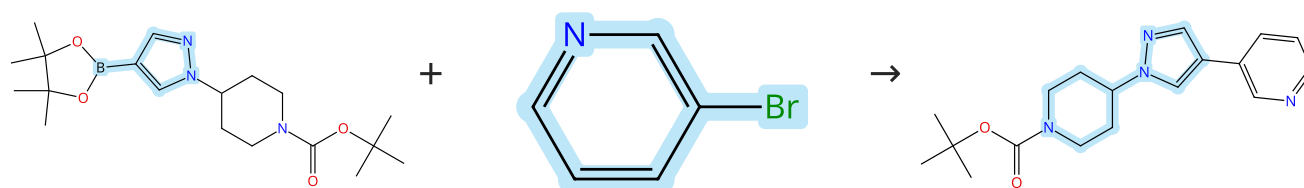


Suppliers (108)

31-614-CAS-36738627	Steps: 1 Yield: 59%	Discovery of C-5 Pyrazole-Substituted Pyrrolopyridine Derivatives as Potent and Selective Inhibitors for Janus Kinase 1
1.1 Reagents: Potassium carbonate Catalysts: (<i>SP</i> 4-1)-Bis[4-[bis(1,1-dimethylethyl)phosphino- κ P]- <i>N,N</i> -dimethylbenzenamine]dichloropalladium Solvents: Isobutanol, Water; 10 h, 60 °C		By: Lang, Jia-Jia; et al Journal of Medicinal Chemistry (2023), 66(10), 6725-6742.
Experimental Protocols		

Scheme 163 (1 Reaction)

Steps: 1 Yield: 59%



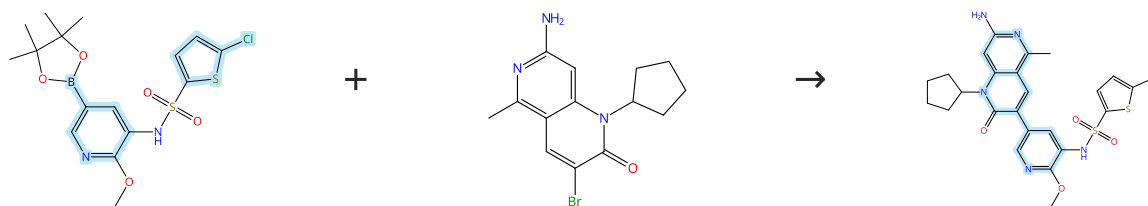
Suppliers (92)

Suppliers (92)

31-614-CAS-44471293	Steps: 1 Yield: 59%	"Naked Nickel"-Catalyzed Heteroaryl-Heteroaryl Suzuki-Miyaura Coupling
1.1 Reagents: Tripotassium phosphate Catalysts: Stereoisomer of tris[1,1'-[η^2 -(1 <i>E</i>)-1,2-ethenediyl]bis[4-(trifluoromethyl)benzene]]nickel Solvents: Dimethylacetamide; 16 h, 60 °C; 60 °C → 25 °C		By: Saeb, Rakan; et al Angewandte Chemie, International Edition (2025), 64(22), e202424051.
1.2 Reagents: Lithium chloride Solvents: <i>tert</i> -Butyl methyl ether, Water; 25 °C		
Experimental Protocols		

Scheme 164 (1 Reaction)

Steps: 1 Yield: 59%



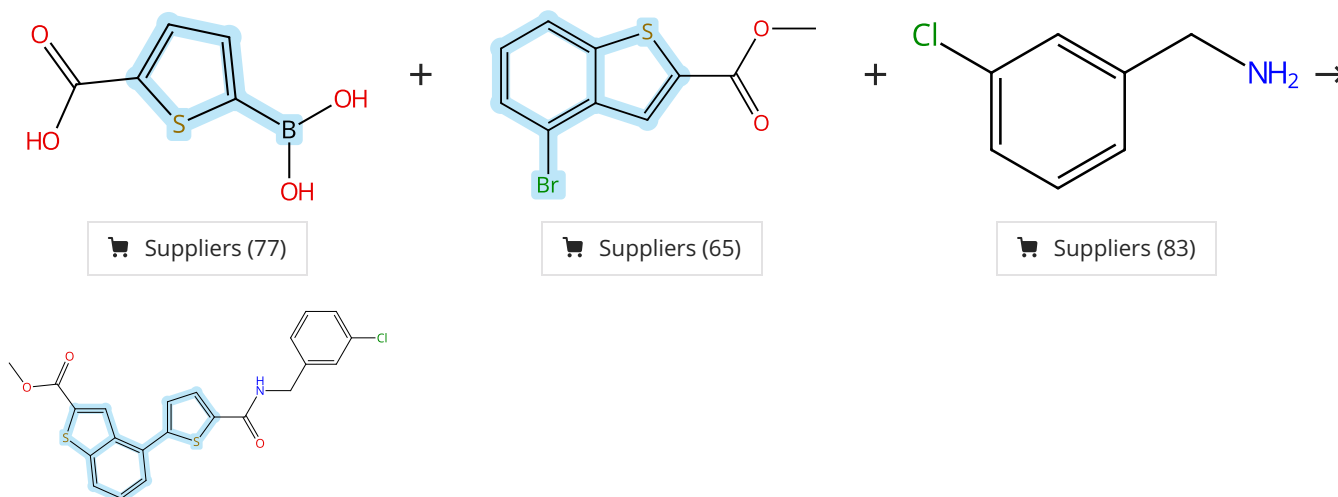
Suppliers (12)

Supplier (1)

31-179-CAS-1358660 Steps: 1 Yield: 59% 1.1 Reagents: Potassium carbonate Catalysts: Dichlorobis(triphenylphosphine)palladium Solvents: Dimethylformamide, Water; 3 h, 100 °C	Identification of novel 7-amino-5-methyl-1,6-naphthyridin-2(1H)-one derivatives as potent PI3K/mTOR dual inhibitors By: Lin, Songwen; et al Bioorganic & Medicinal Chemistry Letters (2014), 24(3), 790-793.
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Scheme 165 (1 Reaction)

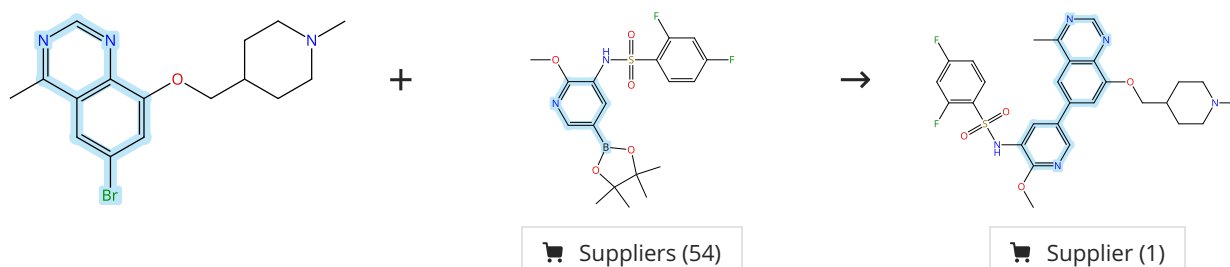
Steps: 1 Yield: 59%



31-614-CAS-32079067 Steps: 1 Yield: 59% 1.1 Reagents: Sodium carbonate, Lithium chloride Catalysts: Dichlorobis(triphenylphosphine)palladium Solvents: 1,4-Dioxane, Water; 2 h, 100 °C 1.2 Reagents: Hydrochloric acid Solvents: Water 1.3 Reagents: Diisopropylethylamine, <i>O</i>-(7-Azabenzotriazol-1-yl)-<i>N,N,N,N</i>-tetramethyluronium hexafluorophosphate Solvents: Dichloromethane; 30 min, rt 1.4 overnight, rt 1.5 Reagents: Sodium bicarbonate Solvents: Water; rt Experimental Protocols	Discovery of novel benzothiophene derivatives as potent and narrow spectrum inhibitors of DYRK1A and DYRK1B By: Segretti, Natanael D.; et al Bioorganic & Medicinal Chemistry Letters (2022), 68, 128764.
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Scheme 166 (1 Reaction)

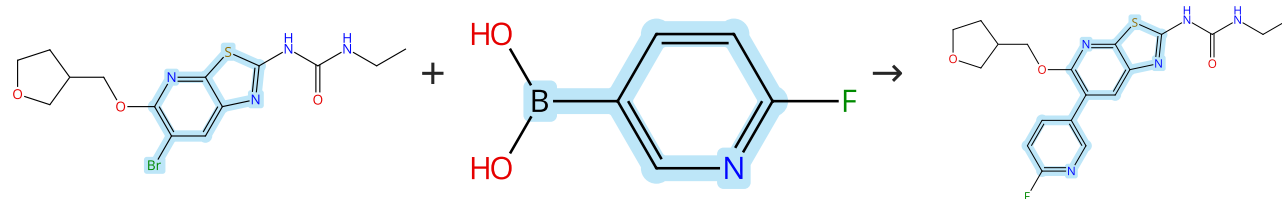
Steps: 1 Yield: 59%



31-179-CAS-20636898 Steps: 1 Yield: 59% 1.1 Reagents: Potassium carbonate Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: 1,4-Dioxane, Water; 5 h, 100 °C; 100 °C → rt 1.2 Reagents: Hydrochloric acid Solvents: Water; pH 5 - 6, rt	Discovery of 4-Methylquinazoline Based PI3K Inhibitors for the Potential Treatment of Idiopathic Pulmonary Fibrosis By: Lin, Songwen; et al Journal of Medicinal Chemistry (2019), 62(19), 8873-8879.
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Scheme 167 (1 Reaction)

Steps: 1 Yield: 59%



Supplier (1)

Suppliers (95)

Suppliers (2)

31-179-CAS-465987

Steps: 1 Yield: 59%

Thiazolopyridine Ureas as Novel Antitubercular Agents Acting through Inhibition of DNA Gyrase B

By: Kale, Manoj G.; et al

Journal of Medicinal Chemistry (2013), 56(21), 8834-8848.

1.1 Reagents: Sodium bicarbonate

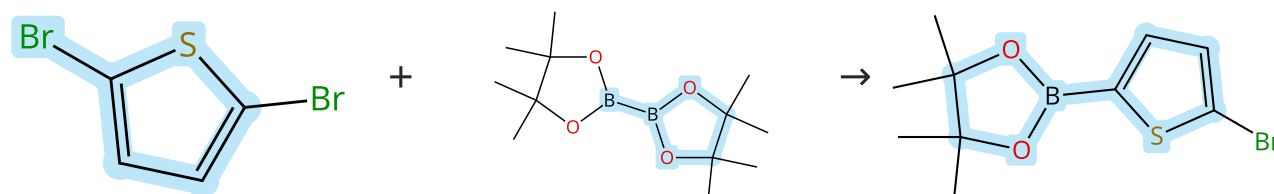
Catalysts: Tetrakis(triphenylphosphine)palladium

Solvents: 1,2-Dimethoxyethane, Water; 80 min, 115 °C

Experimental Protocols

Scheme 168 (1 Reaction)

Steps: 1 Yield: 59%



Suppliers (75)

Suppliers (131)

Suppliers (60)

31-614-CAS-31810526

Steps: 1 Yield: 59%

Synchronous Preparation of Length-Controllable 1D Nanoparticles via Crystallization-Driven In Situ Nanoparticlization of Conjugated Polymers

By: Hwang, Soon-Hyeok; et al

Journal of the American Chemical Society (2022), 144(13), 5921-5929.

1.1 Reagents: Butyllithium

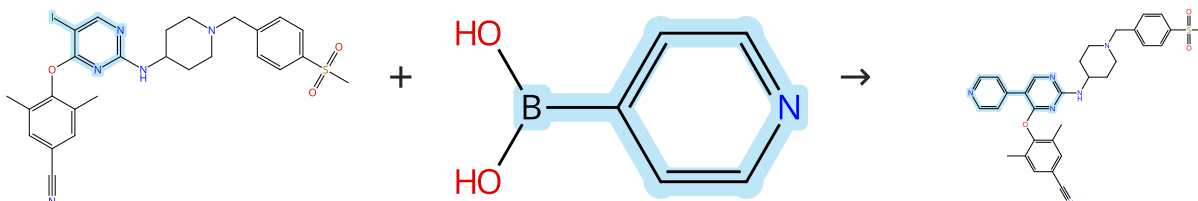
Solvents: Tetrahydrofuran, Hexane; -78 °C; 1 h, -78 °C

1.2 30 min, -78 °C; 3 h, rt

Experimental Protocols

Scheme 169 (1 Reaction)

Steps: 1 Yield: 59%



Suppliers (111)

Supplier (1)

31-614-CAS-31783316

Steps: 1 Yield: 59%

2,4,5-Trisubstituted Pyrimidines as Potent HIV-1 NNRTIs: Rational Design, Synthesis, Activity Evaluation, and Crystallographic Studies

By: Kang, Dongwei; et al

Journal of Medicinal Chemistry (2021), 64(7), 4239-4256.

1.1 Reagents: Sodium carbonate

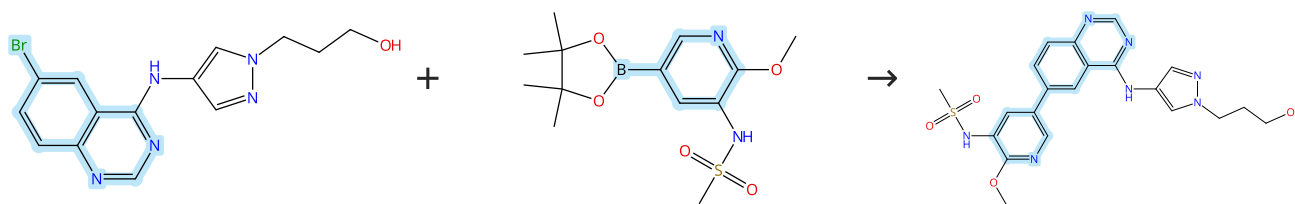
Catalysts: Tetrakis(triphenylphosphine)palladium

Solvents: Dimethylformamide, Water; 6 - 10 h, 100 °C

Experimental Protocols

Scheme 170 (1 Reaction)

Steps: 1 Yield: 59%



Suppliers (60)

Supplier (1)

31-179-CAS-20385960

Steps: 1 Yield: 59%

1.1 Reagents: Cesium carbonate
 Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium
 Solvents: Dimethylformamide; 4 h, 90 °C

Experimental Protocols

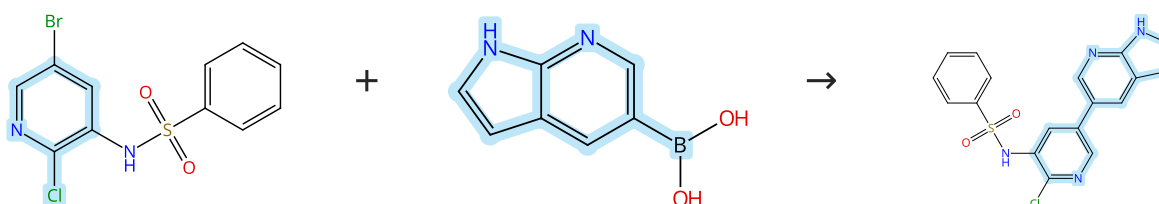
Design, synthesis, and biological evaluation of some novel 4-aminoquinazolines as Pan-PI3K inhibitors

By: Ding, Huai-Wei; et al

Bioorganic & Medicinal Chemistry (2019), 27(13), 2729-2740.

Scheme 171 (1 Reaction)

Steps: 1 Yield: 59%



Suppliers (29)

Suppliers (69)

Supplier (1)

31-614-CAS-24249962

Steps: 1 Yield: 59%

1.1 Reagents: Potassium acetate
 Catalysts: Tetrakis(triphenylphosphine)palladium
 Solvents: 1,4-Dioxane; 8 h, rt → 80 °C

Experimental Protocols

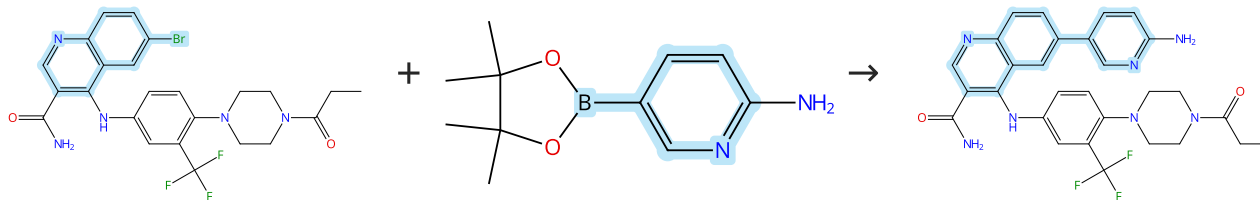
Discovery of 6'-chloro-N-methyl-5'-(phenylsulfonamido)-[3,3'-bipyridine]-5-carboxamide (CHMFL-PI4K-127) as a novel Plasmodium falciparum PI(4)K inhibitor with potent antimalarial activity against both blood and liver stages of Plasmodium

By: Liang, Xiaofei; et al

European Journal of Medicinal Chemistry (2020), 188, 112012.

Scheme 172 (1 Reaction)

Steps: 1 Yield: 59%



Suppliers (93)

Supplier (1)

31-179-CAS-10043489

Steps: 1 Yield: 59%

1.1 Reagents: Potassium carbonate
 Catalysts: Tetrakis(triphenylphosphine)palladium
 Solvents: 1,4-Dioxane, Water; 8 h, 100 °C

Experimental Protocols

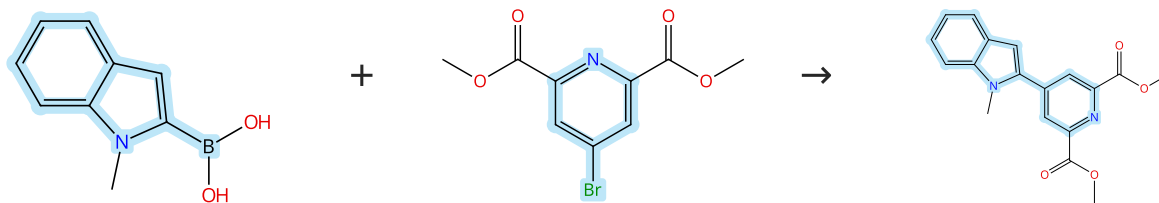
Discovery of novel quinoline-based mTOR inhibitors via introducing intra-molecular hydrogen bonding scaffold (iMHBS): The design, synthesis and biological evaluation

By: Ma, Xiaodong; et al

Bioorganic & Medicinal Chemistry (2015), 23(24), 7585-7596.

Scheme 173 (1 Reaction)

Steps: 1 Yield: 59%



Suppliers (65)

Suppliers (67)

Supplier (1)

31-179-CAS-4361595

Steps: 1 Yield: 59%

STAT6 phosphorylation inhibitors block eotaxin-3 secretion in bronchial epithelial cells

1.1 Reagents: Sodium bicarbonate

Catalysts: Tetrakis(triphenylphosphine)palladium

Solvents: Methanol, Toluene, Water; 19 h, 70 °C; 70 °C → rt

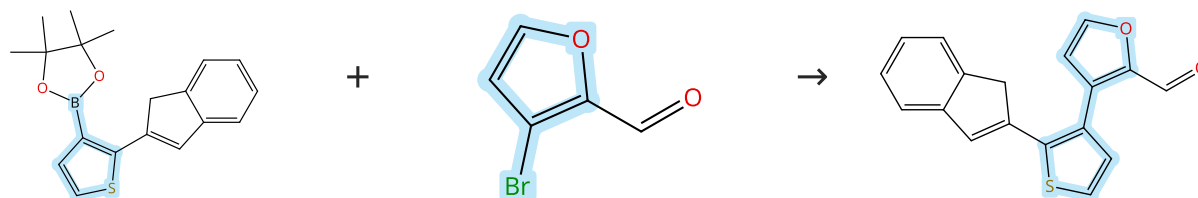
By: Zhou, Li; et al

Bioorganic & Medicinal Chemistry (2012), 20(2), 750-758.

Experimental Protocols

Scheme 174 (1 Reaction)

Steps: 1 Yield: 59%



Suppliers (78)

31-614-CAS-36330036

Steps: 1 Yield: 59%

Modular Synthesis of Structurally Diverse Azulene-Embedded Polycyclic Aromatic Hydrocarbons by Knoevenagel-Type Condensation

1.1 Reagents: Potassium carbonate

Catalysts: Tris(dibenzylideneacetone)dipalladium, Bis[2-(diphenylphosphino)phenyl] ether

Solvents: Toluene, 1,4-Dioxane, Water; 16 h, 85 °C

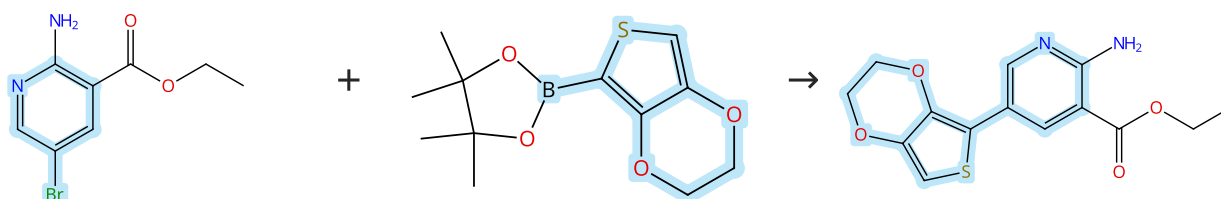
By: Liu, Renxiang; et al

Angewandte Chemie, International Edition (2023), 62(21), e202219091.

Experimental Protocols

Scheme 175 (1 Reaction)

Steps: 1 Yield: 59%



Suppliers (55)

Suppliers (67)

Supplier (1)

31-179-CAS-7606698

Steps: 1 Yield: 59%

Identification and characterisation of 2-aminopyridine inhibitors of checkpoint kinase 2

1.1 Reagents: Sodium carbonate

Solvents: 1,2-Dimethoxyethane, Water; 40 min, 135 °C

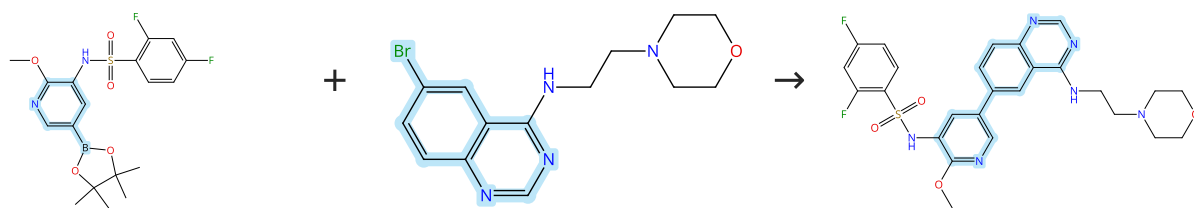
By: Hilton, Stephen; et al

Bioorganic & Medicinal Chemistry (2010), 18(2), 707-718.

Experimental Protocols

Scheme 176 (1 Reaction)

Steps: 1 Yield: 59%



Suppliers (54)

Suppliers (5)

Supplier (1)

31-179-CAS-18935159

Steps: 1 Yield: 59%

Novel 4-aminoquinazoline derivatives induce growth inhibition, cell cycle arrest and apoptosis via P I3Ka inhibition

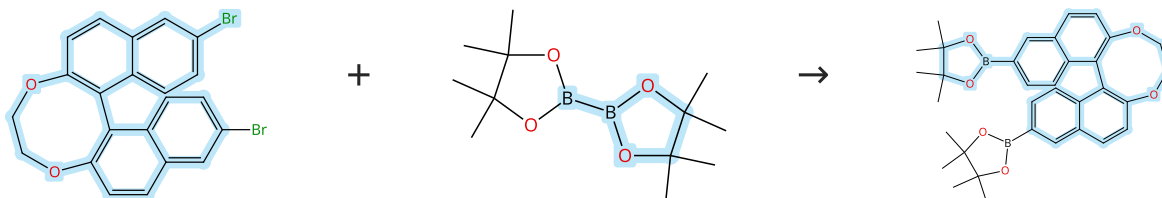
By: Fan, Yan-Hua; et al

Bioorganic & Medicinal Chemistry (2018), 26(8), 1675-1685.

1.1 **Reagents:** Cesium carbonate
Catalysts: Dichloro[1,1'-bis(diphenylphosphino)ferrocene] palladium(II) dichloromethane adduct
Solvents: Dimethylformamide, Water; 8 h, 90 °C

Scheme 177 (1 Reaction)

Steps: 1 Yield: 59%



Suppliers (131)

31-614-CAS-39328467

Steps: 1 Yield: 59%

2,2'-Tethered Binaphthyl-Embedded One-Handed Helical Ladder Polymers: Impact of the Tether Length on Helical Geometry and Chiroptical Property

By: Ikai, Tomoyuki; et al

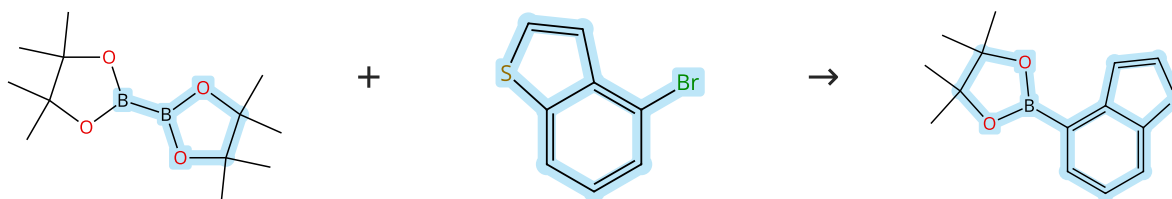
Angewandte Chemie, International Edition (2024), 63(10), e202318712.

1.1 **Reagents:** Potassium acetate
Catalysts: Dichloro[1,1'-bis(diphenylphosphino)ferrocene] palladium(II) dichloromethane adduct
Solvents: 1,4-Dioxane; 22 h, 80 °C

Experimental Protocols

Scheme 178 (1 Reaction)

Steps: 1 Yield: 59%



Suppliers (131)

Suppliers (72)

Suppliers (63)

31-614-CAS-36071296

Steps: 1 Yield: 59%

Heterocyclic Chromophore Amphiphiles and their Supramolecular Polymerization

By: Djordjevic, Luka; et al

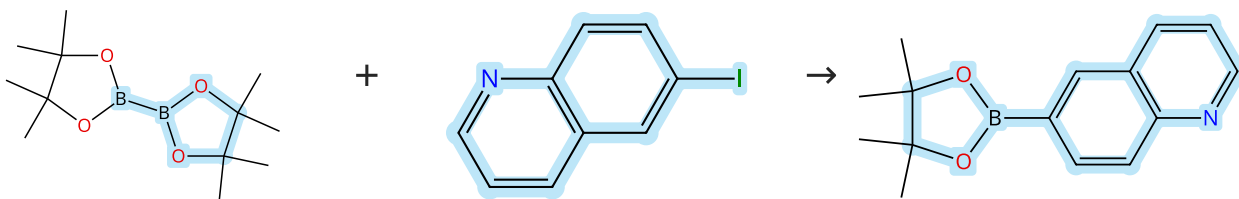
Angewandte Chemie, International Edition (2023), 62(17), e202214997.

1.1 **Reagents:** Potassium acetate
Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium
Solvents: 1,4-Dioxane; 16 h, 80 °C

Experimental Protocols

Scheme 179 (1 Reaction)

Steps: 1 Yield: 58%



Suppliers (131)

Suppliers (92)

Suppliers (74)

31-614-CAS-34396096

Steps: 1 Yield: 58%

Solvent Anions Enable Photoinduced Borylation and Phosphonation of Aryl Halides via EDA Complexes

1.1 Reagents: Dimethylformamide, Sodium hydride

Solvents: Acetonitrile; 24 h, 35 °C

1.2 Reagents: Water; 35 °C

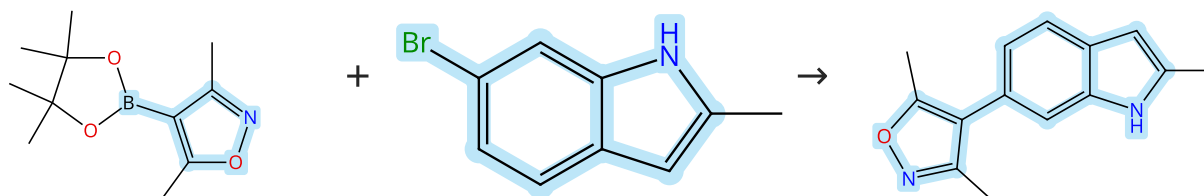
By: Pan, Lei; et al

Organic Letters (2022), 24(35), 6466-6471.

Experimental Protocols

Scheme 180 (1 Reaction)

Steps: 1 Yield: 58%



Suppliers (93)

Suppliers (62)

Suppliers (2)

31-179-CAS-2267622

Steps: 1 Yield: 58%

Discovery and Optimization of Small-Molecule Ligands for the CBP/p300 Bromodomains

1.1 Reagents: Sodium bicarbonate

Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium

Solvents: 1,2-Dimethoxyethane, Water; 24 h, 100 °C

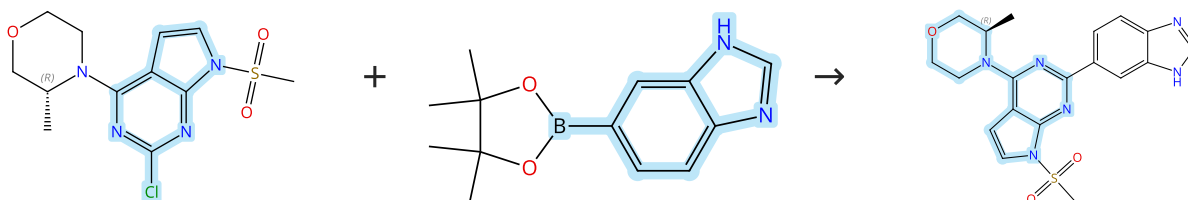
By: Hay, Duncan A.; et al

Journal of the American Chemical Society (2014), 136(26), 9308-9319.

Experimental Protocols

Scheme 181 (1 Reaction)

Steps: 1 Yield: 58%



Absolute stereochemistry shown

Suppliers (77)

Absolute stereochemistry shown

31-614-CAS-41586059

Steps: 1 Yield: 58%

Discovery of Preclinical Candidate AD1058 as a Highly Potent, Selective, and Brain-Penetrant ATR Inhibitor for the Treatment of Advanced Malignancies

1.1 Reagents: Cesium carbonate

Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium

Solvents: 1,4-Dioxane, Water; 4 h, 100 °C

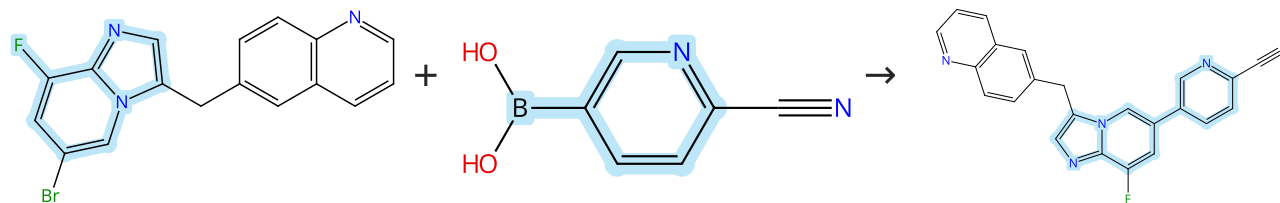
By: Liu, Zhi; et al

Journal of Medicinal Chemistry (2024), 67(15), 12735-12759.

Experimental Protocols

Scheme 182 (1 Reaction)

Steps: 1 Yield: 58%



Suppliers (68)

Supplier (1)

31-179-CAS-5448986

Steps: 1 Yield: 58%

Design, Synthesis, and Biological Evaluation of Novel Imidazo [1,2-a]pyridine Derivatives as Potent c-Met Inhibitors

By: Li, Chunpu; et al

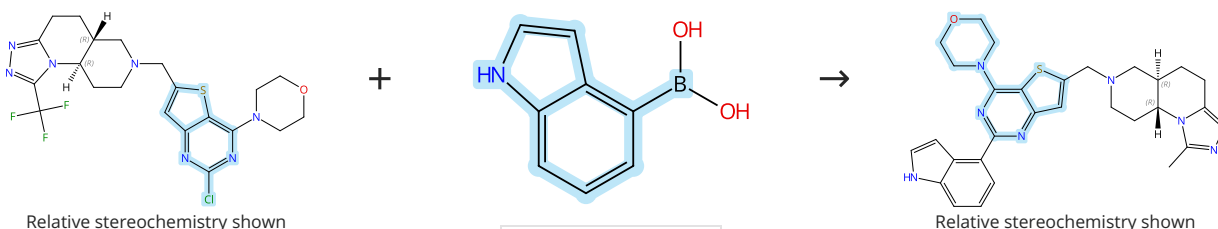
ACS Medicinal Chemistry Letters (2015), 6(5), 507-512.

1.1 **Reagents:** Sodium carbonate
Catalysts: Dichloro[1,1'-bis(diphenylphosphino)ferrocene] palladium(II) dichloromethane adduct
Solvents: Dimethylformamide, Water; 16 h, 80 °C

Experimental Protocols

Scheme 183 (1 Reaction)

Steps: 1 Yield: 58%



Relative stereochemistry shown

Relative stereochemistry shown

Suppliers (79)

Supplier (1)

31-179-CAS-16619302

Steps: 1 Yield: 58%

Design and Elaboration of a Tractable Tricyclic Scaffold To Synthesize Druglike Inhibitors of Dipeptidyl Peptidase-4 (DPP-4), Antagonists of the C-C Chemokine Receptor Type 5 (CCR5), and Highly Potent and Selective Phosphoinositol-3 Kinase δ (PI3Kδ) Inhibitors

By: Schwehm, Carolin; et al

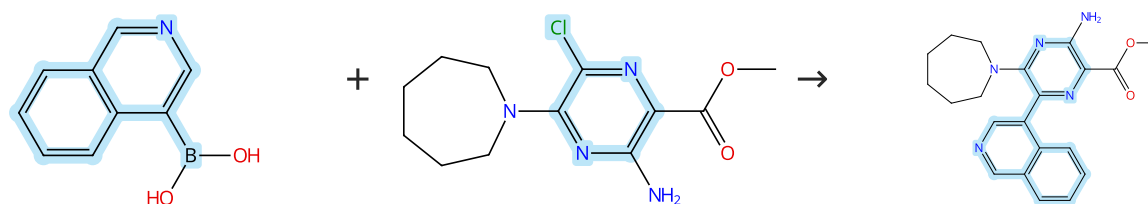
Journal of Medicinal Chemistry (2017), 60(4), 1534-1554.

1.1 **Reagents:** Sodium carbonate
Catalysts: Dichlorobis(triphenylphosphine)palladium
Solvents: Ethanol, Water; 45 min, 125 °C

Experimental Protocols

Scheme 184 (1 Reaction)

Steps: 1 Yield: 58%



Suppliers (75)

Suppliers (2)

31-179-CAS-19303260

Steps: 1 Yield: 58%

6-Substituted Hexamethylene Amiloride (HMA) Derivatives as Potent and Selective Inhibitors of the Human Urokinase Plasminogen Activator for Use in Cancer

By: Buckley, Benjamin J.; et al

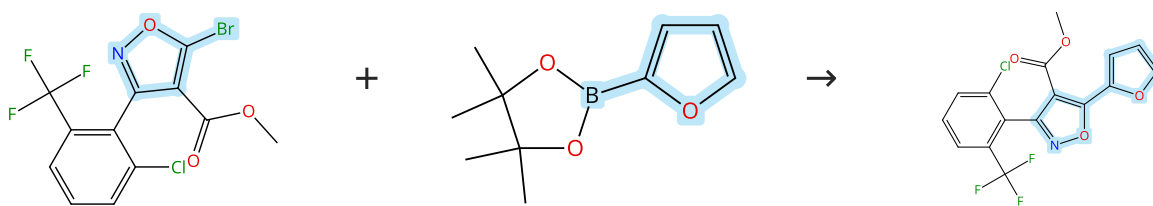
Journal of Medicinal Chemistry (2018), 61(18), 8299-8320.

1.1 **Reagents:** Potassium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Methanol, Toluene; 0.5 - 18 h, reflux

Experimental Protocols

Scheme 185 (1 Reaction)

Steps: 1 Yield: 58%



Suppliers (71)

31-179-CAS-21184044

Steps: 1 Yield: 58%

Ligand-Based Design of Allosteric Retinoic Acid Receptor-Related Orphan Receptor γ t (ROR γ t) Inverse Agonists

By: Meijer, Femke A.; et al

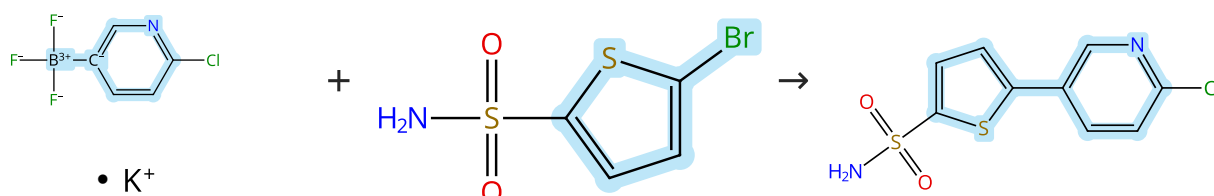
Journal of Medicinal Chemistry (2020), 63(1), 241-259.

- 1.1 **Reagents:** Cesium carbonate
Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium
Solvents: 1,2-Dimethoxyethane; 8 h, 85 °C

Experimental Protocols

Scheme 186 (1 Reaction)

Steps: 1 Yield: 58%



Suppliers (61)

Suppliers (77)

Supplier (1)

31-177-CAS-824955

Steps: 1 Yield: 58%

Arylation and Heteroarylation of Thienylsulfonamides with Organotrifluoroborates

By: Noreen, Mnaza; et al

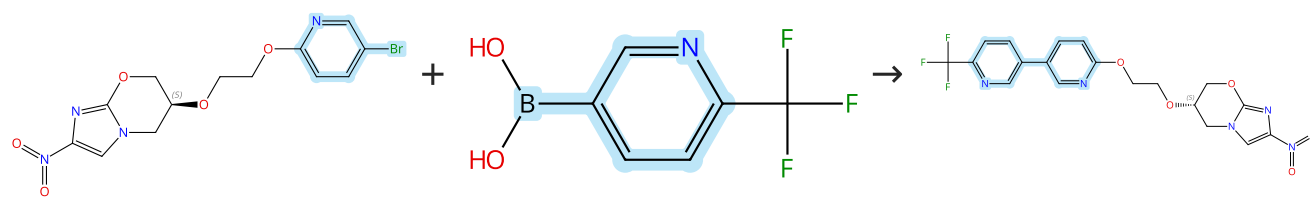
Journal of Organic Chemistry (2014), 79(15), 7243-7249.

- 1.1 **Reagents:** Tripotassium phosphate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: 1,4-Dioxane, Water; 12 h, 100 °C

Experimental Protocols

Scheme 187 (1 Reaction)

Steps: 1 Yield: 58%



Absolute stereochemistry shown

Absolute stereochemistry shown

Suppliers (90)

Supplier (1)

31-179-CAS-1003582

Steps: 1 Yield: 58%

Synthesis and Structure-Activity Relationships of Varied Ether Linker Analogues of the Antitubercular Drug (6S)-2-Nitro-6-{[4-(trifluoromethoxy)benzyl]oxy}-6,7-dihydro-5H-imidazo[2,1-b][1,3]oxazine (PA-824)

By: Thompson, Andrew M.; et al

Journal of Medicinal Chemistry (2011), 54(19), 6563-6585.

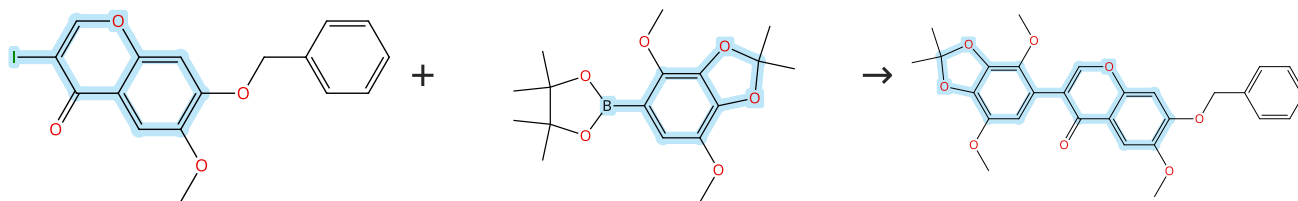
- 1.1 **Catalysts:** [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium
Solvents: Ethanol, Toluene; rt

- 1.2 **Reagents:** Sodium carbonate
Solvents: Water; 90 min, 90 °C; 90 °C → rt

Experimental Protocols

Scheme 188 (1 Reaction)

Steps: 1 Yield: 58%



Suppliers (2)

Suppliers (2)

Supplier (1)

31-614-CAS-24250032

Steps: 1 Yield: 58%

Structure Optimization of Gatastatin for the Development of γ -Tubulin-Specific Inhibitor

By: Shintani, Kana; et al

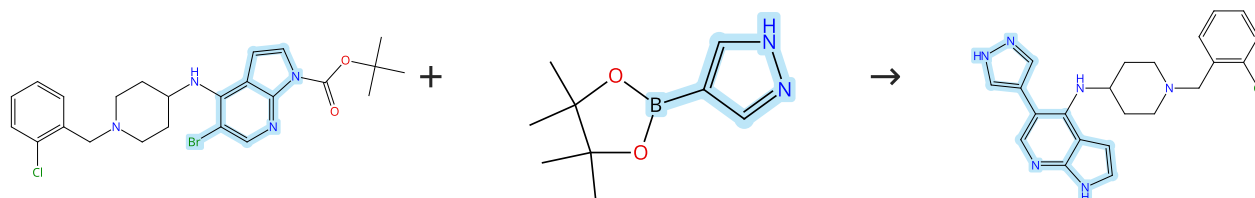
ACS Medicinal Chemistry Letters (2020), 11(6), 1125-1129.

- 1.1 **Reagents:** Sodium carbonate
Catalysts: Dichloro[1,1'-bis(diphenylphosphino)ferrocene]
 palladium(II) dichloromethane adduct
Solvents: 1,4-Dioxane, Water; 15 h, rt

Experimental Protocols

Scheme 189 (1 Reaction)

Steps: 1 Yield: 58%



Suppliers (108)

31-614-CAS-39559338

Steps: 1 Yield: 58%

Design, synthesis and evaluation of C-5 substituted pyrrolopyridine derivatives as potent Janus Kinase 1 inhibitors with excellent selectivity

By: Chen, Limei; et al

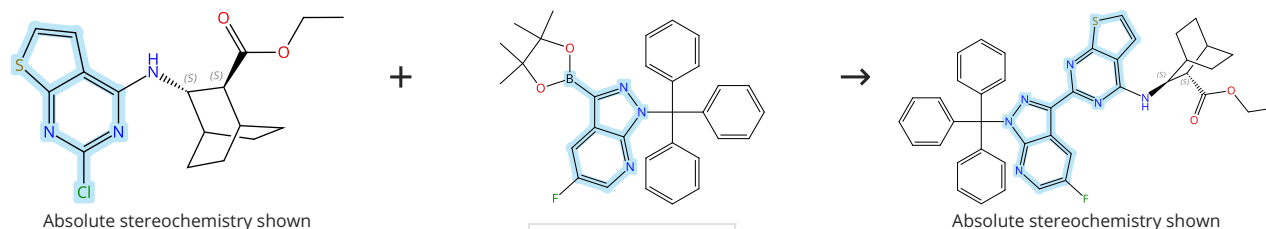
European Journal of Medicinal Chemistry (2024), 267, 116210.

- 1.1 **Reagents:** Potassium carbonate
Catalysts: (*SP*-4-1)-Bis[4-[bis(1,1-dimethylethyl)phosphino- κ P]-*N,N*-dimethylbenzenamine]dichloropalladium
Solvents: Isobutanol, Water; 10 h, 60 °C
- 1.2 **Reagents:** Trifluoroacetic acid
Solvents: Dichloromethane; 1 h, rt
- 1.3 **Reagents:** Sodium hydroxide
Solvents: Water; pH 9 - 10, rt

Experimental Protocols

Scheme 190 (1 Reaction)

Steps: 1 Yield: 58%



Absolute stereochemistry shown

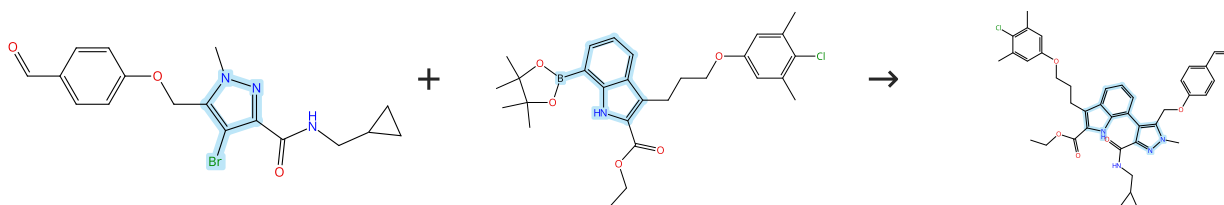
Absolute stereochemistry shown

Suppliers (9)

31-179-CAS-19591051 Steps: 1 Yield: 58%	Design, synthesis and biological evaluation of novel, orally bioavailable pyrimidine-fused heterocycles as influenza PB2 inhibitors By: Xiong, Jian; et al European Journal of Medicinal Chemistry (2019), 162, 249-265.
1.1 Reagents: Tripotassium phosphate Catalysts: Tris(dibenzylideneacetone)dipalladium, 1,1'-(9,9-Dimethyl-9 <i>H</i> -xanthene-4,5-diyl)bis[1,1-diphenylphosphine] Solvents: 2-Methyltetrahydrofuran, Water; 3 h, 80 °C	
Experimental Protocols	

Scheme 191 (1 Reaction)

Steps: 1 Yield: 58%

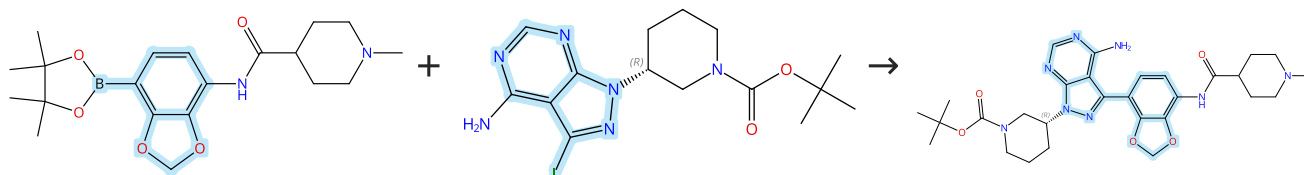


Supplier (1)

31-614-CAS-42046072 Steps: 1 Yield: 58%	Discovery of Novel Mcl-1 Inhibitors with a 3-Substituted-1<i>H</i>-indole-1-yl Moiety Binding to the P1-P3 Pockets to Induce Apoptosis in Acute Myeloid Leukemia Cells By: Deng, Hongguang; et al Journal of Medicinal Chemistry (2024), 67(16), 13925-13958.
1.1 Reagents: Tripotassium phosphate Catalysts: Tris(dibenzylideneacetone)dipalladium, X-Phos Solvents: Toluene, Water; 5 h, 100 °C	
1.2 Reagents: Water Experimental Protocols	

Scheme 192 (1 Reaction)

Steps: 1 Yield: 58%



Absolute stereochemistry shown

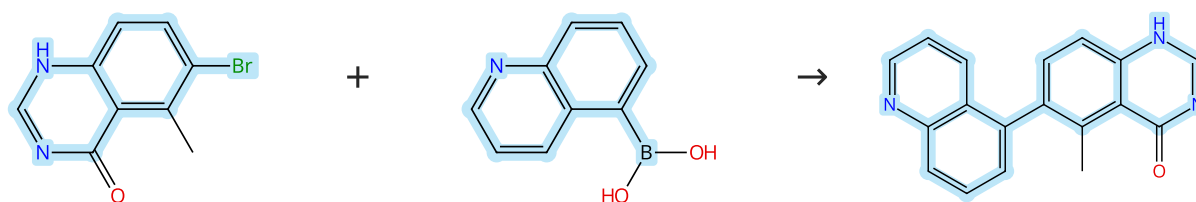
Absolute stereochemistry shown

Suppliers (51)

31-614-CAS-38375008 Steps: 1 Yield: 58%	Discovery of novel dual Bruton's tyrosine kinase (BTK) and Janus kinase 3 (JAK3) inhibitors as a promising strategy for rheumatoid arthritis By: Liang, Tingting; et al Bioorganic & Medicinal Chemistry (2023), 96, 117354.
1.1 Reagents: Sodium carbonate Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: 1,2-Dimethoxyethane; 12 h, 90 °C	
Experimental Protocols	

Scheme 193 (1 Reaction)

Steps: 1 Yield: 58%



Suppliers (41)

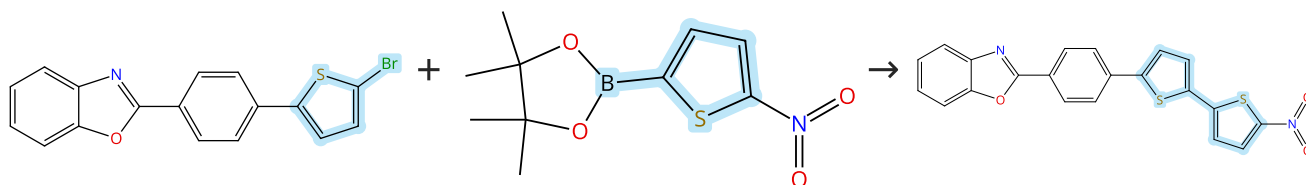
Suppliers (93)

Supplier (1)

31-179-CAS-19229025	Steps: 1 Yield: 58%	Novel Quinazolinone Inhibitors of ALK2 Flip between Alternate Binding Modes: Structure-Activity Relationship, Structural Characterization, Kinase Profiling, and Cellular Proof of Concept
1.1 Reagents: Sodium carbonate Catalysts: Dichlorobis(triphenylphosphine)palladium Solvents: 1,4-Dioxane, Water; 1 h, 120 °C		By: Hudson, Liam; et al
Experimental Protocols		Journal of Medicinal Chemistry (2018), 61(16), 7261-7272.

Scheme 194 (1 Reaction)

Steps: 1 Yield: 58%

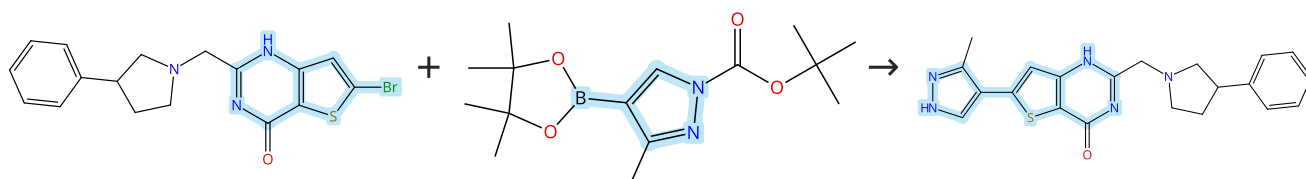


Suppliers (24)

31-614-CAS-44128853	Steps: 1 Yield: 58%	Core-Only Calamitic Liquid Crystals: Molecular Design and Optoelectronic Properties
1.1 Reagents: Potassium carbonate Catalysts: Dichlorobis(triphenylphosphine)palladium Solvents: Toluene; 16 h, reflux		By: Rothera, Jacob G.; et al
		Chemistry - An Asian Journal (2025), 20(8), e202401543.

Scheme 195 (1 Reaction)

Steps: 1 Yield: 58%



Suppliers (60)

Supplier (1)

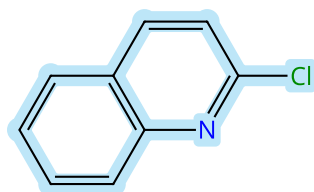
31-179-CAS-17624147	Steps: 1 Yield: 58%	2-Aminomethylthieno[3,2-d]pyrimidin-4(3H)-ones bearing 3-methylpyrazole hinge binding moiety: Highly potent, selective, and time-dependent inhibitors of Cdc7 kinase
1.1 Reagents: Sodium carbonate Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: 1,2-Dimethoxyethane, Water; 1 h, 100 °C		By: Kurasawa, Osamu; et al
Experimental Protocols		Bioorganic & Medicinal Chemistry (2017), 25(14), 3658-3670.

Scheme 196 (1 Reaction)

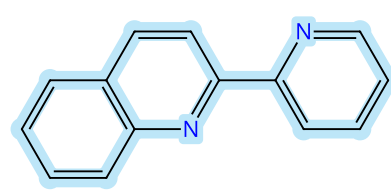
Steps: 1 Yield: 58%

Multi-component
structure image
available in CAS
SciFinder

+



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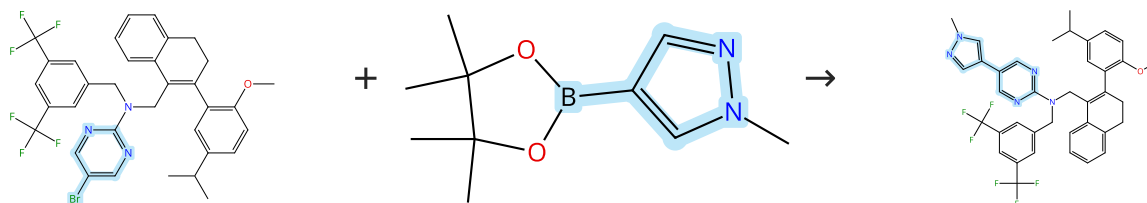
Suppliers (95)

Suppliers (23)

31-177-CAS-7956664 Steps: 1 Yield: 58% 1.1 Catalysts: 2-(Methylamino)ethanol, Cuprous iodide, (<i>SP</i>-4-2)-Dichloro[1,1'-(1,3-propanediyl)bis[1,1-dicyclohexylphosphine-κP]]palladium Solvents: Dimethylformamide; 16 h, 100 °C Experimental Protocols	Tetrabutylammonium 2-Pyridyltriolborate Salts for Suzuki-Miyaura Cross-Coupling Reactions with Aryl Chlorides By: Sakashita, Shohei; et al Organic Letters (2013), 15(17), 4308-4311.
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Scheme 197 (1 Reaction)

Steps: 1 Yield: 58%

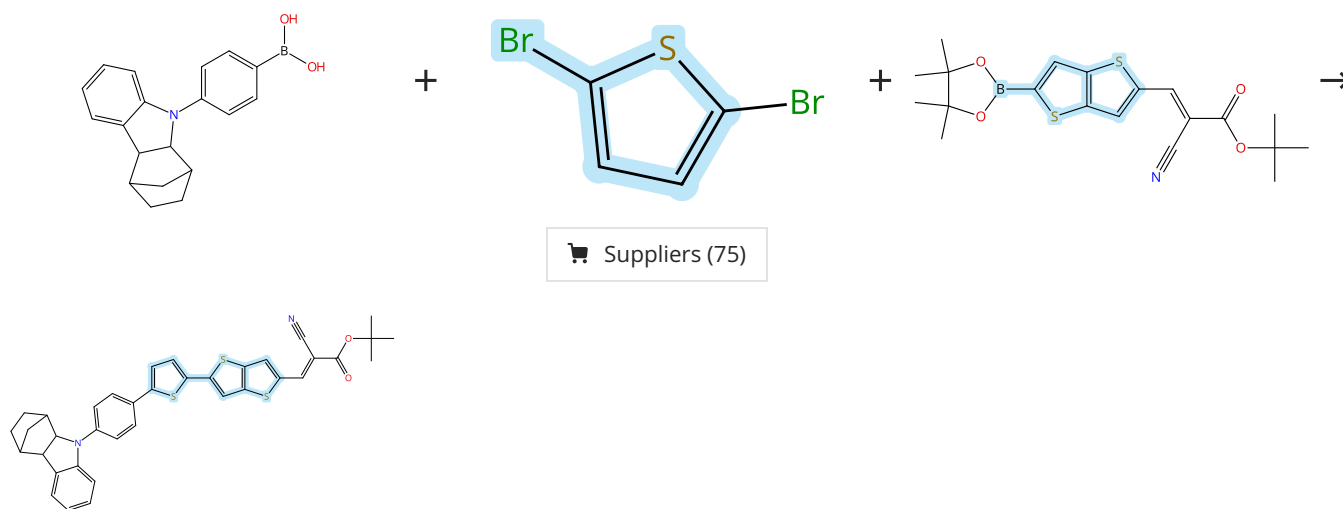


Suppliers (126)

31-179-CAS-16113893 Steps: 1 Yield: 58% 1.1 Reagents: Sodium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: 1,2-Dimethoxyethane, Water; 12 h, 90 °C Experimental Protocols	Design, synthesis and biological evaluation of novel cholesteryl ester transfer protein inhibitors bearing a cycloalkene scaffold By: Liu, Chunchi; et al European Journal of Medicinal Chemistry (2016), 123, 419-430.
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Scheme 198 (1 Reaction)

Steps: 1 Yield: 58%

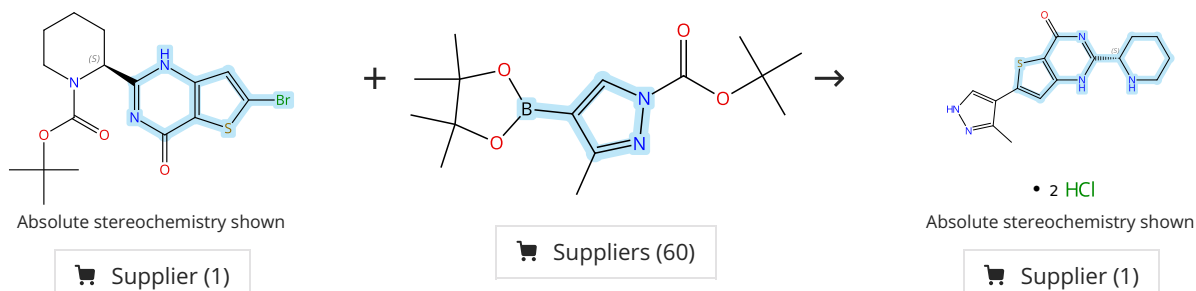


Suppliers (75)

31-179-CAS-10204720 Steps: 1 Yield: 58% 1.1 Reagents: Sodium carbonate Catalysts: Tris(dibenzylideneacetone)dipalladium, 1,1'-(9,9-Dimethyl-9H-xanthene-4,5-diyl)bis[1,1-diphenylphosphine] Solvents: Toluene, Water; 0.5 - 3 h, reflux; reflux → rt 1.2 Reagents: Sodium carbonate Catalysts: Tris(dibenzylideneacetone)dipalladium, Tri-<i>tert</i>-butylphosphonium tetrafluoroborate; 1 - 24 h, rt	Elucidating the Structure-Property Relationships of Donor-π-Acceptor Dyes for Dye-Sensitized Solar Cells (DSSCs) through Rapid Library Synthesis by a One-Pot Procedure By: Fuse, Shinichiro; et al Chemistry - A European Journal (2014), 20(34), 10685-10694.
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Scheme 199 (1 Reaction)

Steps: 1 Yield: 58%



31-179-CAS-21491726

Steps: 1 Yield: 58%

- 1.1 **Reagents:** Cesium carbonate
Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium
Solvents: 1,2-Dimethoxyethane, Water; 2 h, 80 °C
- 1.2 **Reagents:** Hydrochloric acid
Solvents: Methanol, Ethyl acetate, Water; 1.5 h, 50 °C
- 1.3 **Reagents:** Triethylamine
Solvents: Methanol; 1 h, rt

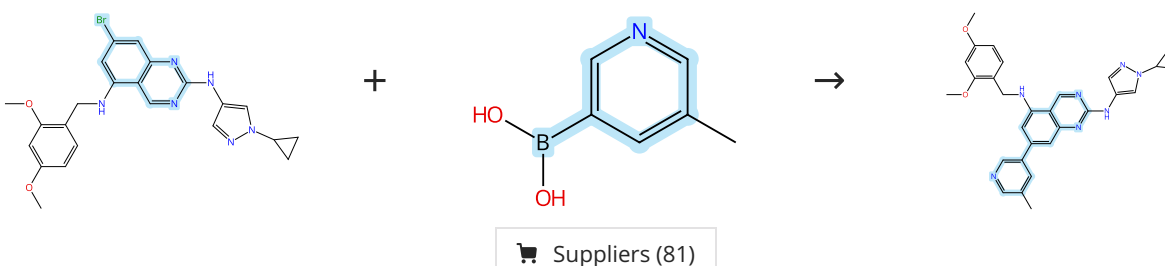
Discovery of a Novel, Highly Potent, and Selective Thieno [3,2-d]pyrimidinone-Based Cdc7 Inhibitor with a Quinuc lidine Moiety (TAK-931) as an Orally Active Investigational Antitumor Agent

By: Kurasawa, Osamu; et al

Journal of Medicinal Chemistry (2020), 63(3), 1084-1104.

Scheme 200 (1 Reaction)

Steps: 1 Yield: 58%



31-614-CAS-35316166

Steps: 1 Yield: 58%

- 1.1 **Reagents:** Potassium carbonate
Catalysts: Dichloro[1,1'-bis(diphenylphosphino)ferrocene]palladium(II) dichloromethane adduct
Solvents: 1,4-Dioxane, Water; rt → 80 °C; overnight, 80 °C

Experimental Protocols

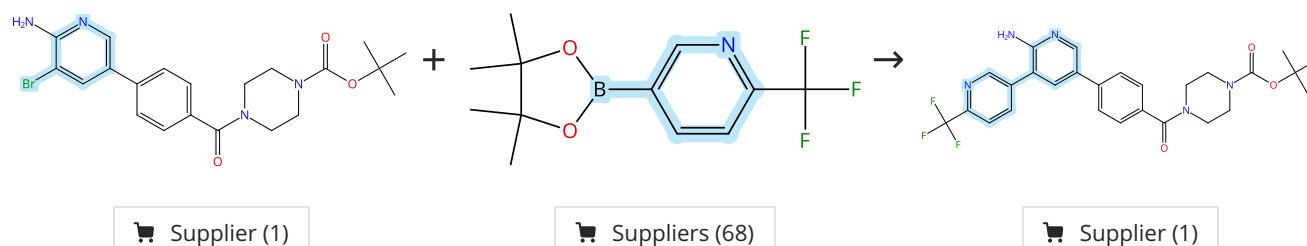
Development of a series of quinazoline-2,5-diamine derivatives as potent hematopoietic progenitor kinase 1 (HPK1) inhibitors

By: Shi, Huanyu; et al

European Journal of Medicinal Chemistry (2023), 248, 115064.

Scheme 201 (1 Reaction)

Steps: 1 Yield: 58%



31-179-CAS-8524181

Steps: 1 Yield: 58%

- 1.1 **Solvents:** 1,4-Dioxane; 15 min
- 1.2 **Reagents:** Potassium carbonate
Catalysts: Dichlorobis(triphenylphosphine)palladium
Solvents: Water; 14 h, 90 °C

Experimental Protocols

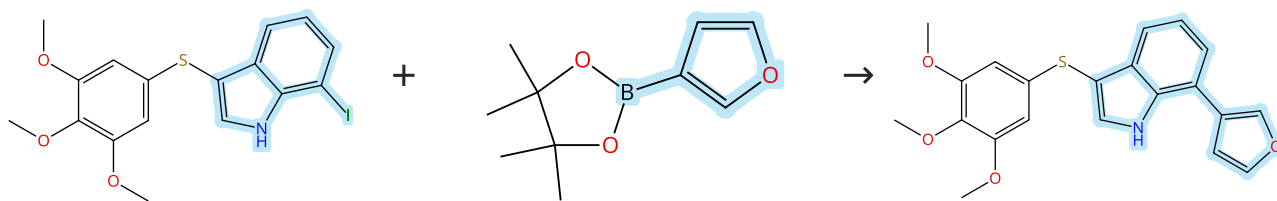
Structure-Activity Relationship Studies of Orally Active Antimalarial 3,5-Substituted 2-Aminopyridines

By: Gonzalez Cabrera, Diego; et al

Journal of Medicinal Chemistry (2012), 55(24), 11022-11030.

Scheme 202 (1 Reaction)

Steps: 1 Yield: 58%



Suppliers (83)

Supplier (1)

31-179-CAS-19098788

Steps: 1 Yield: 58%

New 6- and 7-heterocycl-1H-indole derivatives as potent tubulin assembly and cancer cell growth inhibitors

By: La Regina, Giuseppe; et al

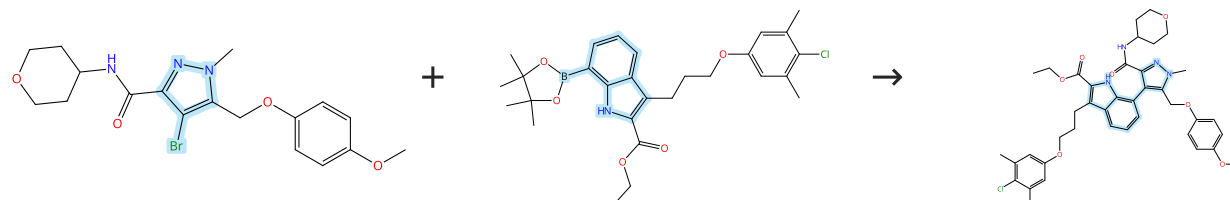
European Journal of Medicinal Chemistry (2018), 152, 283-297.

- 1.1 **Reagents:** Tripotassium phosphate
Catalysts: Tris(dibenzylideneacetone)dipalladium, 2-Dicyclohexylphosphino-2',6'-dimethoxybiphenyl
Solvents: 1-Butanol; 15 h, 100 °C; cooled
- 1.2 **Reagents:** Hydrochloric acid
Solvents: Water; cooled

Experimental Protocols

Scheme 203 (1 Reaction)

Steps: 1 Yield: 58%



Supplier (1)

31-614-CAS-42046084

Steps: 1 Yield: 58%

Discovery of Novel Mcl-1 Inhibitors with a 3-Substituted-1H-indole-1-yl Moiety Binding to the P1-P3 Pockets to Induce Apoptosis in Acute Myeloid Leukemia Cells

By: Deng, Hongguang; et al

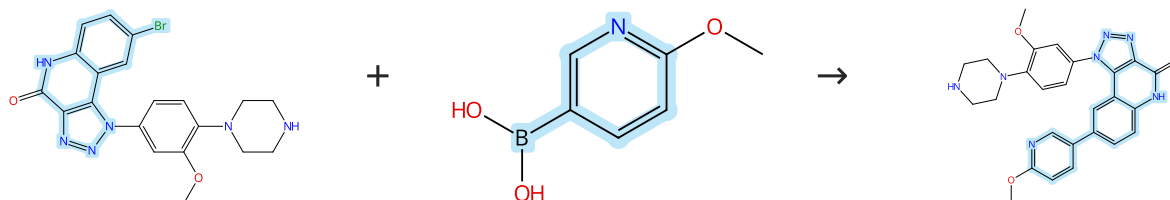
Journal of Medicinal Chemistry (2024), 67(16), 13925-13958.

- 1.1 **Reagents:** Tripotassium phosphate
Catalysts: Tris(dibenzylideneacetone)dipalladium, X-Phos
Solvents: Toluene, Water; 5 h, 100 °C
- 1.2 **Reagents:** Water

Experimental Protocols

Scheme 204 (1 Reaction)

Steps: 1 Yield: 58%



Suppliers (100)

31-614-CAS-32517441

Steps: 1 Yield: 58%

Discovery of 8-(6-Methoxypyridin-3-yl)-1-(4-(piperazin-1-yl)-3-(trifluoromethyl)phenyl)-1,5-dihydro-4H-[1,2,3]triazolo[4,5-c]quinolin-4-one (CQ211) as a Highly Potent and Selective R10 K2 Inhibitor

By: Ouyang, Yifan; et al

Journal of Medicinal Chemistry (2022), 65(11), 7833-7842.

- 1.1 **Reagents:** Hydrochloric acid
Solvents: Ethanol, Water; overnight, 120 °C
- 1.2 **Reagents:** Cesium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Dimethylformamide, Water; overnight, 80 °C
- 1.3 **Reagents:** Water

Experimental Protocols

Steps: **1** Yield: **58%**



Absolute stereochemistry shown

 Suppliers (13)

 Supplier (1)

Steps: **1** Yield: **58%**

Solubility-Driven Optimization of (Pyridin-3-yl) Benzoxazinyl-oxazolidinones Leading to a Promising Antibacterial Agent

By: Guo, Bin; et al

Journal of Medicinal Chemistry (2013), 56(6), 2642-2650.

Experimental Protocols

Steps: **1** Yield: **58%**

 Supplier (1)

Steps: **1** Yield: **58%**

Discovery of 3-(5'-Substituted)-Benzimidazole-5-(1-(3,5-dichloropyridin-4-yl)ethoxy)-1H-indazoles as Potent Fibroblast Growth Factor Receptor Inhibitors: Design, Synthesis, and Biological Evaluation

By: Yan, Wei; et al

Journal of Medicinal Chemistry (2016), 59(14), 6690-6708.

1.2 **Reagents:** Hydrochloric acid
Solvents: Methanol; overnight, rt

Experimental Protocols

Steps: **1** Yield: **58%**

Absolute stereochemistry shown

 Suppliers (4)

 Supplier (1)

Steps: **1** Yield: **58%**

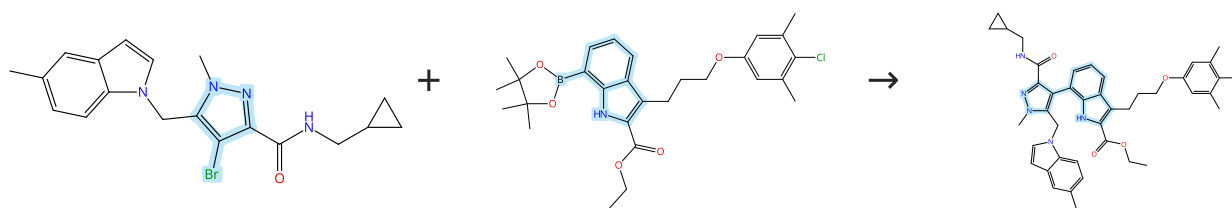
Synthesis and biological evaluation of novel benzoxa zinyloxazolidinones as potential antibacterial agents

By: Guo, Bin; et al

Bioorganic & Medicinal Chemistry Letters (2013), 23(13), 3697-3699.

Scheme 208 (1 Reaction)

Steps: 1 Yield: 58%



Supplier (1)

31-614-CAS-42046076

Steps: 1 Yield: 58%

Discovery of Novel Mcl-1 Inhibitors with a 3-Substituted-1H-indole-1-yl Moiety Binding to the P1-P3 Pockets to Induce Apoptosis in Acute Myeloid Leukemia Cells

By: Deng, Hongguang; et al

Journal of Medicinal Chemistry (2024), 67(16), 13925-13958.

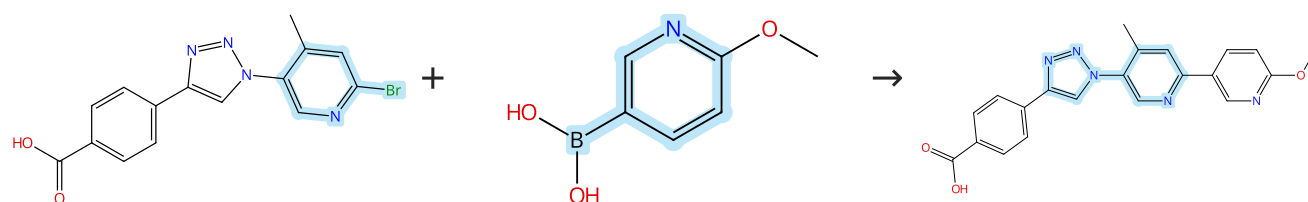
1.1 **Reagents:** Tripotassium phosphate
Catalysts: Tris(dibenzylideneacetone)dipalladium, X-Phos
Solvents: Toluene, Water; 5 h, 100 °C

1.2 **Reagents:** Water

Experimental Protocols

Scheme 209 (1 Reaction)

Steps: 1 Yield: 58%



Suppliers (100)

31-614-CAS-37374559

Steps: 1 Yield: 58%

Disrupting Kaposi's Sarcoma-Associated Herpesvirus (KSHV) Latent Replication with a Small Molecule Inhibitor

By: Berwanger, Aylin; et al

Journal of Medicinal Chemistry (2023), 66(15), 10782-10790.

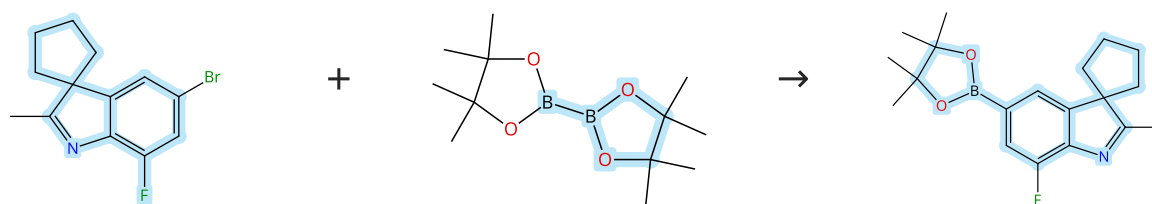
1.1 **Reagents:** Sodium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: 1,4-Dioxane, Water; overnight, 80 °C

1.2 **Reagents:** Hydrochloric acid
Solvents: Water; pH 1

Experimental Protocols

Scheme 210 (1 Reaction)

Steps: 1 Yield: 58%



Suppliers (131)

Supplier (1)

31-614-CAS-24971901

Steps: 1 Yield: 58%

A highly potent CDK4/6 inhibitor was rationally designed to overcome blood brain barrier in glioblastoma therapy

By: Yin, Lei; et al

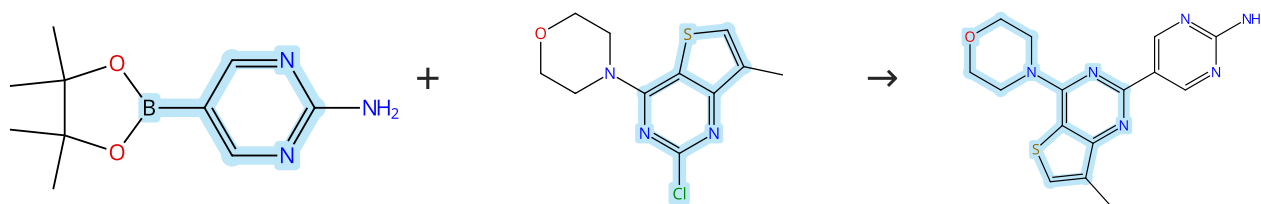
European Journal of Medicinal Chemistry (2018), 144, 1-28.

1.1 **Reagents:** Potassium acetate
Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium
Solvents: 1,4-Dioxane; overnight, rt → 90 °C

Experimental Protocols

Scheme 211 (1 Reaction)

Steps: 1 Yield: 58%



Suppliers (96)

Suppliers (33)

Suppliers (4)

31-179-CAS-11816781

Steps: 1 Yield: 58%

The Design and Identification of Brain Penetrant Inhibitors of Phosphoinositide 3-Kinase α

By: Heffron, Timothy P.; et al

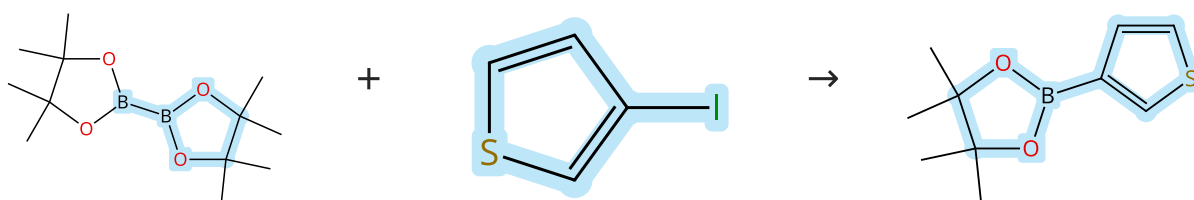
Journal of Medicinal Chemistry (2012), 55(18), 8007-8020.

- 1.1 **Reagents:** Sodium carbonate
Catalysts: (*SP*-4-1)-Bis[4-[bis(1,1-dimethylethyl)phosphino- κ P]-*N,N*-dimethylbenzenamine]dichloropalladium
Solvents: Acetonitrile, Water; 20 min, rt $\rightarrow$ 130 °C

Experimental Protocols

Scheme 212 (1 Reaction)

Steps: 1 Yield: 57%



Suppliers (131)

Suppliers (69)

Suppliers (84)

31-142-CAS-21419222

Steps: 1 Yield: 57%

Pyridine-Catalyzed Radical Borylation of Aryl Halides

By: Zhang, Li; et al

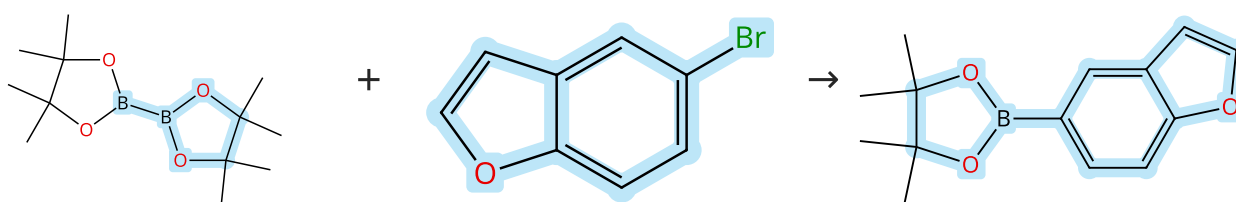
Journal of the American Chemical Society (2017), 139(2), 607-610.

- 1.1 **Reagents:** Potassium methoxide
Catalysts: 4-Phenylpyridine
Solvents: *tert*-Butyl methyl ether; 12 h, 85 °C

Experimental Protocols

Scheme 213 (1 Reaction)

Steps: 1 Yield: 57%



Suppliers (131)

Suppliers (90)

Suppliers (86)

31-614-CAS-28731085

Steps: 1 Yield: 57%

Visible Light-Induced Borylation of C-O, C-N, and C-X Bonds

By: Jin, Shengfei; et al

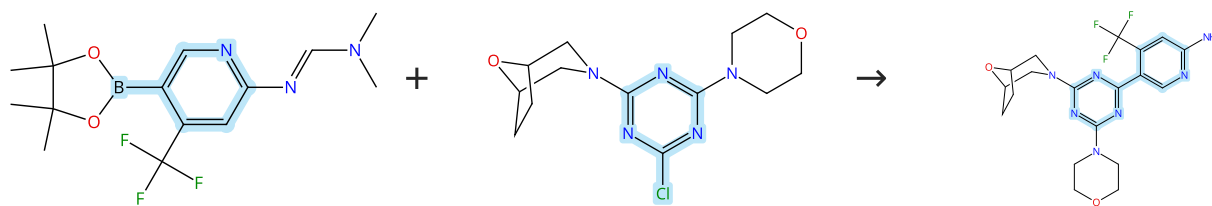
Journal of the American Chemical Society (2020), 142(3), 1603-1613.

- 1.1 **Reagents:** Cesium carbonate
Catalysts: Phenothiazine
Solvents: Acetonitrile; 48 h

Experimental Protocols

Scheme 214 (1 Reaction)

Steps: 1 Yield: 57%



Supplier (1)

Suppliers (2)

Supplier (1)

31-179-CAS-20469764

Steps: 1 Yield: 57%

(S)-4-(Difluoromethyl)-5-(4-(3-methylmorpholino)-6-morpholino-1,3,5-triazin-2-yl)pyridin-2-amine (PQR530), a Potent, Orally Bioavailable, and Brain-Penetrable Dual Inhibitor of Class I PI3K and mTOR Kinase

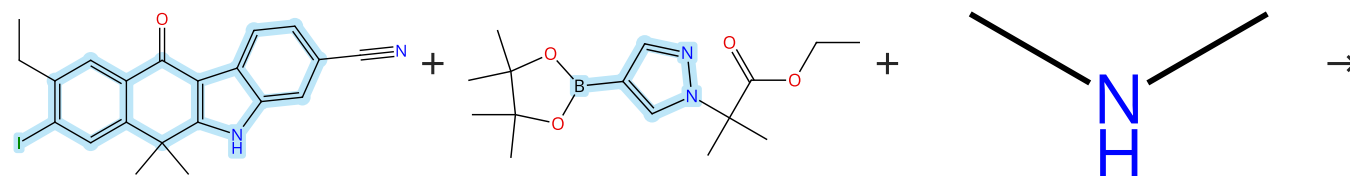
By: Rageot, Denise; et al

Journal of Medicinal Chemistry (2019), 62(13), 6241-6261.

- 1.1 **Reagents:** Tripotassium phosphate
Catalysts: (SP-4-4)-[2'-(Amino-κM)[1,1'-biphenyl]-2-yl-κC]chloro[dicyclohexyl[2',4',6'-tris(1-methylethyl)[1,1'-biphenyl]-2-yl]phosphine]palladium
Solvents: 1,4-Dioxane, Water; 3 - 15 h, 95 °C; 95 °C → rt
- 1.2 **Reagents:** Hydrochloric acid
Solvents: Water; 3 - 15 h, 60 °C; 60 °C → rt
- 1.3 **Reagents:** Sodium hydroxide
Solvents: Water; pH 8 - 9, rt

Scheme 215 (1 Reaction)

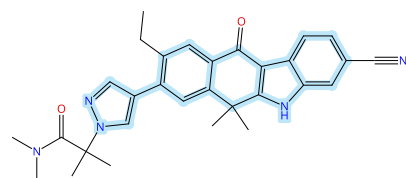
Steps: 1 Yield: 57%



Suppliers (56)

Suppliers (61)

Suppliers (125)



Supplier (1)

31-179-CAS-13960175

Steps: 1 Yield: 57%

Discovery of Inhibitors That Overcome the G1202R Anaplastic Lymphoma Kinase Resistance Mutation

By: Hatcher, John M.; et al

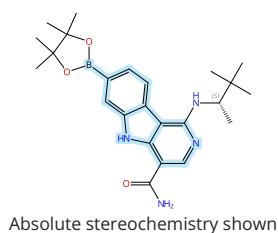
Journal of Medicinal Chemistry (2015), 58(23), 9296-9308.

- 1.1 **Reagents:** Sodium carbonate
Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium, Dicyclohexyl[2',4',6'-tris(1,1-dimethylethyl)[1,1'-biphenyl]-2-yl]phosphine
Solvents: 1,4-Dioxane, Water; rt → 100 °C; 1 h, 100 °C
- 1.2 **Reagents:** Diisopropylethylamine, O-(7-Azabenzotriazol-1-yl)-N,N,N,N-tetramethyluronium hexafluorophosphate
Solvents: Dimethylformamide; rt

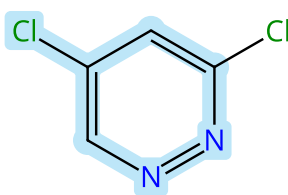
Experimental Protocols

Scheme 216 (1 Reaction)

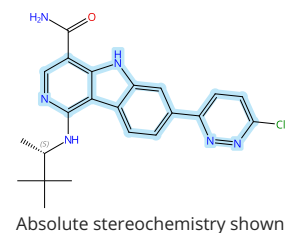
Steps: 1 Yield: 57%



+



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Supplier (1)

Suppliers (81)

Supplier (1)

31-113-CAS-1152876

Steps: 1 Yield: 57%

Discovery of 1-Amino-5H-pyrido[4,3-b]indol-4-carboxamide Inhibitors of Janus Kinase 2 (JAK2) for the Treatment of Myeloproliferative Disorders

By: Lim, Jongwon; et al

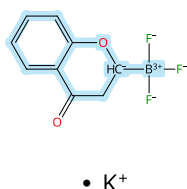
Journal of Medicinal Chemistry (2011), 54(20), 7334-7349.

1.1 **Reagents:** Tripotassium phosphate
Catalysts: Tricyclohexylphosphine, Tris(dibenzylideneacetone) dipalladium
Solvents: 1,4-Dioxane, Water; 1 h, rt → 100 °C

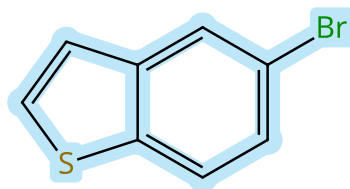
Experimental Protocols

Scheme 217 (1 Reaction)

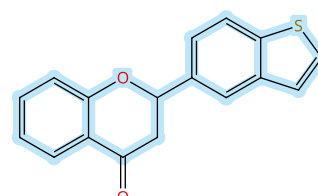
Steps: 1 Yield: 57%



+



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Suppliers (80)

Supplier (1)

31-179-CAS-21331604

Steps: 1 Yield: 57%

Direct α-Arylation/Heteroarylation of 2-Trifluoroboratochromanones via Photoredox/Nickel Dual Catalysis

By: Matsui, Jennifer K.; et al

Organic Letters (2017), 19(3), 436-439.

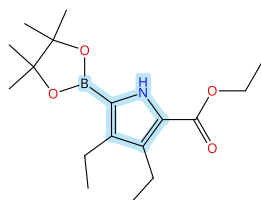
1.1 **Reagents:** Monopotassium phosphate
Catalysts: Dichloro[1,2-di(methoxy-κO)ethane]nickel, 4,4'-Bis(1,1-dimethylethyl)-2,2'-bipyridine, 2,4,5,6-Tetra(9*H*-carbazol-9-yl)isophthalonitrile
Solvents: 1,4-Dioxane; 16 h

1.2 **Reagents:** Sodium bicarbonate
Solvents: Water

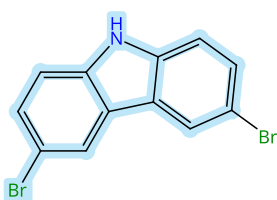
Experimental Protocols

Scheme 218 (1 Reaction)

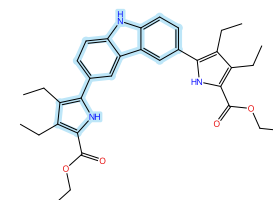
Steps: 1 Yield: 57%



+



→



Suppliers (98)

31-614-CAS-32517335

Steps: 1 Yield: 57%

3,6-Carbazoylene Octaphyrin (1.0.0.0.1.0.0.0) and Its Bis-BF₂ Complex

By: Chen, Hao; et al

Journal of the American Chemical Society (2022), 144(18), 8194-8203.

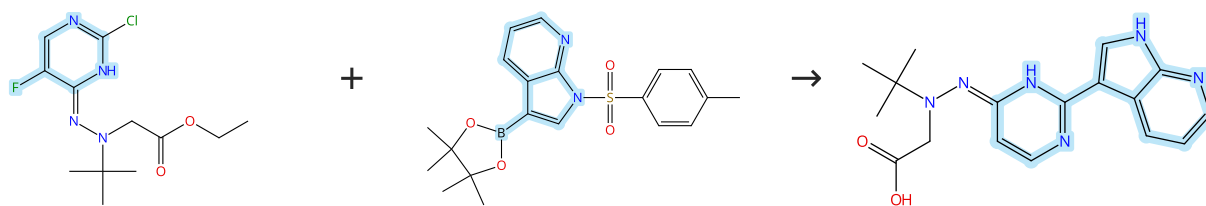
1.1 **Catalysts:** Triphenylphosphine, Palladium diacetate
Solvents: 1,4-Dioxane; 10 min, rt

1.2 **Reagents:** Potassium carbonate
Solvents: Water; rt; 24 h, 90 °C

Experimental Protocols

Scheme 219 (1 Reaction)

Steps: 1 Yield: 57%



Suppliers (63)

31-614-CAS-35654768

Steps: 1 Yield: 57%

Synthesis and structure-activity optimization of 7-azaindoles containing aza-β-amino acids targeting the influenza PB2 subunit

By: Wang, Sihan; et al

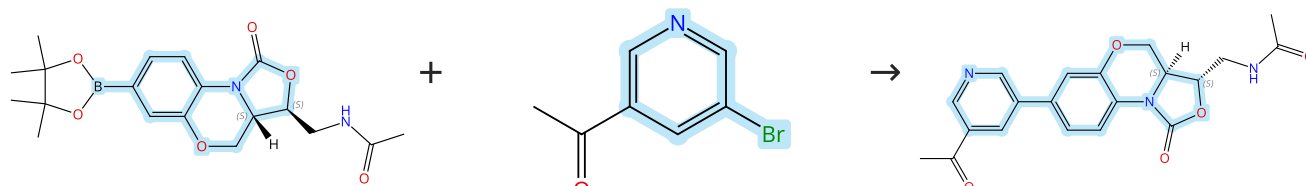
European Journal of Medicinal Chemistry (2023), 250, 115185.

- 1.1 **Reagents:** Potassium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Acetonitrile, Water; 12 h, rt → 60 °C
- 1.2 **Reagents:** Tetrabutylammonium fluoride
Solvents: Tetrahydrofuran; 90 min, rt → 45 °C
- 1.3 **Solvents:** Ethyl acetate; rt
- 1.4 **Reagents:** Lithium hydroxide
Solvents: Tetrahydrofuran, Water; 30 min, rt
- 1.5 **Reagents:** Hydrochloric acid
Solvents: Water; 5 - 6 h, rt

Experimental Protocols

Scheme 220 (1 Reaction)

Steps: 1 Yield: 57%



Absolute stereochemistry shown

Suppliers (86)

Absolute stereochemistry shown

Supplier (1)

31-179-CAS-7632789

Steps: 1 Yield: 57%

Design, Synthesis, and Structure-Activity Relationship Studies of Highly Potent Novel Benzoxazinyl-Oxazolidinone Antibacterial Agents

By: Xin, Qi-Sheng; et al

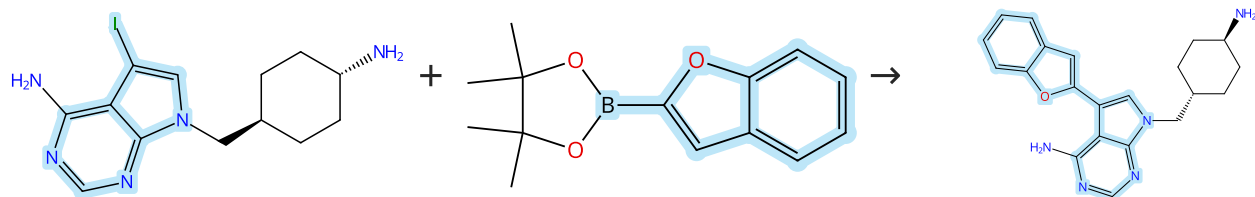
Journal of Medicinal Chemistry (2011), 54(21), 7493-7502.

- 1.1 **Reagents:** Cesium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: 1,4-Dioxane; 2 h, 80 °C

Experimental Protocols

Scheme 221 (1 Reaction)

Steps: 1 Yield: 57%



Relative stereochemistry shown

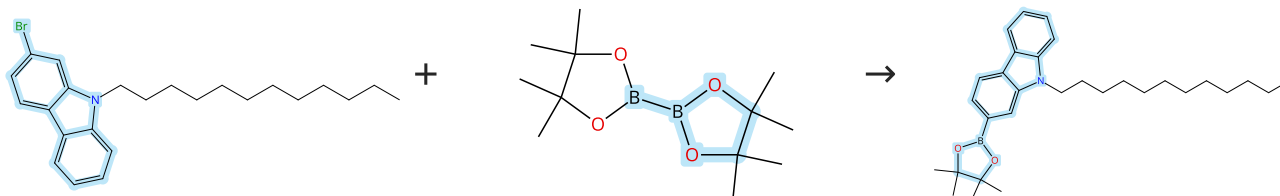
Suppliers (65)

Relative stereochemistry shown

31-614-CAS-38998511 Steps: 1 Yield: 57% 1.1 Reagents: Potassium carbonate Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: Ethanol, 1,4-Dioxane, Water; 2 h, 80 °C Experimental Protocols	Discovery of Potent and Selective Phosphatidylinositol 3-Phosphate 5-Kinase (PIKfyve) Inhibitors as Methuosis Inducers By: Chen, Yong; et al Journal of Medicinal Chemistry (2024), 67(1), 165-179.
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Scheme 222 (1 Reaction)

Steps: 1 Yield: 57%

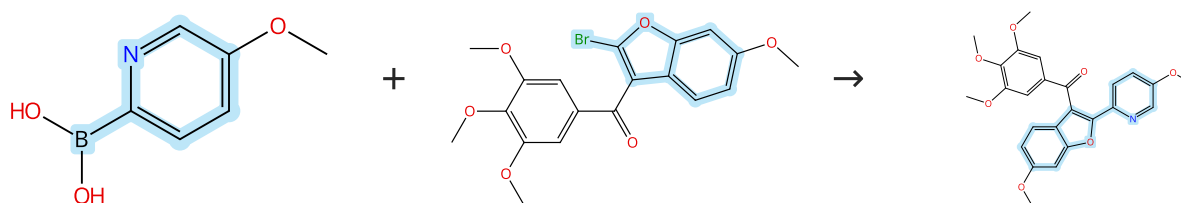


Suppliers (131)

31-614-CAS-28062820 Steps: 1 Yield: 57% 1.1 Reagents: Potassium acetate Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: 1,4-Dioxane; 16 h, 80 °C Experimental Protocols	Catalytic Synthesis of Conjugated Azopolymers from Aromatic Diazides By: Andjaba, John M.; et al Journal of the American Chemical Society (2021), 143(10), 3975-3982.
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Scheme 223 (1 Reaction)

Steps: 1 Yield: 57%



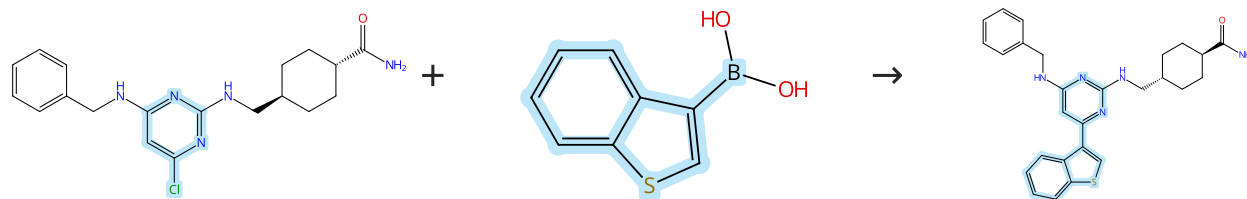
Suppliers (54)

Supplier (1)

31-179-CAS-9922225 Steps: 1 Yield: 57% 1.1 Reagents: Sodium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: 1,4-Dioxane, Water; reflux	Discovery of 7-Hydroxy-6-methoxy-2-methyl-3-(3,4,5-trimethoxybenzoyl)benzo[b]furan (BNC105), a Tubulin Polymerization Inhibitor with Potent Antiproliferative and Tumor Vascular Disrupting Properties By: Flynn, Bernard L.; et al Journal of Medicinal Chemistry (2011), 54(17), 6014-6027.
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Scheme 224 (1 Reaction)

Steps: 1 Yield: 57%



Relative stereochemistry shown

Relative stereochemistry shown

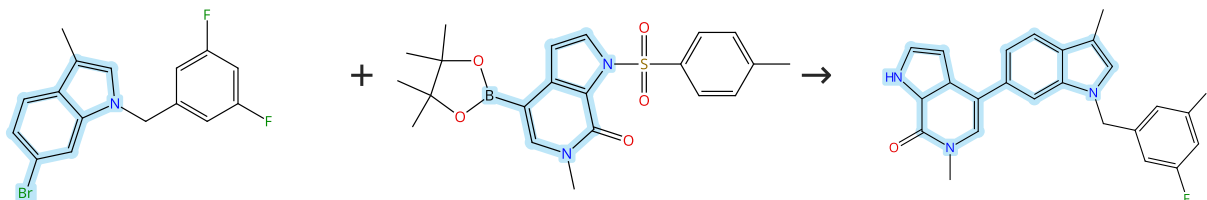
Suppliers (87)

Suppliers (3)

31-179-CAS-5097464	Steps: 1 Yield: 57%	Discovery of 2-(Cyclohexylmethylamino)pyrimidines as a New Class of Reversible Valosine Containing Protein Inhibitors
1.1 Reagents: Cesium carbonate Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: 1,4-Dioxane, Water; 30 min, 100 °C		By: Cervi, Giovanni; et al Journal of Medicinal Chemistry (2014), 57(24), 10443-10454.
Experimental Protocols		

Scheme 225 (1 Reaction)

Steps: 1 Yield: 57%

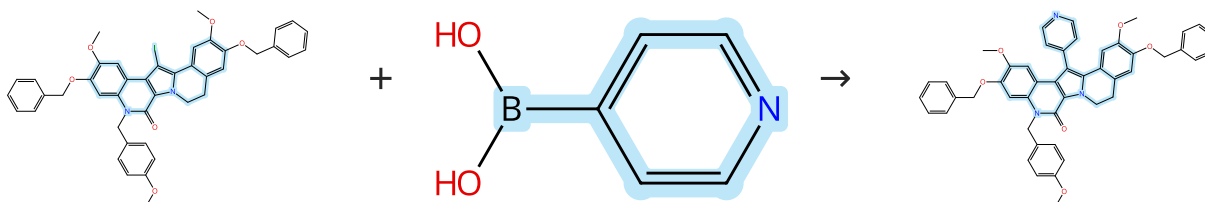


Suppliers (48)

31-614-CAS-31373962	Steps: 1 Yield: 57%	Discovery of indol-6-yl-pyrrolo[2,3-c]pyridin-7-one derivatives as bromodomain-containing protein 4 (BRD4) inhibitors for the treatment of kidney fibrosis
1.1 Reagents: Sodium carbonate Catalysts: Dichloro[1,1'-bis(diphenylphosphino)ferrocene]palladium(II) dichloromethane adduct Solvents: 1,4-Dioxane, Water; 4 h, 110 °C		By: Tao, Sibe; et al European Journal of Medicinal Chemistry (2022), 231, 114153.
Experimental Protocols		

Scheme 226 (1 Reaction)

Steps: 1 Yield: 57%

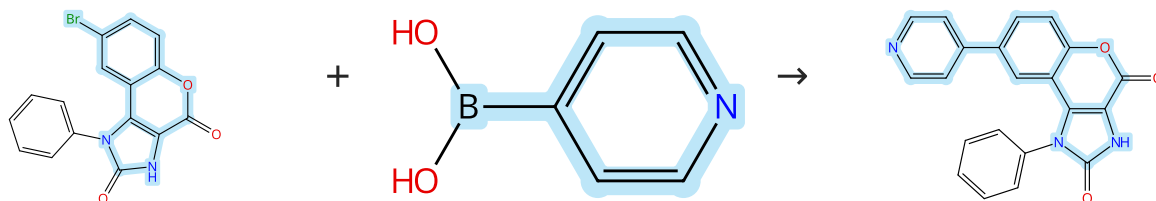


Suppliers (111)

31-614-CAS-25048823	Steps: 1 Yield: 57%	An Expedient Modular Hybrid Strategy for the Diversity-Oriented Synthesis of Lamellarins/Azalamellarins with Anticancer Cytotoxicity
1.1 Reagents: Cesium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: 1,4-Dioxane, Water; 16 h, rt → 110 °C		By: Klumthong, Kanawut; et al Journal of Organic Chemistry (2021), 86(21), 14883-14902.
1.2 Reagents: Water		
Experimental Protocols		

Scheme 227 (1 Reaction)

Steps: 1 Yield: 57%



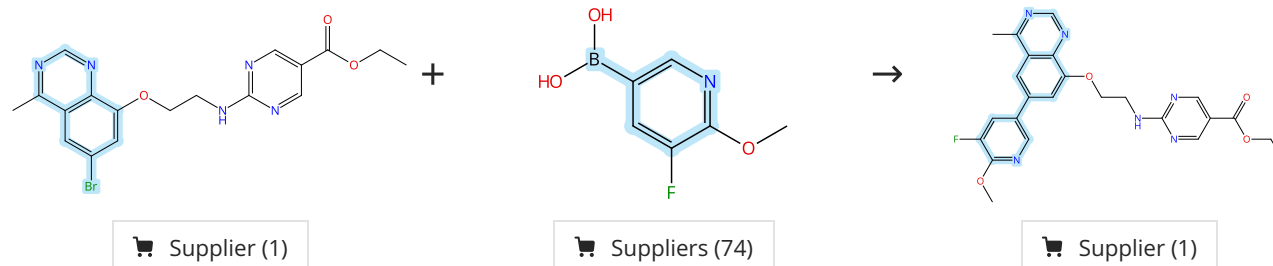
Suppliers (111)

Supplier (1)

31-179-CAS-14434288 Steps: 1 Yield: 57% 1.1 Reagents: Potassium carbonate Catalysts: Dichloro[1,1'-bis(diphenylphosphino)ferrocene] palladium(II) dichloromethane adduct Solvents: 1,4-Dioxane, Water; 5 h, 50 °C Experimental Protocols	Synthesis and biological evaluation of novel 2,3-dihydrocromeno[3,4-d]imidazol-4(1H)-one derivatives as potent anticancer cell proliferation and migration agents By: Han, Xuan; et al European Journal of Medicinal Chemistry (2016), 114, 232-243.
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Scheme 228 (1 Reaction)

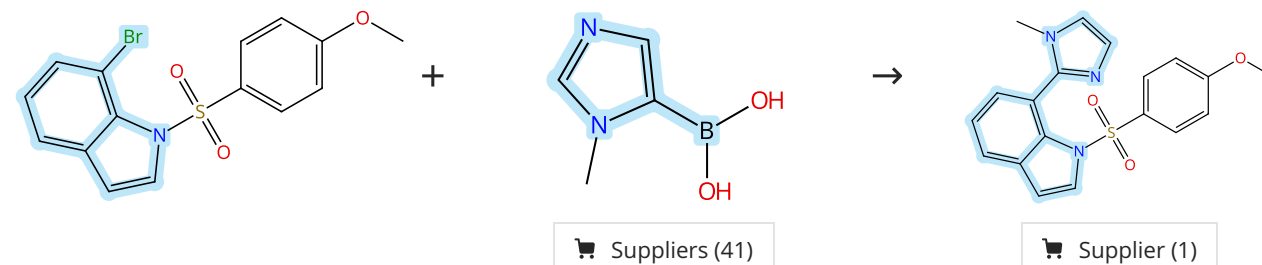
Steps: 1 Yield: 57%



31-179-CAS-20380021 Steps: 1 Yield: 57% 1.1 Reagents: Potassium carbonate Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: Toluene, Water; 8 h, 80 °C	Design, Synthesis, and Biological Evaluation of 4-Methyl Quinazoline Derivatives as Anticancer Agents Simultaneously Targeting Phosphoinositide 3-Kinases and Histone Deacetylases By: Zhang, Kehui; et al Journal of Medicinal Chemistry (2019), 62(15), 6992-7014.
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Scheme 229 (1 Reaction)

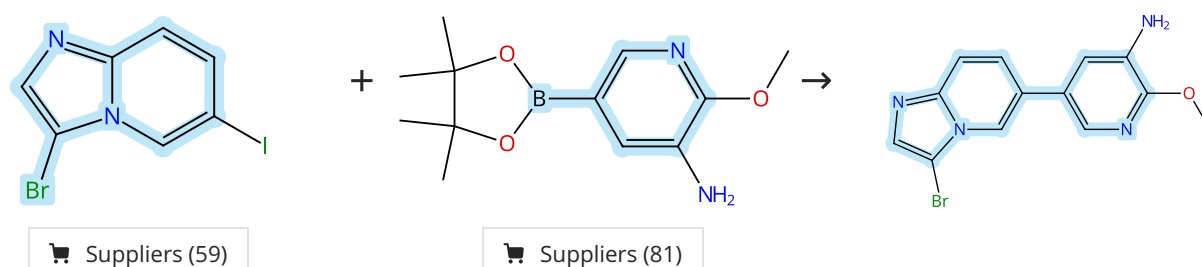
Steps: 1 Yield: 57%



31-614-CAS-26297878 Steps: 1 Yield: 57% 1.1 Reagents: Potassium carbonate Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: 1,2-Dimethoxyethane, Water; 2 h, 86 °C 1.2 Solvents: Water Experimental Protocols	N-Arylsulfonylsubstituted-1H indole derivatives as small molecule dual inhibitors of signal transducer and activator of transcription 3 (STAT3) and tubulin By: Zhou, Qiang; et al Bioorganic & Medicinal Chemistry (2018), 26(1), 96-106.
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Scheme 230 (1 Reaction)

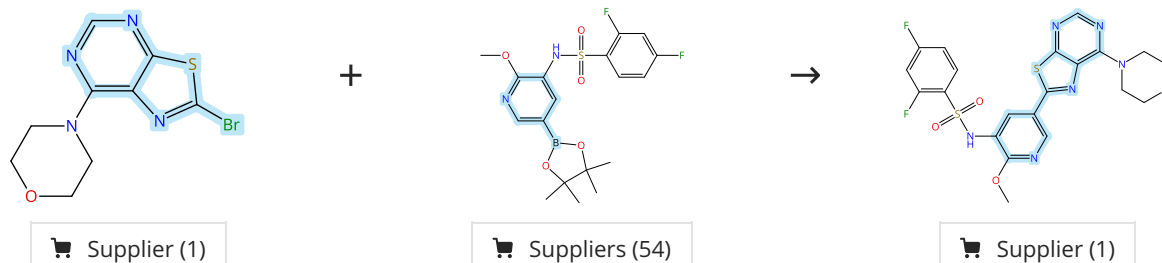
Steps: 1 Yield: 57%



31-614-CAS-24893616 Steps: 1 Yield: 57% 1.1 Reagents: Sodium carbonate Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: 1,2-Dimethoxyethane, Water; 40 min, 110 °C Experimental Protocols	Omipalisib inspired macrocycles as dual PI3K/mTOR inhibitors By: Alvarez, Rosa M.; et al European Journal of Medicinal Chemistry (2021), 211, 113109.
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Scheme 231 (1 Reaction)

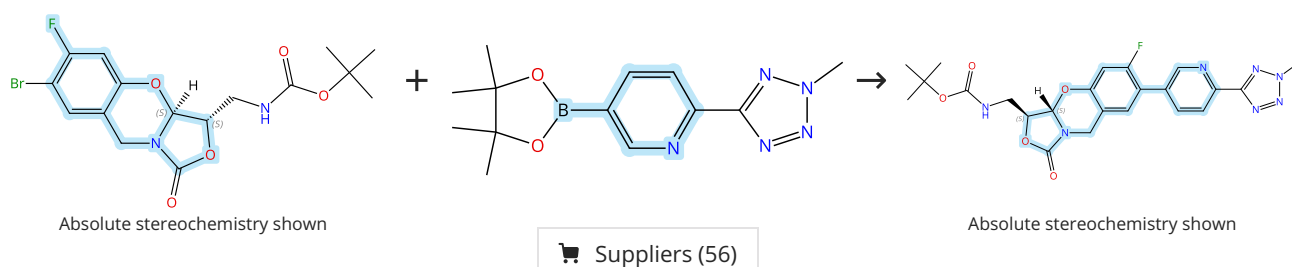
Steps: 1 Yield: 57%



31-614-CAS-24495443 Steps: 1 Yield: 57% 1.1 Reagents: Potassium carbonate Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: 1,4-Dioxane, Water; 4 h, 100 °C Experimental Protocols	Discovery of new thieno[2,3-d]pyrimidine and thiazolo[5,4-d]pyrimidine derivatives as orally active phosphoinositide 3-kinase inhibitors By: Sun, Yan; et al Bioorganic & Medicinal Chemistry (2021), 29, 115890.
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Scheme 232 (1 Reaction)

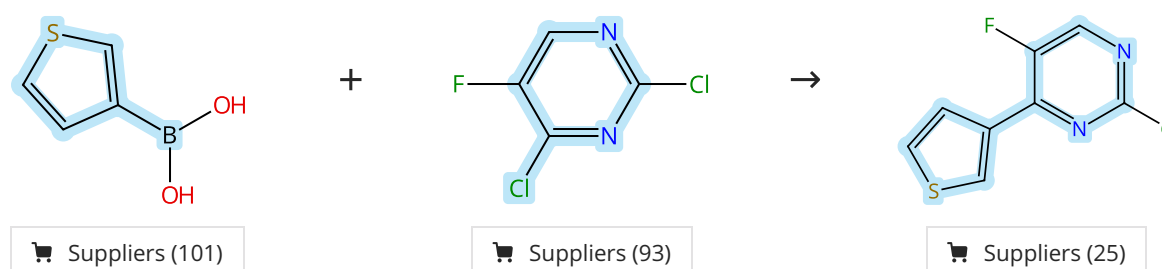
Steps: 1 Yield: 57%



31-614-CAS-31838268 Steps: 1 Yield: 57% 1.1 Reagents: Sodium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Dimethyl sulfoxide, Water; 5 h, 100 °C Experimental Protocols	Identification of Novel Tricyclic Benzo[1,3]oxazinyloxazolidinones as Potent Antibacterial Agents with Excellent Pharmacokinetic Profiles against Drug-Resistant Pathogens By: Wu, Yongqi; et al Journal of Medicinal Chemistry (2021), 64(6), 3234-3248.
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Scheme 233 (1 Reaction)

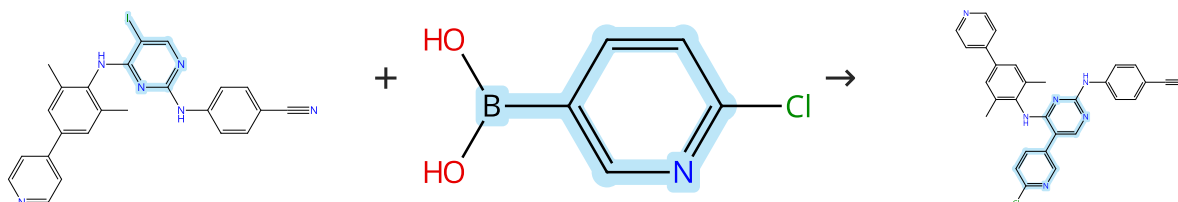
Steps: 1 Yield: 57%



31-179-CAS-16662551	Steps: 1 Yield: 57%	Intramolecular Inverse Electron-Demand [4 + 2] Cycloadditions of Ynamides with Pyrimidines: Scope and Density Functional Theory Insights
1.1 Reagents: Sodium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Acetonitrile, Water; overnight, reflux		By: Duret, Guillaume; et al
Experimental Protocols		Journal of Organic Chemistry (2017), 82(3), 1726-1742.

Scheme 234 (1 Reaction)

Steps: 1 Yield: 57%

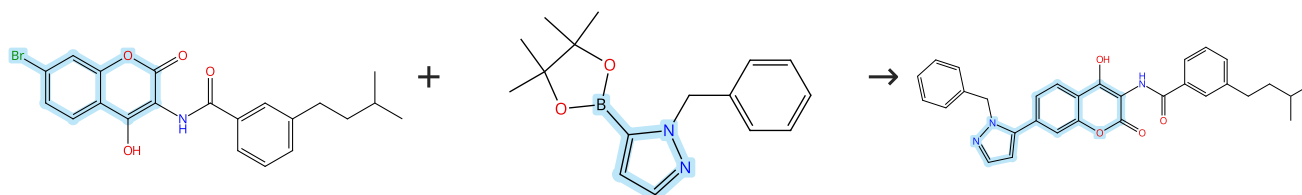


Suppliers (98)

31-614-CAS-42560696	Steps: 1 Yield: 57%	Fragment Addition-Based Design of Heteroaromatic-Biphenyl-DAPYs as Potent and Orally Available Non-nucleoside Reverse Transcriptase Inhibitors Featuring Significantly Enhanced Safety
1.1 Reagents: Cesium carbonate Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: 1,4-Dioxane, Water; 16 h, 110 °C		By: Huang, Wen-Juan; et al
Experimental Protocols		Journal of Medicinal Chemistry (2024), 67(19), 17568-17584.

Scheme 235 (1 Reaction)

Steps: 1 Yield: 57%



Supplier (1)

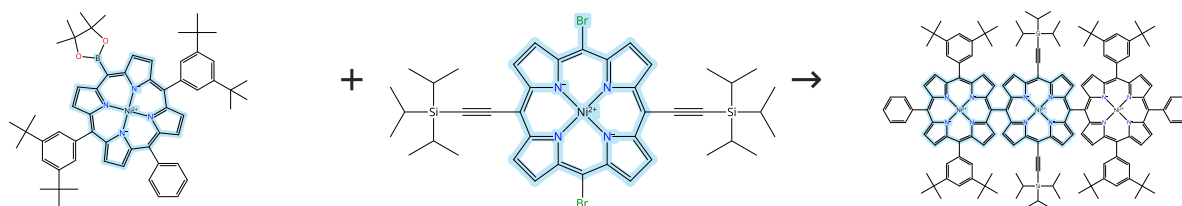
Suppliers (56)

Supplier (1)

31-614-CAS-25136549	Steps: 1 Yield: 57%	Demonstrating Ligandability of the LC3A and LC3B Adapter Interface
1.1 Reagents: Sodium bicarbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Dimethylformamide, Water; 12 h, 80 °C		By: Hartmann, Markus; et al
1.2 Reagents: Hydrochloric acid Solvents: Water		Journal of Medicinal Chemistry (2021), 64(7), 3720-3746.
Experimental Protocols		

Scheme 236 (1 Reaction)

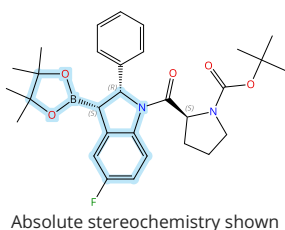
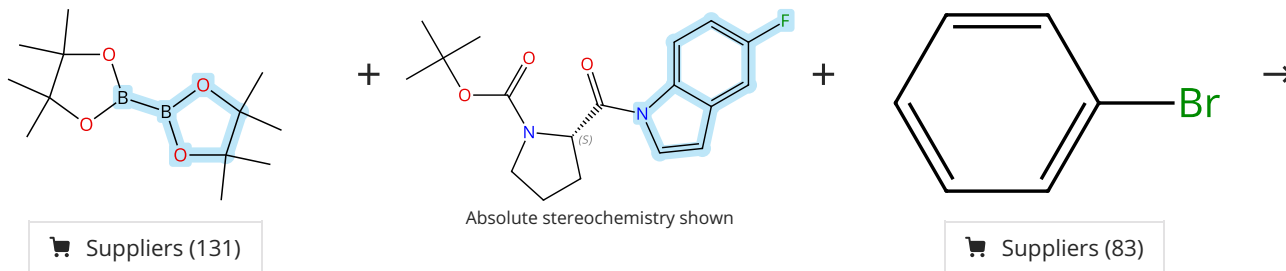
Steps: 1 Yield: 57%



31-177-CAS-878300 Steps: 1 Yield: 57% 1.1 Reagents: Cesium carbonate Catalysts: Triphenylphosphine, Water, Palladium, tris[μ-[(1,2-η:4,5-η)-(1<i>E</i>,4<i>E</i>)-1,5-diphenyl-1,4-pentadien-3-one]]di-, compd. with trichloromethane (1:1) Solvents: Dimethylformamide, Toluene; 3 h, 100 °C Experimental Protocols	Silylethynyl Substituents as Porphyrin Protecting Groups for Solubilization and Selectivity Control By: Oda, Kazuma; et al Organic Letters (2014), 16(6), 1818-1821.
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Scheme 237 (1 Reaction)

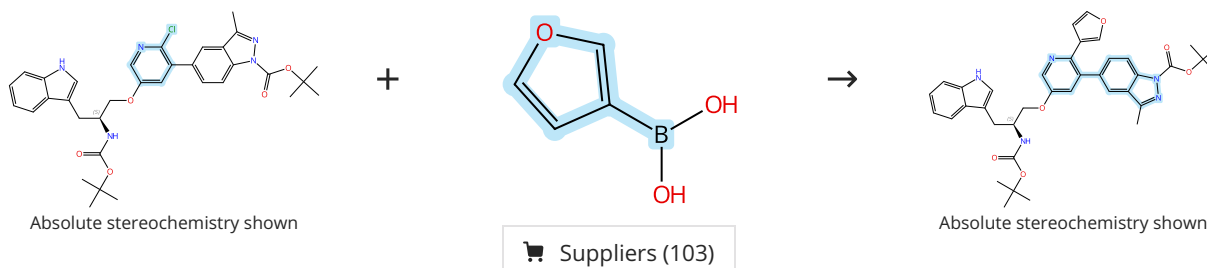
Steps: 1 Yield: 57%



31-614-CAS-24363529 Steps: 1 Yield: 57% 1.1 Reagents: Sodium <i>tert</i>-butoxide Solvents: Tetrahydrofuran; 30 min, 25 °C 1.2 Solvents: Tetrahydrofuran; 25 °C → 0 °C 1.3 Catalysts: Dichloro[1,2-di(methoxy-κO)ethane]nickel Solvents: Dimethylacetamide; 18 h, 2 - 10 °C 1.4 Reagents: Water Experimental Protocols	Nickel-Catalyzed Dearomative Arylboration of Indoles: Regioselective Synthesis of C2- and C3-Borylated Indolines By: Trammel, Grace L.; et al Journal of the American Chemical Society (2021), 143(40), 16502-16511.
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Scheme 238 (1 Reaction)

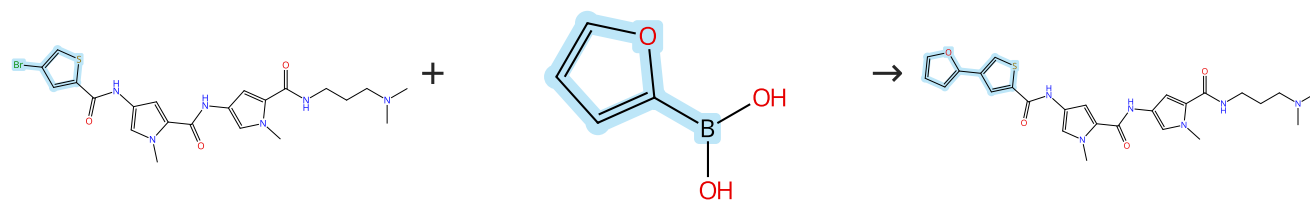
Steps: 1 Yield: 57%



31-179-CAS-11802002 Steps: 1 Yield: 57% 1.1 Reagents: Sodium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: 1,4-Dioxane; 20 min, 160 °C	2,3,5-Trisubstituted pyridines as selective AKT inhibitors-Part I: Substitution at 2-position of the core pyridine for ROCK1 selectivity By: Lin, Hong; et al Bioorganic & Medicinal Chemistry Letters (2010), 20(2), 673-678.
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Scheme 239 (1 Reaction)

Steps: 1 Yield: 57%



Suppliers (93)

Supplier (1)

31-179-CAS-5493896

Steps: 1 Yield: 57%

Synthesis, anti-mycobacterial activity and DNA sequence-selectivity of a library of biaryl- motifs containing polyamides

By: Brucoli, Federico; et al

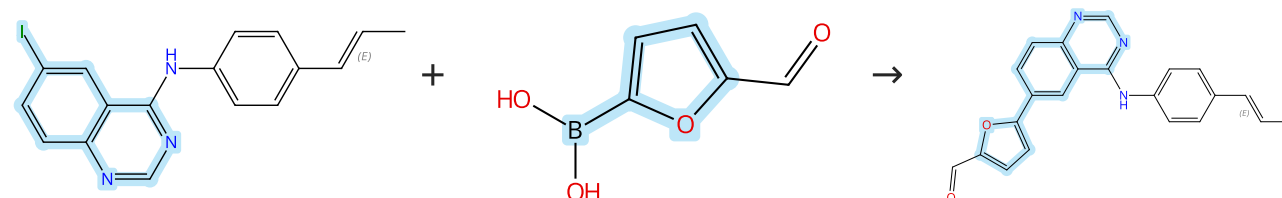
Bioorganic & Medicinal Chemistry (2015), 23(13), 3705-3711.

1.1 **Reagents:** Potassium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Ethanol, Toluene, Water; 8 min, 100 °C

Experimental Protocols

Scheme 240 (1 Reaction)

Steps: 1 Yield: 57%



Double bond geometry shown

Suppliers (97)

Double bond geometry shown

31-179-CAS-16908676

Steps: 1 Yield: 57%

Synthesis and in vitro biological evaluation of novel quinazoline derivatives

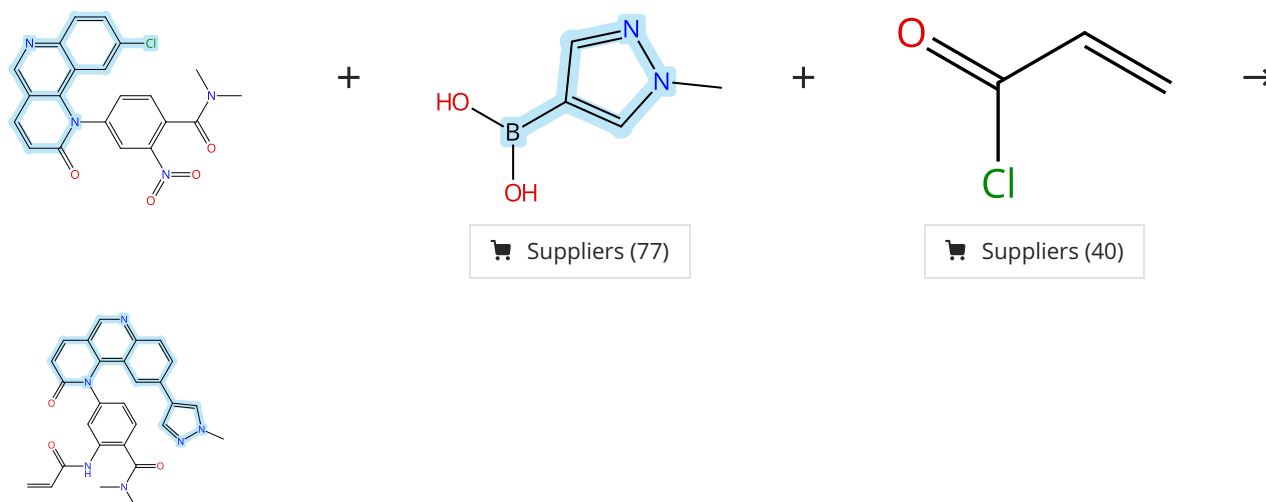
By: Zhang, Yaling; et al

Bioorganic & Medicinal Chemistry Letters (2017), 27(7), 1584-1587.

1.1 **Reagents:** Triethylamine
Catalysts: Palladium
Solvents: Methanol, 1,2-Dimethoxyethane; 0.5 h, rt → 50 °C

Scheme 241 (1 Reaction)

Steps: 1 Yield: 56%



Suppliers (77)

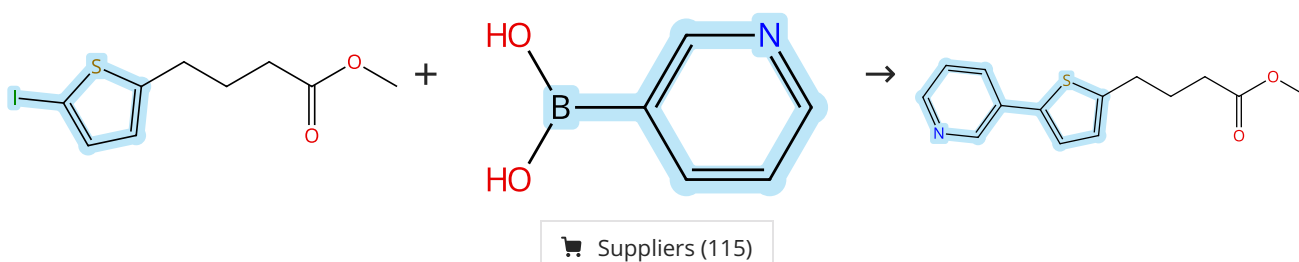
Suppliers (40)

Supplier (1)

31-179-CAS-17828801 Steps: 1 Yield: 56% 1.1 Reagents: Sodium carbonate Catalysts: Dichlorobis(triphenylphosphine)palladium, Bis(1,1-dimethylethyl)[2',4',6'-tris(1-methylethyl)[1,1'-biphenyl]-2-yl] phosphine Solvents: 1,4-Dioxane, Water; overnight, 80 °C 1.2 Reagents: Stannous chloride Solvents: Ethyl acetate; overnight, 70 °C 1.3 Reagents: Sodium bicarbonate Solvents: Tetrahydrofuran, Water; 70 °C → 0 °C Experimental Protocols	Structure-Activity Relationship study of QL47-a Broad-spectrum Antiviral agent By: Liang, Yanke; et al ACS Medicinal Chemistry Letters (2017), 8(3), 344-349.
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Scheme 242 (1 Reaction)

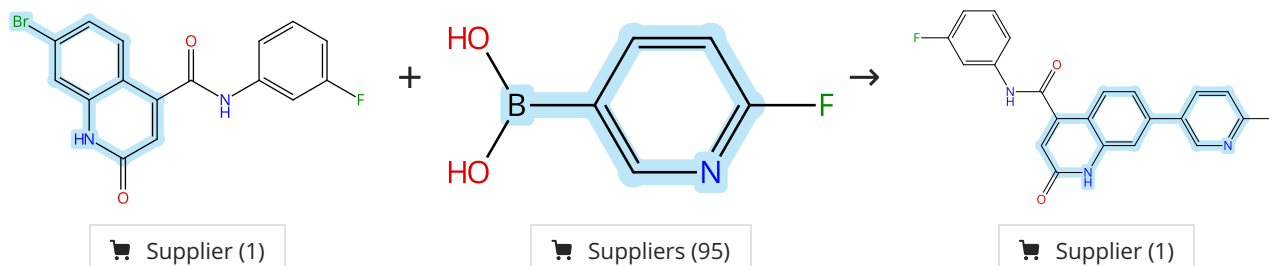
Steps: 1 Yield: 56%



31-614-CAS-36521209 Steps: 1 Yield: 56% 1.1 Reagents: Sodium carbonate Catalysts: Triphenylphosphine, Palladium diacetate Solvents: 1,2-Dimethoxyethane; 3 h, reflux Experimental Protocols	In Silico-Guided Rational Drug Design and Synthesis of Novel 4-(Thiophen-2-yl)butanamides as Potent and Selective TRPV1 Agonists By: Romeo, Isabella; et al Journal of Medicinal Chemistry (2023), 66(10), 6994-7015.
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Scheme 243 (1 Reaction)

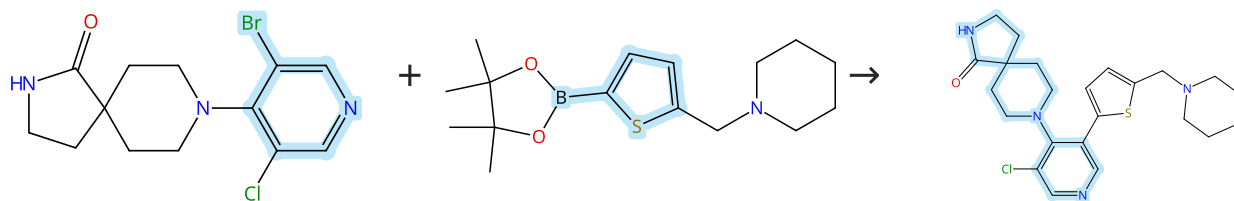
Steps: 1 Yield: 56%



31-179-CAS-18950715 Steps: 1 Yield: 56% 1.1 Reagents: Cesium carbonate Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: 1,4-Dioxane, Water; 24 h, rt → 100 °C	Synthesis and biological evaluation of novel 2-oxo-1,2-dihydroquinoline-4-carboxamide derivatives for the treatment of esophageal squamous cell carcinoma By: Shahin, Mai I.; et al European Journal of Medicinal Chemistry (2018), 155, 516-530.
--	--

Scheme 244 (1 Reaction)

Steps: 1 Yield: 56%



Suppliers (5)

Suppliers (54)

Supplier (1)

31-614-CAS-24921698

Steps: 1 Yield: 56%

Discovery of a potent, highly selective, and orally bioavailable inhibitor of CDK8 through a structure-based optimisation

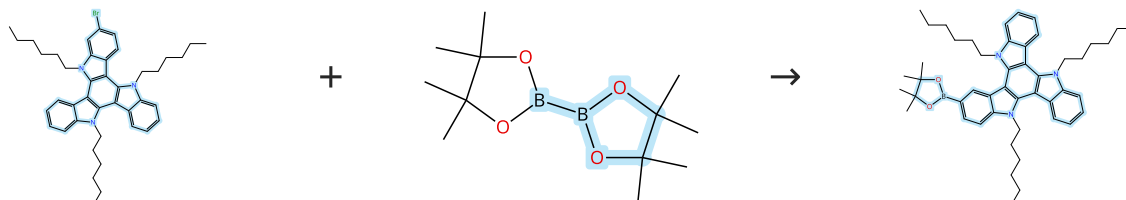
1.1 **Reagents:** Sodium carbonate
Catalysts: Dichloro[1,1'-bis(diphenylphosphino)ferrocene] palladium(II) dichloromethane adduct
Solvents: Acetonitrile, Water; 1 h, 120 °C

By: Yu, Mingfeng; et al

European Journal of Medicinal Chemistry (2021), 218, 113391.

Scheme 245 (1 Reaction)

Steps: 1 Yield: 56%



Suppliers (131)

31-113-CAS-101421

Steps: 1 Yield: 56%

Star-shaped panchromatic absorbing dyes based on a triazole platform with diketopyrrolopyrrole and boron dipyrromethene substituents

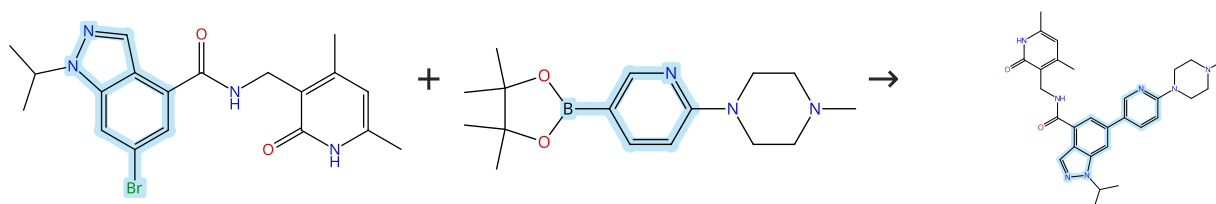
1.1 **Reagents:** Potassium *tert*-butoxide
Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium
Solvents: 1,4-Dioxane

By: Huaulme, Quentin; et al

Tetrahedron Letters (2014), 55(35), 4953-4958.

Scheme 246 (1 Reaction)

Steps: 1 Yield: 56%



Supplier (1)

Suppliers (88)

Suppliers (8)

31-179-CAS-1735841

Steps: 1 Yield: 56%

Identification of Potent, Selective, Cell-Active Inhibitors of the Histone Lysine Methyltransferase EZH2

1.1 **Reagents:** Sodium bicarbonate
Catalysts: Dichloro[1,1'-bis(diphenylphosphino)ferrocene] palladium(II) dichloromethane adduct
Solvents: 1,2-Dimethoxyethane, Water; 30 min, 130 °C

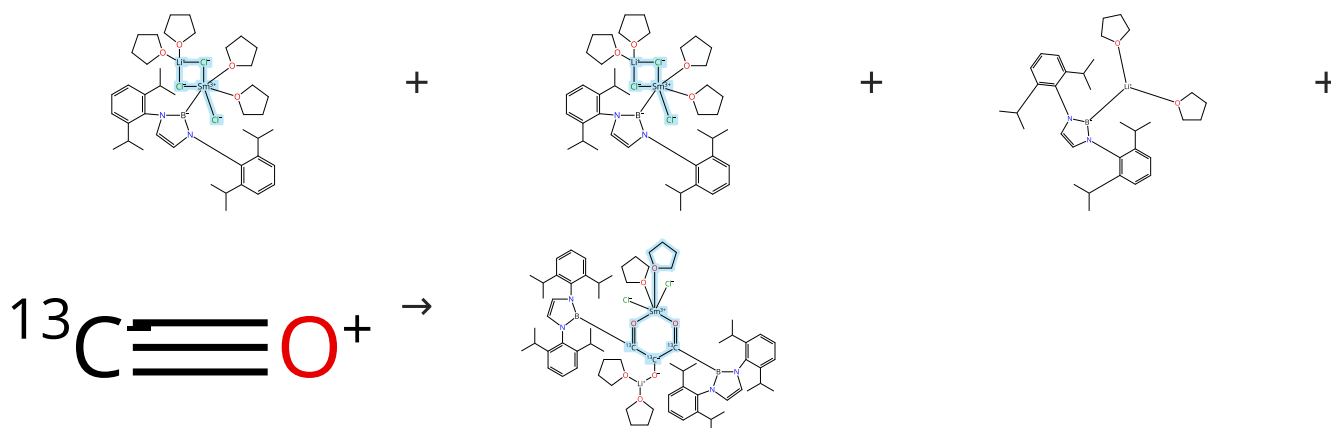
By: Verma, Sharad K.; et al

ACS Medicinal Chemistry Letters (2012), 3(12), 1091-1096.

Experimental Protocols

Scheme 247 (1 Reaction)

Steps: 1 Yield: 56%



Suppliers (10)

31-562-CAS-18359115

Steps: 1 Yield: 56%

Cooperative Trimerization of Carbon Monoxide by Lithium and Samarium Boryls

1.1 Solvents: Tetrahydrofuran; 5 min, 1 atm, 25 °C

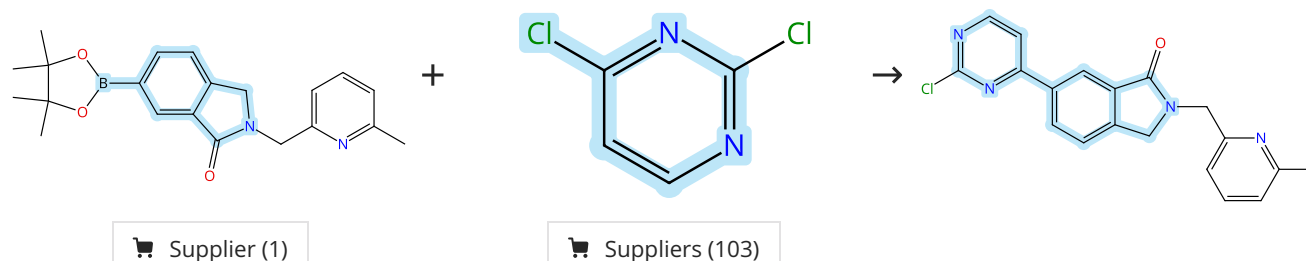
By: Wang, Baoli; et al

Experimental Protocols

Journal of the American Chemical Society (2017), 139(46), 16967-16973.

Scheme 248 (1 Reaction)

Steps: 1 Yield: 56%



31-179-CAS-20224469

Steps: 1 Yield: 56%

Discovery of potent, orally bioavailable ERK1/2 inhibitors with isoindolin-1-one structure by structure-based drug design

1.1 Reagents: Potassium acetate

By: Ji, Dezhong; et al

Catalysts: Tetrakis(triphenylphosphine)palladium

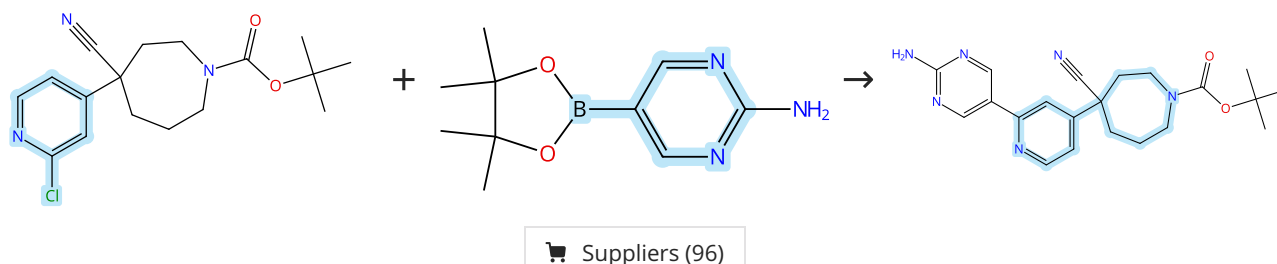
Solvents: 1,4-Dioxane, Water; 12 h, 25 °C → 80 °C

Experimental Protocols

European Journal of Medicinal Chemistry (2019), 164, 334-341.

Scheme 249 (1 Reaction)

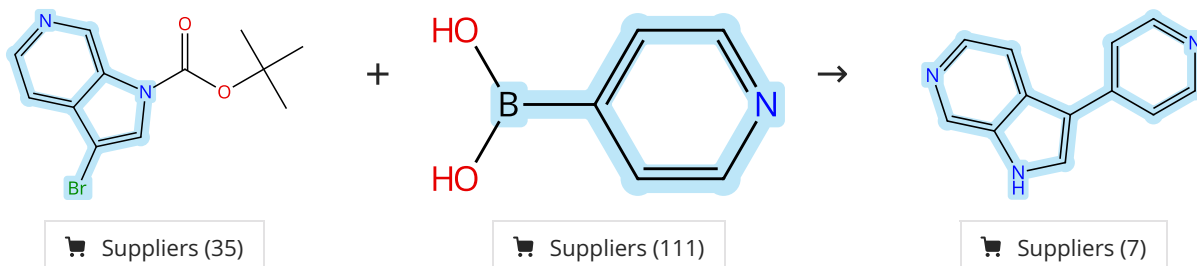
Steps: 1 Yield: 56%



31-179-CAS-8706206	Steps: 1 Yield: 56%	Cyanide anion as a leaving group in nucleophilic aromatic substitution: synthesis of quaternary centers at azine hetero cycles
1.1 Reagents: Potassium carbonate Catalysts: (SP-4-1)-Bis[4-[bis(1,1-dimethylethyl)phosphino-κP]-N,N-dimethylbenzamine]dichloropalladium Solvents: Toluene, Water; 19 h, 100 °C		By: Thompson, Alexander D.; et al Journal of Organic Chemistry (2013), 78(2), 762-769.
Experimental Protocols		

Scheme 250 (1 Reaction)

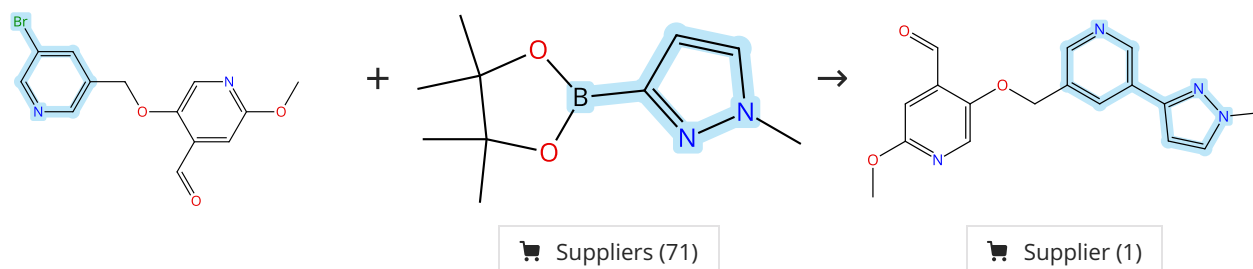
Steps: 1 Yield: 56%



31-614-CAS-32020047	Steps: 1 Yield: 56%	Selective DYRK1A Inhibitor for the Treatment of Type 1 Diabetes: Discovery of 6-Azaindole Derivative GNF2133
1.1 Reagents: Potassium carbonate Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: 1,4-Dioxane, Water; 1 h, 150 °C		By: Liu, Yahu A.; et al Journal of Medicinal Chemistry (2020), 63(6), 2958-2973.
Experimental Protocols		

Scheme 251 (1 Reaction)

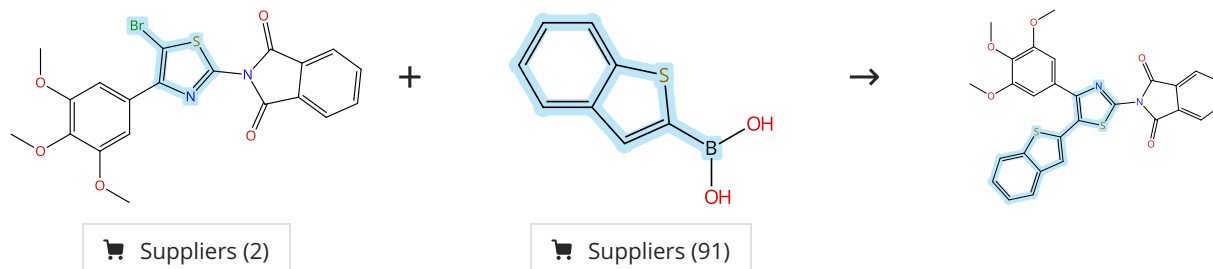
Steps: 1 Yield: 56%



31-179-CAS-17004524	Steps: 1 Yield: 56%	Discovery of GBT440, an Orally Bioavailable R-State Stabilizer of Sickle Cell Hemoglobin
1.1 Reagents: Potassium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Dimethylformamide; 30 min, 125 °C		By: Metcalf, Brian; et al ACS Medicinal Chemistry Letters (2017), 8(3), 321-326.
Experimental Protocols		

Scheme 252 (1 Reaction)

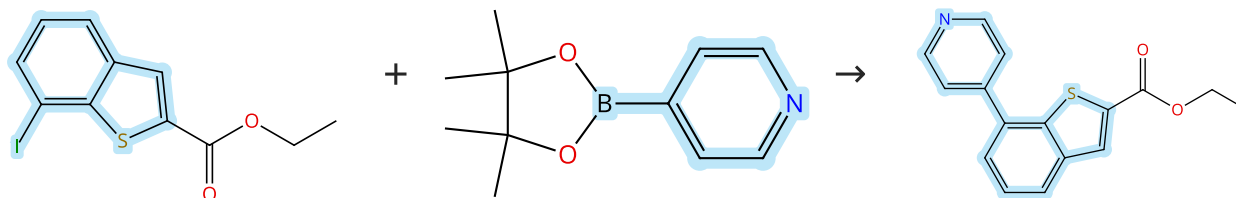
Steps: 1 Yield: 56%



31-179-CAS-12797052 Steps: 1 Yield: 56% 1.1 Reagents: Cesium fluoride Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: 1,4-Dioxane, Water; 30 min, 45 °C; 5 h, 75 °C Experimental Protocols	Convergent Synthesis and Biological Evaluation of 2-Amino-4-(3',4',5'-trimethoxyphenyl)-5-aryl Thiazoles as Microtubule Targeting Agents By: Romagnoli, Romeo; et al Journal of Medicinal Chemistry (2011), 54(14), 5144-5153.
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Scheme 253 (1 Reaction)

Steps: 1 Yield: 56%



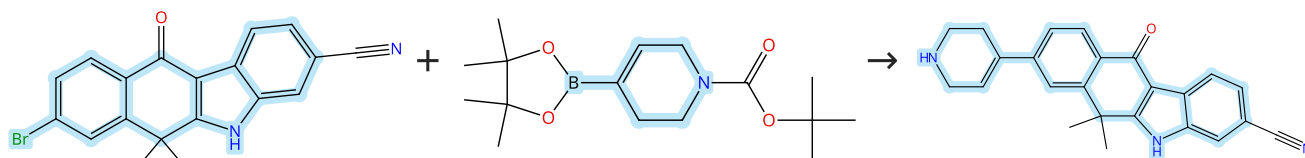
Suppliers (5)

Suppliers (94)

31-179-CAS-20571279 Steps: 1 Yield: 56% 1.1 Reagents: Sodium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Ethanol, 1,2-Dimethoxyethane, Water; 2 d, 80 °C	Benzothiophene derivatives as phosphodiesterase 10A (PD E10A) inhibitors: Hit-to-lead studies By: Kawamoto, Yoshito; et al Bioorganic & Medicinal Chemistry Letters (2019), 29(11), 1419-1422.
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Scheme 254 (1 Reaction)

Steps: 1 Yield: 56%

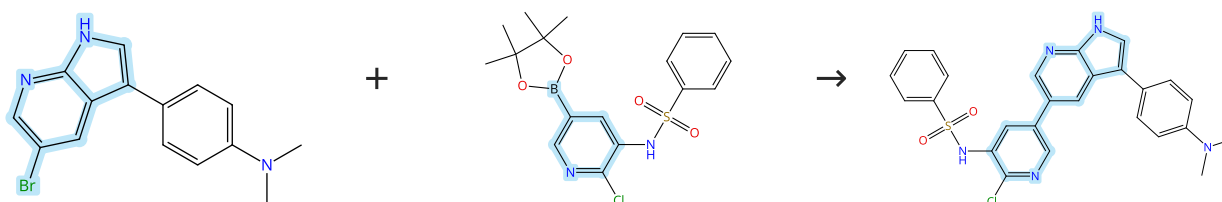


Suppliers (127)

31-177-CAS-9263997 Steps: 1 Yield: 56% 1.1 Reagents: Sodium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: 1,4-Dioxane, Water; 110 °C 1.2 Reagents: Trifluoroacetic acid Solvents: Dichloromethane; rt Experimental Protocols	Novel tetracyclic benzo[b]carbazolones as highly potent and orally bioavailable ALK inhibitors: Design, synthesis, and structure-activity relationship study By: Jiang, Xiaolong; et al European Journal of Medicinal Chemistry (2015), 105, 39-56.
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Scheme 255 (1 Reaction)

Steps: 1 Yield: 56%

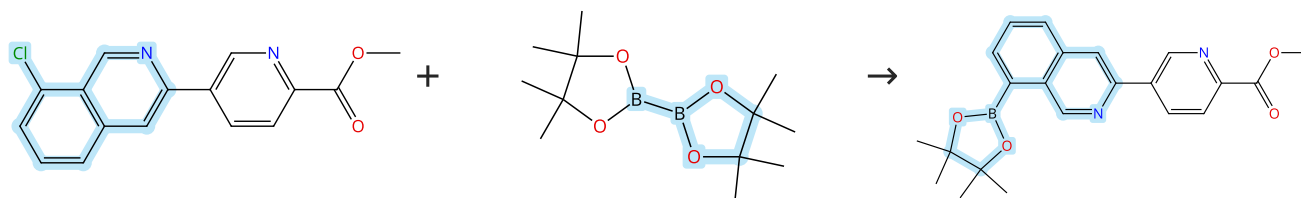


Suppliers (18)

31-614-CAS-32017302 Steps: 1 Yield: 56% 1.1 Reagents: Potassium carbonate Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: 1,4-Dioxane, Water; 12 h, 100 °C Experimental Protocols	Development of anti-breast cancer PI3K inhibitors based on 7-azaindole derivatives through scaffold hopping: Design, synthesis and in vitro biological evaluation By: Yang, Chengbin; et al Bioorganic Chemistry (2021), 117, 105405.
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Scheme 256 (1 Reaction)

Steps: 1 Yield: 56%

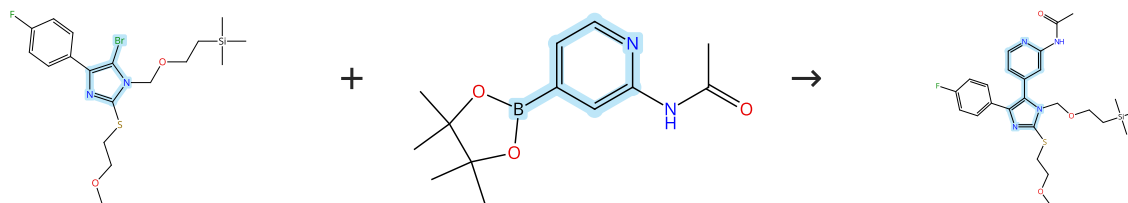


Suppliers (131)

31-614-CAS-40329155 Steps: 1 Yield: 56% 1.1 Reagents: Potassium acetate Catalysts: X-Phos, [2'-(Amino-κM)[1,1'-biphenyl]-2-yl-κC][[2',6'-bis(1-methylethoxy)[1,1'-biphenyl]-2-yl]dicyclohexylphosphine-κP]chloropalladium Solvents: 1,4-Dioxane; 1 h, rt → 106 °C Experimental Protocols	Discovery of Exceptionally Potent, Selective, and Efficacious P ROTAC Degraders of CBP and p300 Proteins By: Thomas, Junius Eugene II; et al Journal of Medicinal Chemistry (2023), 66(12), 8178-8199.
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Scheme 257 (1 Reaction)

Steps: 1 Yield: 56%

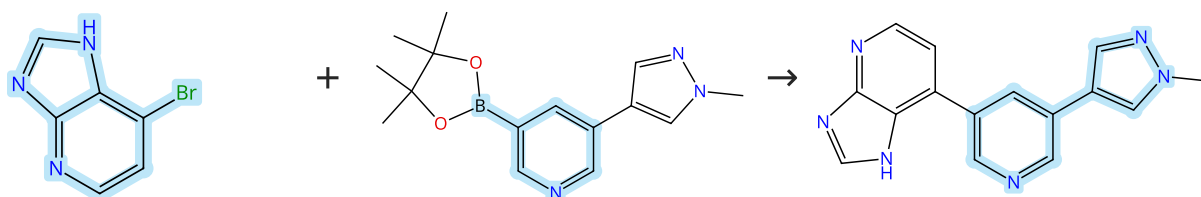


Suppliers (66)

31-179-CAS-22191288 Steps: 1 Yield: 56% 1.1 Reagents: Tripotassium phosphate Catalysts: [2'-(Amino-κM)[1,1'-biphenyl]-2-yl-κC](methanesulfonato-κO)[tris(1,1-dimethylethyl)phosphine]palladium Solvents: 1,4-Dioxane, Water; 5 h, 50 °C Experimental Protocols	Structural Basis for EGFR Mutant Inhibition by Trisubstituted Imidazole Inhibitors By: Heppner, David E.; et al Journal of Medicinal Chemistry (2020), 63(8), 4293-4305.
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Scheme 258 (1 Reaction)

Steps: 1 Yield: 56%



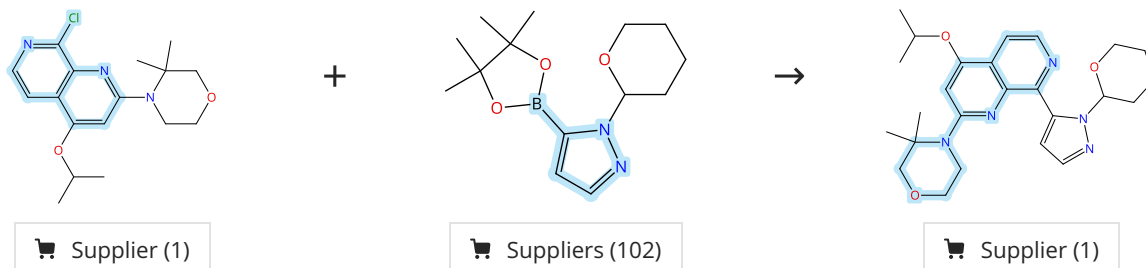
Suppliers (69)

Suppliers (29)

31-614-CAS-39099445 Steps: 1 Yield: 56% 1.1 Reagents: Cesium carbonate Catalysts: Dichloro[1,1'-bis(diphenylphosphino)ferrocene] palladium(II) dichloromethane adduct Solvents: 1,4-Dioxane, Water; 7 h, 100 °C Experimental Protocols	Discovery of a brain-permeable bromodomain and extra terminal domain (BET) inhibitor with selectivity for BD1 for the treatment of multiple sclerosis By: Chen, Xuetao; et al European Journal of Medicinal Chemistry (2024), 265, 116080.
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Scheme 259 (1 Reaction)

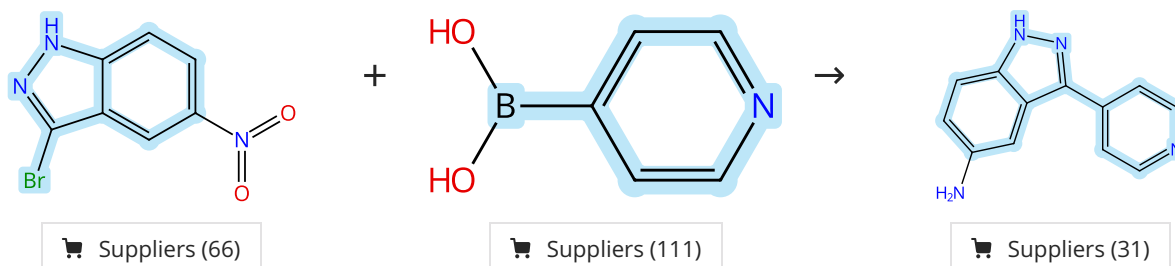
Steps: 1 Yield: 56%



31-179-CAS-22678443 Steps: 1 Yield: 56% 1.1 Reagents: Cesium carbonate Catalysts: Dichloro[1,1'-bis(diphenylphosphino)ferrocene] palladium(II) dichloromethane adduct Solvents: 1,4-Dioxane; 16 h, 90 °C	Damage Incorporated: Discovery of the Potent, Highly Selective, Orally Available ATR Inhibitor BAY 1895344 with Favorable Pharmacokinetic Properties and Promising Efficacy in Monotherapy and in Combination Treatments in Preclinical Tumor Models By: Luecking, Ulrich; et al Journal of Medicinal Chemistry (2020), 63(13), 7293-7325.
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Scheme 260 (1 Reaction)

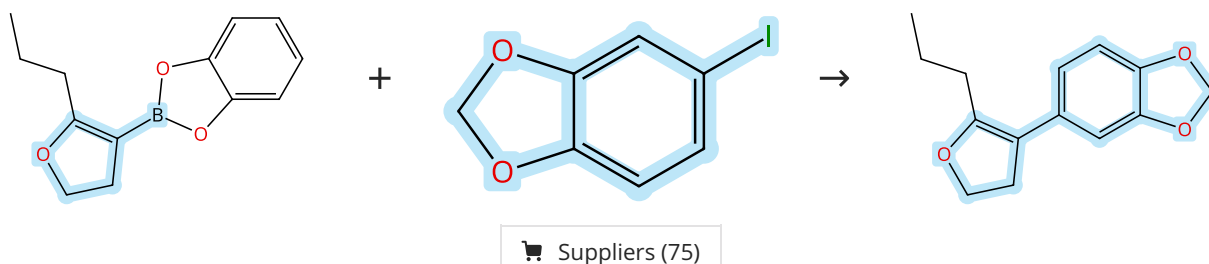
Steps: 1 Yield: 56%



31-179-CAS-23648036 Steps: 1 Yield: 56% 1.1 Reagents: Potassium acetate Catalysts: (<i>SP</i>-4-1)-Bis[4-[bis(1,1-dimethylethyl)phosphino-κ<i>P</i>]-<i>N,N</i>-dimethylbenzenamine]dichloropalladium Solvents: Ethanol, Water; 16 h, 100 °C 1.2 Reagents: Zinc, Ammonium chloride Solvents: Ethanol, Water; 12 h, 80 °C Experimental Protocols	Selective Inhibitors of G2019S-LRRK2 Kinase Activity By: Garofalo, Albert W.; et al Journal of Medicinal Chemistry (2020), 63(23), 14821-14839.
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Scheme 261 (1 Reaction)

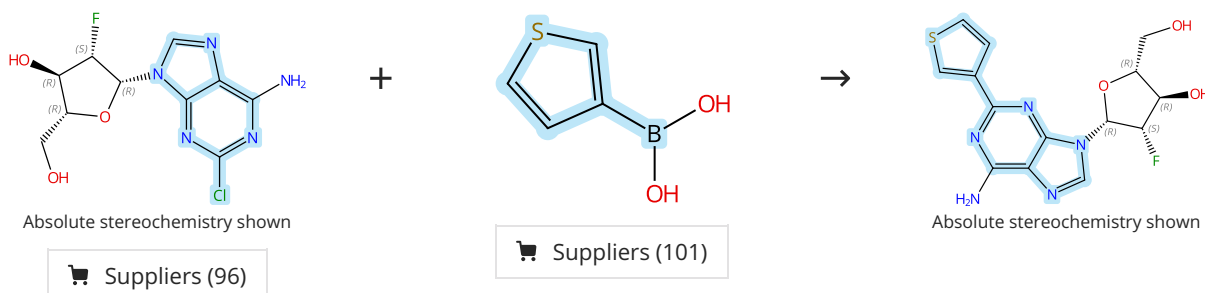
Steps: 1 Yield: 56%



31-177-CAS-19169799 Steps: 1 Yield: 56% 1.1 Reagents: Cesium hydroxide Catalysts: (SP-4-3)-[2'-(Amino-κM)[1,1'-biphenyl]-2-yl-κC][dicyclohexyl[2',4',6'-tris(1-methylethyl)[1,1'-biphenyl]-2-yl]phosphine](methanesulfonato-κO)palladium Solvents: Tetrahydrofuran; 2.5 d, 25 °C Experimental Protocols	An Oxyboration Route to a Single Regioisomer of Borylated Dihydrofurans and Isochromenes By: Tu, Kim N.; et al Journal of Organic Chemistry (2018), 83(18), 11204-11217.
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Scheme 262 (1 Reaction)

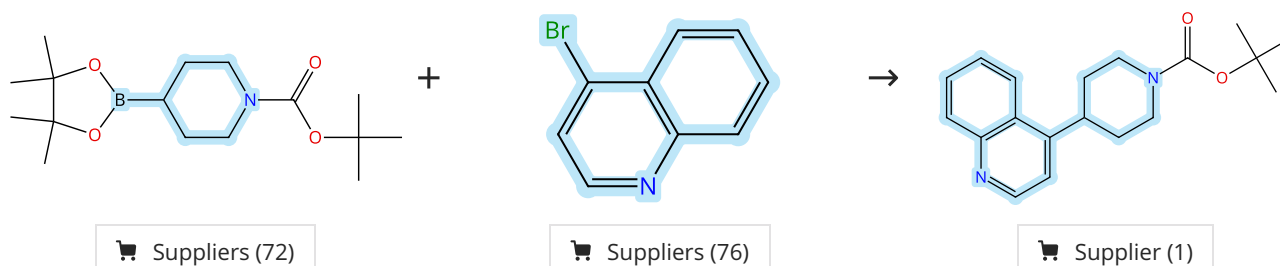
Steps: 1 Yield: 56%



31-614-CAS-39759932 Steps: 1 Yield: 56% 1.1 Reagents: Potassium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Tetrahydrofuran, Water; 20 h, rt → 100 °C Experimental Protocols	Design, synthesis and biological evaluation of Nucleosidic C D99 inhibitors that selectively reduce Ewing Sarcoma viability By: Balaraman, Kaluvu; et al European Journal of Medicinal Chemistry (2023), 251, 115244.
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Scheme 263 (1 Reaction)

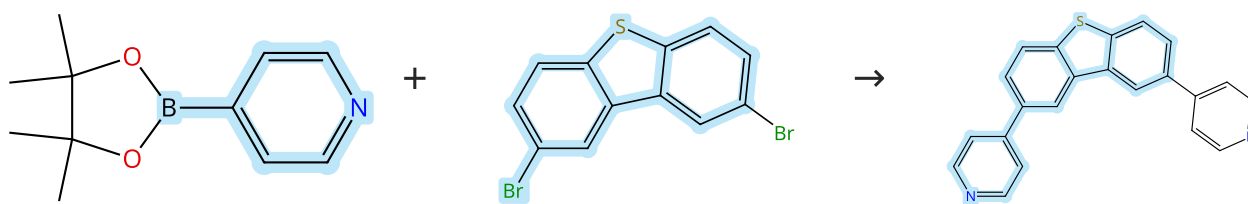
Steps: 1 Yield: 56%



31-614-CAS-32067937 Steps: 1 Yield: 56% 1.1 Reagents: Morpholine Catalysts: Iridium(1+), [4,4'-bis(1,1-dimethylethyl)-2,2'-bipyridine-κM¹,κM¹][bis[3,5-difluoro-2-[5-(trifluoromethyl)-2-pyridinyl-κM]phenyl-κC]-, (OC-6-33)-, hexafluorophosphate(1-)(1:1) Solvents: Dimethylformamide; rt 1.2 Reagents: Dichloro[1,2-di(methoxy-κO)ethane]nickel, 4,4'-Bis(1,1-dimethylethyl)-2,2'-bipyridine Solvents: Dimethylformamide; 30 s; 100 °C 1.3 2 h Experimental Protocols	ART - An Amino Radical Transfer Strategy for C (sp²)-C(sp³) Coupling Reactions, Enabled by Dual Photo/Nickel Catalysis By: Speckmeier, Elisabeth; et al Journal of the American Chemical Society (2022), 144(22), 9997-10005.
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Scheme 264 (1 Reaction)

Steps: 1 Yield: 56%



Suppliers (94)

Suppliers (84)

Suppliers (6)

31-614-CAS-41764047

Steps: 1 Yield: 56%

Non-Templated Assembly of D_{5h} -Symmetric Pd_5L_{10} Rings by Precise Ligand Angle Adjustment

- 1.1 **Reagents:** Potassium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: 1,4-Dioxane, Water; 22 h, 90 °C

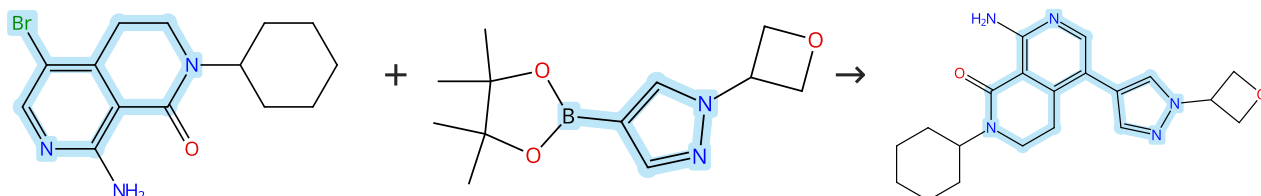
By: Neukirch, Laura; et al

Chemistry - A European Journal (2024), 30(27), e202400132.

Experimental Protocols

Scheme 265 (1 Reaction)

Steps: 1 Yield: 56%



Suppliers (61)

31-614-CAS-32756539

Steps: 1 Yield: 56%

Inhibition of ALK2 with bicyclic pyridyl lactams

- 1.1 **Reagents:** Tripotassium phosphate
Catalysts: (*SP*-4-4)-[2'-(Amino- κ M)[1,1'-biphenyl]-2-yl- κ C]chloro [dicyclohexyl[2',4',6'-tris(1-methylethyl)[1,1'-biphenyl]-2-yl] phosphine]palladium
Solvents: 1,4-Dioxane, Water; rt $\rightarrow$ 90 °C; 12 h, 90 °C

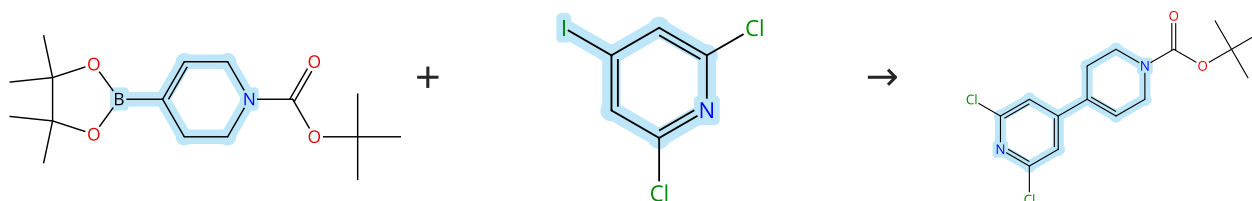
By: Witten, Michael R.; et al

Bioorganic & Medicinal Chemistry Letters (2022), 55, 128452.

Experimental Protocols

Scheme 266 (1 Reaction)

Steps: 1 Yield: 56%



Suppliers (127)

Suppliers (82)

Suppliers (36)

31-177-CAS-13944061

Steps: 1 Yield: 56%

Discovery of Dual Leucine Zipper Kinase (DLK, MAP3K12) Inhibitors with Activity in Neurodegeneration Models

- 1.1 **Reagents:** Potassium carbonate
Catalysts: Dichloro[1,1'-bis(diphenylphosphino)ferrocene] palladium(II) dichloromethane adduct
Solvents: 1,4-Dioxane, Water; 16 h, 90 °C

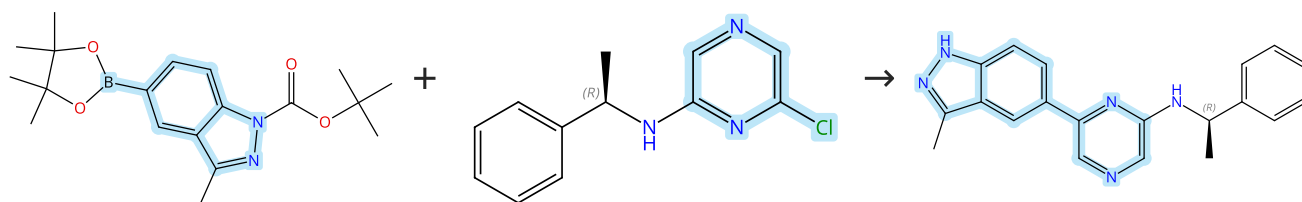
By: Patel, Snahel; et al

Journal of Medicinal Chemistry (2015), 58(1), 401-418.

Experimental Protocols

Scheme 267 (1 Reaction)

Steps: 1 Yield: 56%



Suppliers (60)

Suppliers (5)

Supplier (1)

31-614-CAS-31341831

Steps: 1 Yield: 56%

Rational Design and Development of Novel CDK9 Inhibitors for the Treatment of Acute Myeloid Leukemia

By: Han, Xu; et al

Journal of Medicinal Chemistry (2021), 64(19), 14647-14663.

1.1 **Reagents:** Potassium carbonate
Catalysts: Dichloro[1,1'-bis(diphenylphosphino)ferrocene] palladium(II) dichloromethane adduct
Solvents: 1,4-Dioxane, Water; rt → 90 °C; 12 h, 90 °C; 90 °C → rt

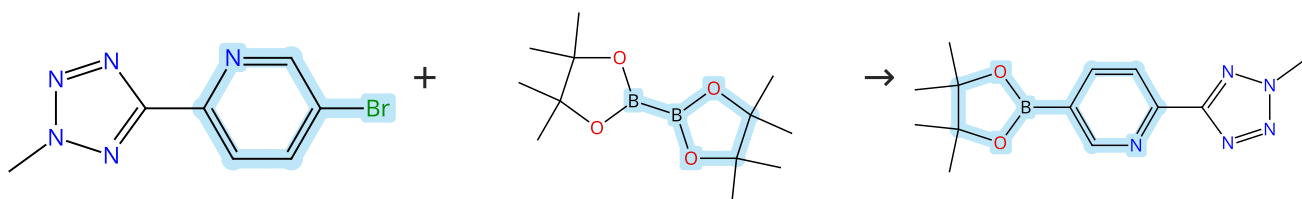
1.2 **Reagents:** Trifluoroacetic acid
Solvents: Dichloromethane; 1 - 2 h, rt

1.3 **Reagents:** Sodium bicarbonate
Solvents: Ethyl acetate, Water; > pH 7, rt

Experimental Protocols

Scheme 268 (1 Reaction)

Steps: 1 Yield: 56%



Suppliers (66)

Suppliers (131)

Suppliers (56)

31-614-CAS-28941634

Steps: 1 Yield: 56%

Copper Catalyzed Assembly of N-Aryloxazolidinones: Synthesis of Linezolid, Tedizolid, and Rivaroxaban

By: Mahy, William; et al

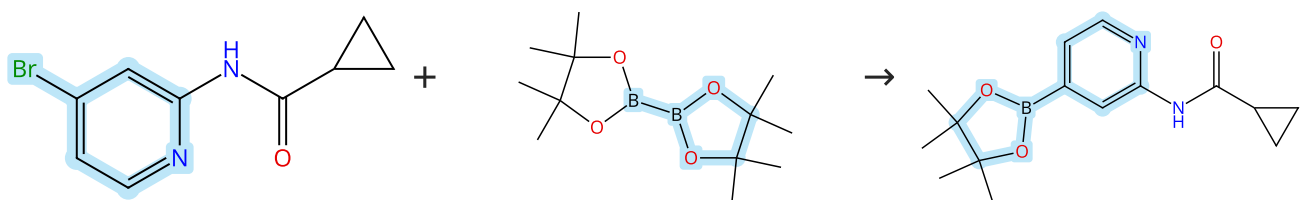
European Journal of Organic Chemistry (2016), 2016(7), 1305-1313.

1.1 **Reagents:** Potassium acetate
Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium
Solvents: Dimethyl sulfoxide; 2 h, 80 °C

Experimental Protocols

Scheme 269 (1 Reaction)

Steps: 1 Yield: 55%



Suppliers (32)

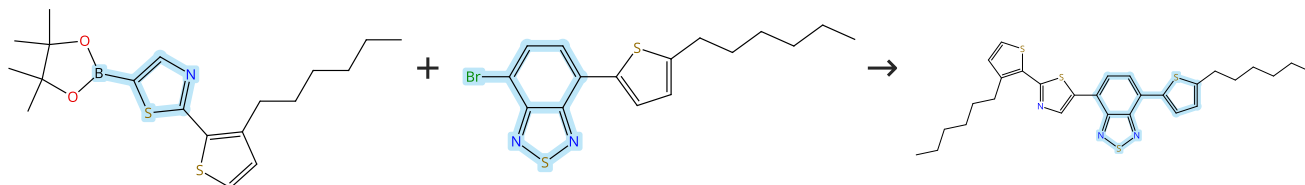
Suppliers (131)

Suppliers (47)

31-142-CAS-18130602 Steps: 1 Yield: 55% 1.1 Reagents: Potassium acetate Catalysts: Dichloro[1,1'-bis(diphenylphosphino)ferrocene] palladium(II) dichloromethane adduct Solvents: Dimethylformamide; 2 h, 80 °C	Discovery of new acylaminopyridines as GSK-3 inhibitors by a structure guided in-depth exploration of chemical space around a pyrrolopyridinone core By: Sivaprakasam, Prasanna; et al Bioorganic & Medicinal Chemistry Letters (2015), 25(9), 1856-1863.
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Scheme 270 (1 Reaction)

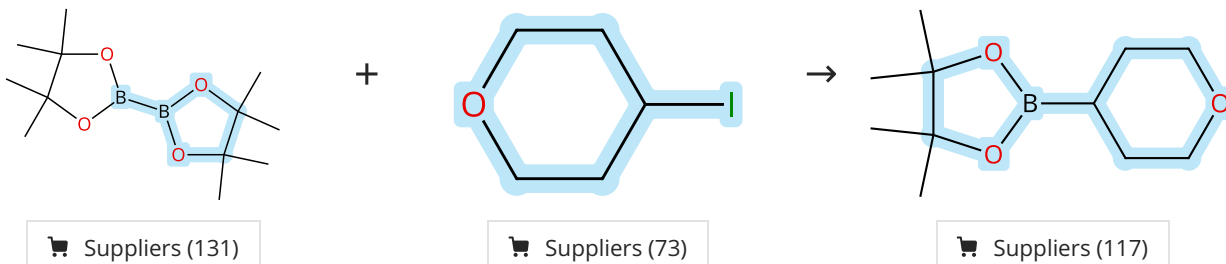
Steps: 1 Yield: 55%



31-179-CAS-10361896 Steps: 1 Yield: 55% 1.1 Reagents: Pivalic acid, Potassium carbonate Catalysts: Palladium diacetate, Phosphine, bis(1,1-dimethylethyl)methyl-, tetrafluoroborate(1-) (1:1) Solvents: Dimethylacetamide; 5 h, 100 °C Experimental Protocols	Synthesis of Benzothiadiazole Derivatives by Applying C-C Cross-Couplings By: Heiskanen, Juha P.; et al Journal of Organic Chemistry (2016), 81(4), 1535-1546.
--	--

Scheme 271 (1 Reaction)

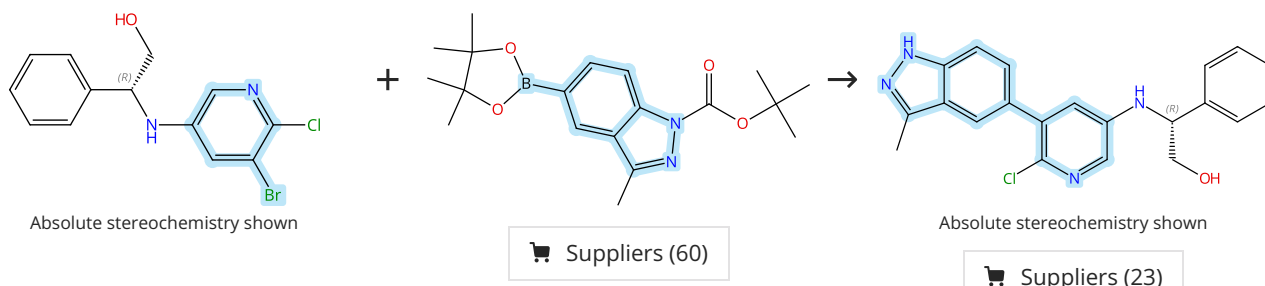
Steps: 1 Yield: 55%



31-614-CAS-30870870 Steps: 1 Yield: 55% 1.1 Reagents: Lithium <i>tert</i>-butoxide Solvents: Methanol, Water; 48 h, 50 °C	Transition-Metal-Free Borylation of Alkyl Iodides via a Radical Mechanism By: Liu, Qianyi; et al Organic Letters (2019), 21(17), 6597-6602.
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Scheme 272 (1 Reaction)

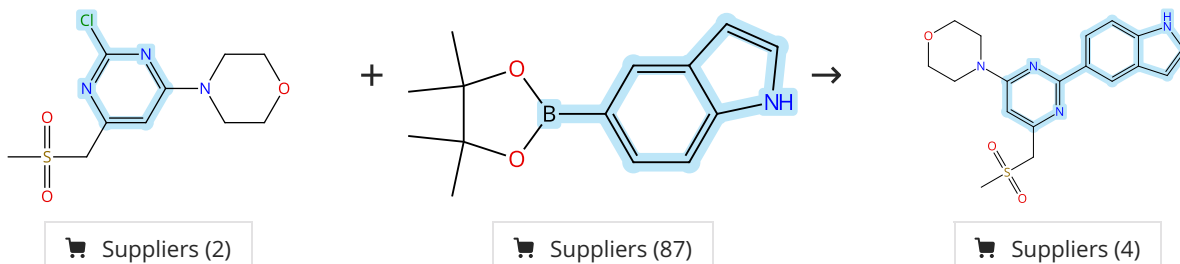
Steps: 1 Yield: 55%



31-179-CAS-17519080 Steps: 1 Yield: 55% 1.1 Reagents: Sodium carbonate Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: 1,4-Dioxane, Water; overnight, 80 °C; 80 °C → rt 1.2 Reagents: Trifluoroacetic acid Solvents: Dichloromethane; 6 h, rt 1.3 Reagents: Sodium hydroxide Solvents: Water; neutralized, cooled Experimental Protocols	Design, Synthesis, and Pharmacological Evaluation of Novel Multisubstituted Pyridin-3-amine Derivatives as Multitargeted Protein Kinase Inhibitors for the Treatment of Non-Small Cell Lung Cancer By: Zhu, Wei; et al Journal of Medicinal Chemistry (2017), 60(14), 6018-6035.
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Scheme 273 (1 Reaction)

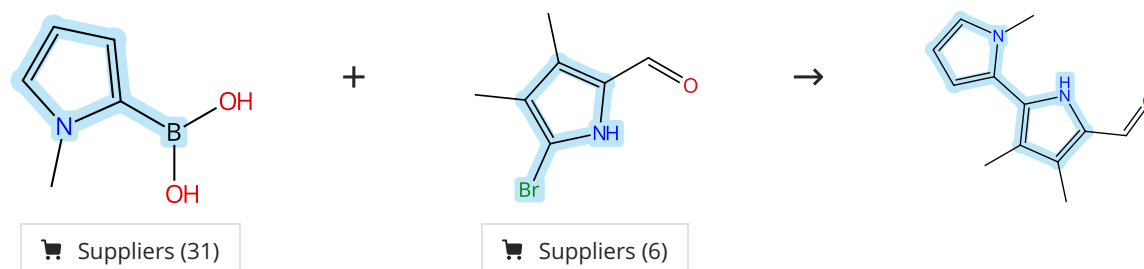
Steps: 1 Yield: 55%



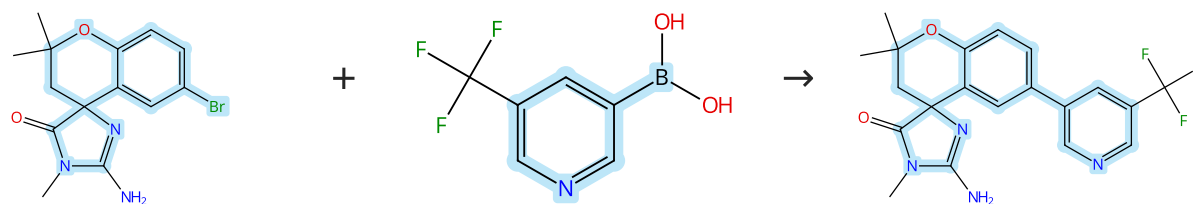
31-179-CAS-5967143 Steps: 1 Yield: 55% 1.1 Reagents: Sodium carbonate Catalysts: Dichlorobis(triphenylphosphine)palladium Solvents: Ethanol, Dimethylformamide, 1,2-Dimethoxyethane, Water; 1 h, 95 °C Experimental Protocols	Discovery of 4-{4-[(3R)-3-Methylmorpholin-4-yl]-6-[1-(methylsulfonyl)cyclopropyl]pyrimidin-2-yl}-1H-indole (AZ20): A Potent and Selective Inhibitor of ATR Protein Kinase with Monotherapy in Vivo Antitumor Activity By: Foote, Kevin M.; et al Journal of Medicinal Chemistry (2013), 56(5), 2125-2138.
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Scheme 274 (1 Reaction)

Steps: 1 Yield: 55%



31-179-CAS-12607790 Steps: 1 Yield: 55% 1.1 Reagents: Sodium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: 1,4-Dioxane, Water; 3 h, 100 °C 1.2 Reagents: Hydrochloric acid Solvents: Water; pH 7, rt Experimental Protocols	Synthesis and Structure-Activity Relationships of Tambjamine and B-Ring Functionalized Prodigins as Potent Antimalarials By: Kancharla, Papireddy; et al Journal of Medicinal Chemistry (2015), 58(18), 7286-7309.
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Steps: **1** Yield: **55%**

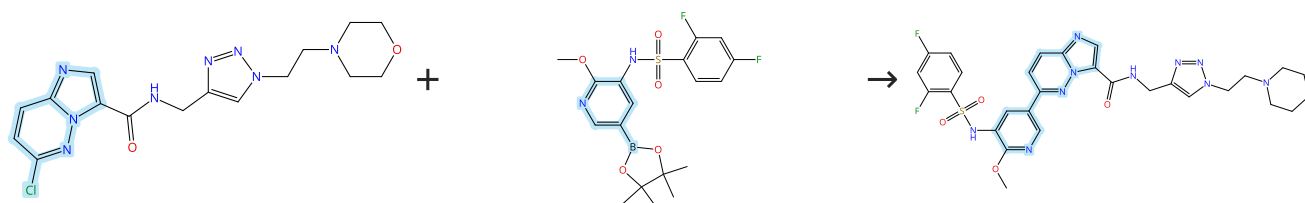
 Supplier (1)

Spirocyclic β -Site Amyloid Precursor Protein Cleaving Enzyme 1 (BACE1) Inhibitors: From Hit to Lowering of Cerebrospinal Fluid (CSF) Amyloid β in a Higher Species

By: Hunt, Kevin W.; et al

Journal of Medicinal Chemistry (2013), 56(8), 3379-3403.

Steps: **1** Yield: **55%**



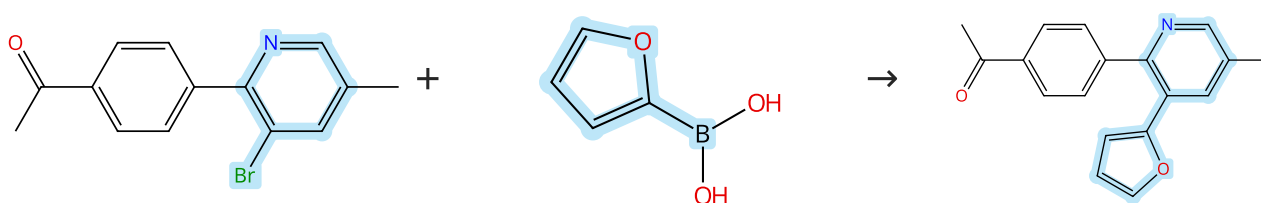
 Suppliers (54)

Function-oriented synthesis of Imidazo[1,2-a]pyrazine and Imidazo[1,2-b]pyridazine derivatives as potent PI3K/mTOR dual inhibitors

By: Li, Chuchu; et al

European Journal of Medicinal Chemistry (2023), 247, 115030.

Steps: **1** Yield: **55%**



 Suppliers (93)

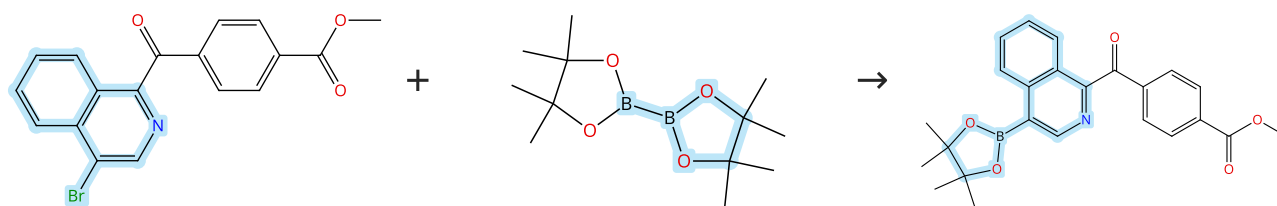
Syntheses of Benzofuranoquinolines and Analogues via Photoinduced Acceptorless Dehydrogenative Annulation of o-Phenylfuranylpiperidines

By: Fan, Jinming; et al

Organic Letters (2019), 21(22), 9183-9187.

Scheme 278 (2 Reactions)

Steps: 1 Yield: 55%



Suppliers (11)

Suppliers (131)

Suppliers (10)

31-142-CAS-4991086

Steps: 1 Yield: 55%

Isoquinoline derivatives as potent CRTH2 antagonists: design, synthesis and SAR

By: Nishikawa-Shimono, Rie; et al

Bioorganic & Medicinal Chemistry (2013), 21(24), 7674-7685.

1.1 **Reagents:** Potassium acetate
Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium
Solvents: Dimethylformamide; 2 h, 80 °C

Experimental Protocols

31-142-CAS-10224669

Steps: 1 Yield: 55%

Isoquinoline derivatives as potent CRTH2 receptor antagonists: Synthesis and SAR

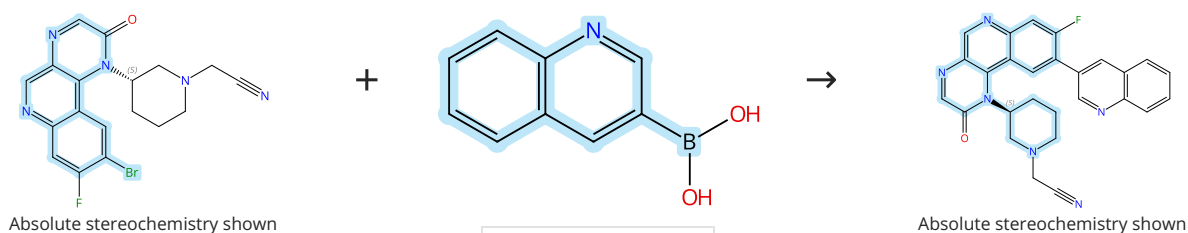
By: Nishikawa-Shimono, Rie; et al

Bioorganic & Medicinal Chemistry Letters (2012), 22(9), 3305-3310.

1.1 **Reagents:** Potassium acetate
Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium; 80 °C

Scheme 279 (1 Reaction)

Steps: 1 Yield: 55%



Absolute stereochemistry shown

Absolute stereochemistry shown

Supplier (1)

Suppliers (107)

Supplier (1)

31-179-CAS-18736821

Steps: 1 Yield: 55%

Highly Selective, Potent, and Oral mTOR Inhibitor for Treatment of Cancer as Autophagy Inducer

By: Guo, Qingxiang; et al

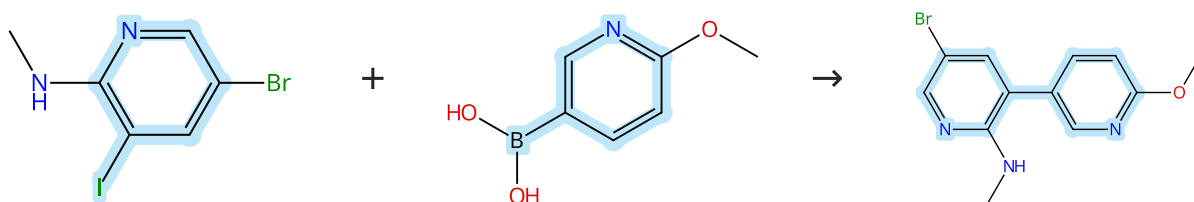
Journal of Medicinal Chemistry (2018), 61(3), 881-904.

1.1 **Reagents:** Sodium carbonate
Catalysts: Dichlorobis(triphenylphosphine)palladium
Solvents: 1,4-Dioxane; overnight, rt → 85 °C

Experimental Protocols

Scheme 280 (1 Reaction)

Steps: 1 Yield: 55%



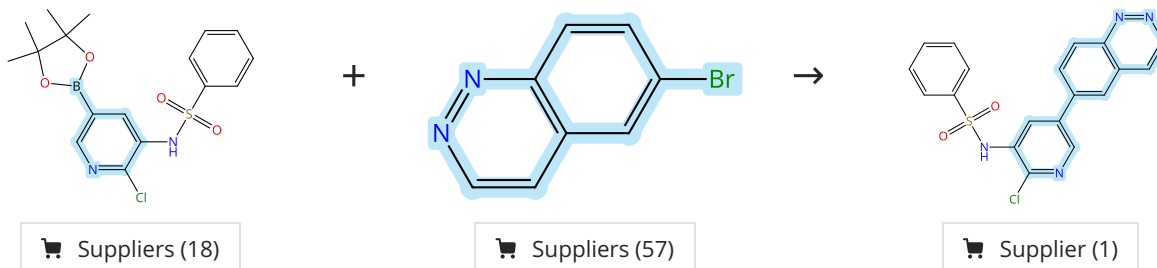
Suppliers (7)

Suppliers (100)

31-179-CAS-15107313	Steps: 1 Yield: 55%	Structure-Activity-Relationship Studies around the 2 Amino Group and Pyridine Core of Antimalarial 3,5-Diarylaminopyridines Lead to a Novel Series of Pyrazine Analogues with Oral in Vivo Activity By: Younis, Yassir; et al Journal of Medicinal Chemistry (2013), 56(21), 8860-8871.
1.1 Reagents: Potassium carbonate Solvents: 1,4-Dioxane, Water; 16 h, 110 °C		
Experimental Protocols		

Scheme 281 (1 Reaction)

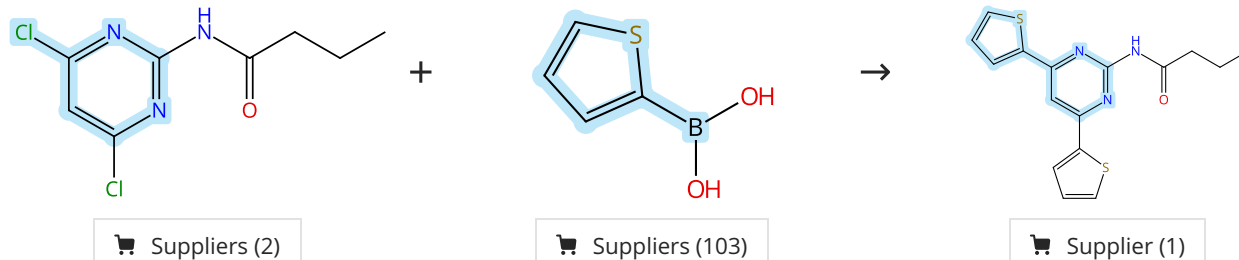
Steps: 1 Yield: 55%



31-614-CAS-32106598	Steps: 1 Yield: 55%	Discovery of cinnoline derivatives as potent PI3K inhibitors with antiproliferative activity By: Tian, Chengze; et al Bioorganic & Medicinal Chemistry Letters (2021), 48, 128271.
1.1 Reagents: Potassium carbonate Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: 1,4-Dioxane, Water; overnight, 55 °C		
Experimental Protocols		

Scheme 282 (1 Reaction)

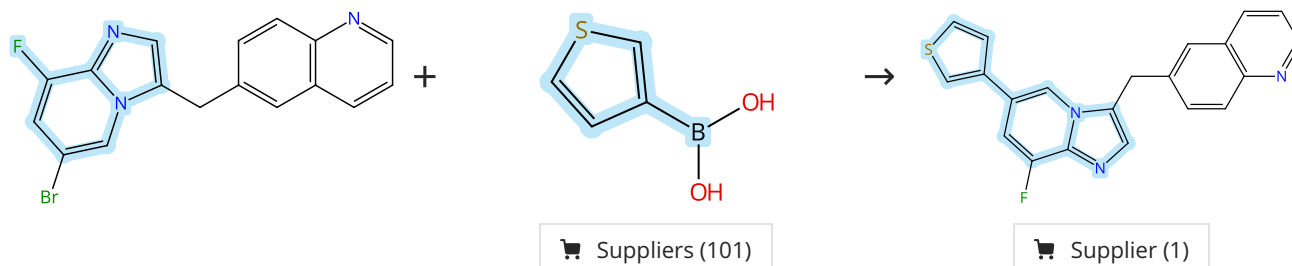
Steps: 1 Yield: 55%



31-179-CAS-5150103	Steps: 1 Yield: 55%	Pyrimidine derivatives as potent and selective A₃ adenosine receptor antagonists By: Yaziji, Vicente; et al Journal of Medicinal Chemistry (2011), 54(2), 457-471.
1.1 Reagents: Sodium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: 1,2-Dimethoxyethane, Water; 12 h, 110 °C		
Experimental Protocols		

Scheme 283 (1 Reaction)

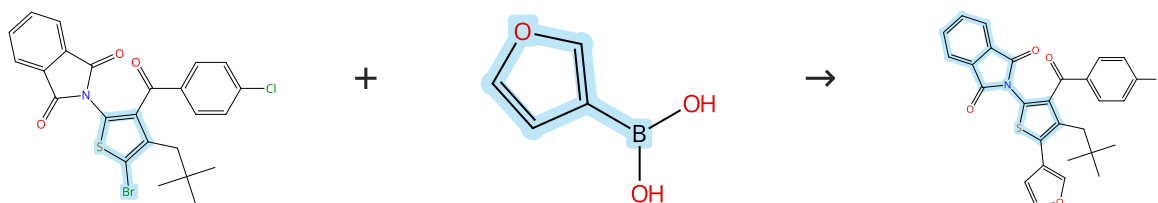
Steps: 1 Yield: 55%



31-179-CAS-633393	Steps: 1 Yield: 55%	Design, Synthesis, and Biological Evaluation of Novel Imidazo [1,2-a]pyridine Derivatives as Potent c-Met Inhibitors
1.1 Reagents: Sodium carbonate Catalysts: Dichloro[1,1'-bis(diphenylphosphino)ferrocene] palladium(II) dichloromethane adduct Solvents: Dimethylformamide, Water; 16 h, 80 °C		By: Li, Chunpu; et al ACS Medicinal Chemistry Letters (2015), 6(5), 507-512.
Experimental Protocols		

Scheme 284 (1 Reaction)

Steps: 1 Yield: 55%



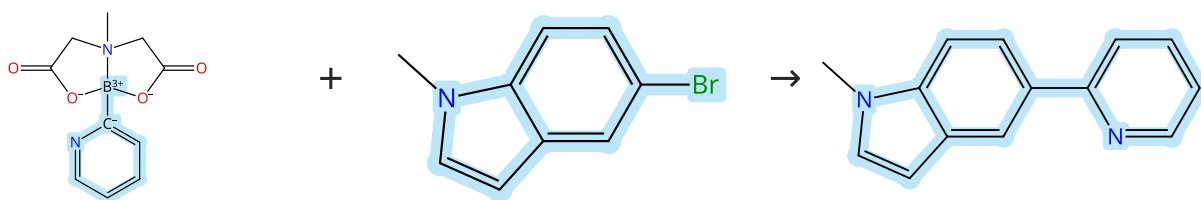
Supplier (1)

Suppliers (103)

31-179-CAS-11407853	Steps: 1 Yield: 55%	Synthesis and biological evaluation of novel 2- amino-3-aryl-4-neopentyl-5-substituted thiophene derivatives as allosteric enhancers of the A₁ adenosine receptor
1.1 Reagents: Cesium fluoride Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: 1,4-Dioxane, Water; 30 min, 45 °C; 6 h, 65 °C		By: Romagnoli, Romeo; et al Bioorganic & Medicinal Chemistry (2014), 22(1), 148-166.
Experimental Protocols		

Scheme 285 (1 Reaction)

Steps: 1 Yield: 55%



Suppliers (36)

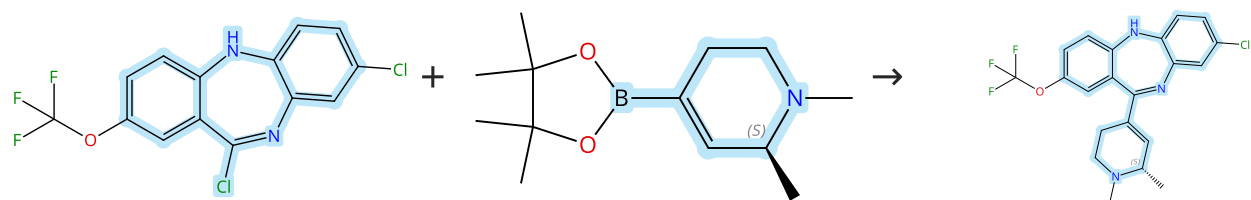
Suppliers (82)

Suppliers (8)

31-177-CAS-14435836	Steps: 1 Yield: 55%	A General Solution for the 2-Pyridyl Problem
1.1 Reagents: Diethanolamine, Cupric acetate, Tripotassium phosphate Catalysts: (SP-4-4)-[2-[2-(Amino-κMethyl)phenyl-κC]chloro[dicyclohexyl[2',4',6'-tris(1-methylethyl)[1,1'-biphenyl]-2-yl]phosphine]palladium Solvents: Dimethylformamide; 24 h, 100 °C; 0.5 h, 100 °C → 23 °C		By: Dick, Graham R.; et al Angewandte Chemie, International Edition (2012), 51(11), 2667-2672, S2667/1-S2667/80.
Experimental Protocols		

Scheme 286 (1 Reaction)

Steps: 1 Yield: 55%



Absolute stereochemistry shown

Absolute stereochemistry shown

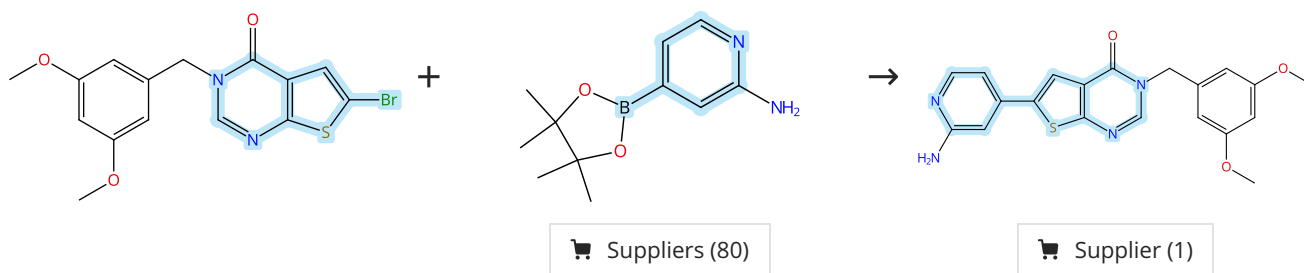
Suppliers (4)

Supplier (1)

31-614-CAS-29181075	Synthesis and pharmacological evaluation of 11-(1,6-dimethyl-1,2,3,6-tetrahydropyridin-4-yl)-5H-dibenzo[b,e][1,4]diazepines with clozapine-like receptor occupancy at dopamine D₁/D₂ receptor
1.1 Reagents: Sodium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Tetrahydrofuran, Water; 1 h, 75 °C	By: Watanabe, Hitoshi; et al Bioorganic & Medicinal Chemistry Letters (2020), 30(21), 127563.
Experimental Protocols	

Scheme 287 (1 Reaction)

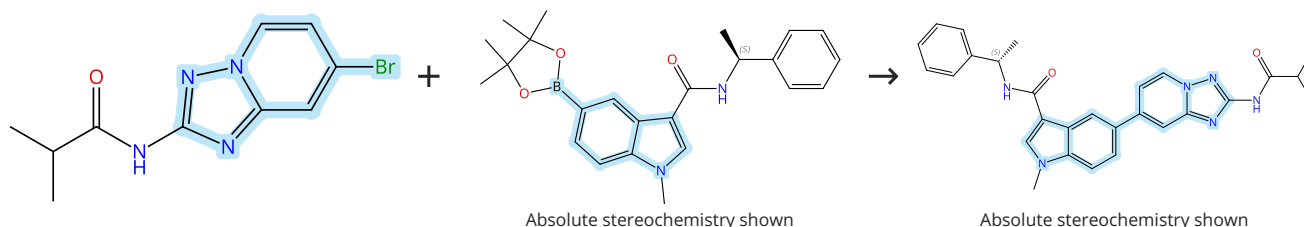
Steps: 1 Yield: 55%



31-614-CAS-32027366	Discovery of thieno[2,3-d]pyrimidin-4(3H)-one derivatives as a new class of ROCK inhibitors
1.1 Reagents: Potassium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: 1,4-Dioxane, Water; 12 h, 95 °C	By: Miao, Zhuang; et al Bioorganic & Medicinal Chemistry Letters (2020), 30(6), 126966.
Experimental Protocols	

Scheme 288 (1 Reaction)

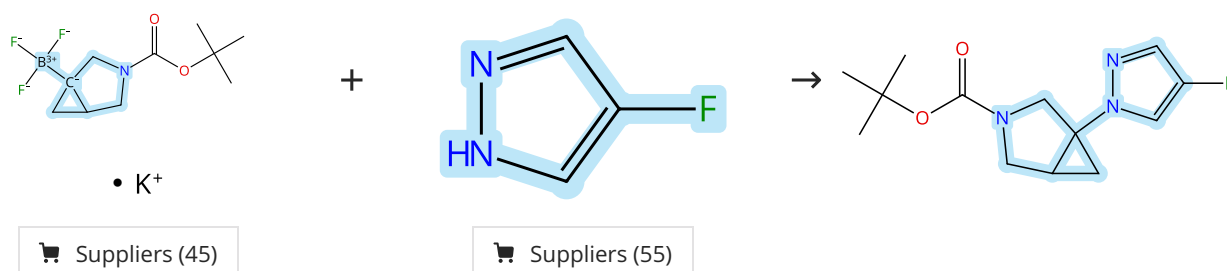
Steps: 1 Yield: 55%



31-614-CAS-39000315	From Hit to Lead: Structure-Based Optimization of Novel Selective Inhibitors of Receptor-Interacting Protein Kinase 1 (RIPK1) for the Treatment of Inflammatory Diseases
1.1 Reagents: Cesium carbonate Catalysts: Tricyclohexylphosphine, Dichloro[1,1'-bis(diphenylphosphino)ferrocene]palladium(II) dichloromethane adduct Solvents: 1,4-Dioxane, Water; 8 h, 100 °C	By: Zhang, Liting; et al Journal of Medicinal Chemistry (2024), 67(1), 754-773.
Experimental Protocols	

Scheme 289 (1 Reaction)

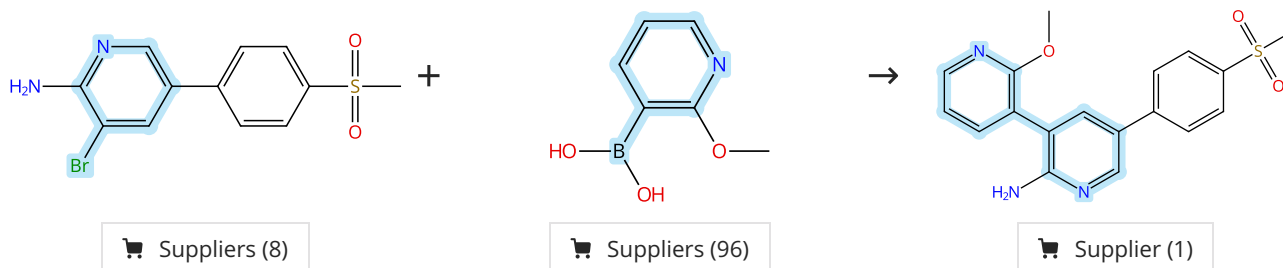
Steps: 1 Yield: 55%



31-614-CAS-26879290 Steps: 1 Yield: 55% 1.1 Catalysts: 1,10-Phenanthroline, Cupric acetate Solvents: 1,2-Dichloroethane; 20 min, 60 °C 1.2 Reagents: Tripotassium phosphate Solvents: 1,2-Dichloroethane, Water; 18 h, rt → 80 °C; 80 °C → rt; 18 h, 60 °C Experimental Protocols	Construction of 1-Heteroaryl-3-azabicyclo[3.1.0]hexanes by sp^3-sp^2 Suzuki-Miyaura and Chan-Evans-Lam Coupling Reactions of Tertiary Trifluoroborates By: Harris, Michael R.; et al Organic Letters (2017), 19(9), 2450-2453.
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Scheme 290 (1 Reaction)

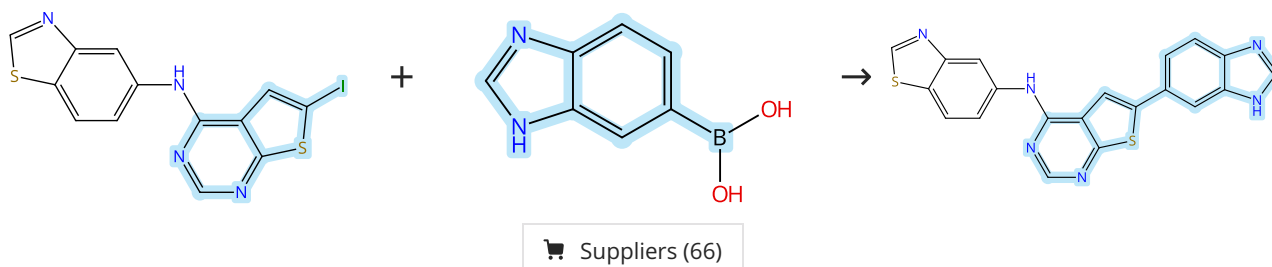
Steps: 1 Yield: 55%



31-179-CAS-13291004 Steps: 1 Yield: 55% 1.1 Reagents: Potassium carbonate Catalysts: Dichlorobis(triphenylphosphine)palladium Solvents: 1,4-Dioxane, Water; 16 h, 110 °C Experimental Protocols	3,5-Diaryl-2-aminopyridines as a Novel Class of Orally Active Antimalarials Demonstrating Single Dose Cure in Mice and Clinical Candidate Potential By: Younis, Yassir; et al Journal of Medicinal Chemistry (2012), 55(7), 3479-3487.
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Scheme 291 (1 Reaction)

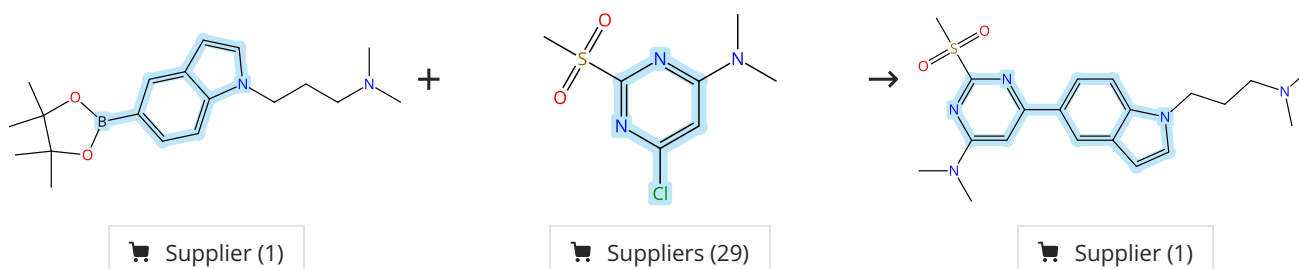
Steps: 1 Yield: 55%



31-614-CAS-38842749 Steps: 1 Yield: 55% 1.1 Reagents: Tripotassium phosphate Catalysts: Tris(dibenzylideneacetone)dipalladium, 1,1'-(9,9-Dimethyl-9H-xanthene-4,5-diyl)bis[1,1-diphenylphosphine] Solvents: 1,4-Dioxane, Water; 24 h, 95 °C Experimental Protocols	Design, synthesis and evaluation of novel thieno [2,3d] pyrimidine derivatives as potent and specific RIPK2 inhibitors By: Misehe, Mbilo; et al Bioorganic & Medicinal Chemistry Letters (2023), 97, 129567.
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Scheme 292 (1 Reaction)

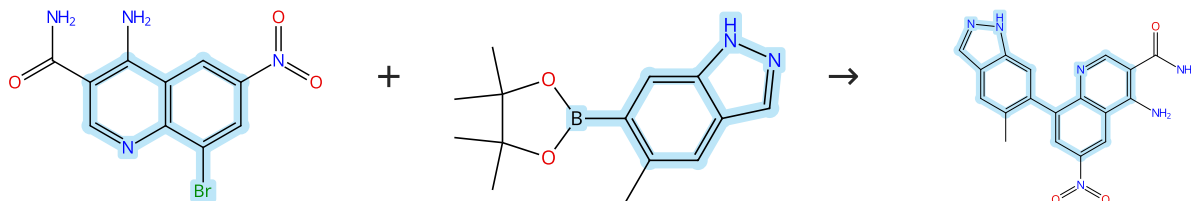
Steps: 1 Yield: 55%



31-614-CAS-31059949	Steps: 1 Yield: 55%	Discovery of High-Affinity Inhibitors of the BPTF Bromodomain By: Lu, Tian; et al Journal of Medicinal Chemistry (2021), 64(16), 12075-12088.
1.1 Reagents: Sodium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: 1,4-Dioxane, Water; 12 h, 90 °C		
Experimental Protocols		

Scheme 293 (1 Reaction)

Steps: 1 Yield: 55%



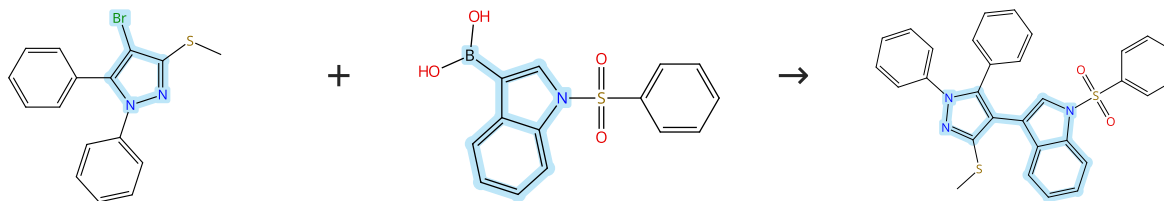
Suppliers (18)

Supplier (1)

31-179-CAS-20601373	Steps: 1 Yield: 55%	Discovery of 4-Aminoquinoline-3-carboxamide Derivatives as Potent Reversible Bruton's Tyrosine Kinase Inhibitors for the Treatment of Rheumatoid Arthritis By: Yao, Xia; et al Journal of Medicinal Chemistry (2019), 62(14), 6561-6574.
1.1 Reagents: Sodium bicarbonate Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: 1,4-Dioxane, Water; 1 - 2 h, 140 °C		
Experimental Protocols		

Scheme 294 (1 Reaction)

Steps: 1 Yield: 55%



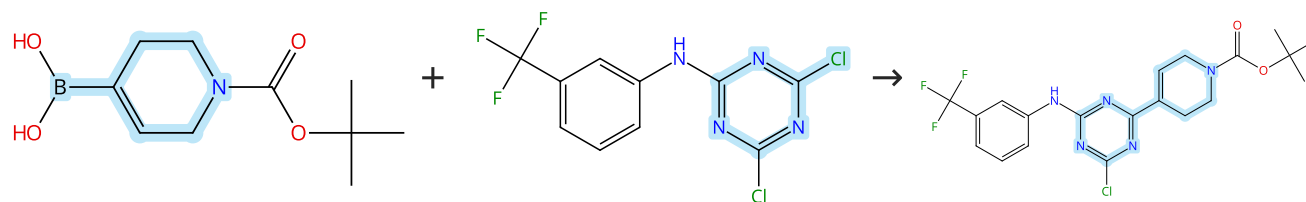
Supplier (1)

Suppliers (65)

31-179-CAS-5332661	Steps: 1 Yield: 55%	Regioselective rapid analog synthesis of 1,3-(or 1,5-)diphenyl-4-aryl/heteroaryl-5-(or 3-)(methylthio)pyrazoles via Suzuki cross-coupling By: Khan, Taukeer A.; et al Tetrahedron (2011), 67(16), 2961-2968.
1.1 Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Ethanol, Toluene, Water; 10 - 12 h, 90 - 120 °C		
Experimental Protocols		

Scheme 295 (1 Reaction)

Steps: 1 Yield: 55%



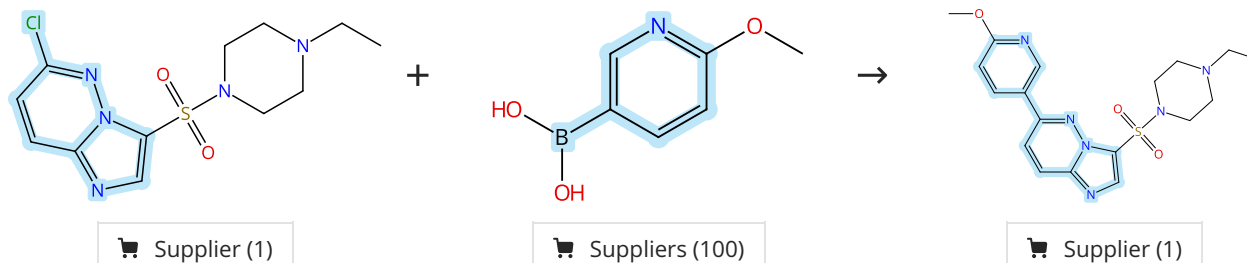
Suppliers (63)

Suppliers (28)

31-614-CAS-40632098 Steps: 1 Yield: 55% 1.1 Reagents: Potassium carbonate Catalysts: Dichloro[1,1'-bis(diphenylphosphino)ferrocene] palladium(II) dichloromethane adduct Solvents: 1,4-Dioxane, Water; 5 h, 80 °C Experimental Protocols	Discovery of CDC42 inhibitors with a favorable pharmacokinetic profile and anticancer in vivo efficacy By: Brindani, Nicoletta; et al Journal of Medicinal Chemistry (2024), 67(12), 10401-10424.
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Scheme 296 (1 Reaction)

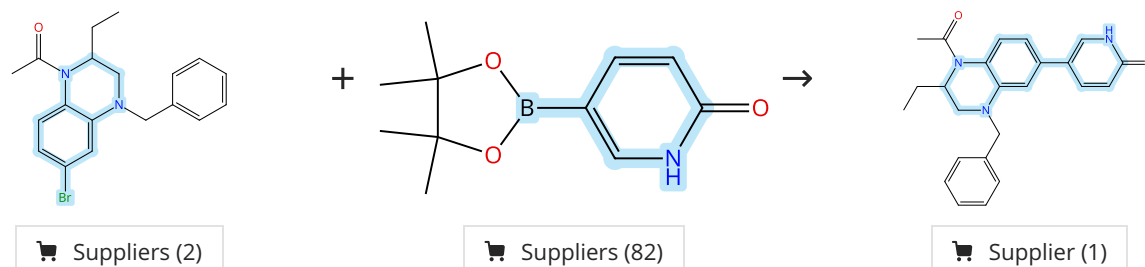
Steps: 1 Yield: 55%



31-179-CAS-18399001 Steps: 1 Yield: 55% 1.1 Reagents: Potassium carbonate Catalysts: Dichlorobis(triphenylphosphine)palladium Solvents: Dimethylformamide, Water; 3 - 5 h, rt → 120 °C	Synthesis and in vitro evaluations of 6-(hetero)-aryl-imidazo [1,2-b]pyridazine-3-sulfonamide's as an inhibitor of TNF-α production By: Pandit, Shivaji S.; et al Bioorganic & Medicinal Chemistry Letters (2018), 28(1), 24-30.
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Scheme 297 (1 Reaction)

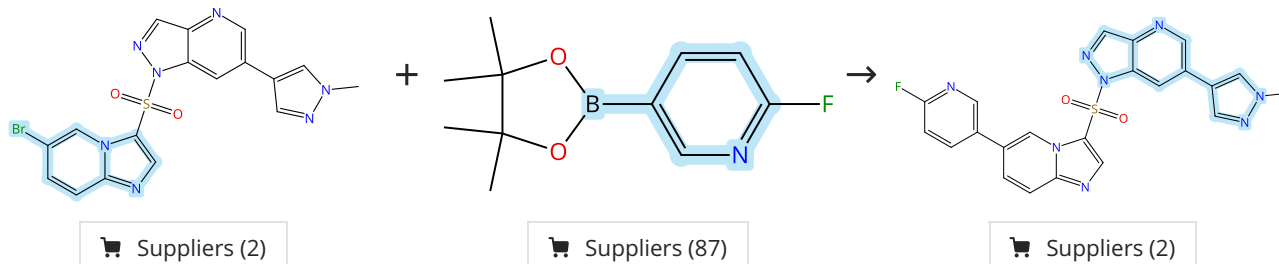
Steps: 1 Yield: 55%



31-179-CAS-18808903 Steps: 1 Yield: 55% 1.1 Reagents: Cesium carbonate Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: 1,4-Dioxane, Water; 2 h, 110 °C Experimental Protocols	Discovery of Tetrahydroquinoxalines as Bromodomain and Extra-Terminal Domain (BET) Inhibitors with Selectivity for the Second Bromodomain By: Law, Robert P.; et al Journal of Medicinal Chemistry (2018), 61(10), 4317-4334.
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Scheme 298 (1 Reaction)

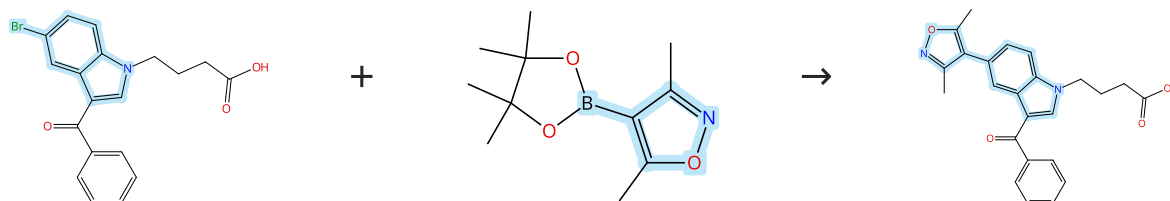
Steps: 1 Yield: 55%



31-179-CAS-8108174	Steps: 1 Yield: 55%	Design and Optimization of a Series of 1-Sulfonylpyrazolo[4,3-b]pyridines as Selective c-Met Inhibitors
1.1 Reagents: Potassium carbonate Catalysts: Palladium chloride Solvents: 1,4-Dioxane, Water; 5 min, rt; 30 min, 90 °C		By: Ma, Yuchi; et al Journal of Medicinal Chemistry (2015), 58(5), 2513-2529.

Scheme 299 (1 Reaction)

Steps: 1 Yield: 55%

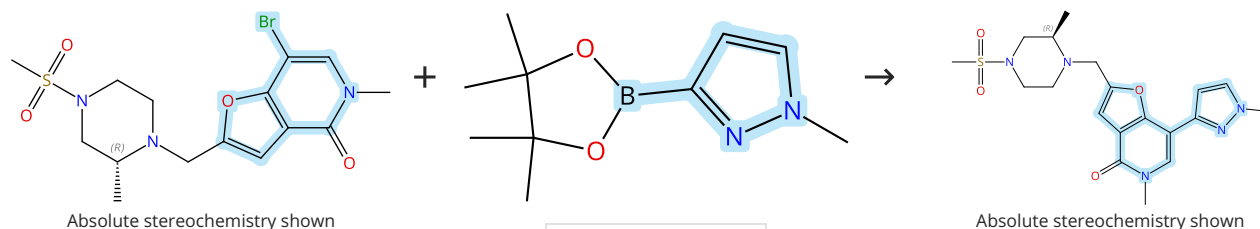


Suppliers (93)

31-179-CAS-434406	Steps: 1 Yield: 55%	Discovery and Optimization of Small-Molecule Ligands for the CBP/p300 Bromodomains
1.1 Reagents: Sodium bicarbonate Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: 1,2-Dimethoxyethane, Water; 48 h, 100 °C		By: Hay, Duncan A.; et al Journal of the American Chemical Society (2014), 136(26), 9308-9319.
Experimental Protocols		

Scheme 300 (1 Reaction)

Steps: 1 Yield: 55%



Absolute stereochemistry shown

Absolute stereochemistry shown

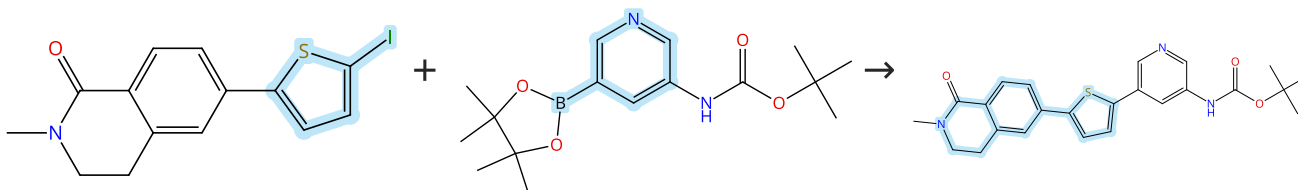
Supplier (1)

Suppliers (71)

31-614-CAS-41199290	Steps: 1 Yield: 55%	Discovery of a potent, orally available furopyridine derivative as a novel selective bromodomain and extra-terminal domain (BET)-first bromodomain (BD1) inhibitor
1.1 Reagents: Cesium carbonate Catalysts: (<i>SP</i> -4-1)-Bis[4-[bis(1,1-dimethylethyl)phosphino- κP]- <i>N,N</i> -dimethylbenzenamine]dichloropalladium Solvents: 1,4-Dioxane, Water; 5 h, 100 °C		By: Hagihara, Shuichi; et al Bioorganic & Medicinal Chemistry Letters (2024), 109, 129848.
Experimental Protocols		

Scheme 301 (1 Reaction)

Steps: 1 Yield: 55%

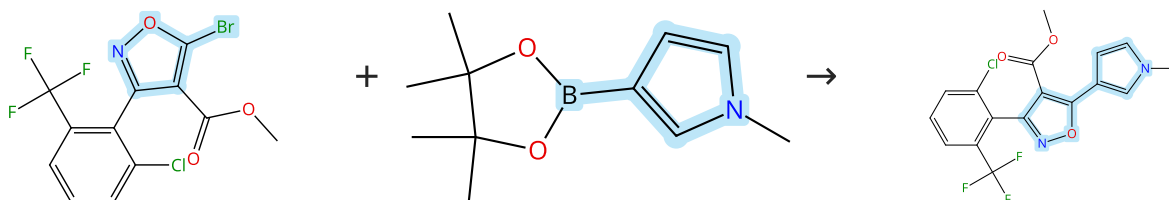


Suppliers (64)

31-179-CAS-17456561 Steps: 1 Yield: 55% 1.1 Reagents: Sodium carbonate Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: Ethanol, Toluene, Water; 2 h, reflux; overnight, reflux Experimental Protocols	Benzenesulphonamide inhibitors of the cytolytic protein perforin By: Spicer, Julie A.; et al Bioorganic & Medicinal Chemistry Letters (2017), 27(4), 1050-1054.
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Scheme 302 (1 Reaction)

Steps: 1 Yield: 55%

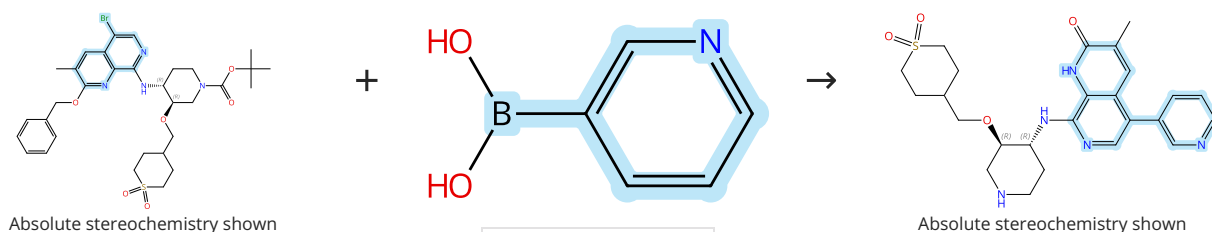


Suppliers (63)

31-614-CAS-32679843 Steps: 1 Yield: 55% 1.1 Reagents: Sodium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: 1,2-Dimethoxyethane, Water; 8 h, 85 °C Experimental Protocols	Structure-Activity Relationship Studies of Trisubstituted Isoxazoles as Selective Allosteric Ligands for the Retinoic-Acid-Receptor-Related Orphan Receptor γt By: Meijer, Femke A.; et al Journal of Medicinal Chemistry (2021), 64(13), 9238-9258.
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Scheme 303 (1 Reaction)

Steps: 1 Yield: 55%



Absolute stereochemistry shown

Absolute stereochemistry shown

Supplier (1)

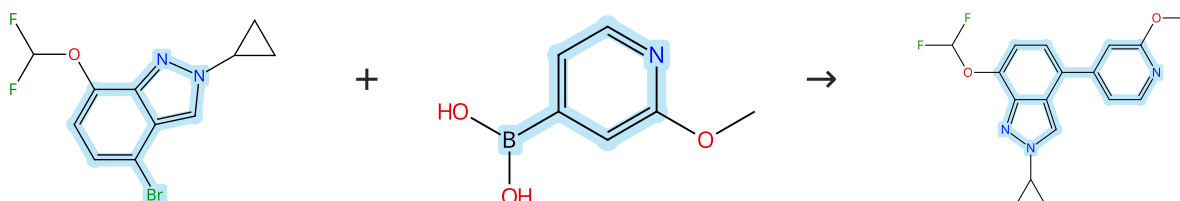
Suppliers (115)

Supplier (1)

31-179-CAS-10186033 Steps: 1 Yield: 55% 1.1 Reagents: Potassium carbonate Catalysts: Palladium diacetate, CataCXium A Solvents: 1,4-Dioxane, Water; 20 min, rt; 30 min, 100 °C; 100 °C → rt 1.2 Reagents: Trifluoroacetic acid; 16 h, reflux; reflux → rt Experimental Protocols	Structure-Based Optimization of Naphthyridones into Potent ATAD2 Bromodomain Inhibitors By: Bamborough, Paul; et al Journal of Medicinal Chemistry (2015), 58(15), 6151-6178.
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Scheme 304 (1 Reaction)

Steps: 1 Yield: 55%



Suppliers (99)

Scheme 307 (1 Reaction)

Steps: **1** Yield: **54%**

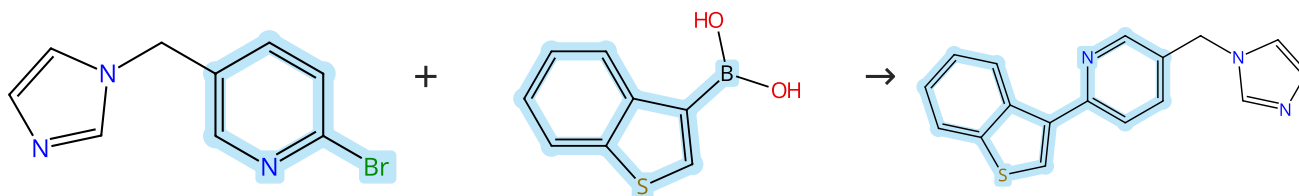
COc1ccc(cc1)c2nc(Cl)c(nc2N3CCN(C3CC4CCOCC4)c5cccnc5) + OBO(O)O >> COc1ccc(cc1)c2nc3c(ncn3N4CCN(C4CC5CCOCC5)c6cccnc6)C

Suppliers (115)

31-179-CAS-15736999	Steps: 1 Yield: 54%	Trisubstituted Pyrimidines as Efficacious and Fast-Acting Antimalarials
1.1 Reagents: Tripotassium phosphate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: 1,2-Dimethoxyethane, Water; 20 min, 120 °C		By: Norcross, Neil R.; et al
Experimental Protocols		Journal of Medicinal Chemistry (2016), 59(13), 6101-6120.

Scheme 308 (1 Reaction)

Steps: 1 Yield: 54%



Suppliers (22)

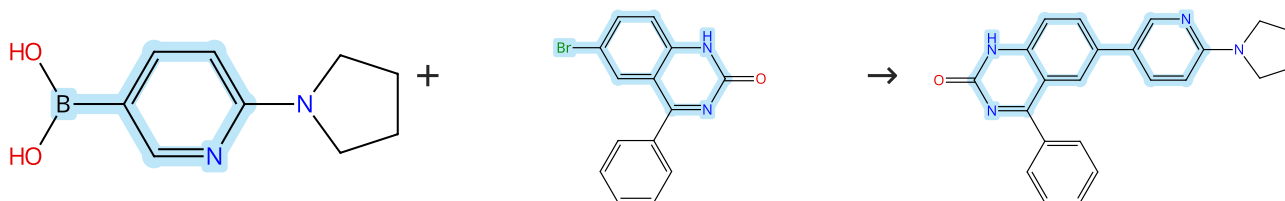
Suppliers (87)

Suppliers (3)

31-179-CAS-9003086	Steps: 1 Yield: 54%	Optimization of the First Selective Steroid-11β-hydroxylase (CYP11B1) Inhibitors for the Treatment of Cortisol Dependent Diseases
1.1 Reagents: Sodium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Methanol, Toluene, Water; 8 h, reflux		By: Hille, Ulrike E.; et al
Experimental Protocols		ACS Medicinal Chemistry Letters (2011), 2(8), 559-564.

Scheme 309 (1 Reaction)

Steps: 1 Yield: 54%



Suppliers (71)

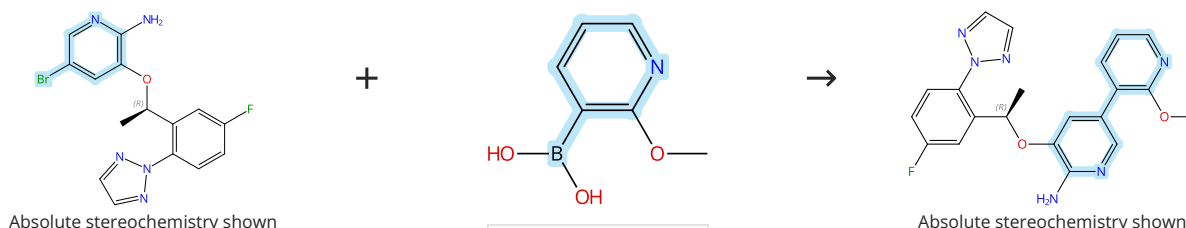
Suppliers (33)

Suppliers (2)

31-179-CAS-17699146	Steps: 1 Yield: 54%	Novel 1,2-dihydroquinazolin-2-ones: Design, synthesis, and biological evaluation against Trypanosoma brucei
1.1 Reagents: Sodium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Dimethylformamide, Water; 4 - 24 min, 140 °C		By: Pham, Thanh Truc; et al
Experimental Protocols		Bioorganic & Medicinal Chemistry Letters (2017), 27(16), 3629-3635.

Scheme 310 (1 Reaction)

Steps: 1 Yield: 54%



Absolute stereochemistry shown

Suppliers (2)

Suppliers (96)

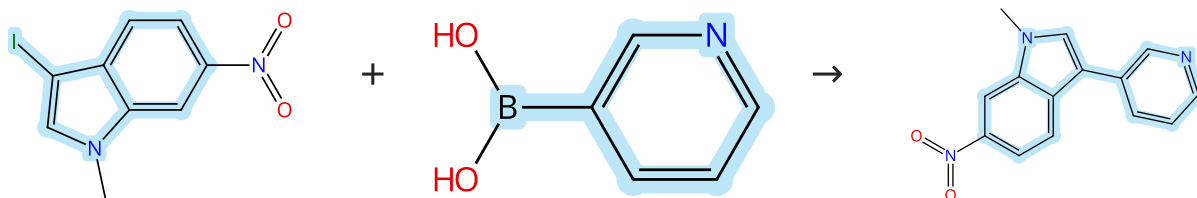
Absolute stereochemistry shown

Supplier (1)

31-179-CAS-14485689	Steps: 1 Yield: 54%	Design of Potent and Selective Inhibitors to Overcome Clinical Anaplastic Lymphoma Kinase Mutations Resistant to Crizotinib By: Huang, Qinhua; et al Journal of Medicinal Chemistry (2014), 57(4), 1170-1187.
1.1 Reagents: Cesium fluoride Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: Methanol; 1 h, 120 °C		
Experimental Protocols		

Scheme 311 (1 Reaction)

Steps: 1 Yield: 54%

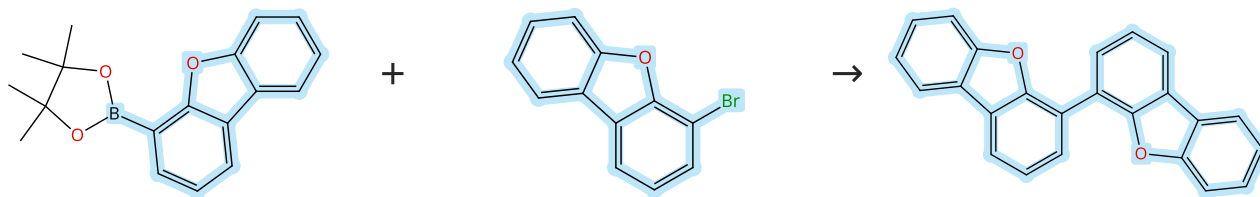


Suppliers (115)

31-614-CAS-41112671	Steps: 1 Yield: 54%	Discovery of Novel Receptor Tyrosine Kinase-like Orphan Receptor 1 (ROR1) Inhibitors for Cancer Treatment By: Luo, Dongdong; et al Journal of Medicinal Chemistry (2024), 67(13), 10655-10686.
1.1 Reagents: Sodium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: 1,4-Dioxane, Water; 9 h, 80 °C		
Experimental Protocols		

Scheme 312 (1 Reaction)

Steps: 1 Yield: 54%



Suppliers (65)

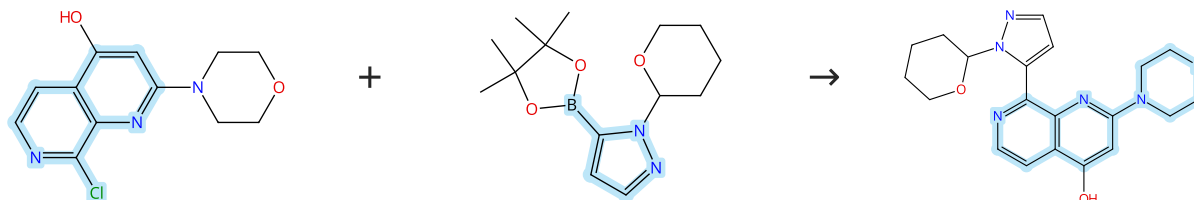
Suppliers (75)

Suppliers (5)

31-614-CAS-34800183	Steps: 1 Yield: 54%	HAT Catalysts to Facilitate Acid Mediated Cleavage of Biaryl C-C Bonds in Binaphthols By: Sojda, Christopher A.; et al Organic Letters (2022), 24(45), 8326-8330.
1.1 Reagents: Potassium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Tetrahydrofuran, Water; 17 h, 80 °C		
Experimental Protocols		

Scheme 313 (1 Reaction)

Steps: 1 Yield: 54%

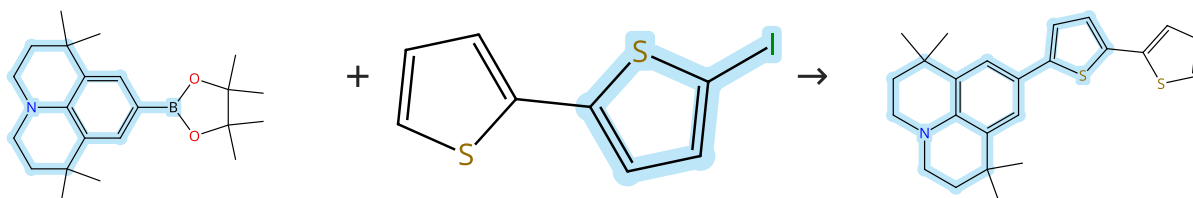


Suppliers (102)

31-179-CAS-22678400 Steps: 1 Yield: 54%	Damage Incorporated: Discovery of the Potent, Highly Selective, Orally Available ATR Inhibitor BAY 1895344 with Favorable Pharmacokinetic Properties and Promising Efficacy in Monotherapy and in Combination Treatments in Preclinical Tumor Models By: Luecking, Ulrich; et al Journal of Medicinal Chemistry (2020), 63(13), 7293-7325.
1.1 Reagents: Cesium carbonate Catalysts: Dichloro[1,1'-bis(diphenylphosphino)ferrocene] palladium(II) dichloromethane adduct Solvents: 1,4-Dioxane; 16 h, 80 °C	

Scheme 314 (1 Reaction)

Steps: 1 Yield: 54%



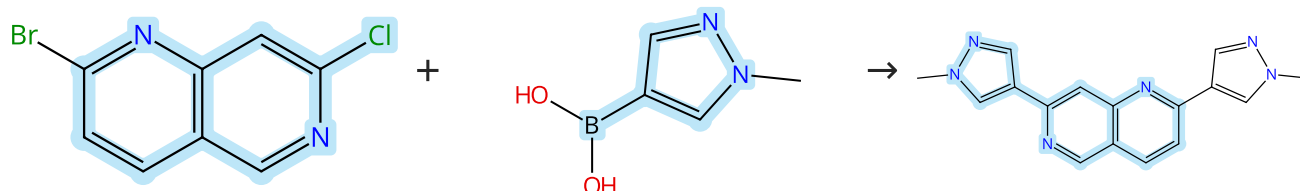
Supplier (1)

Suppliers (12)

31-179-CAS-13966261 Steps: 1 Yield: 54%	D-π-A Triarylboron Compounds with Tunable Push-Pull Character Achieved by Modification of Both the Donor and Acceptor Moieties By: Zhang, Zuolun; et al Chemistry - A European Journal (2015), 21(1), 177-190.
1.1 Reagents: Silver carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Tetrahydrofuran; 24 h, reflux Experimental Protocols	

Scheme 315 (1 Reaction)

Steps: 1 Yield: 54%



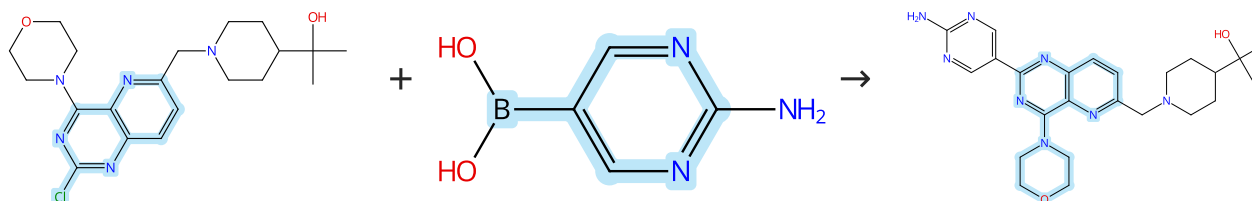
Suppliers (16)

Suppliers (77)

31-614-CAS-46384624 Steps: 1 Yield: 54%	Switchable and Ring-Selective Hydrogenation of Naphthyridine Isomers By: Viereck, Peter Organic Letters (2025), 27(25), 6635-6640.
1.1 Reagents: Sodium carbonate Catalysts: Dichloro[1,1'-bis(diphenylphosphino)ferrocene] palladium(II) dichloromethane adduct Solvents: 1,4-Dioxane, Water; 18 h, 80 °C Experimental Protocols	

Scheme 316 (1 Reaction)

Steps: 1 Yield: 54%



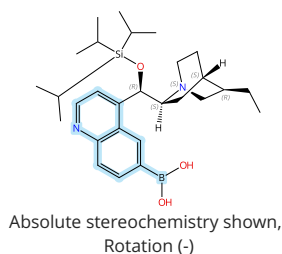
Suppliers (85)

Supplier (1)

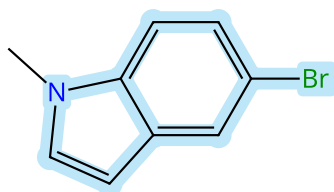
31-179-CAS-1377921	Potent and selective inhibitors of PI3Kδ: Obtaining isoform selectivity from the affinity pocket and tryptophan shelf
Steps: 1 Yield: 54% 1.1 Reagents: Sodium carbonate Catalysts: Dichlorobis(triphenylphosphine)palladium Solvents: Acetonitrile, Water; 0.5 h, 140 °C	By: Sutherlin, Daniel P.; et al Bioorganic & Medicinal Chemistry Letters (2012), 22(13), 4296-4302.

Scheme 317 (1 Reaction)

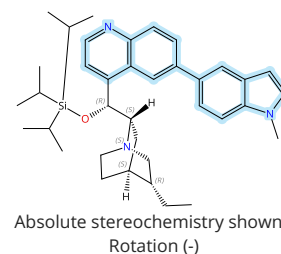
Steps: 1 Yield: 54%



+



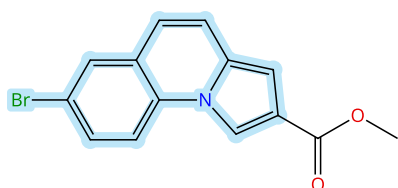
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 Suppliers (82)

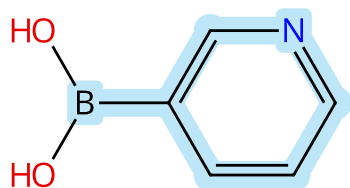
31-614-CAS-37170684	A Palladium-Catalyzed Borylation/Silica Gel Promoted Hydrolysis Sequence for the Synthesis of Hydroquinine-6'-Boric Acid and Its Applications
1.1 Reagents: Potassium carbonate Catalysts: Palladium diacetate, CataCXium A Solvents: Butyl ether; 3 h, rt → 120 °C	By: Liang, Wei; et al
Experimental Protocols	Journal of Organic Chemistry (2023), 88(15), 11284-11298.

Scheme 318 (1 Reaction)

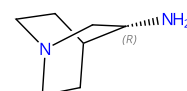
Steps: 1 Yield: 54%



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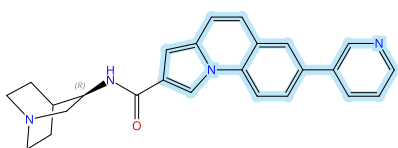


→

• 2 HCl

 Absolute stereochemistry shown,
 Rotation (+)

 Suppliers (115)

 Suppliers (87)


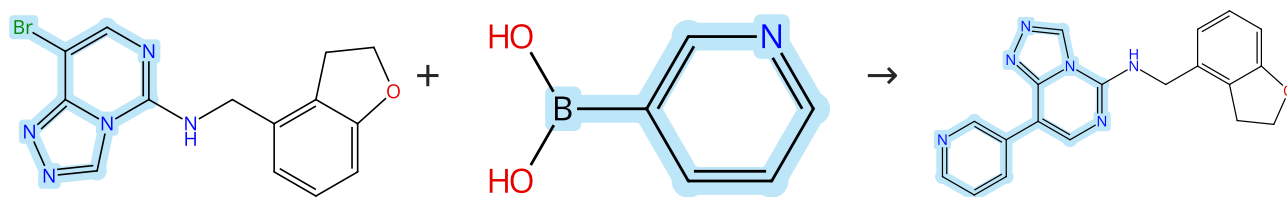
Absolute stereochemistry shown

 Supplier (1)

31-179-CAS-20624521 Steps: 1 Yield: 54% 1.1 Reagents: Sodium carbonate, Selenium dioxide Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Ethanol, Toluene; 5 h, 85 °C 1.2 Reagents: Potassium hydroxide Solvents: Methanol, Tetrahydrofuran, Water; 2 h, reflux; reflux → rt 1.3 Reagents: Hydrochloric acid Solvents: Water; 0.5 h, acidified, 0 °C 1.4 Reagents: Diisopropylethylamine, <i>O</i>-(7-Azabenzotriazol-1-yl)-<i>N,N,N,N</i>-tetramethyluronium hexafluorophosphate Solvents: Dichloromethane; 0 °C; 0 °C → rt; 10 h, rt	Discovery of fused heterocyclic carboxamide derivatives as novel $\alpha 7$-nAChR agonists: synthesis, preliminary SAR and biological evaluation By: Xue, Yu; et al European Journal of Medicinal Chemistry (2019), 182, 111618.
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Scheme 319 (1 Reaction)

Steps: 1 Yield: 54%

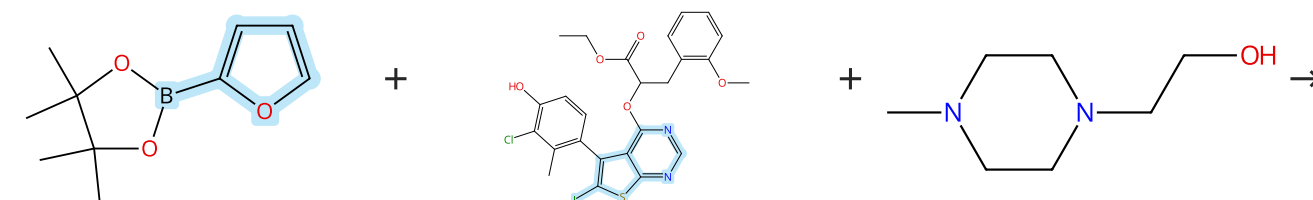


Suppliers (115)

31-614-CAS-39884759 Steps: 1 Yield: 54% 1.1 Reagents: Potassium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Acetonitrile, 1,4-Dioxane, Water; 3 h, 110 °C Experimental Protocols	Discovery of the Clinical Candidate MAK683: An EED-Directed, Allosteric, and Selective PRC2 Inhibitor for the Treatment of Advanced Malignancies By: Huang, Ying; et al Journal of Medicinal Chemistry (2022), 65(7), 5317-5333.
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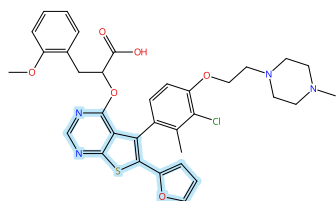
Scheme 320 (1 Reaction)

Steps: 1 Yield: 54%



Suppliers (71)

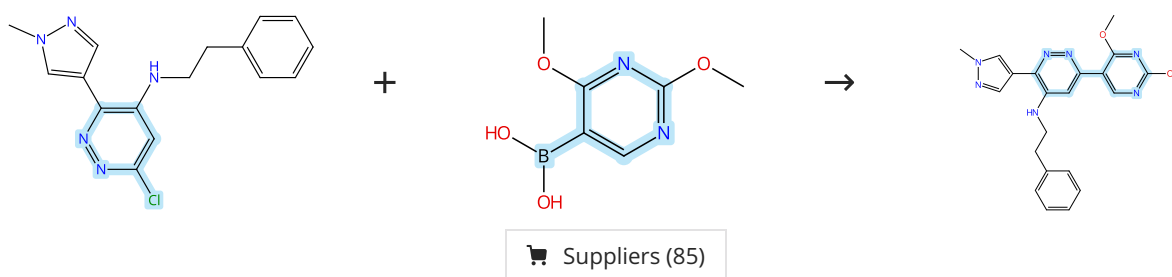
Suppliers (82)



31-012-CAS-23905256 Steps: 1 Yield: 54% 1.1 Reagents: Pinacol Solvents: 1,4-Dioxane; overnight, rt 1.2 Reagents: Cesium carbonate Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: Water; 110 °C 1.3 Reagents: Hydrochloric acid, Sodium chloride Solvents: Dichloromethane, Water; neutralized 1.4 Reagents: Triphenylphosphine, Di-<i>tert</i>-butyl azodicarboxylate Solvents: Toluene; 50 °C 1.5 Reagents: Lithium hydroxide Solvents: 1,4-Dioxane, Water; rt 1.6 Reagents: Hydrochloric acid, Sodium chloride Solvents: Water; neutralized, rt Experimental Protocols	Discovery of S64315, a Potent and Selective Mcl-1 Inhibitor By: Szlavik, Zoltan; et al Journal of Medicinal Chemistry (2020), 63(22), 13762-13795.
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Scheme 321 (1 Reaction)

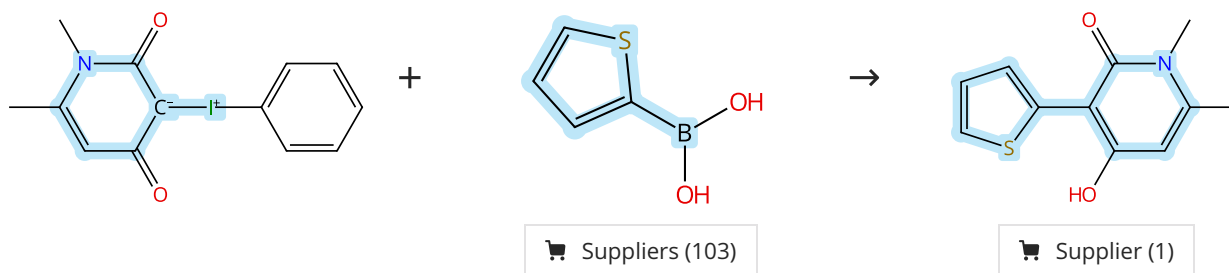
Steps: 1 Yield: 54%



31-614-CAS-42060184 Steps: 1 Yield: 54% 1.1 Reagents: Potassium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: 1,4-Dioxane, Water; overnight, 90 °C Experimental Protocols	Discovery of Novel 5-(pyridazin-3-yl)pyrimidine-2,4(1H,3H)-dione Derivatives as Potent and Orally Bioavailable Inhibitors Targeting Ecto-5'-Nucleotidase By: Xu, Yu; et al Journal of Medicinal Chemistry (2024), 67(20), 18491-18511.
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Scheme 322 (1 Reaction)

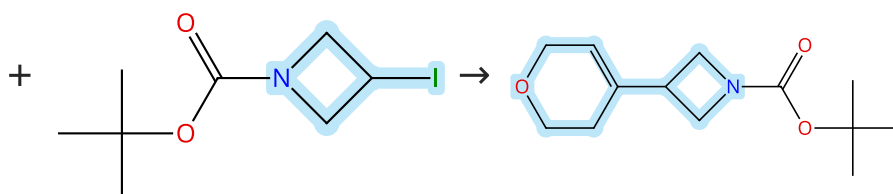
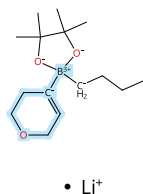
Steps: 1 Yield: 54%



31-614-CAS-39973583 Steps: 1 Yield: 54% 1.1 Catalysts: Bis[dichloro[η⁵-(pentamethylcyclopentadienyl)]rhodium] Solvents: 1,4-Dioxane, Water; 30 min, 50 °C Experimental Protocols	Catalytic and Base-free Suzuki-type α-Arylation of Cyclic 1,3-Dicarbonyls via a Cyclic Iodonium Ylide Strategy By: Liang, Mingxuan; et al Angewandte Chemie, International Edition (2024), 63(17), e202400741.
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Scheme 323 (1 Reaction)

Steps: 1 Yield: 54%



Suppliers (93)

Supplier (1)

31-614-CAS-31059357

Steps: 1 Yield: 54%

Merging Halogen-Atom Transfer (XAT) and Copper Catalysis for the Modular Suzuki-Miyaura-Type Cross-Coupling of Alkyl Iodides and Organoborons

By: Zhang, Zhenhua; et al

Journal of the American Chemical Society (2022), 144(4), 1986-1992.

1.1 **Reagents:** Triethylamine, 1-Methyl-1-phenylethyl trimethylsilyl peroxide

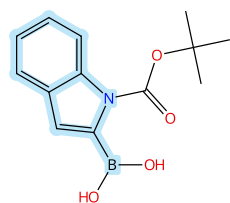
Catalysts: 2,2,6,6-Tetramethyl-3,5-heptanedione, Tetrakis (acetonitrile)copper(1+) hexafluorophosphate

Solvents: Dimethyl sulfoxide, Tetrahydrofuran; 1 h, rt

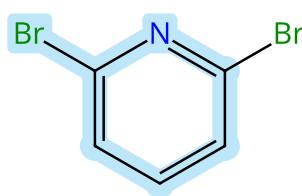
Experimental Protocols

Scheme 324 (1 Reaction)

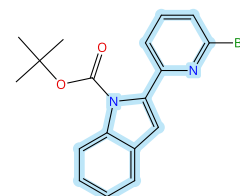
Steps: 1 Yield: 54%



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→



Suppliers (90)

Suppliers (91)

31-179-CAS-16373919

Steps: 1 Yield: 54%

Further investigation of inhibitors of MRSA pyruvate kinase: Towards the conception of novel antimicrobial agents

By: Labriere, Christophe; et al

European Journal of Medicinal Chemistry (2017), 125, 1-13.

1.1 **Reagents:** Potassium carbonate

Catalysts: Tetrakis(triphenylphosphine)palladium

Solvents: Tetrahydrofuran, Water; 6 h, reflux; reflux → rt

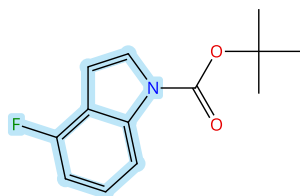
1.2 **Reagents:** Sodium chloride

Solvents: Water; rt

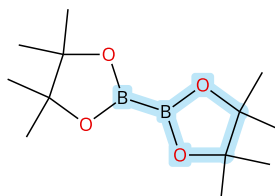
Experimental Protocols

Scheme 325 (1 Reaction)

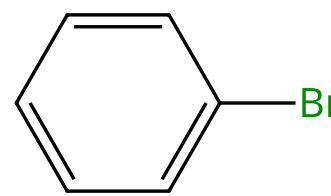
Steps: 1 Yield: 54%



+



+

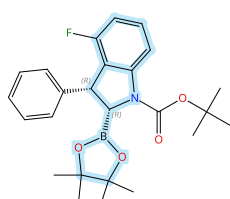


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Suppliers (58)

Suppliers (131)

Suppliers (83)

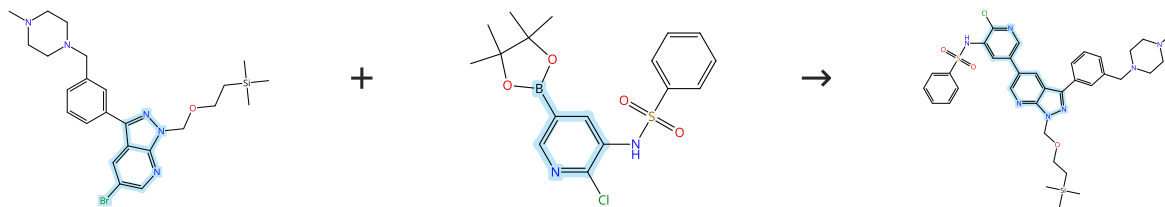


Relative stereochemistry shown

31-614-CAS-24363496	Steps: 1 Yield: 54%	Nickel-Catalyzed Dearomative Arylboration of Indoles: Regioselective Synthesis of C2- and C3-Borylated Indolines By: Trammel, Grace L.; et al Journal of the American Chemical Society (2021), 143(40), 16502-16511.
1.1 Reagents: Sodium <i>tert</i> -butoxide Solvents: Tetrahydrofuran; 25 °C → 0 °C		
1.2 Catalysts: Dichloro[1,2-di(methoxy-κO)ethane]nickel Solvents: Dimethylacetamide; 24 h, 2 - 10 °C		
1.3 Reagents: Water		
Experimental Protocols		

Scheme 326 (1 Reaction)

Steps: 1 Yield: 54%

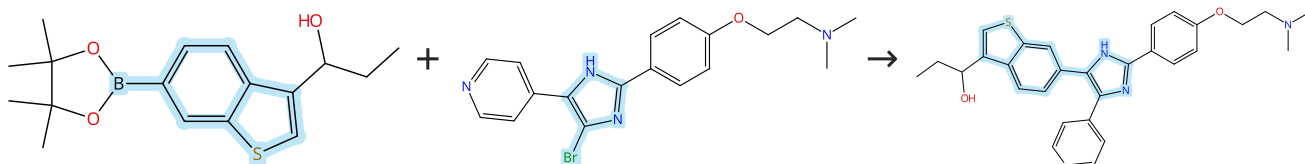


Suppliers (18)

31-614-CAS-37102649	Steps: 1 Yield: 54%	Discovery of N-(2-chloro-5-(3-(pyridin-4-yl)-1H-pyrazolo[3,4-b]pyridin-5-yl)pyridin-3-yl)-4-fluorobenzenesulfonamide (F D274) as a highly potent PI3K/mTOR dual inhibitor for the treatment of acute myeloid leukemia By: Yang, Chengbin; et al European Journal of Medicinal Chemistry (2023), 258, 115543.
1.1 Reagents: Potassium carbonate Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: 1,4-Dioxane, Water; 12 h, 100 °C		
Experimental Protocols		

Scheme 327 (1 Reaction)

Steps: 1 Yield: 54%



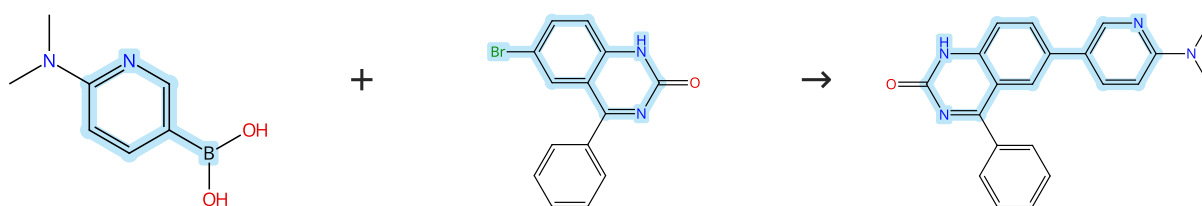
Supplier (1)

Supplier (1)

31-179-CAS-6394550	Steps: 1 Yield: 54%	Potent BRAF kinase inhibitors based on 2,4,5-trisubstituted imidazole with naphthyl and benzothiophene 4-substituents By: Niculescu-Duvaz, Dan; et al Bioorganic & Medicinal Chemistry (2013), 21(5), 1284-1304.
1.1 Reagents: Potassium carbonate Catalysts: Triphenylphosphine, Palladium diacetate Solvents: 1,2-Dimethoxyethane, Water; 2 - 16 h, reflux; reflux → rt		
1.2 Reagents: Hydrochloric acid Solvents: Water; pH 1		
Experimental Protocols		

Scheme 328 (1 Reaction)

Steps: 1 Yield: 54%



Suppliers (71)

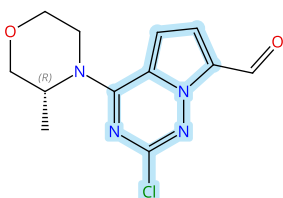
Suppliers (33)

Supplier (1)

31-179-CAS-17699134	Steps: 1 Yield: 54%	Novel 1,2-dihydroquinazolin-2-ones: Design, synthesis, and biological evaluation against <i>Trypanosoma brucei</i>
1.1 Reagents: Sodium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Dimethylformamide, Water; 4 - 24 min, 140 °C		By: Pham, Thanh Truc; et al
Experimental Protocols		Bioorganic & Medicinal Chemistry Letters (2017), 27(16), 3629-3635.

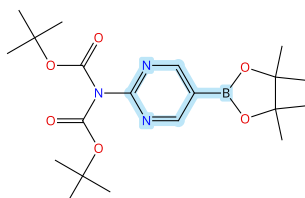
Scheme 329 (1 Reaction)

Steps: 1 Yield: 54%

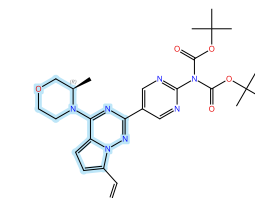


Absolute stereochemistry shown

+



→



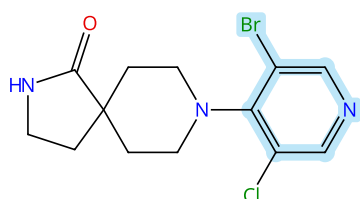
Absolute stereochemistry shown

Suppliers (53)

31-179-CAS-7796430	Steps: 1 Yield: 54%	Synthesis and evaluation of pyrrolotriazine based molecules as PI3 kinase inhibitors
1.1 Reagents: Sodium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Ethanol, Toluene, Water; 4 h, 140 °C		By: Dugar, Sundee; et al
Experimental Protocols		Bioorganic & Medicinal Chemistry Letters (2015), 25(16), 3142-3146.

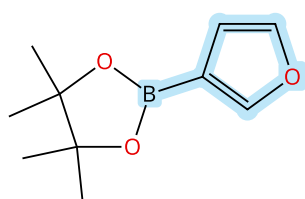
Scheme 330 (1 Reaction)

Steps: 1 Yield: 54%



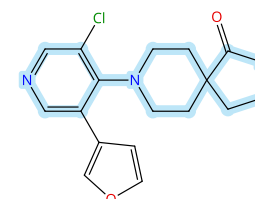
Suppliers (5)

+



Suppliers (83)

→

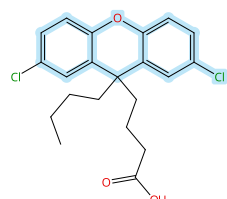


Supplier (1)

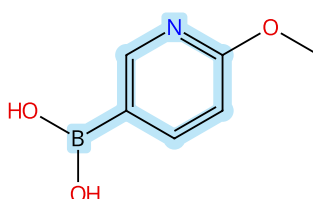
31-614-CAS-24921673	Steps: 1 Yield: 54%	Discovery of a potent, highly selective, and orally bioavailable inhibitor of CDK8 through a structure-based optimisation
1.1 Reagents: Sodium carbonate Catalysts: Dichloro[1,1'-bis(diphenylphosphino)ferrocene] palladium(II) dichloromethane adduct Solvents: Acetonitrile, Water; 1 h, 120 °C		By: Yu, Mingfeng; et al
		European Journal of Medicinal Chemistry (2021), 218, 113391.

Scheme 331 (1 Reaction)

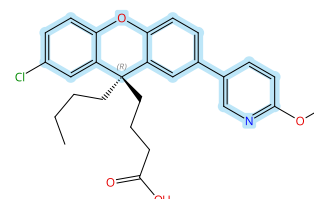
Steps: 1 Yield: 54%



+



→

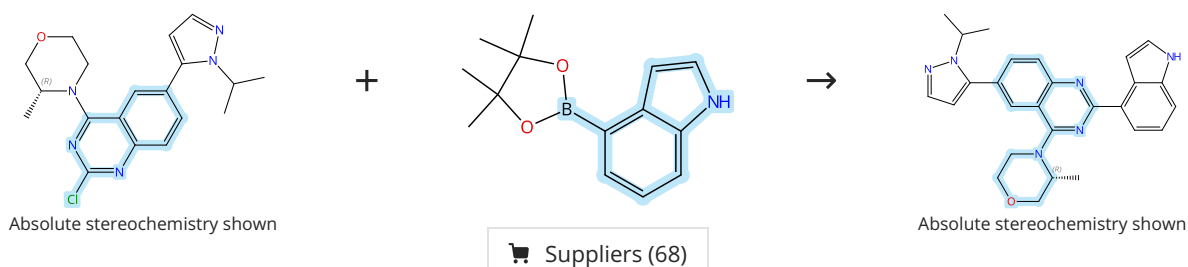
Absolute stereochemistry shown,
Rotation (-)

Suppliers (100)

31-614-CAS-31680401 Steps: 1 Yield: 54% 1.1 Catalysts: Tris(dibenzylideneacetone)dipalladium, Phosphonic acid, <i>P</i>-[(1<i>S</i>)-2'-(dicyclohexylphosphino)-5-methyl-2,6-bis(1-methylethoxy)[1,1'-biphenyl]-3-yl]-, monoethyl ester Solvents: 2-Methyltetrahydrofuran; 20 min, rt 1.2 Reagents: Tripotassium phosphate Solvents: Water; 18 h, 60 °C 1.3 Solvents: Water 1.4 Reagents: Hydrochloric acid Solvents: Water; pH 3 - 5 Experimental Protocols	Distal Ionic Substrate-Catalyst Interactions Enable Long-Range Stereocontrol: Access to Remote Quaternary Stereocenters through a Desymmetrizing Suzuki-Miyaura Reaction By: Lou, Yazhou; et al Journal of the American Chemical Society (2022), 144(1), 123-129.
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Scheme 332 (1 Reaction)

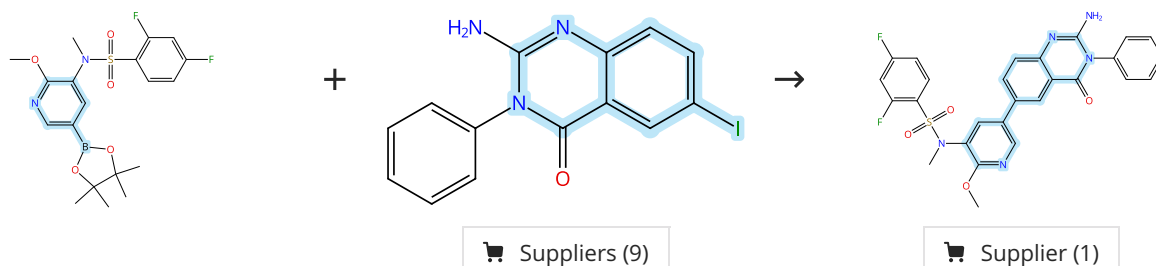
Steps: 1 Yield: 54%



31-614-CAS-33811417 Steps: 1 Yield: 54% 1.1 Reagents: Potassium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: 1,4-Dioxane, Water; 6 h, 105 °C Experimental Protocols	Discovery of a potent and highly selective inhibitor of ataxia telangiectasia mutated and Rad3-Related (ATR) kinase: Structural activity relationship and antitumor activity both in vitro and in vivo By: Bin, Huachao; et al European Journal of Medicinal Chemistry (2022), 232, 114187.
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Scheme 333 (1 Reaction)

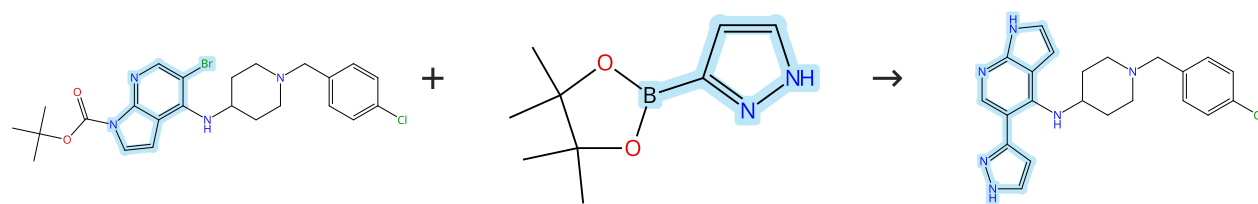
Steps: 1 Yield: 54%



31-179-CAS-4236972 Steps: 1 Yield: 54% 1.1 Reagents: Cesium carbonate Catalysts: Dichloro[1,1'-bis(diphenylphosphino)ferrocene]palladium(II) dichloromethane adduct Solvents: 1,4-Dioxane, Water; 17 h, rt Experimental Protocols	Discovery of Selective Small Molecule Type III Phosphatidylinositol 4-Kinase Alpha (PI4KIIIα) Inhibitors as Anti Hepatitis C (HCV) Agents By: Leivers, Anna L.; et al Journal of Medicinal Chemistry (2014), 57(5), 2091-2106.
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Scheme 334 (1 Reaction)

Steps: 1 Yield: 54%



Suppliers (128)

31-614-CAS-39559347

Steps: 1 Yield: 54%

Design, synthesis and evaluation of C-5 substituted pyrrolopyridine derivatives as potent Janus Kinase 1 inhibitors with excellent selectivity

By: Chen, Limei; et al

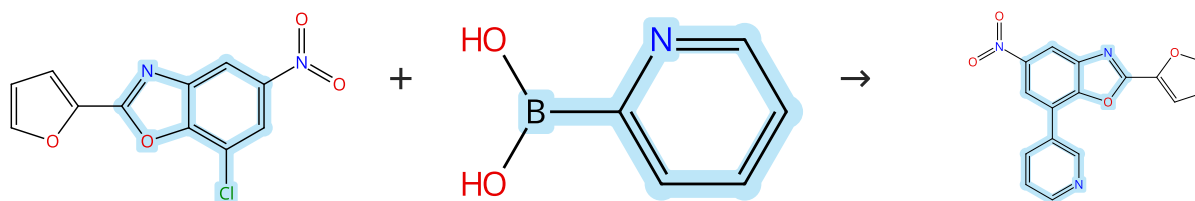
European Journal of Medicinal Chemistry (2024), 267, 116210.

- 1.1 **Reagents:** Potassium carbonate
Catalysts: (SP-4-1)-Bis[4-[bis(1,1-dimethylethyl)phosphino-κP]-N,N-dimethylbenzenamine]dichloropalladium
Solvents: Isobutanol, Water; 10 h, 60 °C
- 1.2 **Reagents:** Trifluoroacetic acid
Solvents: Dichloromethane; 1 h, rt
- 1.3 **Reagents:** Sodium hydroxide
Solvents: Water; pH 9 - 10, rt

Experimental Protocols

Scheme 335 (1 Reaction)

Steps: 1 Yield: 54%



Supplier (1)

Suppliers (68)

31-179-CAS-22105810

Steps: 1 Yield: 54%

Antagonists of the adenosine A_{2A} receptor based on a 2-arylbenzoxazole scaffold: Investigation of the C5- and C7-positions to enhance affinity

By: Duroux, Romain; et al

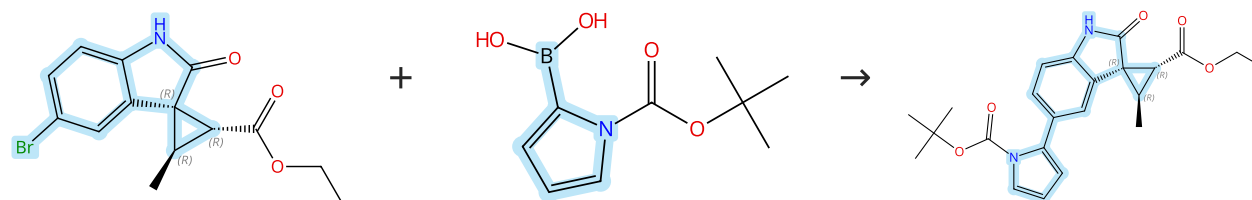
European Journal of Medicinal Chemistry (2018), 144, 151-163.

- 1.1 **Reagents:** Tripotassium phosphate
Catalysts: Triphenylphosphine, Palladium diacetate
Solvents: 1,4-Dioxane; 4 h, 110 °C

Experimental Protocols

Scheme 336 (1 Reaction)

Steps: 1 Yield: 54%



Relative stereochemistry shown

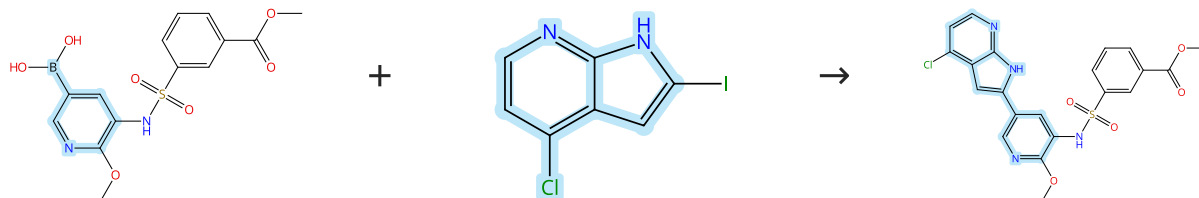
Relative stereochemistry shown

Suppliers (103)

31-179-CAS-21868347 Steps: 1 Yield: 54% 1.1 Reagents: Cesium carbonate Catalysts: Palladium diacetate, CataCXium A Solvents: Butyl ether; 6 h, 110 °C	DABCO-Catalyzed Michael/Alkylation Cascade Reactions Involving α-Substituted Ammonium Ylides for the Construction of Spirocyclopropyl Oxindoles: Access to the Powerful Chemical Leads against HIV-1 By: Chen, Lin; et al Journal of Organic Chemistry (2020), 85(8), 5203-5219.
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Scheme 337 (1 Reaction)

Steps: 1 Yield: 54%

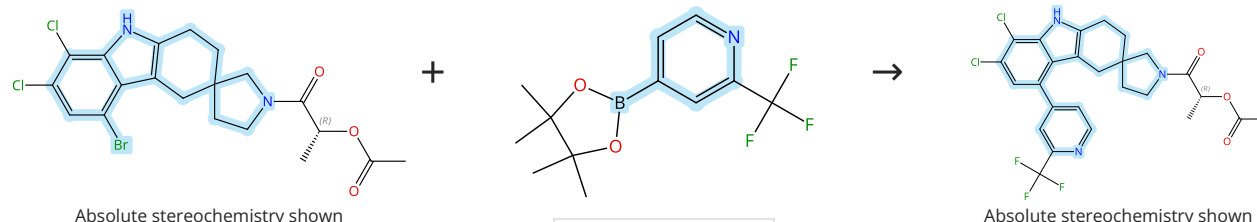


Suppliers (68)

31-614-CAS-24893620 Steps: 1 Yield: 54% 1.1 Reagents: Sodium carbonate Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: 1,2-Dimethoxyethane, Water; 28 h, reflux Experimental Protocols	Omipalisib inspired macrocycles as dual PI3K/mTOR inhibitors By: Alvarez, Rosa M.; et al European Journal of Medicinal Chemistry (2021), 211, 113109.
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Scheme 338 (1 Reaction)

Steps: 1 Yield: 54%

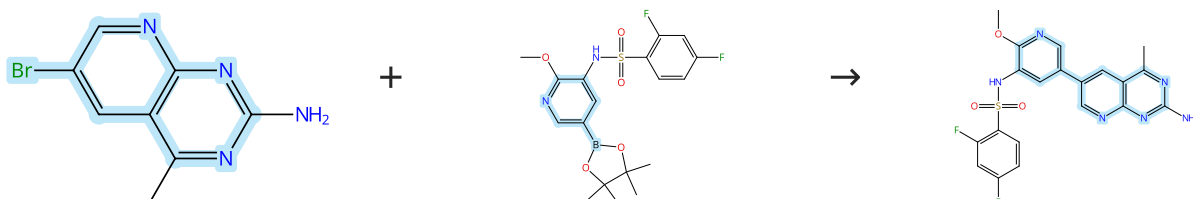


Suppliers (85)

31-614-CAS-40113430 Steps: 1 Yield: 54% 1.1 Reagents: Potassium acetate Catalysts: Dichloro[1,1'-bis(diphenylphosphino)ferrocene]palladium(II) dichloromethane adduct Solvents: 1,4-Dioxane; rt → 100 °C; 10 h, 100 °C Experimental Protocols	Design, Synthesis, and Pharmacological Evaluation of Spiro [carbazole-3,3'-pyrrolidine] Derivatives as cGAS Inhibitors for Treatment of Acute Lung Injury By: Chen, Mingjie; et al Journal of Medicinal Chemistry (2024), 67(8), 6268-6291.
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Scheme 339 (1 Reaction)

Steps: 1 Yield: 54%



Suppliers (19)

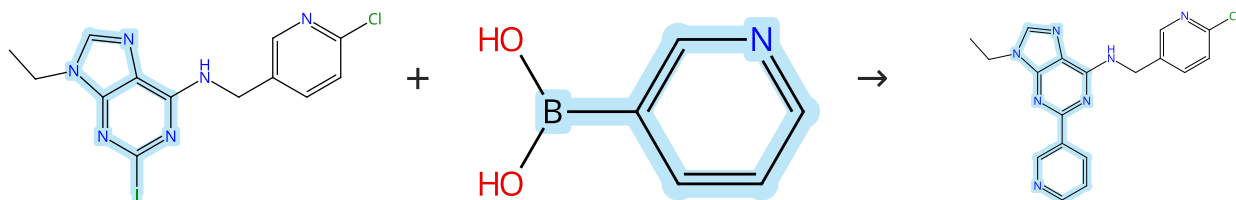
Suppliers (54)

Supplier (1)

31-179-CAS-2349394	Steps: 1 Yield: 54%	Identification of novel 7-amino-5-methyl-1,6-naphthyridin-2(1H)-one derivatives as potent PI3K/mTOR dual inhibitors
1.1 Reagents: Potassium carbonate Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: Dimethylformamide, Water; 100 °C		By: Lin, Songwen; et al Bioorganic & Medicinal Chemistry Letters (2014), 24(3), 790-793.

Scheme 340 (1 Reaction)

Steps: 1 Yield: 54%

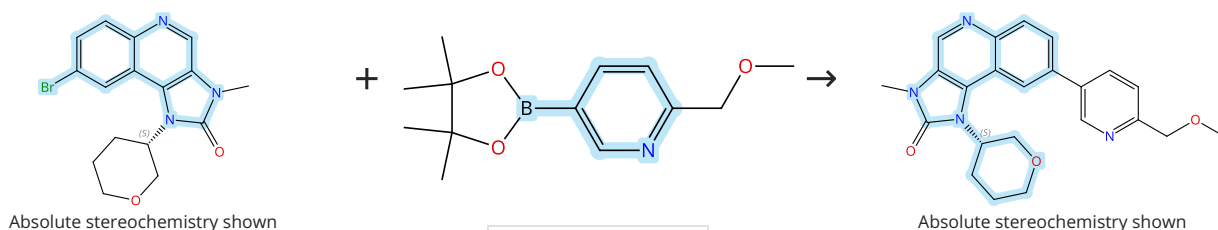


Suppliers (115)

31-614-CAS-38825359	Steps: 1 Yield: 54%	Identification of a novel potent CDK inhibitor degrading cyclin K with a superb activity to reverse trastuzumab-resistance in HER2-positive breast cancer in vivo
1.1 Reagents: Potassium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: 1,4-Dioxane, Water; 6 h, 100 °C		By: Kuchukulla, Ratnakar Reddy; et al European Journal of Medicinal Chemistry (2024), 264, 116014.
Experimental Protocols		

Scheme 341 (1 Reaction)

Steps: 1 Yield: 54%



Absolute stereochemistry shown

Absolute stereochemistry shown

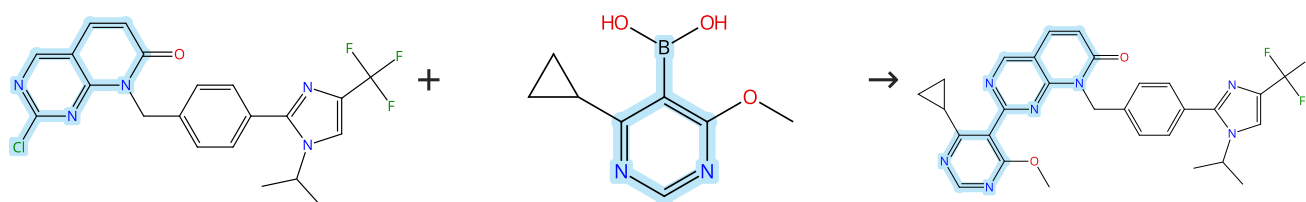
Supplier (1)

Suppliers (35)

31-614-CAS-39405117	Steps: 1 Yield: 54%	Identification of Novel, Selective Ataxia-Telangiectasia Mutated Kinase Inhibitors with the Ability to Penetrate the Blood-Brain Barrier: The Discovery of AZD1390
1.1 Reagents: Potassium carbonate Catalysts: Bis[3-[bis(1,1-dimethylethyl)phosphino-κP]-1-propanesulfonato]dichloropalladate(2-) Solvents: 1,4-Dioxane, Water; 3 h, rt → reflux		By: Pike, Kurt G.; et al Journal of Medicinal Chemistry (2024), 67(4), 3090-3111.
Experimental Protocols		

Scheme 342 (1 Reaction)

Steps: 1 Yield: 53%

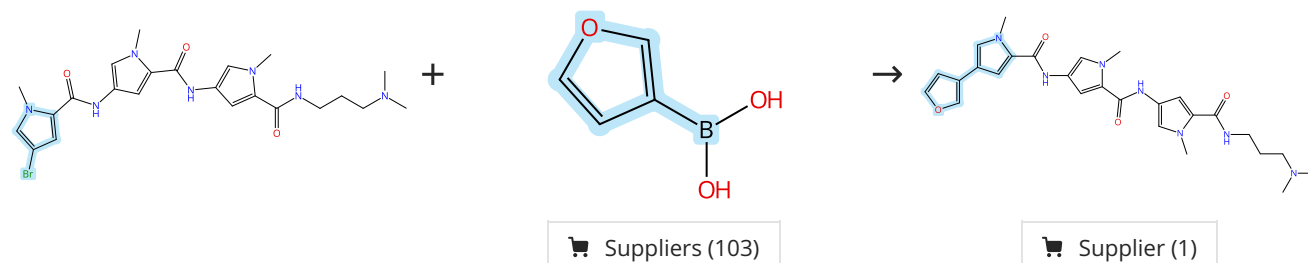


Suppliers (49)

31-614-CAS-40821614 Steps: 1 Yield: 53% 1.1 Reagents: Potassium carbonate Catalysts: (<i>SP</i>-4-4)-[2'-(Amino-κ<i>M</i>)[1,1'-biphenyl]-2-yl-κ<i>C</i>]chloro[dicyclohexyl[2',4',6'-tris(1-methylethyl)[1,1'-biphenyl]-2-yl]phosphine]palladium Solvents: 1,4-Dioxane, Water; 4 h, rt → 80 °C Experimental Protocols	Design, synthesis, and biological evaluation of pyrido [2,3-<i>d</i>]pyrimidin-7(8<i>H</i>)-one derivatives as potent USP1 inhibitors By: Li, Hongrui; et al European Journal of Medicinal Chemistry (2024), 275, 116568.
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Scheme 343 (1 Reaction)

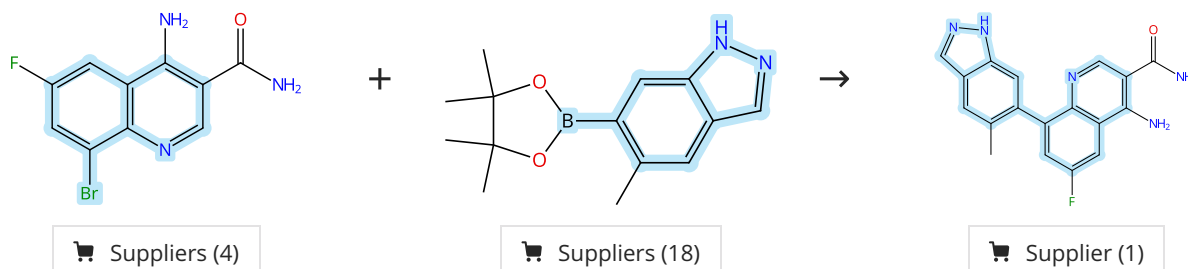
Steps: 1 Yield: 53%



31-179-CAS-6776746 Steps: 1 Yield: 53% 1.1 Reagents: Potassium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Ethanol, Toluene, Water; 9 min, 100 °C Experimental Protocols	Synthesis, anti-mycobacterial activity and DNA sequence-selectivity of a library of biaryl- motifs containing polyamides By: Brucoli, Federico; et al Bioorganic & Medicinal Chemistry (2015), 23(13), 3705-3711.
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Scheme 344 (1 Reaction)

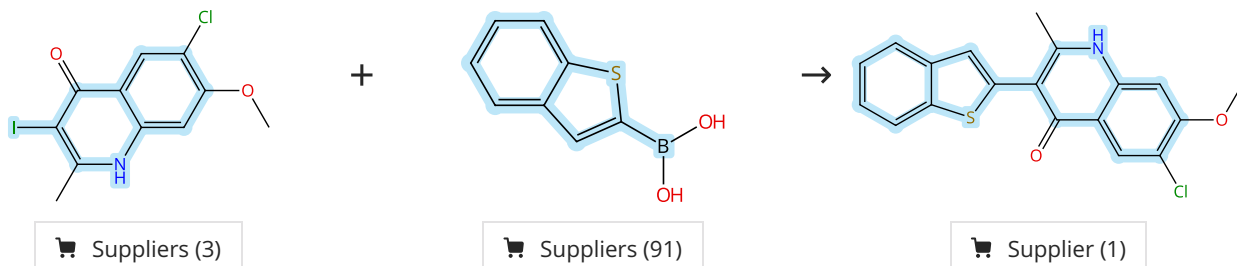
Steps: 1 Yield: 53%



31-179-CAS-20601372 Steps: 1 Yield: 53% 1.1 Reagents: Sodium bicarbonate Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: 1,4-Dioxane, Water; 1 - 2 h, 140 °C	Discovery of 4-Aminoquinoline-3-carboxamide Derivatives as Potent Reversible Bruton's Tyrosine Kinase Inhibitors for the Treatment of Rheumatoid Arthritis By: Yao, Xia; et al Journal of Medicinal Chemistry (2019), 62(14), 6561-6574.
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Scheme 345 (1 Reaction)

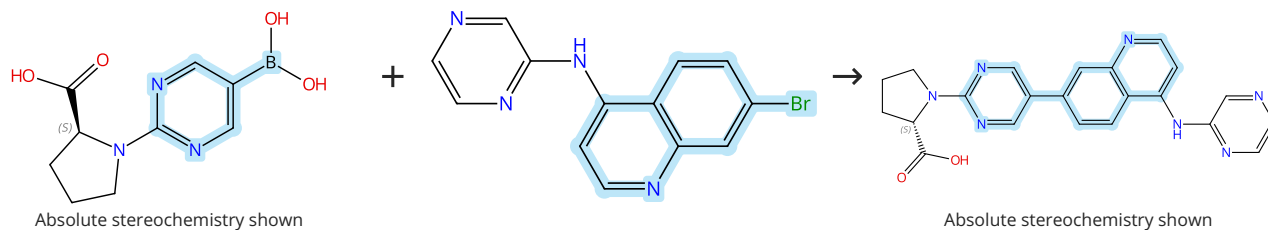
Steps: 1 Yield: 53%



31-179-CAS-4480491	Steps: 1 Yield: 53%	Orally Bioavailable 6-Chloro-7-methoxy-4(1H)-quinolones Efficacious against Multiple Stages of Plasmodium
1.1 Reagents: Tripotassium phosphate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Toluene, Water; 0.5 - 48 h, 110 °C		By: Cross, R. Matthew; et al
Experimental Protocols		Journal of Medicinal Chemistry (2014), 57(21), 8860-8879.

Scheme 346 (1 Reaction)

Steps: 1 Yield: 53%

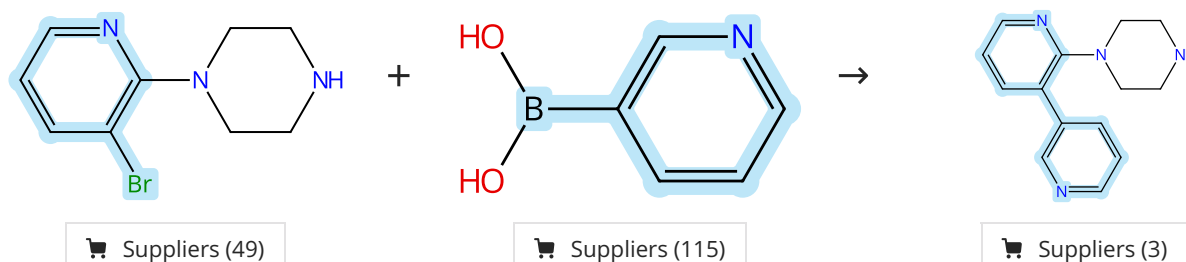


Supplier (1)

31-179-CAS-20117358	Steps: 1 Yield: 53%	Improvement of aqueous solubility of lapatinib-derived analogues: identification of a quinolinimine lead for human African trypanosomiasis drug development
1.1 Reagents: Cesium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Ethanol, Water; 10 min, 130 °C		By: Bachovchin, Kelly A.; et al
		Journal of Medicinal Chemistry (2019), 62(2), 665-687.

Scheme 347 (1 Reaction)

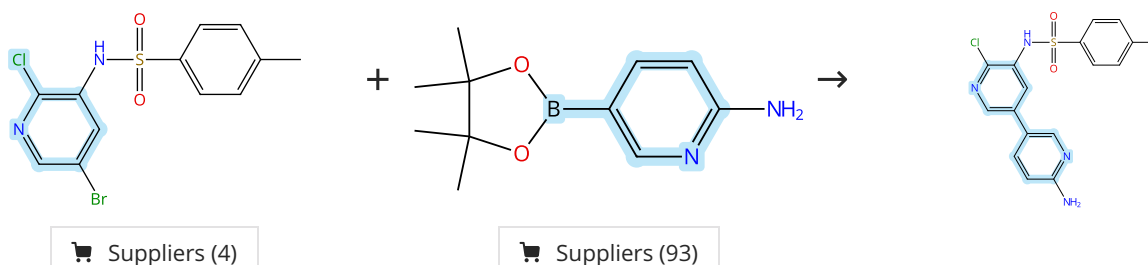
Steps: 1 Yield: 53%



31-179-CAS-15526329	Steps: 1 Yield: 53%	Synthesis, radiolabeling and in vivo evaluation of [¹¹C](R)-1-[4-[2-(4-methoxyphenyl)phenyl]piperazin-1-yl]-3-(2-pyridinyloxy)-2-propanol, a potential PET radioligand for the 5-HT₇ receptor
1.1 Reagents: Sodium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: 1,2-Dimethoxyethane, Water; 18 - 20 h, reflux		By: Hansen, Hanne D.; et al
Experimental Protocols		European Journal of Medicinal Chemistry (2014), 79, 152-163.

Scheme 348 (1 Reaction)

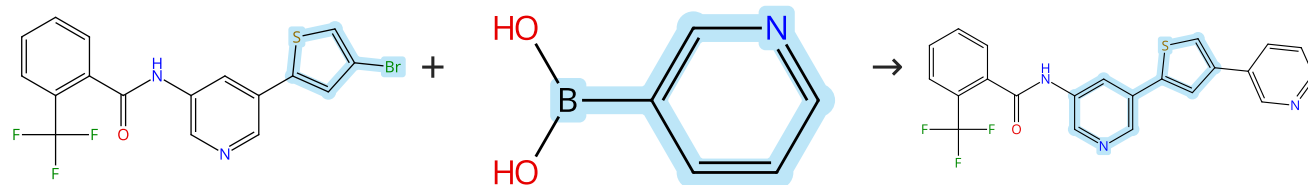
Steps: 1 Yield: 53%



31-614-CAS-35533401	Steps: 1 Yield: 53%	Design, synthesis and in vitro biological evaluation of 2-aminopyridine derivatives as novel PI3Kδ inhibitors for hematological cancer By: Yang, Chengbin; et al Bioorganic & Medicinal Chemistry Letters (2023), 82, 129152.
1.1 Reagents: Potassium carbonate Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: 1,4-Dioxane, Water; 12 h, 100 °C		

Scheme 349 (1 Reaction)

Steps: 1 Yield: 53%



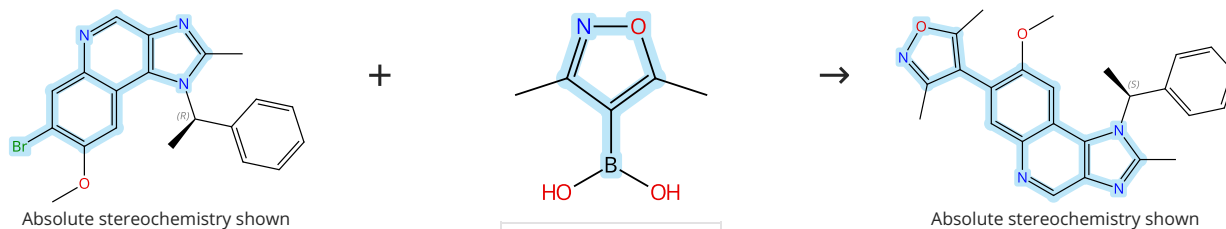
Suppliers (115)

Supplier (1)

31-179-CAS-19108272	Steps: 1 Yield: 53%	Development of novel 2,4-bispyridyl thiophene-based compounds as highly potent and selective Dyrk1A inhibitors. Part I: Benzamide and benzylamide derivatives By: Darwish, Sarah S.; et al European Journal of Medicinal Chemistry (2018), 157, 1031-1050.
1.1 Reagents: Sodium carbonate Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: 1,4-Dioxane, Water; 2 h, rt → reflux		
Experimental Protocols		

Scheme 350 (1 Reaction)

Steps: 1 Yield: 53%

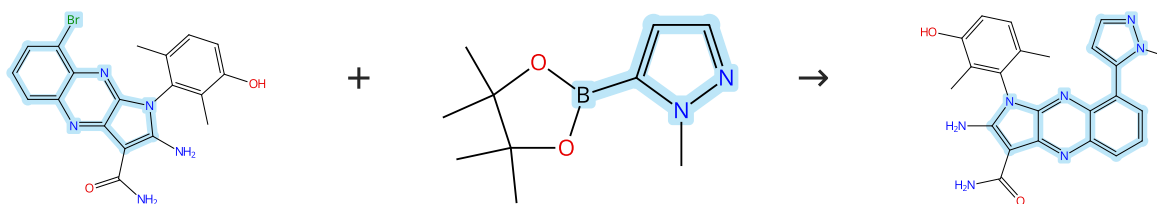


Suppliers (88)

31-614-CAS-38841427	Steps: 1 Yield: 53%	Discovery of (R)-4-(8-methoxy-2-methyl-1-(1-phenylethyl)-1H-imidazo[4,5-c]quinnolin-7-yl)-3,5-dimethylisoxazole as a potent and selective BET inhibitor for treatment of acute myeloid leukemia (AML) guided by FEP calculation By: Yu, Su; et al European Journal of Medicinal Chemistry (2024), 263, 115924.
1.1 Reagents: Potassium acetate Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: 1,4-Dioxane, Water; overnight, 100 °C		
Experimental Protocols		

Scheme 351 (1 Reaction)

Steps: 1 Yield: 53%

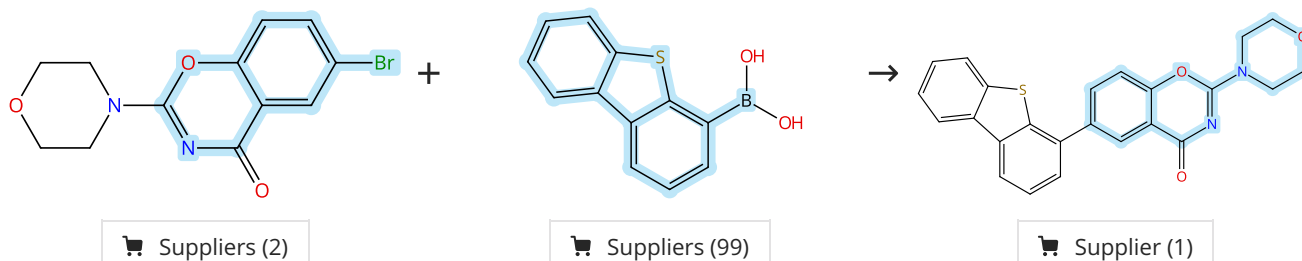


Suppliers (93)

31-614-CAS-40132770	Steps: 1 Yield: 53%	Discovery of an Orally Bioavailable and Selective PKMYT1 Inhibitor, RP-6306
1.1 Reagents: Potassium carbonate Catalysts: Dichloro[1,1'-bis(diphenylphosphino)ferrocene] palladium(II) dichloromethane adduct Solvents: Dimethylformamide, Water; 2 h, 80 °C		By: Szychowski, Janek; et al Journal of Medicinal Chemistry (2022), 65(15), 10251-10284.
Experimental Protocols		

Scheme 352 (1 Reaction)

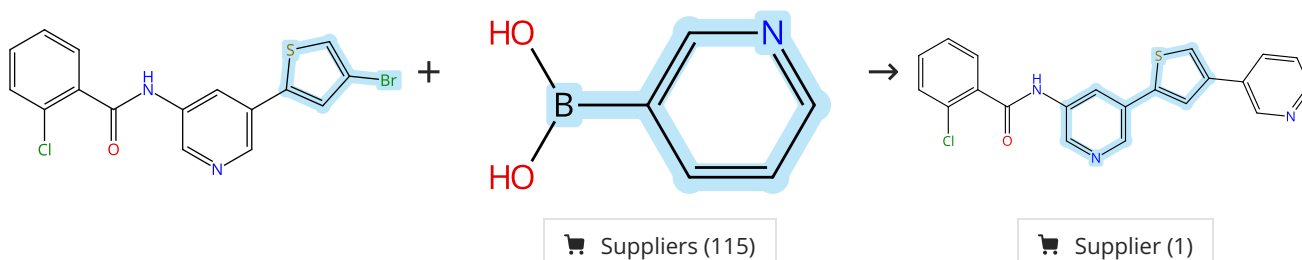
Steps: 1 Yield: 53%



31-179-CAS-58765	Steps: 1 Yield: 53%	Synthesis, structure elucidation, DNA-PK and PI3K and anti-cancer activity of 8- and 6- aryl-substituted-1-3-benzoxazines
1.1 Reagents: Potassium carbonate Catalysts: Dichlorobis(triphenylphosphine)palladium Solvents: 1,4-Dioxane, Water; 16 h, 50 °C		By: Morrison, Rick; et al European Journal of Medicinal Chemistry (2016), 110, 326-339.
Experimental Protocols		

Scheme 353 (1 Reaction)

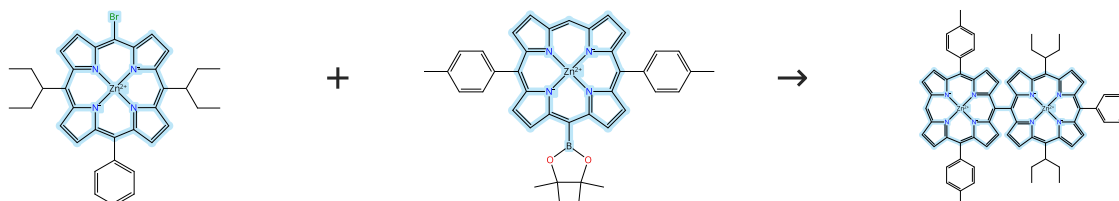
Steps: 1 Yield: 53%



31-179-CAS-19108266	Steps: 1 Yield: 53%	Development of novel 2,4-bispyridyl thiophene-based compounds as highly potent and selective Dyrk1A inhibitors. Part I: Benzamide and benzylamide derivatives
1.1 Reagents: Sodium carbonate Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: 1,4-Dioxane, Water; 2 h, rt → reflux		By: Darwish, Sarah S.; et al European Journal of Medicinal Chemistry (2018), 157, 1031-1050.
Experimental Protocols		

Scheme 354 (1 Reaction)

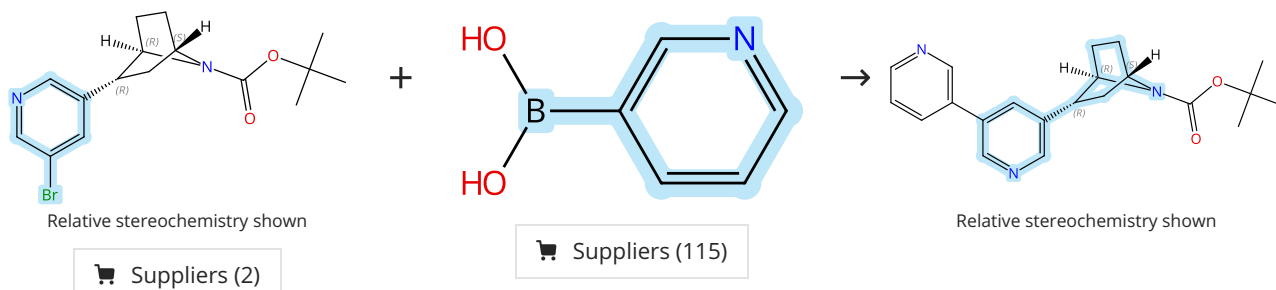
Steps: 1 Yield: 53%



31-614-CAS-25922683	Synthesis and Functionalization of Triply Fused Porphyrin Dimers
Steps: 1 Yield: 53% 1.1 Reagents: Cesium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Dimethylformamide, Toluene; 20 h, 90 °C 1.2 Reagents: Water Experimental Protocols	By: Ryan, Aoife A.; et al European Journal of Organic Chemistry (2013), 2013(18), 3700-3711.

Scheme 355 (1 Reaction)

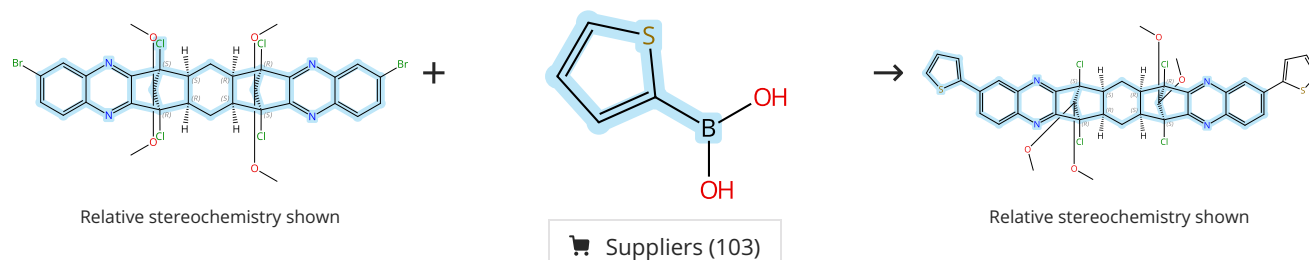
Steps: 1 Yield: 53%



31-179-CAS-14665165	Synthesis, nicotinic acetylcholine receptor binding, in vitro and in vivo pharmacology properties of 3'-(substituted pyridinyl)-deschloroepibatidine analogs
Steps: 1 Yield: 53% 1.1 Reagents: Potassium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: 1,4-Dioxane, Water; 24 h, 110 °C Experimental Protocols	By: Ondachi, Pauline W.; et al Bioorganic & Medicinal Chemistry (2015), 23(17), 5693-5701.

Scheme 356 (1 Reaction)

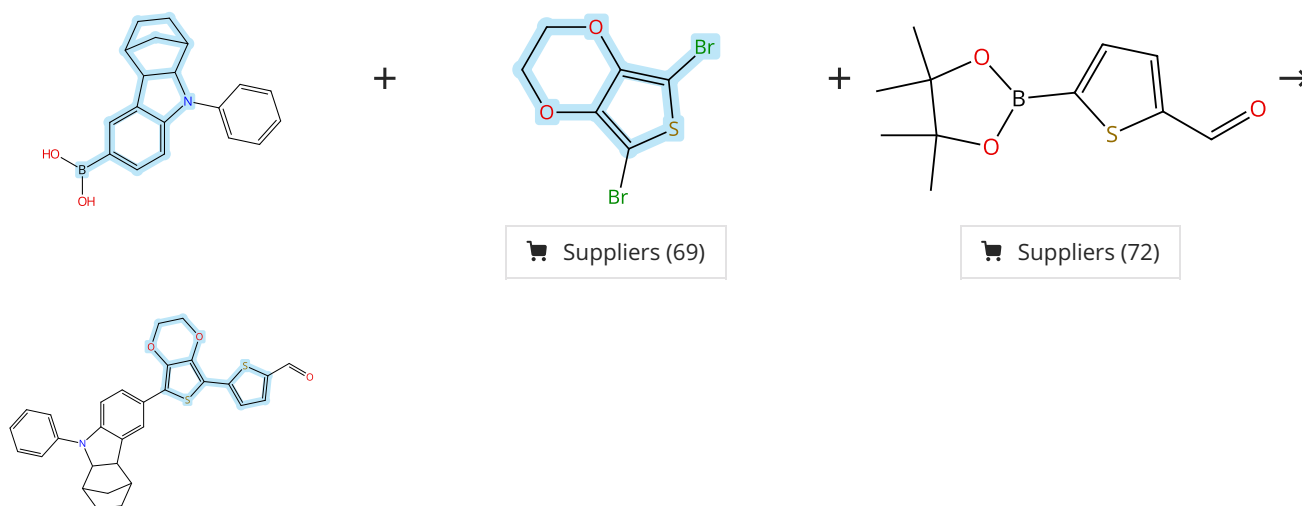
Steps: 1 Yield: 53%



31-179-CAS-10796258	C-Shaped Diastereomers Containing Cofacial Thiophene-Substituted Quinoxaline Rings: Synthesis, Photophysical Properties, and X-ray Crystallography
Steps: 1 Yield: 53% 1.1 Reagents: Sodium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Ethanol, Toluene; 30 min, 120 °C Experimental Protocols	By: DeBlase, Catherine R.; et al Journal of Organic Chemistry (2014), 79(10), 4312-4321.

Scheme 357 (1 Reaction)

Steps: 1 Yield: 53%



31-179-CAS-6274581

Steps: 1 Yield: 53%

Elucidating the Structure-Property Relationships of Donor- π -Acceptor Dyes for Dye-Sensitized Solar Cells (DSSCs) through Rapid Library Synthesis by a One-Pot Procedure

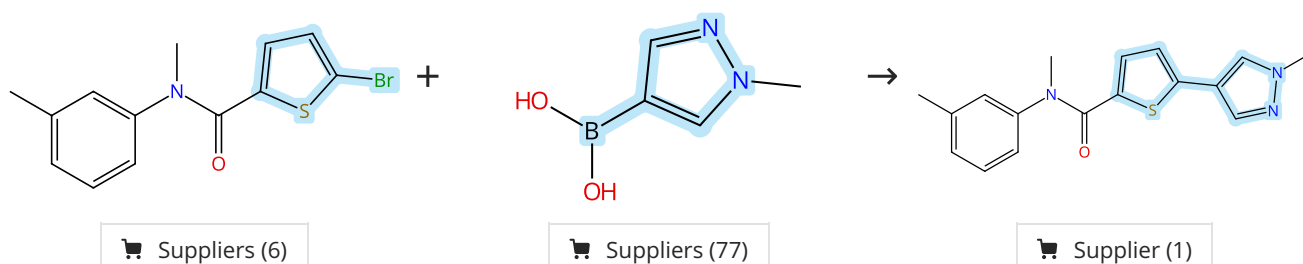
By: Fuse, Shinichiro; et al

Chemistry - A European Journal (2014), 20(34), 10685-10694.

- 1.1 **Reagents:** Sodium carbonate
Catalysts: Tris(dibenzylideneacetone)dipalladium, 1,1'-(9,9-Dimethyl-9H-xanthene-4,5-diyl)bis[1,1-diphenylphosphine]
Solvents: Toluene, Water; 0.5 - 3 h, reflux; reflux $\rightarrow$ rt
- 1.2 **Reagents:** Sodium carbonate
Catalysts: Tris(dibenzylideneacetone)dipalladium, Tri-*tert*-butylphosphonium tetrafluoroborate; 1 - 24 h, rt

Scheme 358 (1 Reaction)

Steps: 1 Yield: 53%



31-179-CAS-3792944

Steps: 1 Yield: 53%

Metabolic stability optimization and metabolite identification of 2,5-thiophene amide 17 β -hydroxysteroid dehydrogenase type 2 inhibitors

By: Gargano, Emanuele M.; et al

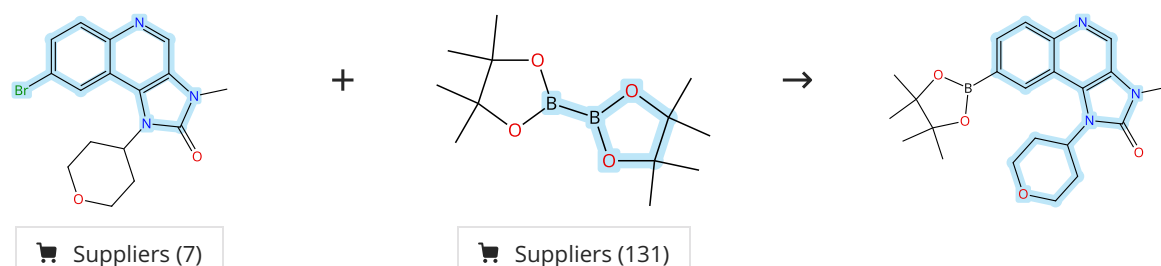
European Journal of Medicinal Chemistry (2014), 87, 203-219.

- 1.1 **Reagents:** Cesium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Ethanol, 1,2-Dimethoxyethane, Water; 20 min, 150 $^{\circ}$ C; 150 $^{\circ}$ C $\rightarrow$ rt
- 1.2 **Reagents:** Ethyl acetate, Water

Experimental Protocols

Scheme 359 (1 Reaction)

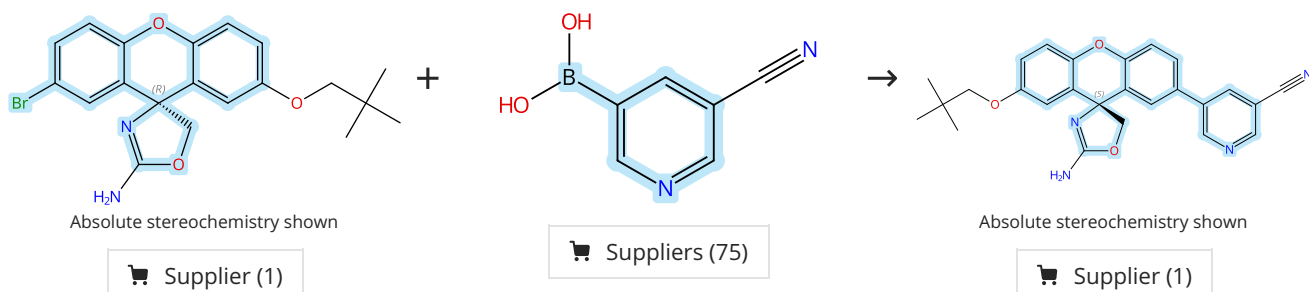
Steps: 1 Yield: 53%



31-614-CAS-34087981 Steps: 1 Yield: 53% 1.1 Reagents: Potassium acetate Catalysts: Dichloro[1,1'-bis(diphenylphosphino)ferrocene] palladium(II) dichloromethane adduct Solvents: 1,4-Dioxane; 24 h, rt → 75 °C Experimental Protocols	Development of novel urea-based ATM kinase inhibitors with subnanomolar cellular potency and high kinome selectivity By: Dimitrov, Teodor; et al European Journal of Medicinal Chemistry (2022), 235, 114234.
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Scheme 360 (1 Reaction)

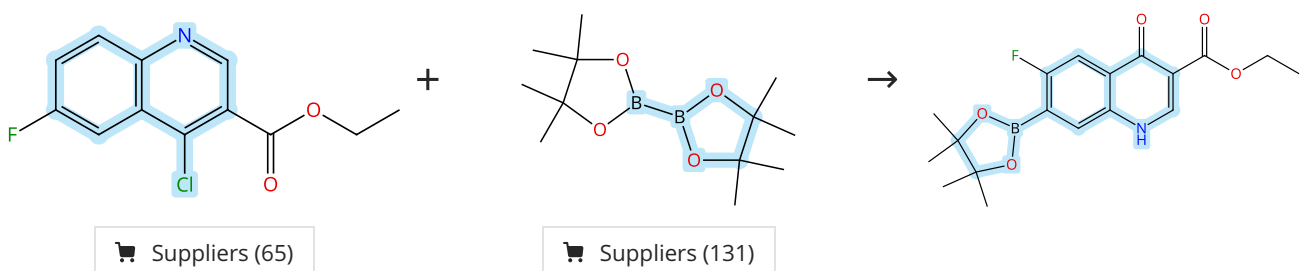
Steps: 1 Yield: 53%



31-179-CAS-2852467 Steps: 1 Yield: 53% 1.1 Reagents: Potassium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: 1,4-Dioxane, Water; overnight, 100 °C Experimental Protocols	Lead Optimization and Modulation of hERG Activity in a Series of Aminooxazoline Xanthene β-Site Amyloid Precursor Protein Cleaving Enzyme (BACE1) Inhibitors By: Epstein, Oleg; et al Journal of Medicinal Chemistry (2014), 57(23), 9796-9810.
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Scheme 361 (1 Reaction)

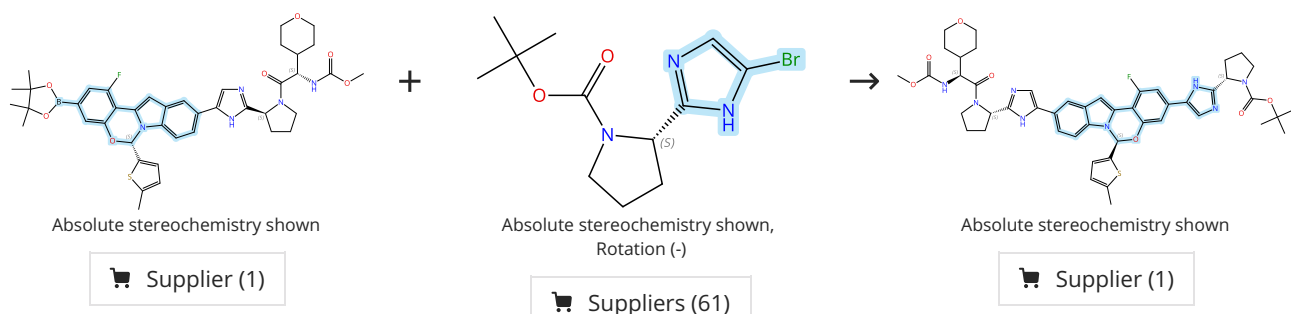
Steps: 1 Yield: 53%



31-614-CAS-34106217 Steps: 1 Yield: 53% 1.1 Catalysts: Bis[(1,2,5,6-η)-1,5-cyclooctadiene]di-μ-methoxydiiridium, 4,4'-Bis(1,1-dimethylethyl)-2,2'-bipyridine Solvents: Tetrahydrofuran; 14 h, rt → 80 °C; 80 °C → rt 1.2 Reagents: Sodium acetate Catalysts: Acetic acid; 2 h, 120 °C Experimental Protocols	Iridium-Catalyzed Borylation of 6-Fluoroquinolines: Access to 6-Fluoroquinolones By: Hickey, Aobha; et al Journal of Organic Chemistry (2022), 87(15), 9977-9987.
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Scheme 362 (1 Reaction)

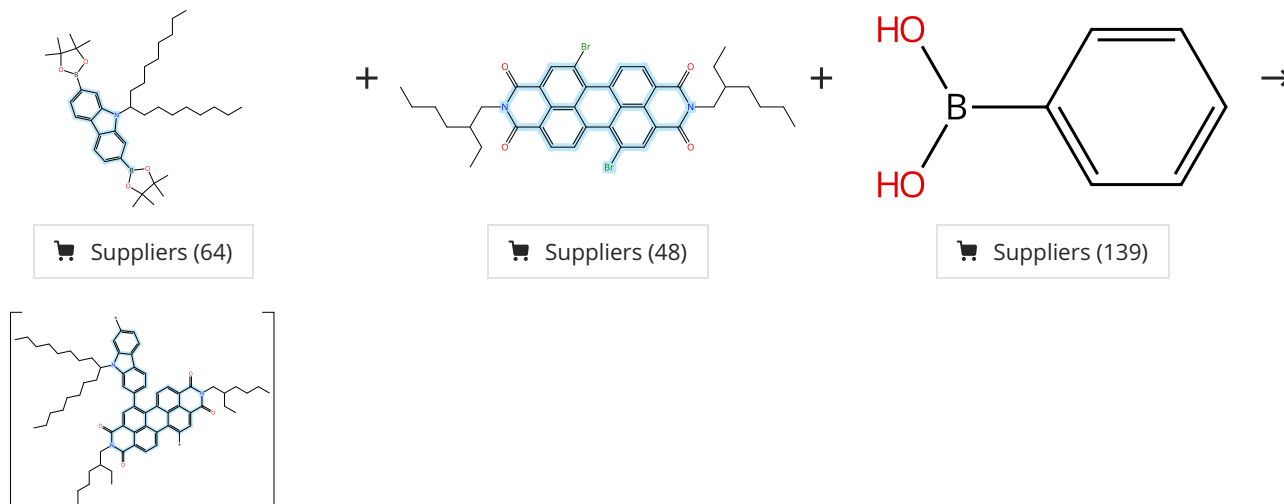
Steps: 1 Yield: 53%



31-179-CAS-18796017	Steps: 1 Yield: 53%	Discovery of MK-6169, a Potent Pan-Genotype Hepatitis C Virus NS5A Inhibitor with Optimized Activity against Common Resistance-Associated Substitutions
1.1 Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: Tetrahydrofuran, Water; overnight, 100 °C; cooled	By: Yu, Wensheng; et al	Journal of Medicinal Chemistry (2018), 61(9), 3984-4003.
1.2 Reagents: Sodium bicarbonate Solvents: Ethyl acetate, Water; rt		

Scheme 363 (1 Reaction)

Steps: 1 Yield: 53%



Suppliers (64)

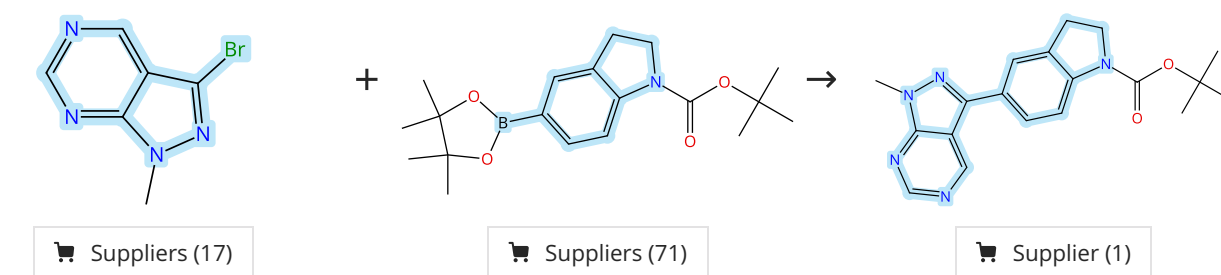
Suppliers (48)

Suppliers (139)

31-614-CAS-25720998	Steps: 1 Yield: 53%	All-polymer solar cells from perylene diimide based copolymers: material design and phase separation control
1.1 Reagents: Sodium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Tetrahydrofuran, Water; 48 h, reflux	By: Zhou, Erjun; et al	Angewandte Chemie, International Edition (2011), 50(12), 2799-2803, S2799/1-S2799/16.
1.2 2 h, reflux		
1.3 Reagents: Bromobenzene; 2 h, reflux		
Experimental Protocols		

Scheme 364 (1 Reaction)

Steps: 1 Yield: 53%



Suppliers (17)

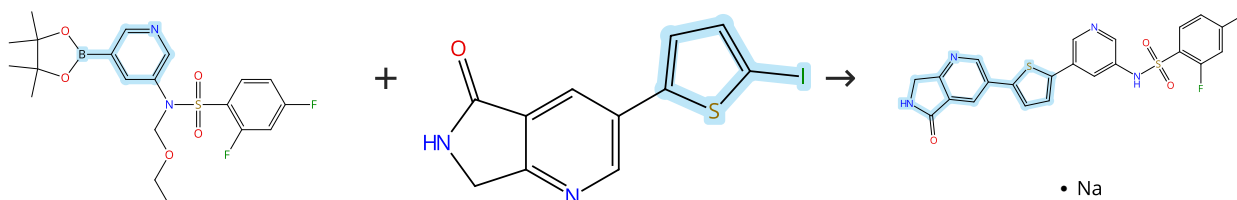
Suppliers (71)

Supplier (1)

31-179-CAS-23653187	Steps: 1 Yield: 53%	Identification of 5-(2,3-Dihydro-1H-indol-5-yl)-7H-pyrrolo[2,3-d]pyrimidin-4-amine derivatives as a new class of receptor-interacting protein kinase 1 (RIPK1) inhibitors, which showed potent activity in a tumor metastasis model
1.1 Reagents: Cesium carbonate Catalysts: Dichloro[1,1'-bis(diphenylphosphino)ferrocene]palladium(II) dichloromethane adduct Solvents: 1,4-Dioxane, Water; 18 h, 95 °C	By: Li, Yueshan; et al	Journal of Medicinal Chemistry (2018), 61(24), 11398-11414.
Experimental Protocols		

Scheme 365 (1 Reaction)

Steps: 1 Yield: 53%



31-179-CAS-21660232

Steps: 1 Yield: 53%

Inhibition of the cytolytic protein Perforin prevents rejection of transplanted bone marrow stem cells in vivo

By: Spicer, Julie A.; et al

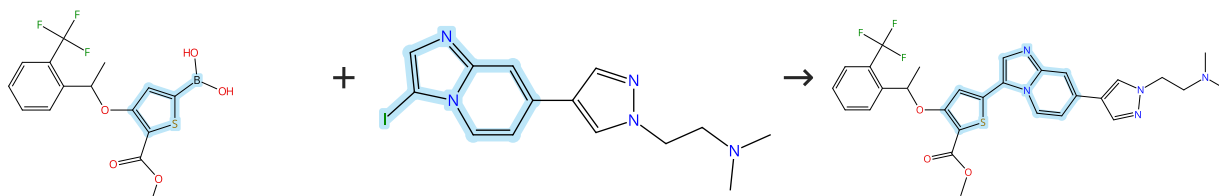
Journal of Medicinal Chemistry (2020), 63(5), 2229-2239.

- 1.1 **Reagents:** Sodium carbonate
Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium
Solvents: Ethanol, Toluene; 2 h, reflux
- 1.2 **Reagents:** Sodium hydroxide
Solvents: Ethanol, Water

Experimental Protocols

Scheme 366 (1 Reaction)

Steps: 1 Yield: 53%



31-179-CAS-22738207

Steps: 1 Yield: 53%

Design, synthesis, and structure activity relationship (SAR) studies of novel imidazo[1,2-a]pyridine derivatives as Nek2 inhibitors

By: Wang, Haili; et al

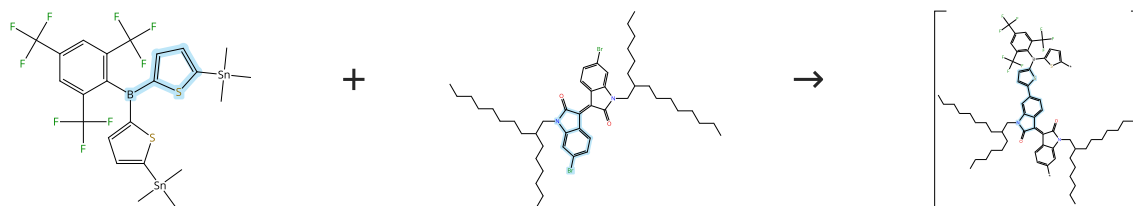
Bioorganic & Medicinal Chemistry (2020), 28(23), 115775.

- 1.1 **Reagents:** Sodium carbonate
Catalysts: Dichloro[1,1'-bis(diphenylphosphino)ferrocene]palladium(II) dichloromethane adduct
Solvents: Tetrahydrofuran, Water; overnight, 80 °C

Experimental Protocols

Scheme 367 (1 Reaction)

Steps: 1 Yield: 53%



Suppliers (41)

31-614-CAS-28951459

Steps: 1 Yield: 53%

p-π Conjugated polymers based on stable triaryl borane with n-type behavior in optoelectronic devices

By: Meng, Bin; et al

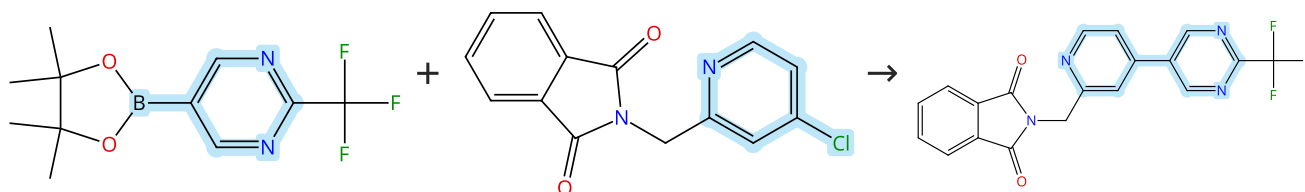
Angewandte Chemie, International Edition (2018), 57(8), 2183-2187.

- 1.1 **Catalysts:** Tri-*o*-tolylphosphine, Palladium, tris[μ-[(1,2-η:4,5-η)-(1*E*,4*E*)-1,5-diphenyl-1,4-pentadien-3-one]]di-, compd. with trichloromethane (1:1)
Solvents: Dimethylformamide, Toluene; 36 h, 120 °C
- 1.2 **Reagents:** 2-(Tributylstannyl)thiophene
Solvents: Toluene; 4 h
- 1.3 **Reagents:** 2-Bromothiophene; 4 h

Experimental Protocols

Scheme 368 (1 Reaction)

Steps: 1 Yield: 53%



Suppliers (57)

Suppliers (7)

31-179-CAS-18939723

Steps: 1 Yield: 53%

1.1 **Catalysts:** [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium
Solvents: 1,4-Dioxane; 12 h, 110 °C

Experimental Protocols

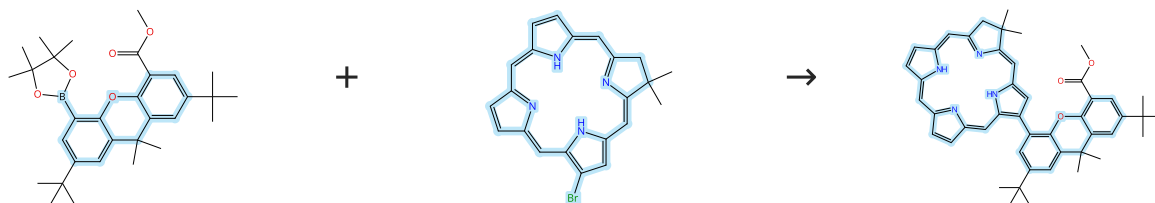
Discovery of a Potent (4R,5S)-4-Fluoro-5-methylproline Sulfonamide Transient Receptor Potential Ankyrin 1 Antagonist and Its Methylene Phosphate Prodrug Guided by Molecular Modeling

By: Chen, Huifen; et al

Journal of Medicinal Chemistry (2018), 61(8), 3641-3659.

Scheme 369 (1 Reaction)

Steps: 1 Yield: 53%



31-179-CAS-21706108

Steps: 1 Yield: 53%

1.1 **Reagents:** Potassium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Dimethylformamide, Toluene; 20 h, 90 °C

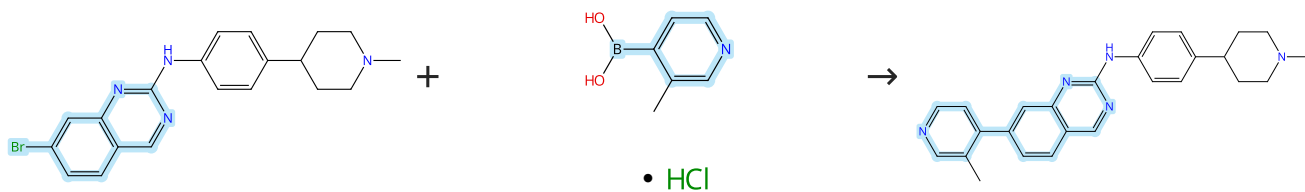
Synthesis of Hangman Chlorins

By: Liu, Mengran; et al

Journal of Organic Chemistry (2020), 85(7), 5065-5072.

Scheme 370 (1 Reaction)

Steps: 1 Yield: 53%



Suppliers (50)

31-614-CAS-37427352

Steps: 1 Yield: 53%

1.1 **Reagents:** Potassium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Ethanol, Tetrahydrofuran, Water; 1 h, 100 °C

Experimental Protocols

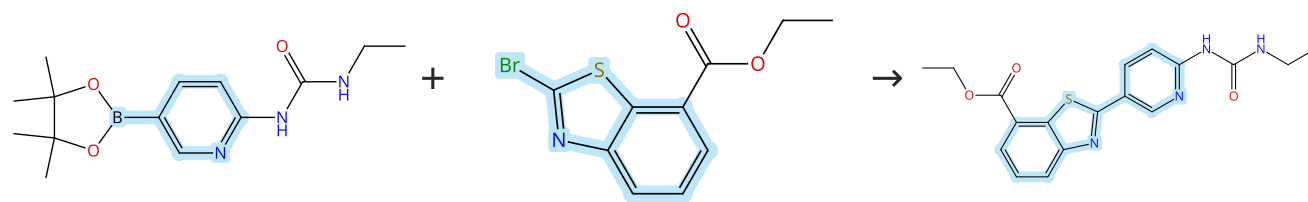
Discovery of quinazoline HPK1 inhibitors with high cellular potency

By: Toure, Momar; et al

Bioorganic & Medicinal Chemistry (2023), 92, 117423.

Scheme 371 (1 Reaction)

Steps: 1 Yield: 53%



Suppliers (45)

Suppliers (47)

31-179-CAS-1894845

Steps: 1 Yield: 53%

Fragment-to-Hit-to-Lead Discovery of a Novel Pyridylurea Scaffold of ATP Competitive Dual Targeting Type II Topoisomerase Inhibiting Antibacterial Agents

By: Basarab, Gregory S.; et al

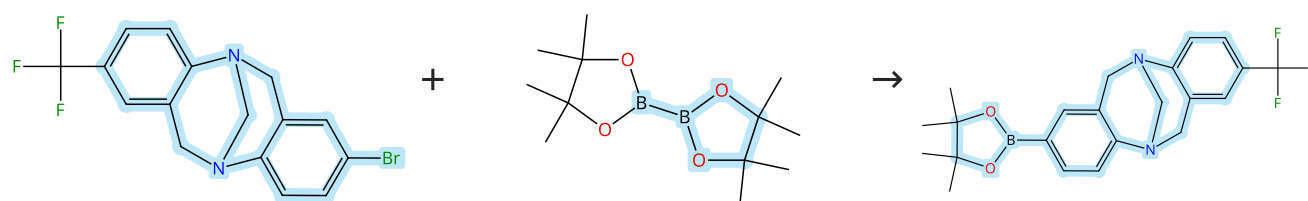
Journal of Medicinal Chemistry (2013), 56(21), 8712-8735.

1.1 **Reagents:** Cesium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: 1,4-Dioxane, Water; 30 min, 110 °C

Experimental Protocols

Scheme 372 (1 Reaction)

Steps: 1 Yield: 52%



Suppliers (131)

31-142-CAS-18138993

Steps: 1 Yield: 52%

Partitioning Solvophobic and Dispersion Forces in Alkyl and Perfluoroalkyl Cohesion

By: Adam, Catherine; et al

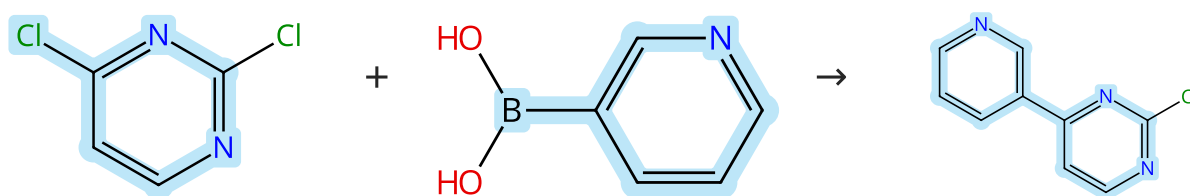
Angewandte Chemie, International Edition (2015), 54(4), 1164-1167.

1.1 **Reagents:** Potassium acetate
Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium
Solvents: Dimethyl sulfoxide; 3 d, reflux

Experimental Protocols

Scheme 373 (1 Reaction)

Steps: 1 Yield: 52%



Suppliers (103)

Suppliers (115)

Suppliers (63)

31-179-CAS-17598938

Steps: 1 Yield: 52%

Introduction of fluorine to phenyl group of 4-(2-pyrimidinylamino)benzamides leading to a series of potent hedgehog signaling pathway inhibitors

By: Xin, Minhang; et al

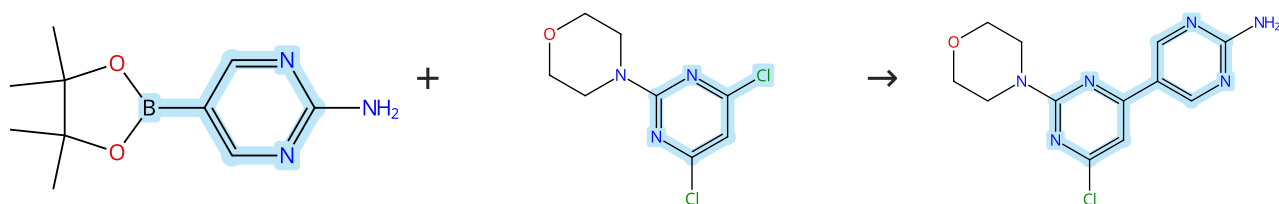
Bioorganic & Medicinal Chemistry Letters (2017), 27(15), 3259-3263.

1.1 **Reagents:** Sodium carbonate
Catalysts: Dichlorobis(triphenylphosphine)palladium
Solvents: 1,4-Dioxane, Water; 14 h, reflux

Experimental Protocols

Scheme 374 (1 Reaction)

Steps: 1 Yield: 52%



Suppliers (96)

Suppliers (77)

Suppliers (2)

31-179-CAS-10309437

Steps: 1 Yield: 52%

Synthesis and in Vitro and in Vivo Evaluation of Phosphoinositide-3-kinase Inhibitors

1.1 **Reagents:** Sodium carbonate
Catalysts: Triphenylphosphine, Palladium diacetate
Solvents: 1,2-Dimethoxyethane, Water; 1 h, 95 °C

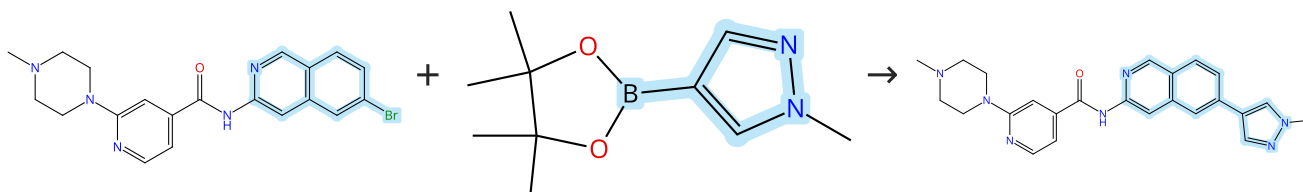
By: Burger, Matthew T.; et al

ACS Medicinal Chemistry Letters (2011), 2(1), 34-38.

Experimental Protocols

Scheme 375 (1 Reaction)

Steps: 1 Yield: 52%



Suppliers (126)

Suppliers (33)

31-614-CAS-40641090

Steps: 1 Yield: 52%

Comparative efficacy and selectivity of pharmacological inhibitors of DYRK and CLK protein kinases

1.1 **Reagents:** Cesium carbonate
Catalysts: (*SP*-4-1)-[1,3-Bis[2,6-bis(1-methylethyl)phenyl]-1,3-dihydro-2*H*-imidazol-2-ylidene]dichloro(3-chloropyridine-*κN*) palladium
Solvents: 1,4-Dioxane, Water; 5.5 h, 95 °C

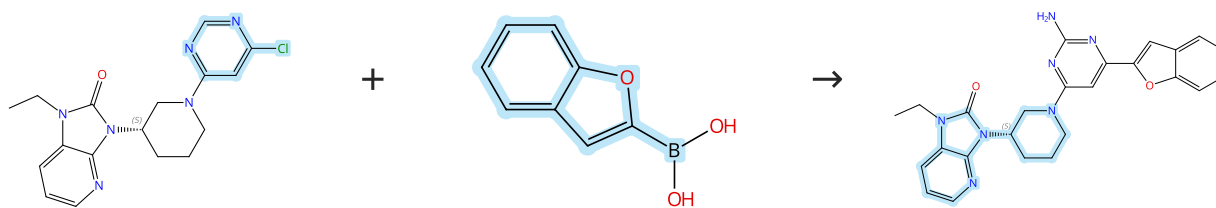
By: Lindberg, Mattias F.; et al

Journal of Medicinal Chemistry (2023), 66(6), 4106-4130.

Experimental Protocols

Scheme 376 (1 Reaction)

Steps: 1 Yield: 52%



Absolute stereochemistry shown

Suppliers (92)

Absolute stereochemistry shown

Supplier (1)

31-080-CAS-9024730

Steps: 1 Yield: 52%

Identification of potent, noncovalent fatty acid amide hydrolase (FAAH) inhibitors

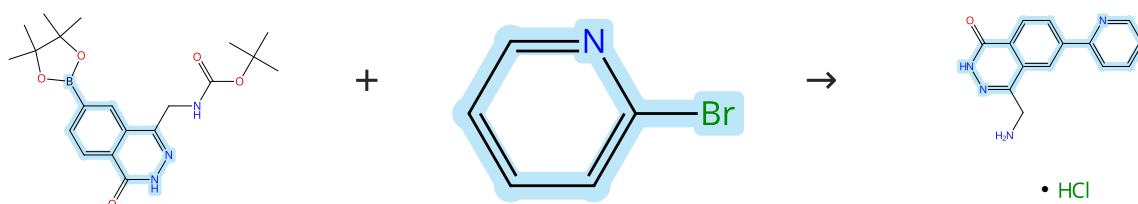
1.1 **Reagents:** Sodium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Methanol, Benzene, Water; 85 °C

By: Gustin, Darin J.; et al

Bioorganic & Medicinal Chemistry Letters (2011), 21(8), 2492-2496.

Scheme 377 (1 Reaction)

Steps: 1 Yield: 52%



Suppliers (15)

Suppliers (89)

31-614-CAS-40642502

Steps: 1 Yield: 52%

Fragment-Based Discovery of MRTX1719, a Synthetic Lethal Inhibitor of the PRMT5•MTA Complex for the Treatment of M TAP-Deleted Cancers

By: Smith, Christopher R.; et al

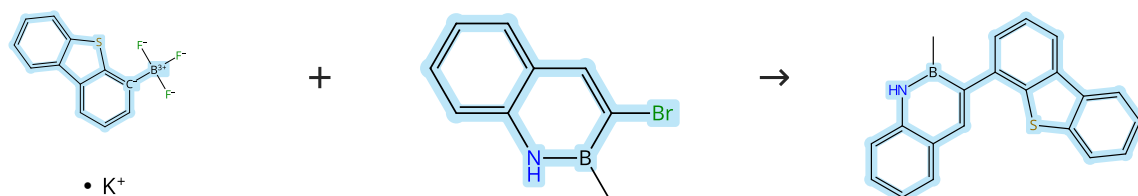
Journal of Medicinal Chemistry (2022), 65(3), 1749-1766.

- 1.1 **Reagents:** Sodium carbonate
Catalysts: (*SP*-4-2)-[1,1'-Bis[bis(1,1-dimethylethyl)phosphino- κ ']ferrocene]dichloropalladium
Solvents: 1,4-Dioxane, Water; 2 h, 80 °C
- 1.2 **Reagents:** Trifluoroacetic acid
Solvents: Dichloromethane; 1 h, 20 °C
- 1.3 **Reagents:** Hydrochloric acid

Experimental Protocols

Scheme 378 (1 Reaction)

Steps: 1 Yield: 52%

• K⁺

Suppliers (16)

Suppliers (2)

31-614-CAS-30028290

Steps: 1 Yield: 52%

Accessing Molecularly Complex Azaborines: Palladium-Catalyzed Suzuki-Miyaura Cross-Couplings of Brominated 2,1-Borazaronaphthalenes and Potassium Organotrifluoroborates

By: Molander, Gary A.; et al

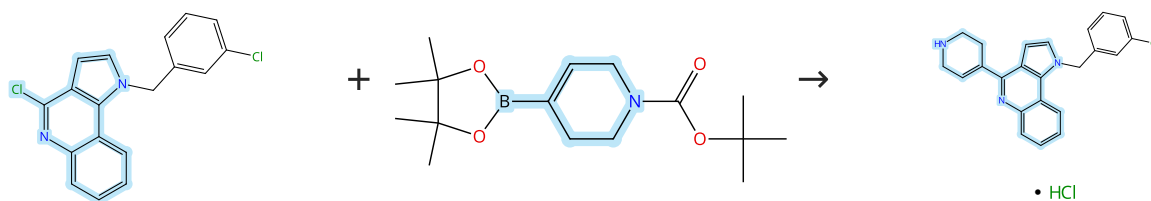
Journal of Organic Chemistry (2014), 79(14), 6663-6678.

- 1.1 **Reagents:** Cesium carbonate
Catalysts: (*SP*-4-3)-[2'-(Amino- κ M)[1,1'-biphenyl]-2-yl- κ C]chloro[tris(1,1-dimethylethyl)phosphine]palladium
Solvents: Cyclopentyl methyl ether, Water; 18 h, 60 °C

Experimental Protocols

Scheme 379 (1 Reaction)

Steps: 1 Yield: 52%



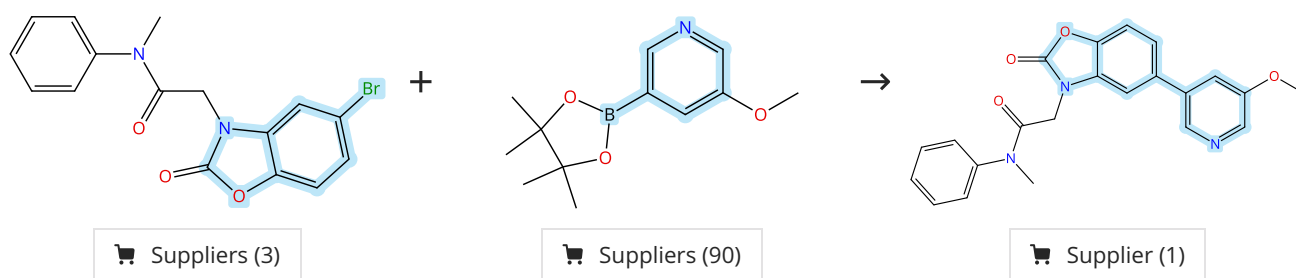
• HCl

Suppliers (127)

31-614-CAS-38120442 Steps: 1 Yield: 52% 1.1 Reagents: Potassium carbonate Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: 1,4-Dioxane, Water; 6 h, 80 °C 1.2 Reagents: Hydrochloric acid Solvents: Methanol; 12 h, rt Experimental Protocols	Superiority of the Triple-Acting 5-HT₆R/5-HT₃R Antagonist and MAO-B Reversible Inhibitor PZ-1922 over 5-HT₆R Antagonist Intepirdine in Alleviation of Cognitive Deficits in Rats By: Grychowska, Katarzyna; et al Journal of Medicinal Chemistry (2023), 66(21), 14928-14947.
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Scheme 380 (1 Reaction)

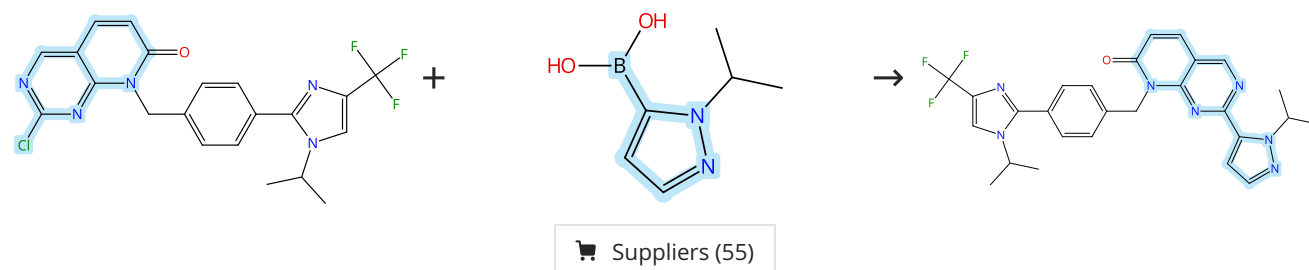
Steps: 1 Yield: 52%



31-179-CAS-17321604 Steps: 1 Yield: 52% 1.1 Reagents: Potassium carbonate Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: 1,2-Dimethoxyethane, Water; 7 h, 80 °C Experimental Protocols	Development of a ¹⁸F-Labeled Radiotracer with Improved Brain Kinetics for Positron Emission Tomography Imaging of Translocator Protein (18 kDa) in Ischemic Brain and Glioma By: Fujinaga, Masayuki; et al Journal of Medicinal Chemistry (2017), 60(9), 4047-4061.
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Scheme 381 (1 Reaction)

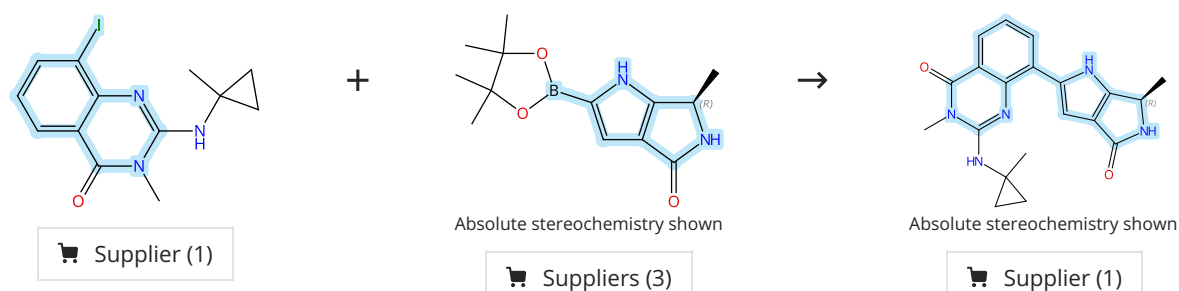
Steps: 1 Yield: 52%



31-614-CAS-40821629 Steps: 1 Yield: 52% 1.1 Reagents: Potassium carbonate Catalysts: (SP4-4)-[2'-(Amino-κM)[1,1'-biphenyl]-2-yl-κC]chloro[dicyclohexyl[2',4',6'-tris(1-methylethyl)[1,1'-biphenyl]-2-yl]phosphine]palladium Solvents: 1,4-Dioxane, Water; 4 h, rt → 80 °C Experimental Protocols	Design, synthesis, and biological evaluation of pyrido [2,3-d] pyrimidin-7(8H)-one derivatives as potent USP1 inhibitors By: Li, Hongrui; et al European Journal of Medicinal Chemistry (2024), 275, 116568.
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Scheme 382 (1 Reaction)

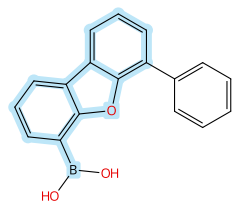
Steps: 1 Yield: 52%



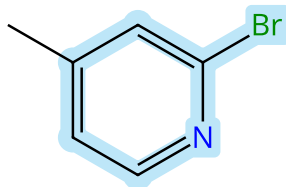
31-179-CAS-15737224	Steps: 1 Yield: 52%	Discovery and Optimization of Quinazolinone-pyrrolopyrrolones as Potent and Orally Bioavailable Pan-Pim Kinase Inhibitors By: Pettus, Liping H.; et al Journal of Medicinal Chemistry (2016), 59(13), 6407-6430.
1.1 Reagents: Tripotassium phosphate Catalysts: Tris(dibenzylideneacetone)dipalladium, X-Phos Solvents: 1,4-Dioxane, Water; 35 min, 105 °C		

Scheme 383 (1 Reaction)

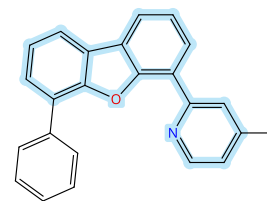
Steps: 1 Yield: 52%



+



→



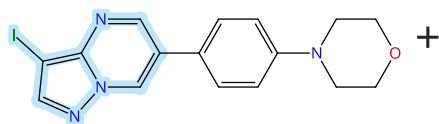
Suppliers (50)

Suppliers (103)

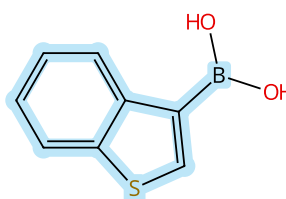
31-614-CAS-37044377	Steps: 1 Yield: 52%	Transition-metal-free synthesis of functional 2-arylphenols by an intermolecular S_NAr reaction between dibenzofurans and various nucleophiles By: Lin, Yueping; et al Organic Chemistry Frontiers (2023), 10(15), 3781-3785.
1.1 Reagents: Sodium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Ethanol, 1,2-Dimethoxyethane, Water; 24 h, 90 °C		
Experimental Protocols		

Scheme 384 (1 Reaction)

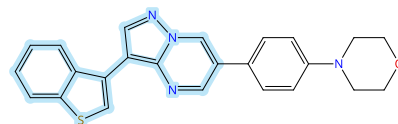
Steps: 1 Yield: 52%



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→



Suppliers (2)

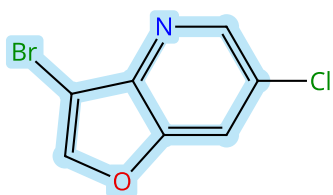
Suppliers (87)

Supplier (1)

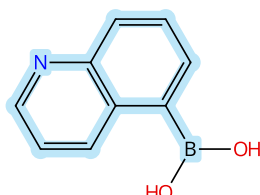
31-179-CAS-7388047	Steps: 1 Yield: 52%	Synthesis and structure-activity relationships of a novel and selective bone morphogenetic protein receptor (BMP) inhibitor derived from the pyrazolo[1.5-a]pyrimidine scaffold of Dorsomorphin: The discovery of ML347 as an ALK2 versus ALK3 selective MLPCN probe By: Engers, Darren W.; et al Bioorganic & Medicinal Chemistry Letters (2013), 23(11), 3248-3252.
1.1 Reagents: Tripotassium phosphate Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: 1,4-Dioxane, Water; 30 min, 120 °C		

Scheme 385 (1 Reaction)

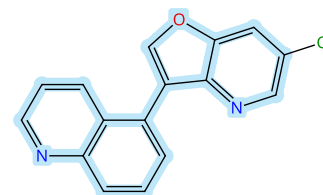
Steps: 1 Yield: 52%



+



→



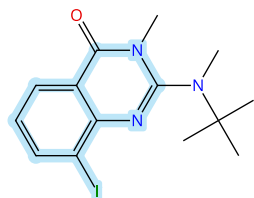
Suppliers (3)

Suppliers (93)

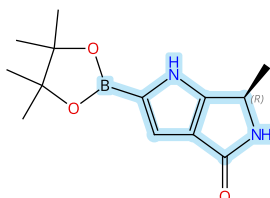
31-614-CAS-41986951 Steps: 1 Yield: 52% 1.1 Reagents: Tripotassium phosphate Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: 1,4-Dioxane, Water; 2 h, 90 °C Experimental Protocols	Discovery of Two Highly Selective Structurally Orthogonal Chemical Probes for Activin Receptor-like Kinases 1 and 2 By: Nemeč, Vaclav; et al Journal of Medicinal Chemistry (2024), 67(15), 12632-12659.
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Scheme 386 (1 Reaction)

Steps: 1 Yield: 52%

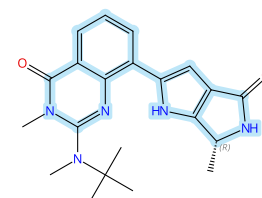


+



Absolute stereochemistry shown

→



Absolute stereochemistry shown

Supplier (1)

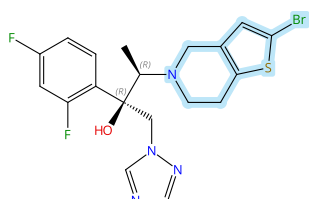
Suppliers (3)

Supplier (1)

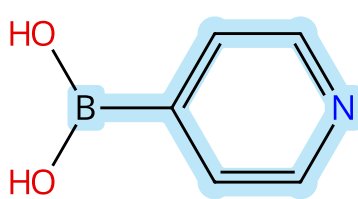
31-179-CAS-15737251 Steps: 1 Yield: 52% 1.1 Reagents: Tripotassium phosphate Catalysts: (SP-4-4)-[2'-(Amino-κM)[1,1'-biphenyl]-2-yl-κC]chloro[dicyclohexyl[2',4',6'-tris(1-methylethyl)[1,1'-biphenyl]-2-yl]phosphine]palladium Solvents: 1,4-Dioxane, Water; 35 min, 50 °C	Discovery and Optimization of Quinazolinone-pyrrolopyrrolones as Potent and Orally Bioavailable Pan-Pim Kinase Inhibitors By: Pettus, Liping H.; et al Journal of Medicinal Chemistry (2016), 59(13), 6407-6430.
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Scheme 387 (1 Reaction)

Steps: 1 Yield: 52%

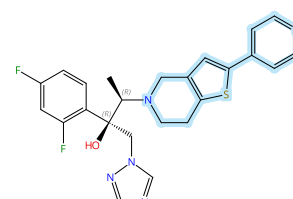
Absolute stereochemistry shown,
Rotation (-)

+



Suppliers (111)

→

Absolute stereochemistry shown,
Rotation (-)

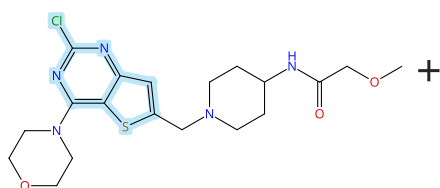
Supplier (1)

Supplier (1)

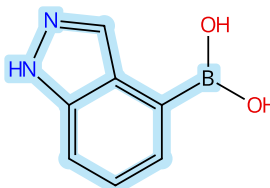
31-179-CAS-10835589 Steps: 1 Yield: 52% 1.1 Reagents: Cesium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: 1,4-Dioxane, Water; 10 h, 80 °C Experimental Protocols	Design, Synthesis, and Structure-Activity Relationship Studies of Novel Fused Heterocycles-Linked Triazoles with Good Activity and Water Solubility By: Cao, Xufeng; et al Journal of Medicinal Chemistry (2014), 57(9), 3687-3706.
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Scheme 388 (1 Reaction)

Steps: 1 Yield: 52%

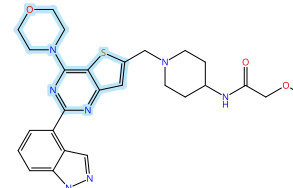


+



Suppliers (86)

→

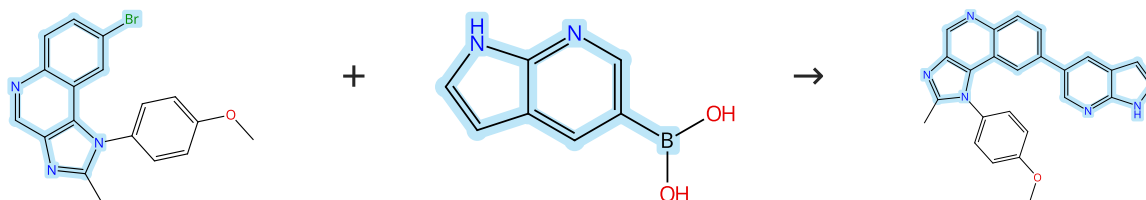


Supplier (1)

31-179-CAS-22386582	Steps: 1 Yield: 52%	Free-Wilson Analysis of Comprehensive Data on Phosphoinositide-3-kinase (PI3K) Inhibitors Reveals Importance of N-Methylation for PI3Kδ Activity
1.1 Reagents: Tripotassium phosphate Catalysts: [Bis(bicyclo[2.2.1]hept-2-yl)phosphine]chloro[2-[(dimethylamino-κM)methyl]ferrocenyl-κC]palladium Solvents: Ethanol, Water; 0.5 h, rt → 140 °C		By: Barnes, Lydia; et al Journal of Medicinal Chemistry (2019), 62(22), 10402-10422.
Experimental Protocols		

Scheme 389 (1 Reaction)

Steps: 1 Yield: 52%

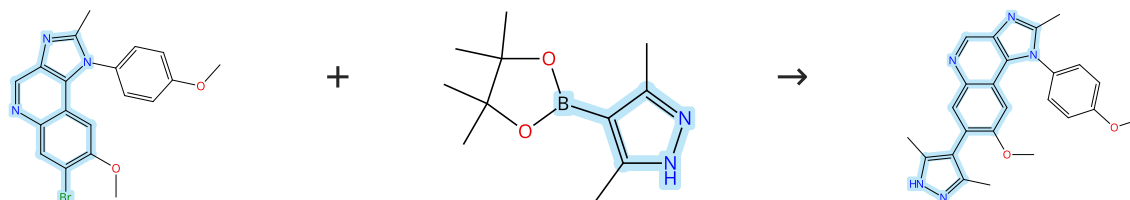


Suppliers (69)

31-614-CAS-34329925	Steps: 1 Yield: 52%	Discovery of 2-Methyl-2-(4-(2-methyl-8-(1H-pyrrolo[2,3-b]pyridin-6-yl)-1H-naphtho[1,2-d]imidazol-1-yl)phenyl)propane nitrile as a Novel PI3K/mTOR Inhibitor with Enhanced Antitumor Efficacy In Vitro and In Vivo
1.1 Reagents: Potassium acetate Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: 1,4-Dioxane, Water; overnight, rt → 100 °C		By: Yang, Jie; et al Journal of Medicinal Chemistry (2022), 65(19), 12781-12801.
Experimental Protocols		

Scheme 390 (1 Reaction)

Steps: 1 Yield: 52%

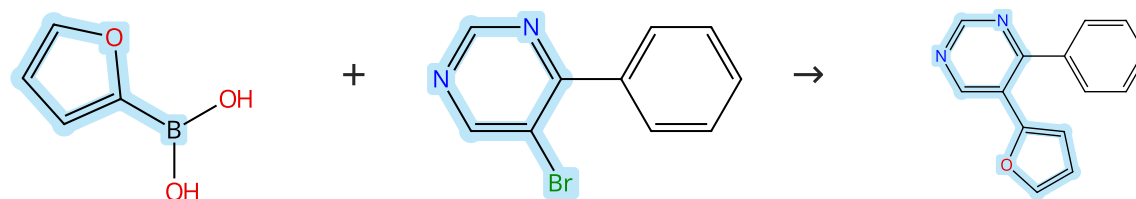


Suppliers (77)

31-614-CAS-38841406	Steps: 1 Yield: 52%	Discovery of (R)-4-(8-methoxy-2-methyl-1-(1-phenylethyl)-1H-imidazo[4,5-c]quinnolin-7-yl)-3,5-dimethylisoxazole as a potent and selective BET inhibitor for treatment of acute myeloid leukemia (AML) guided by FEP calculation
1.1 Reagents: Potassium acetate Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: 1,4-Dioxane, Water; overnight, 100 °C		By: Yu, Su; et al European Journal of Medicinal Chemistry (2024), 263, 115924.
Experimental Protocols		

Scheme 391 (1 Reaction)

Steps: 1 Yield: 52%



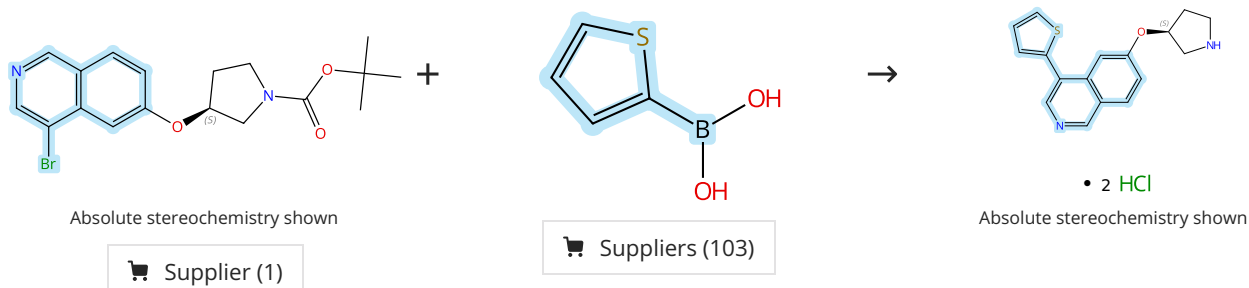
Suppliers (93)

Suppliers (52)

31-179-CAS-20726227	Steps: 1 Yield: 52%	Syntheses of Benzofuranoquinolines and Analogues via Photoinduced Acceptorless Dehydrogenative Annulation of o-Phenylfuranylpyridines
1.1 Reagents: Cesium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: 1,4-Dioxane, Water; 6 - 12 h, 85 °C		By: Fan, Jinming; et al
Experimental Protocols		Organic Letters (2019), 21(22), 9183-9187.

Scheme 392 (1 Reaction)

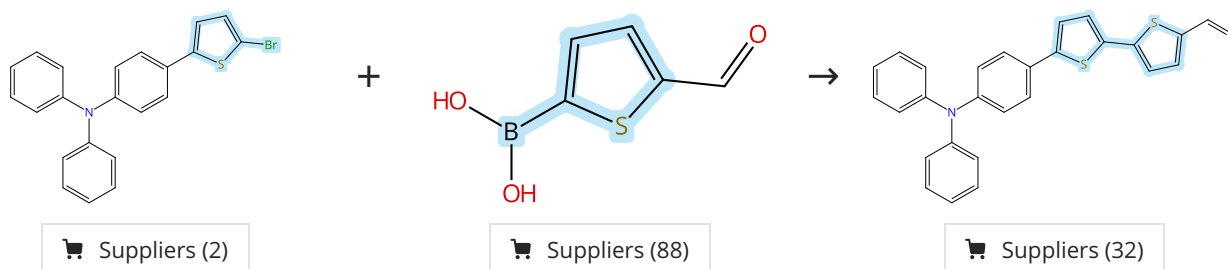
Steps: 1 Yield: 52%



31-179-CAS-22732954	Steps: 1 Yield: 52%	Discovery of 4,6- and 5,7-Disubstituted Isoquinoline Derivatives as a Novel Class of Protein Kinase C ζ Inhibitors with Fragment-Merging Strategy
1.1 Reagents: Tripotassium phosphate Catalysts: Palladium diacetate, 2-Dicyclohexylphosphino-2',6'-dimethoxybiphenyl Solvents: 1-Butanol; rt; 18 h, 110 °C		By: Atobe, Masakazu; et al
1.2 Reagents: Hydrochloric acid Solvents: Methanol, Water; 4 h, rt		Journal of Medicinal Chemistry (2020), 63(13), 7143-7162.
1.3 Reagents: Hydrochloric acid Solvents: Methanol; rt		
Experimental Protocols		

Scheme 393 (1 Reaction)

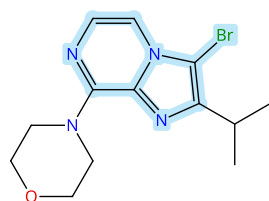
Steps: 1 Yield: 52%



31-179-CAS-4260512	Steps: 1 Yield: 52%	Rapid and structurally diverse synthesis of multi-substituted β-keto amide derivatives based on a dioxinone scaffold
1.1 Reagents: Potassium carbonate Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: Methanol, Toluene; 12 h, 80 °C		By: Fuse, Shinichiro; et al
Experimental Protocols		European Journal of Organic Chemistry (2014), 2014(22), 4854-4860.

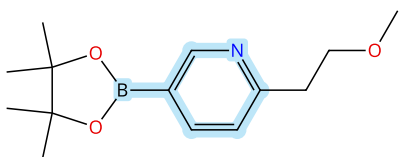
Scheme 394 (1 Reaction)

Steps: 1 Yield: 52%

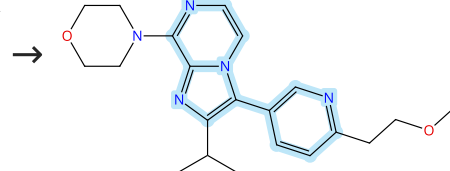


Suppliers (2)

+



Suppliers (12)



Supplier (1)

31-179-CAS-11762113

Steps: 1 Yield: 52%

Identification of a Novel Orally Bioavailable Phosphodiesterase 10A (PDE10A) Inhibitor with Efficacy in Animal Models of Schizophrenia.

By: Bartolome-Nebreda, Jose Manuel; et al

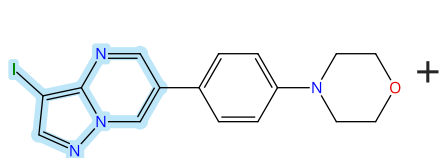
Journal of Medicinal Chemistry (2015), 58(2), 978-993.

1.1 **Reagents:** Sodium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: 1,4-Dioxane; 15 min, 150 °C

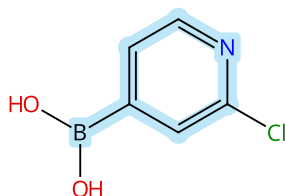
Experimental Protocols

Scheme 395 (1 Reaction)

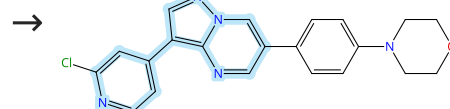
Steps: 1 Yield: 52%



Suppliers (2)



Suppliers (97)



Supplier (1)

31-179-CAS-907125

Steps: 1 Yield: 52%

Synthesis and structure-activity relationships of a novel and selective bone morphogenetic protein receptor (BMP) inhibitor derived from the pyrazolo[1.5-a]pyrimidine scaffold of Dorsomorphin: The discovery of ML347 as an ALK2 versus ALK3 selective MLPCN probe

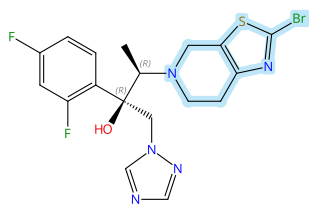
By: Engers, Darren W.; et al

Bioorganic & Medicinal Chemistry Letters (2013), 23(11), 3248-3252.

1.1 **Reagents:** Tripotassium phosphate
Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium
Solvents: 1,4-Dioxane, Water; 30 min, 120 °C

Scheme 396 (1 Reaction)

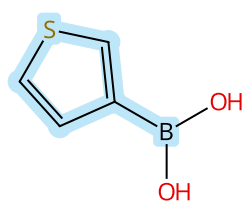
Steps: 1 Yield: 52%



Absolute stereochemistry shown,
Rotation (-)

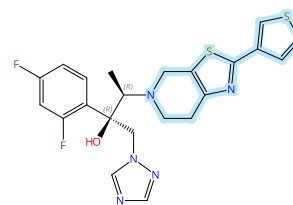
Supplier (1)

+



Suppliers (101)

→



Absolute stereochemistry shown,
Rotation (-)

Supplier (1)

31-179-CAS-3524440

Steps: 1 Yield: 52%

Design, Synthesis, and Structure-Activity Relationship Studies of Novel Fused Heterocycles-Linked Triazoles with Good Activity and Water Solubility

By: Cao, Xufeng; et al

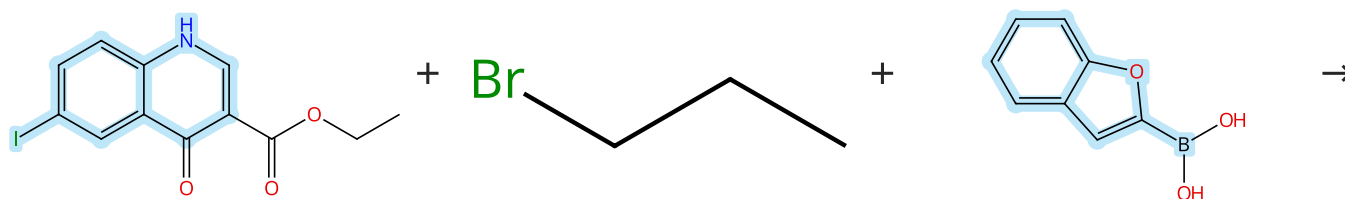
Journal of Medicinal Chemistry (2014), 57(9), 3687-3706.

1.1 **Reagents:** Cesium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: 1,4-Dioxane, Water; 10 h, 80 °C

Experimental Protocols

Scheme 397 (1 Reaction)

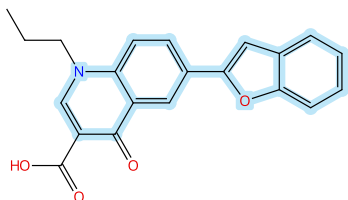
Steps: 1 Yield: 52%



Suppliers (44)

Suppliers (66)

Suppliers (92)



Suppliers (2)

31-031-CAS-4081268

Steps: 1 Yield: 52%

C-6 aryl substituted 4-quinolone-3-carboxylic acids as inhibitors of hepatitis C virus

By: Chen, Yue-Lei; et al

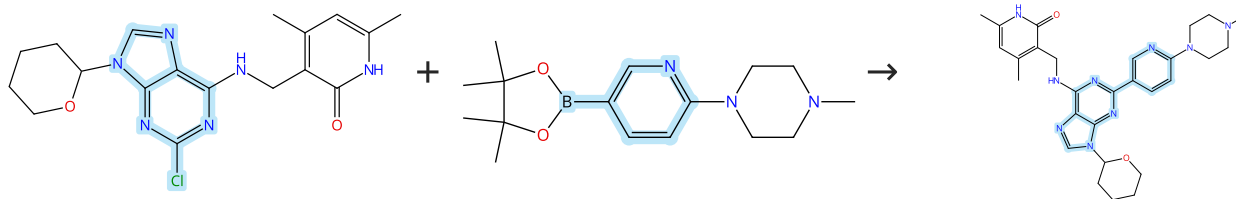
Bioorganic & Medicinal Chemistry (2012), 20(15), 4790-4800.

- 1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 30 min, 100 °C; 100 °C → rt
- 1.2 rt; 10000 s, 110 °C
- 1.3 **Reagents:** Potassium carbonate
Catalysts: Palladium chloride
Solvents: Dimethylformamide; 35 min, 130 °C
- 1.4 **Reagents:** Potassium carbonate
Solvents: Water; 35 min, 130 °C
- 1.5 **Reagents:** Hydrochloric acid
Solvents: Water; pH 1

Experimental Protocols

Scheme 398 (1 Reaction)

Steps: 1 Yield: 52%



Supplier (1)

Suppliers (88)

Supplier (1)

31-179-CAS-20725057

Steps: 1 Yield: 52%

Discovery of 3-(((9H-purin-6-yl)amino)methyl)-4,6-dimethylpyridin-2(1H)-one derivatives as novel tubulin polymerization inhibitors for treatment of cancer

By: Zhang, Qiangsheng; et al

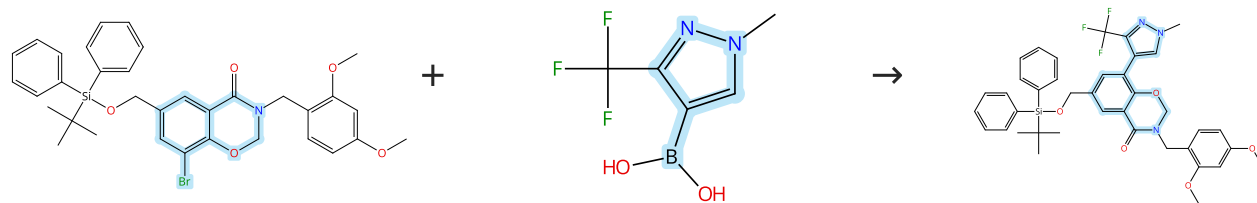
European Journal of Medicinal Chemistry (2019), 184, 111728.

- 1.1 **Reagents:** Sodium carbonate
Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium
Solvents: 1,4-Dioxane, Water; 4 h, 100 °C

Experimental Protocols

Scheme 399 (1 Reaction)

Steps: 1 Yield: 52%



Suppliers (61)

31-614-CAS-38624754

Steps: 1 Yield: 52%

Structure-Based Discovery of Potent Orally Bioavailable Benzoxazepinone-Based WD Repeat Domain 5 Inhibitors

By: Teuscher, Kevin B.; et al

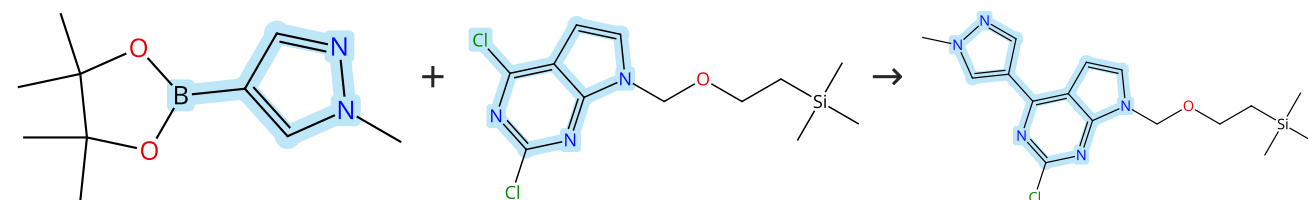
Journal of Medicinal Chemistry (2023), 66(24), 16783-16806.

- 1.1 **Reagents:** Potassium carbonate
Catalysts: Dichloro[1,1'-bis(diphenylphosphino)ferrocene] palladium(II) dichloromethane adduct
Solvents: 1,4-Dioxane, Water; 1 - 14 h, 80 °C

Experimental Protocols

Scheme 400 (1 Reaction)

Steps: 1 Yield: 52%



Suppliers (126)

Suppliers (46)

31-614-CAS-34771361

Steps: 1 Yield: 52%

Design, synthesis and biological evaluation of a new class of 7 H-pyrrolo[2,3-d]pyrimidine derivatives as Mps1 inhibitors for the treatment of breast cancer

By: Li, Xinyue; et al

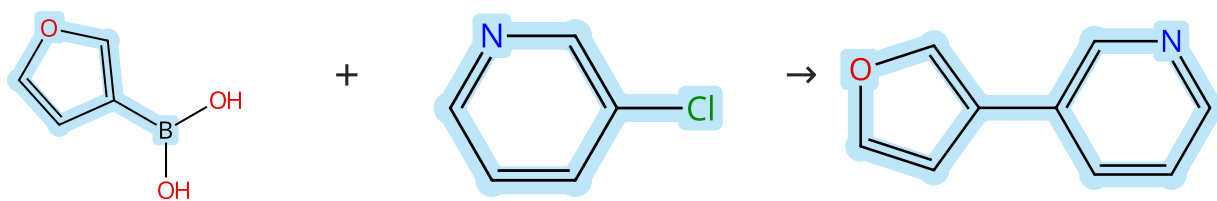
European Journal of Medicinal Chemistry (2023), 245(Part_1), 114887.

- 1.1 **Reagents:** Potassium carbonate
Catalysts: Dichloro[1,1'-bis(diphenylphosphino)ferrocene] palladium(II) dichloromethane adduct
Solvents: 1,4-Dioxane, Water; rt → 70 °C; 8 h, 70 °C

Experimental Protocols

Scheme 401 (1 Reaction)

Steps: 1 Yield: 51%



Suppliers (103)

Suppliers (78)

Suppliers (13)

31-179-CAS-21883968

Steps: 1 Yield: 51%

Ni(COD)(DQ): An Air-Stable 18-Electron Nickel(0)-Olefin Precatalyst

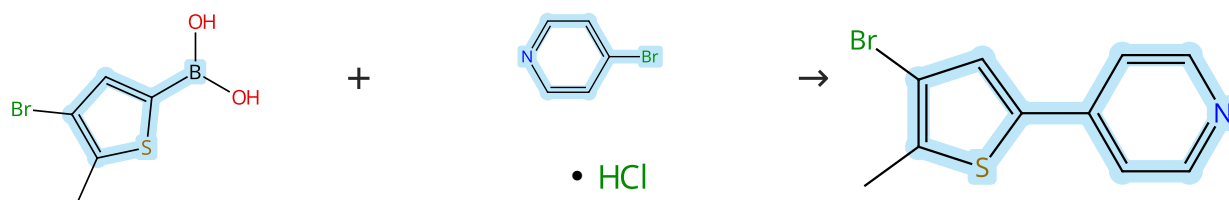
By: Tran, Van T.; et al

Angewandte Chemie, International Edition (2020), 59(19), 7409-7413.

- 1.1 **Reagents:** Tripotassium phosphate
Catalysts: 1,1-Bis(diphenylphosphino)ferrocene, [(1,2,5,6-η)-1,5-Cyclooctadiene][(2,3,5,6-η)-2,3,5,6-tetramethyl-2,5-cyclohexadiene-1,4-dione]nickel
Solvents: 1,4-Dioxane; 8 h, 80 °C; 80 °C → rt
- 1.2 **Solvents:** Water; rt

Scheme 402 (1 Reaction)

Steps: 1 Yield: 51%



Suppliers (65)

Suppliers (101)

Suppliers (13)

31-614-CAS-34476239

Steps: 1 Yield: 51%

Development of High-Performance Pyrimidine Nucleoside and Oligonucleotide Diarylethene Photoswitches

By: Kolmar, Theresa; et al

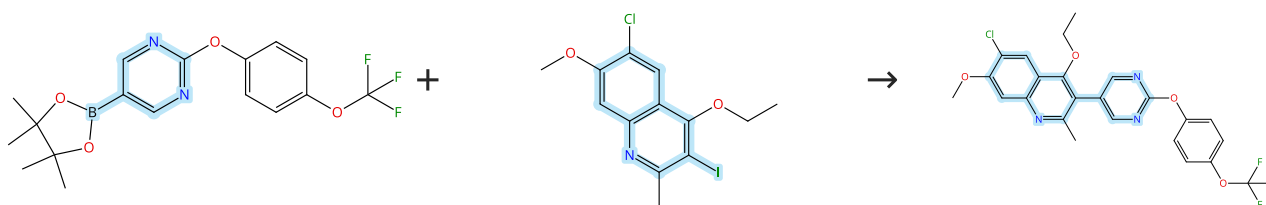
Angewandte Chemie, International Edition (2021), 60(15), 8164-8173.

1.1 **Reagents:** Sodium carbonate
Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium
Solvents: Tetrahydrofuran, Water; overnight, 80 °C

Experimental Protocols

Scheme 403 (1 Reaction)

Steps: 1 Yield: 51%



Suppliers (19)

31-179-CAS-7296433

Steps: 1 Yield: 51%

Discovery, Synthesis, and Optimization of Antimalarial 4(1H)-Quinolone-3-Diarylethers

By: Nilsen, Aaron; et al

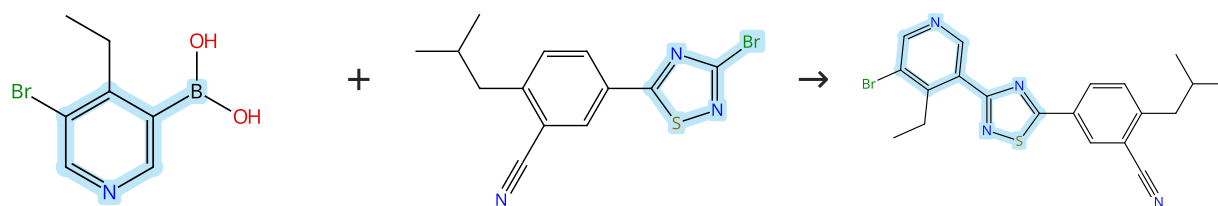
Journal of Medicinal Chemistry (2014), 57(9), 3818-3834.

1.1 **Reagents:** Potassium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Dimethylformamide, Water; 18 h, 85 °C

Experimental Protocols

Scheme 404 (1 Reaction)

Steps: 1 Yield: 51%



Suppliers (29)

Suppliers (2)

Supplier (1)

31-179-CAS-486933

Steps: 1 Yield: 51%

Discovery of novel 1,2,4-thiadiazole derivatives as potent, orally active agonists of sphingosine 1-phosphate receptor subtype 1 (S1P₁)

By: Ren, Feng; et al

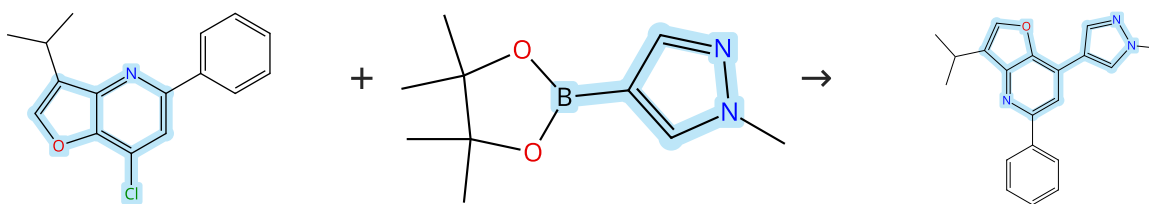
Journal of Medicinal Chemistry (2012), 55(9), 4286-4296.

1.1 **Reagents:** Tripotassium phosphate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Dimethylformamide, Water; 20 min, 110 °C

Experimental Protocols

Scheme 405 (1 Reaction)

Steps: 1 Yield: 51%



Suppliers (126)

31-179-CAS-19761229

Steps: 1 Yield: 51%

Furo[3,2-b]pyridine: A Privileged Scaffold for Highly Selective Kinase Inhibitors and Effective Modulators of the Hedgehog Pathway

By: Nemec, Vaclav; et al

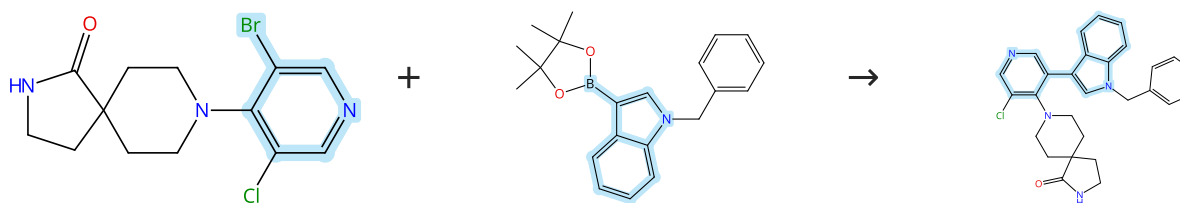
Angewandte Chemie, International Edition (2019), 58(4), 1062-1066.

1.1 **Reagents:** Triethylamine
Catalysts: Palladium diacetate, 2-Dicyclohexylphosphino-2',6'-dimethoxybiphenyl
Solvents: 1-Butanol, Water; 25 min, 100 °C

Experimental Protocols

Scheme 406 (1 Reaction)

Steps: 1 Yield: 51%



Suppliers (5)

Suppliers (5)

Supplier (1)

31-179-CAS-23224004

Steps: 1 Yield: 51%

Potent and orally bioavailable CDK8 inhibitors: Design, synthesis, structure-activity relationship analysis and biological evaluation

By: Yu, Mingfeng; et al

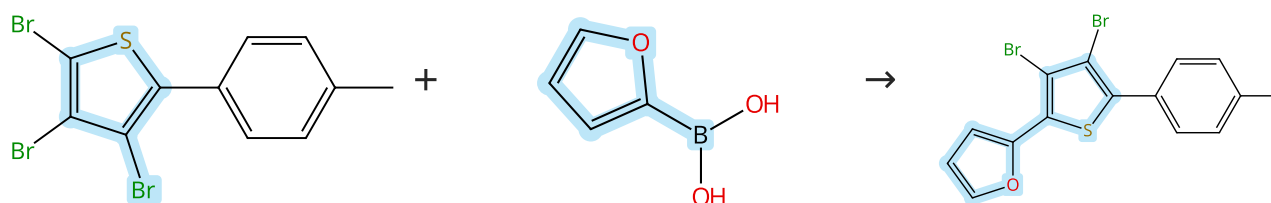
European Journal of Medicinal Chemistry (2021), 214, 113248.

1.1 **Reagents:** Sodium carbonate
Catalysts: Dichloro[1,1'-bis(diphenylphosphino)ferrocene] palladium(II) dichloromethane adduct
Solvents: Acetonitrile, Water; 1 h, 120 °C

Experimental Protocols

Scheme 407 (1 Reaction)

Steps: 1 Yield: 51%



Suppliers (93)

31-179-CAS-13202917

Steps: 1 Yield: 51%

Selective Mono- to Perarylations of Tetrabromothiophene by a Cyclobutene-1,2-diylbisimidazolium Preligand

By: Rahimi, Alireza; et al

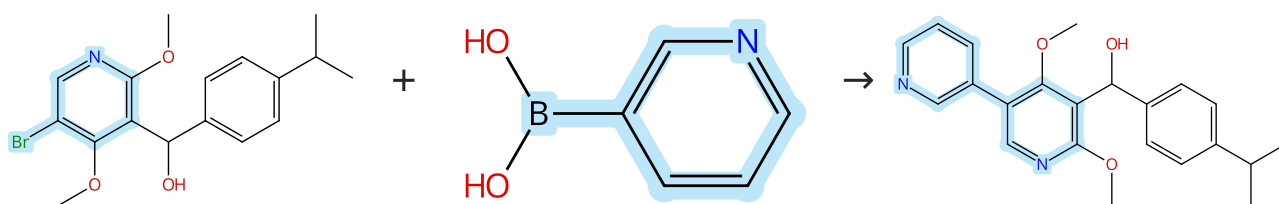
Journal of Organic Chemistry (2011), 76(18), 7316-7325.

1.1 **Reagents:** Tripotassium phosphate
Catalysts: Palladium diacetate, 1*H*-Imidazolium, 3,3'-[3,4-bis(dichloromethylene)-1-cyclobutene-1,2-diyl]bis[1-methyl-, tetrafluoroborate(1-)] (1:2)
Solvents: Toluene, 1,4-Dioxane; 16 h, 80 °C

Experimental Protocols

Scheme 408 (1 Reaction)

Steps: 1 Yield: 51%



Suppliers (115)

31-179-CAS-19607231

Steps: 1 Yield: 51%

Discovery of 2,4-dimethoxypyridines as novel autophagy inhibitors

By: Robke, Lucas; et al

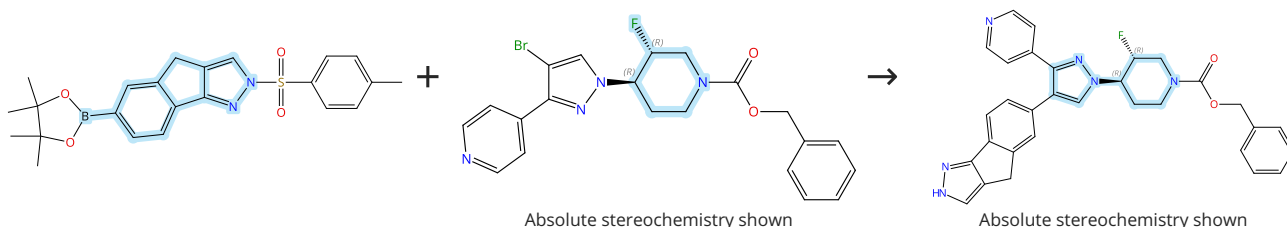
Tetrahedron (2018), 74(35), 4531-4537.

1.1 **Reagents:** Sodium carbonate
Catalysts: 1,1-Bis(diphenylphosphino)ferrocene, Tetrakis(triphenylphosphine)palladium
Solvents: Ethanol, Toluene, Water; 16 h, 110 °C

Experimental Protocols

Scheme 409 (1 Reaction)

Steps: 1 Yield: 51%



31-179-CAS-255184

Steps: 1 Yield: 51%

Non-oxime pyrazole based inhibitors of B-Raf kinase

By: Newhouse, Bradley J.; et al

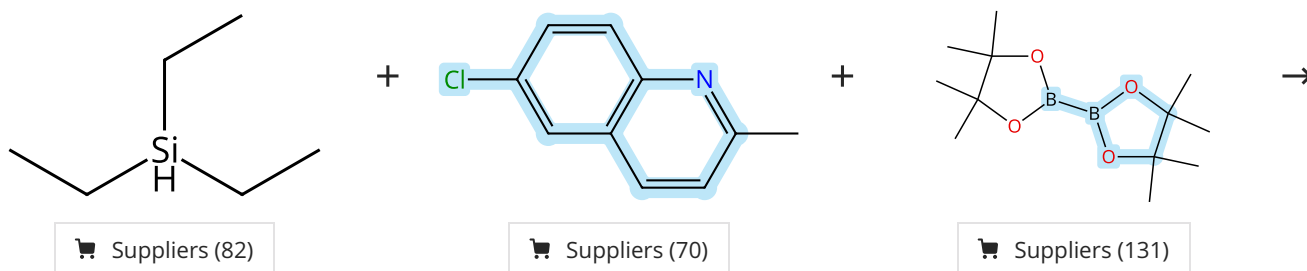
Bioorganic & Medicinal Chemistry Letters (2011), 21(11), 3488-3492.

1.1 **Reagents:** Potassium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: 1,2-Dimethoxyethane, Water; 80 °C

1.2 **Reagents:** Sodium hydroxide
Solvents: Methanol, Water; 60 °C

Scheme 410 (1 Reaction)

Steps: 1 Yield: 51%



Suppliers (82)

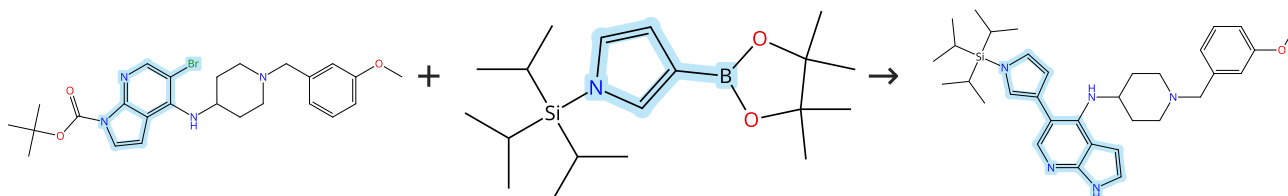
Suppliers (70)

Suppliers (131)

31-142-CAS-18793448 Steps: 1 Yield: 51% 1.1 Reagents: 3,3-Dimethyl-1-butene Catalysts: Bis[(1,2,5,6-η)-1,5-cyclooctadiene]di-μ-methoxydiiridium Solvents: Tetrahydrofuran; 2 h, 70 °C 1.2 Catalysts: 5,6-Dimethyl-1,10-phenanthroline; 30 min, 25 °C 1.3 5 h, 70 °C Experimental Protocols	Iridium-Catalyzed Sequential Silylation and Borylation of Heteroarenes Based on Regioselective C-H Bond Activation By: Murai, Masahito; et al Angewandte Chemie, International Edition (2018), 57(20), 5843-5847.
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Scheme 411 (1 Reaction)

Steps: 1 Yield: 51%

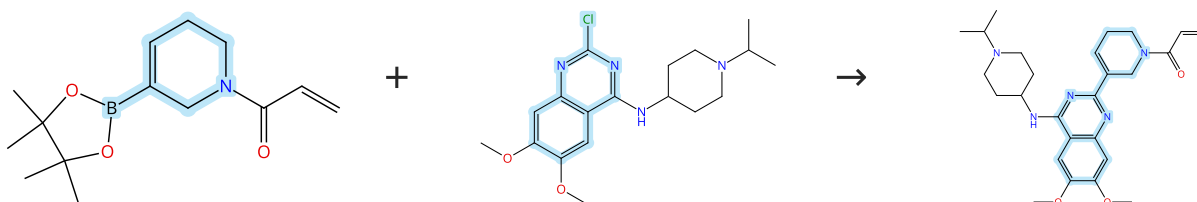


Suppliers (62)

31-614-CAS-39559365 Steps: 1 Yield: 51% 1.1 Reagents: Potassium carbonate Catalysts: (<i>SP</i>-4-1)-Bis[4-[bis(1,1-dimethylethyl)phosphino-κ^P]-<i>N,N</i>-dimethylbenzenamine]dichloropalladium Solvents: Isobutanol, Water; 10 h, 60 °C 1.2 Reagents: Trifluoroacetic acid Solvents: Dichloromethane; 1 h, rt 1.3 Reagents: Sodium hydroxide Solvents: Water; pH 9 - 10, rt Experimental Protocols	Design, synthesis and evaluation of C-5 substituted pyrrolopyridine derivatives as potent Janus Kinase 1 inhibitors with excellent selectivity By: Chen, Limei; et al European Journal of Medicinal Chemistry (2024), 267, 116210.
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Scheme 412 (1 Reaction)

Steps: 1 Yield: 51%



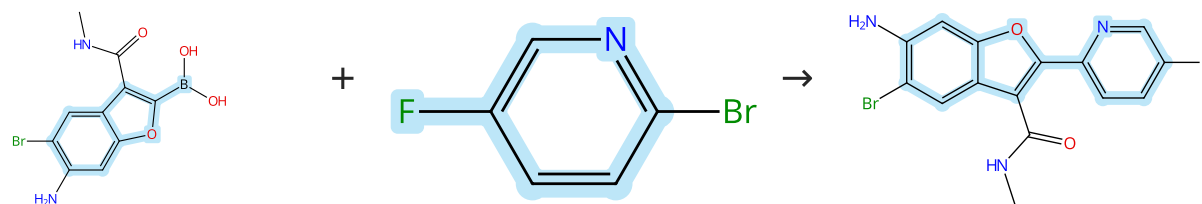
Suppliers (4)

Supplier (1)

31-614-CAS-38047159 Steps: 1 Yield: 51% 1.1 Reagents: Sodium carbonate Catalysts: Dichloro[1,1'-bis(diphenylphosphino)ferrocene]palladium(II) dichloromethane adduct Solvents: 1,4-Dioxane, Water; 4 h, 100 °C Experimental Protocols	Discovery of novel G9a/GLP covalent inhibitors for the treatment of triple-negative breast cancer By: Zhang, Qiangsheng; et al European Journal of Medicinal Chemistry (2023), 261, 115841.
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Scheme 413 (1 Reaction)

Steps: 1 Yield: 51%



Suppliers (2)

Suppliers (92)

31-179-CAS-4826297

Steps: 1 Yield: 51%

Facile functionalization at the C2 position of a highly substituted benzofuran

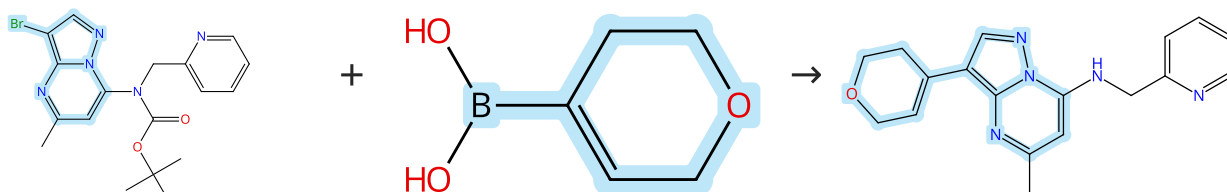
By: He, Shuwen; et al

Tetrahedron Letters (2014), 55(14), 2212-2216.

- 1.1 **Reagents:** Tripotassium phosphate
Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium
Solvents: 1,4-Dioxane, Water; 10 h, 50 °C

Scheme 414 (1 Reaction)

Steps: 1 Yield: 51%



Suppliers (60)

Supplier (1)

31-177-CAS-11783576

Steps: 1 Yield: 51%

Aminopyrazolo[1,5-a]pyrimidines as potential inhibitors of Mycobacterium tuberculosis: Structure activity relationships and ADME characterization

By: Soares de Melo, Candice; et al

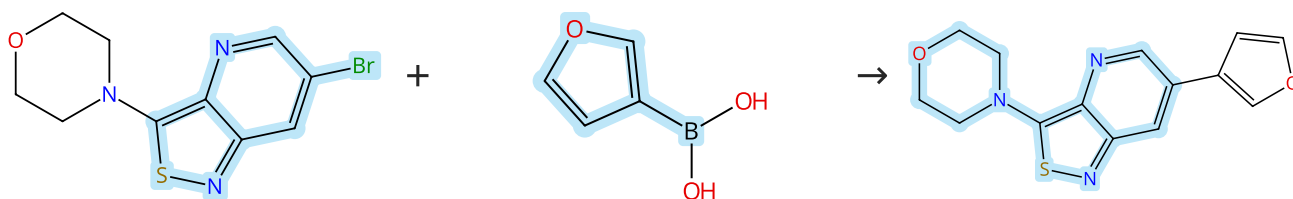
Bioorganic & Medicinal Chemistry (2015), 23(22), 7240-7250.

- 1.1 **Reagents:** Cesium carbonate
Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium
Solvents: 1,4-Dioxane, Water; overnight, 80 °C
 1.2 **Reagents:** Trifluoroacetic acid
Solvents: Dichloromethane; 4 h, rt

Experimental Protocols

Scheme 415 (1 Reaction)

Steps: 1 Yield: 51%



Suppliers (103)

Suppliers (2)

31-179-CAS-11770270

Steps: 1 Yield: 51%

Selective Inhibitors of Cyclin G Associated Kinase (GAK) as Anti-Hepatitis C Agents

By: Kovackova, Sona; et al

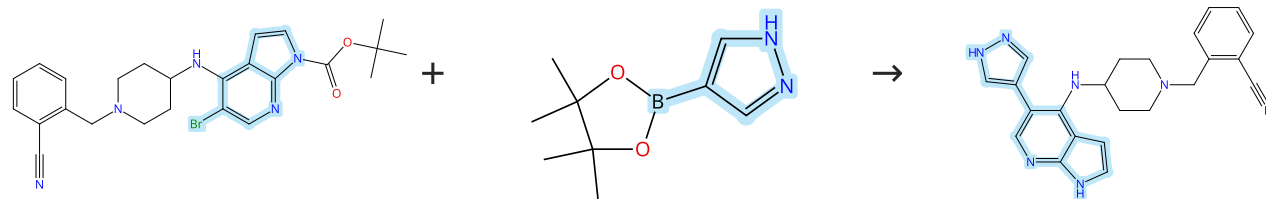
Journal of Medicinal Chemistry (2015), 58(8), 3393-3410.

- 1.1 **Reagents:** Potassium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: 1,2-Dimethoxyethane, Water; 80 °C

Experimental Protocols

Scheme 416 (1 Reaction)

Steps: 1 Yield: 51%



Suppliers (108)

31-614-CAS-39559326

Steps: 1 Yield: 51%

- 1.1 **Reagents:** Potassium carbonate
Catalysts: (*SP*-4-1)-Bis[4-[bis(1,1-dimethylethyl)phosphino- κ P]-*N,N*-dimethylbenzenamine]dichloropalladium
Solvents: Isobutanol, Water; 10 h, 60 °C
- 1.2 **Reagents:** Trifluoroacetic acid
Solvents: Dichloromethane; 1 h, rt
- 1.3 **Reagents:** Sodium hydroxide
Solvents: Water; pH 9 - 10, rt

Experimental Protocols

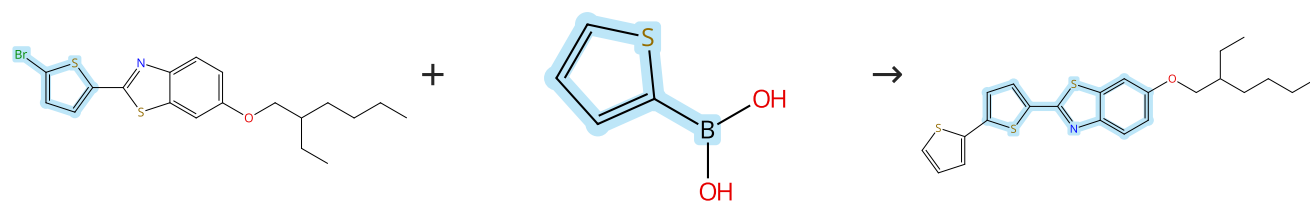
Design, synthesis and evaluation of C-5 substituted pyrrolopyridine derivatives as potent Janus Kinase 1 inhibitors with excellent selectivity

By: Chen, Limei; et al

European Journal of Medicinal Chemistry (2024), 267, 116210.

Scheme 417 (1 Reaction)

Steps: 1 Yield: 51%



Suppliers (103)

31-179-CAS-11960651

Steps: 1 Yield: 51%

- 1.1 **Reagents:** Cesium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Tetrahydrofuran, Water; 6 h, 110 °C

Experimental Protocols

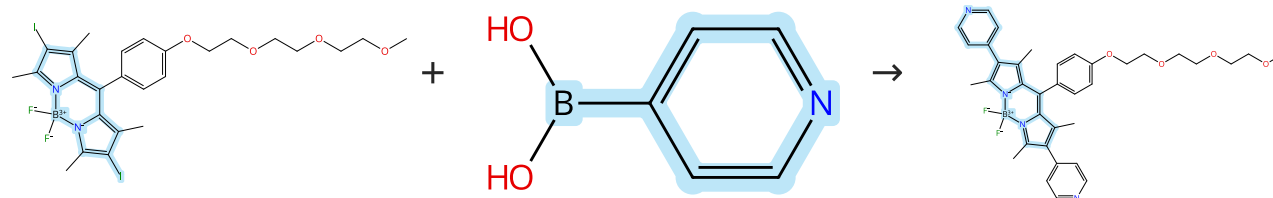
Synthesis of highly soluble fluorescent π -extended 2-(2-thienyl)benzothiazole derivatives via oxidative cyclization of 2-thienylthioanilide as the key step

By: Ji, Sanghoon; et al

Tetrahedron Letters (2013), 54(52), 7103-7106.

Scheme 418 (1 Reaction)

Steps: 1 Yield: 51%



Supplier (1)

Suppliers (111)

31-614-CAS-35628435

Steps: 1 Yield: 51%

- 1.1 **Reagents:** Cesium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: 1,4-Dioxane, Water; 2 h, 110 °C

Experimental Protocols

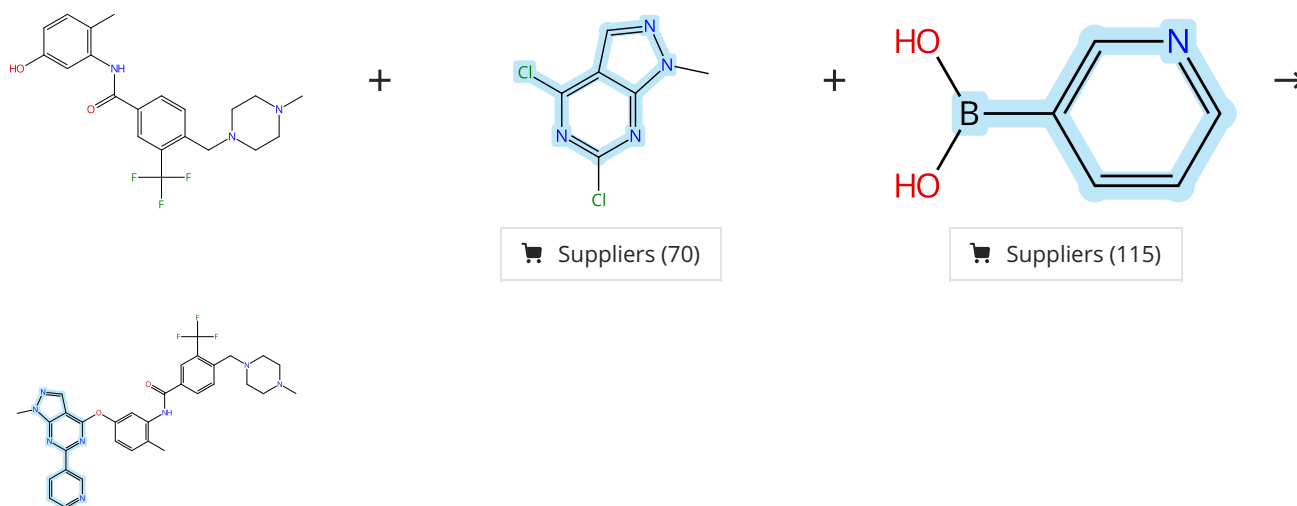
Glycosylated BODIPY- Incorporated Pt(II) Metallacycles for Targeted and Synergistic Chemo-Photodynamic Therapy

By: Duran-Sampedro, Gonzalo; et al

Journal of Medicinal Chemistry (2023), 66(5), 3448-3459.

Scheme 419 (1 Reaction)

Steps: 1 Yield: 51%



31-170-CAS-23362447

Steps: 1 Yield: 51%

- 1.1 Reagents: Potassium carbonate
Solvents: Dimethylformamide; 10 h, rt
- 1.2 Reagents: Sodium carbonate
Catalysts: Dichloro[1,1'-bis(diphenylphosphino)ferrocene] palladium(II) dichloromethane adduct
Solvents: 1,4-Dioxane, Water; 5 h, 110 °C

Experimental Protocols

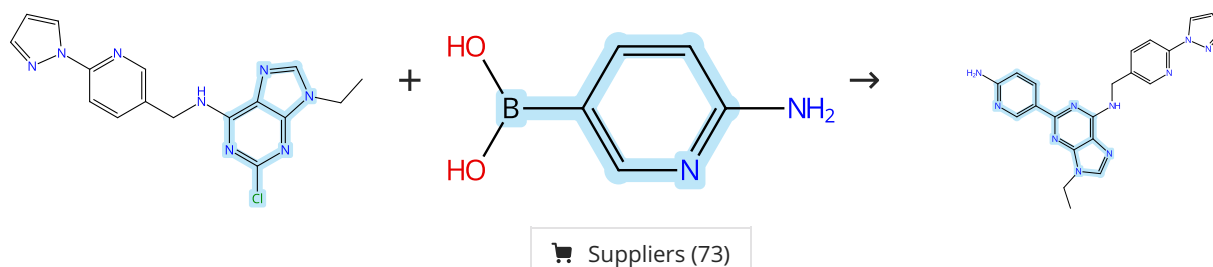
Discovery of 4-methyl-N-(4-((4-methylpiperazin-1-yl)methyl)-3-(trifluoromethyl)phenyl)-3-((6-(pyridin-3-yl)-1H-pyrazolo[3,4-d]pyrimidin-4-yl)-oxy)benzamide as a potent inhibitor of RET and its gatekeeper mutant

By: Li, Xiaoyang; et al

European Journal of Medicinal Chemistry (2020), 207, 112755.

Scheme 420 (1 Reaction)

Steps: 1 Yield: 51%



31-614-CAS-38825423

Steps: 1 Yield: 51%

- 1.1 Reagents: Potassium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: 1,4-Dioxane, Water; 12 h, rt → 100 °C

Experimental Protocols

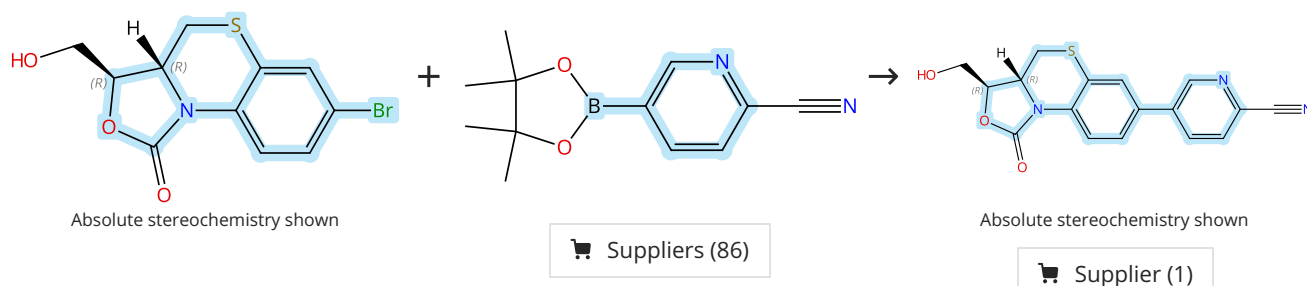
Identification of a novel potent CDK inhibitor degrading cyclin K with a superb activity to reverse trastuzumab-resistance in HER2-positive breast cancer in vivo

By: Kuchukulla, Ratnakar Reddy; et al

European Journal of Medicinal Chemistry (2024), 264, 116014.

Scheme 421 (1 Reaction)

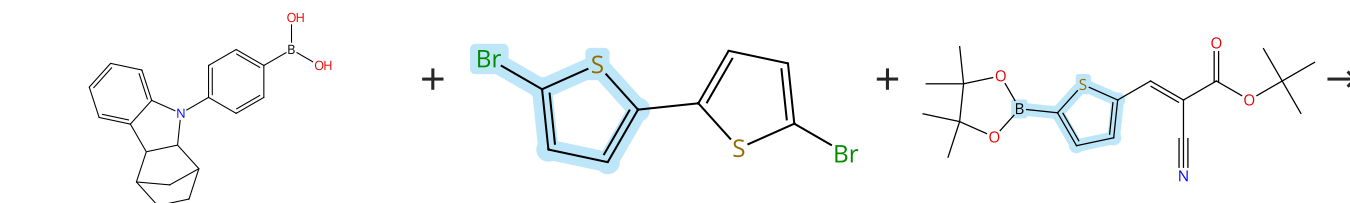
Steps: 1 Yield: 51%



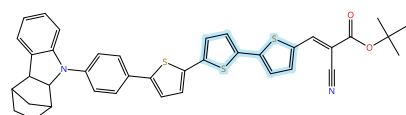
31-179-CAS-5022675	Steps: 1 Yield: 51%	Synthesis and structure-activity relationship studies of novel [6,6,5] tricyclic oxazolidinone derivatives as potential antibacterial agents
1.1 Reagents: Cesium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: 1,4-Dioxane, Water; overnight, 80 °C		By: Xue, Tao; et al
Experimental Protocols		Bioorganic & Medicinal Chemistry Letters (2015), 25(10), 2203-2210.

Scheme 422 (1 Reaction)

Steps: 1 Yield: 51%



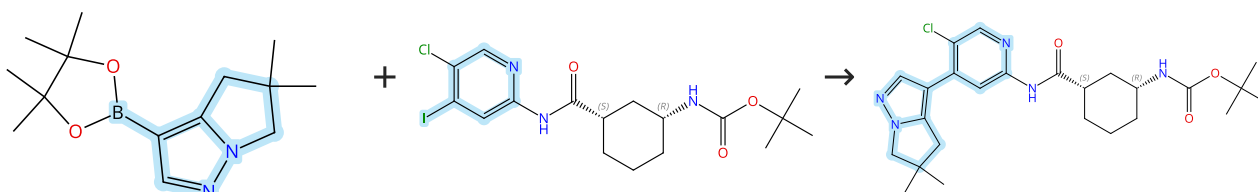
Suppliers (74)



31-179-CAS-14793247	Steps: 1 Yield: 51%	Elucidating the Structure-Property Relationships of Donor-π-Acceptor Dyes for Dye-Sensitized Solar Cells (DSSCs) through Rapid Library Synthesis by a One-Pot Procedure
1.1 Reagents: Sodium carbonate Catalysts: Tris(dibenzylideneacetone)dipalladium, 1,1'-(9,9-Dimethyl-9H-xanthene-4,5-diyl)bis[1,1-diphenylphosphine] Solvents: Toluene, Water; 0.5 - 3 h, reflux; reflux $\rightarrow$ rt		By: Fuse, Shinichiro; et al
1.2 Reagents: Sodium carbonate Catalysts: Tris(dibenzylideneacetone)dipalladium, Tri- <i>tert</i> -butylphosphonium tetrafluoroborate; 1 - 24 h, rt		Chemistry - A European Journal (2014), 20(34), 10685-10694.

Scheme 423 (1 Reaction)

Steps: 1 Yield: 51%



Suppliers (37)

Absolute stereochemistry shown

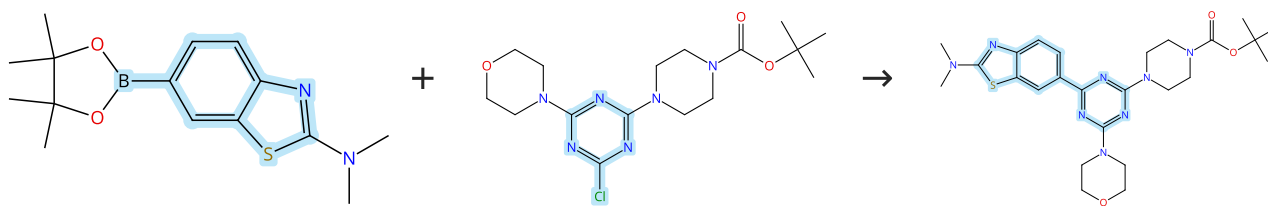
Suppliers (5)

Absolute stereochemistry shown

31-614-CAS-31188361	Steps: 1 Yield: 51%	Discovery of AZD4573, a Potent and Selective Inhibitor of C D K9 That Enables Short Duration of Target Engagement for the Treatment of Hematological Malignancies
1.1 Reagents: Tripotassium phosphate Catalysts: (<i>SP</i> -4-4)-[2'-(Amino- κ M)[1,1'-biphenyl]-2-yl- κ C]chloro[dicyclohexyl[2',4',6'-tris(1-methylethyl)[1,1'-biphenyl]-2-yl]phosphine]palladium Solvents: 1,4-Dioxane, Water; 5 h, 50 °C		By: Barlaam, Bernard; et al
Experimental Protocols		Journal of Medicinal Chemistry (2020), 63(24), 15564-15590.

Scheme 424 (1 Reaction)

Steps: 1 Yield: 51%



Suppliers (2)

Suppliers (16)

31-614-CAS-40064334

Steps: 1 Yield: 51%

Discovery of Novel Phosphoinositide-3-Kinase α Inhibitors with High Selectivity, Excellent Bioavailability, and Long-Acting Efficacy for Gastric Cancer

By: Hou, Yi; et al

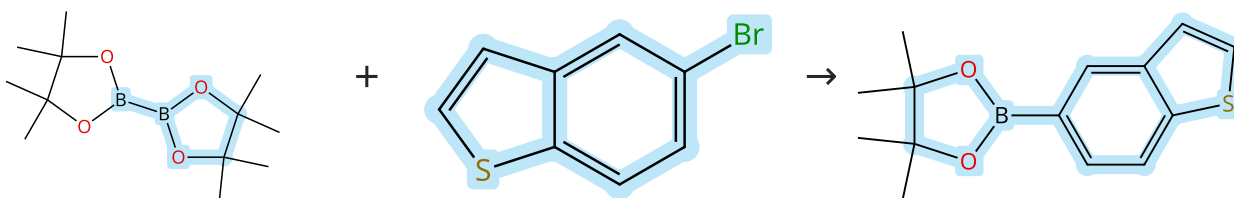
Journal of Medicinal Chemistry (2022), 65(14), 9873-9892.

1.1 **Reagents:** Sodium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: 1,2-Dimethoxyethane, Water; 24 h, rt $\rightarrow$ 92 $^{\circ}$ C

Experimental Protocols

Scheme 425 (1 Reaction)

Steps: 1 Yield: 50%



Suppliers (131)

Suppliers (80)

Suppliers (75)

31-614-CAS-29178273

Steps: 1 Yield: 50%

Photoinduced Miyaura Borylation by a Rare-Earth-Metal Photoreductant: the Hexachloroacetate(III) Anion

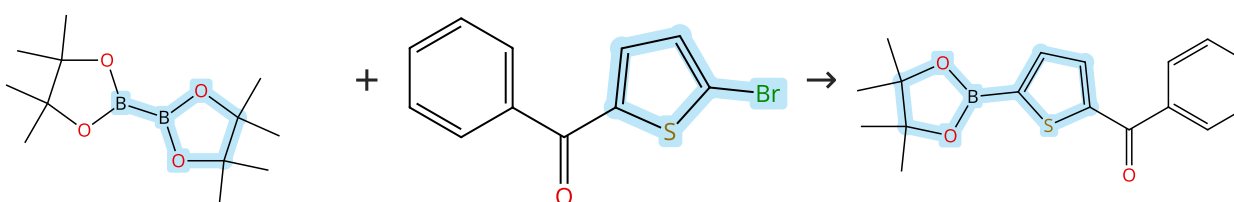
By: Qiao, Yusen; et al

Angewandte Chemie, International Edition (2018), 57(34), 10999-11003.

1.1 **Reagents:** Tetraethylammonium chloride
Catalysts: Cerium trichloride
Solvents: Acetonitrile; 24 h, rt

Scheme 426 (1 Reaction)

Steps: 1 Yield: 50%



Suppliers (131)

Suppliers (62)

Suppliers (7)

31-142-CAS-871807

Steps: 1 Yield: 50%

Phenacyl-Thiophene and Quinone Semiconductors Designed for Solution Processability and Air-Stability in High Mobility n-Channel Field-Effect Transistors

By: Letizia, Joseph A.; et al

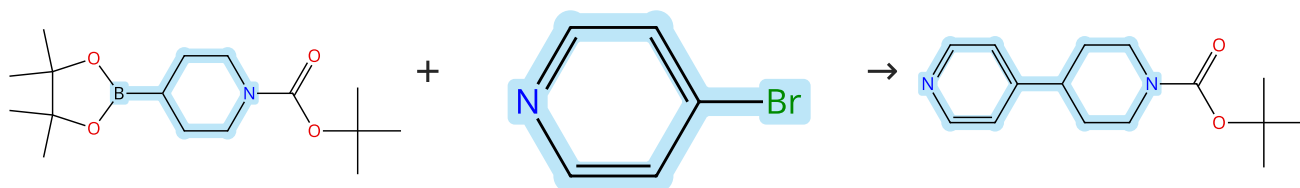
Chemistry - A European Journal (2010), 16(6), 1911-1928, S1911/1-S1911/9.

1.1 **Reagents:** Potassium acetate
Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium
Solvents: Toluene; rt; rt $\rightarrow$ 160 $^{\circ}$ C; 20 min, 160 $^{\circ}$ C

Experimental Protocols

Scheme 427 (1 Reaction)

Steps: 1 Yield: 50%



Suppliers (72)

Suppliers (39)

Suppliers (42)

31-614-CAS-33630967

Steps: 1 Yield: 50%

Ni/Photoredox-Catalyzed C(sp²)-C(sp³) Cross-Coupling of Alkyl Pinacolboronates and (Hetero)Aryl Bromides

By: Cauley, Anthony N.; et al

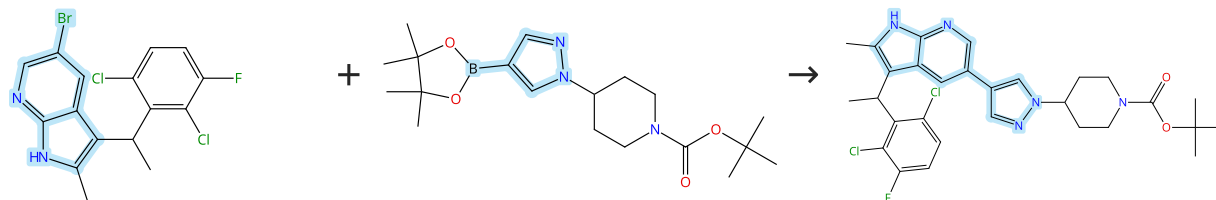
Organic Letters (2022), 24(31), 5663-5668.

1.1 **Reagents:** 2-*tert*-Butyl-1,1,3,3-tetramethylguanidine
Catalysts: Quinoline, Tris[2-[4-(1,1-dimethylethyl)-2-pyridinyl- κ N]-3,5-difluorophenyl- κ C]iridium, Nickel, [4,4'-bis(1,1-dimethylethyl)-2,2'-bipyridine- κ N¹, κ N^{1'}]dibromo-, (7-4)-
Solvents: Tetrahydrofuran; 24 h, rt

Experimental Protocols

Scheme 428 (1 Reaction)

Steps: 1 Yield: 50%



Suppliers (92)

31-179-CAS-569703

Steps: 1 Yield: 50%

Design, synthesis and biological evaluation of 1 H-pyrrolo[2,3-b]pyridine and 1H-pyrazolo[3,4-b]pyridine derivatives as c-Met inhibitors

By: Liu, Na; et al

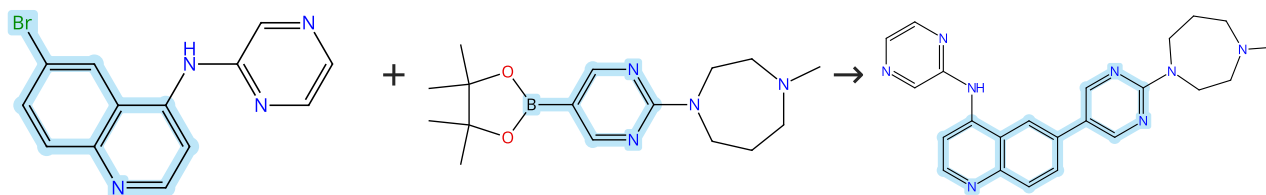
Bioorganic Chemistry (2016), 65, 146-158.

1.1 **Reagents:** Potassium carbonate
Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium
Solvents: 1,4-Dioxane, Water; 12 h, 100 °C

Experimental Protocols

Scheme 429 (1 Reaction)

Steps: 1 Yield: 50%



Suppliers (17)

31-614-CAS-23925086

Steps: 1 Yield: 50%

Scaffold and Parasite Hopping: Discovery of New Protozoal Proliferation Inhibitors

By: Singh, Baljinder; et al

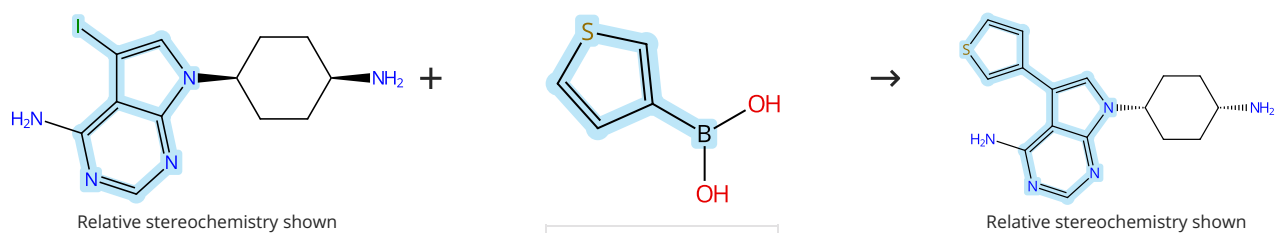
ACS Medicinal Chemistry Letters (2020), 11(3), 249-257.

1.1 **Reagents:** Cesium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Ethanol, Water; 12 h, 85 °C

Experimental Protocols

Scheme 430 (1 Reaction)

Steps: 1 Yield: 50%



Suppliers (101)

31-614-CAS-33275919

Steps: 1 Yield: 50%

- 1.1 **Reagents:** Potassium carbonate
Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium
Solvents: Ethanol, 1,4-Dioxane, Water; 2 h, 80 °C

- 1.2 **Reagents:** Water

Experimental Protocols

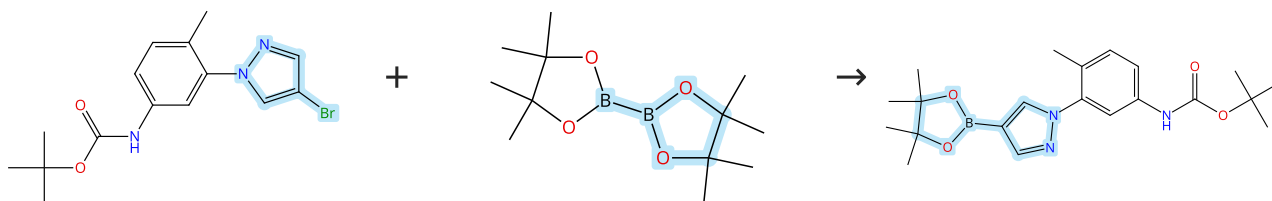
Discovery of Potent and Selective Receptor-Interacting Serine/Threonine Protein Kinase 2 (RIPK2) Inhibitors for the Treatment of Inflammatory Bowel Diseases (IBDs)

By: Yuan, Xue; et al

Journal of Medicinal Chemistry (2022), 65(13), 9312-9327.

Scheme 431 (1 Reaction)

Steps: 1 Yield: 50%



Suppliers (131)

31-614-CAS-30480850

Steps: 1 Yield: 50%

- 1.1 **Reagents:** Potassium acetate
Catalysts: Dichlorobis(triphenylphosphine)palladium
Solvents: 1,4-Dioxane; 15 - 18 h, rt → reflux

Experimental Protocols

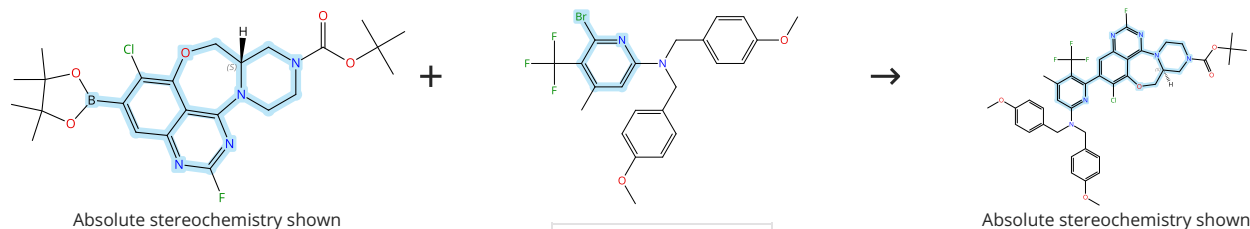
Design, synthesis, and biological activity of 4-(imidazo[1,2-b]pyridazin-3-yl)-1H-pyrazol-1-yl-phenylbenzamide derivatives as BCR-ABL kinase inhibitors

By: Hu, Liming; et al

Bioorganic & Medicinal Chemistry Letters (2016), 26(23), 5830-5835.

Scheme 432 (1 Reaction)

Steps: 1 Yield: 50%



Suppliers (45)

31-614-CAS-38626922

Steps: 1 Yield: 50%

- 1.1 **Reagents:** Sodium carbonate
Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium
Solvents: Water; rt; 2 h, 90 °C

Experimental Protocols

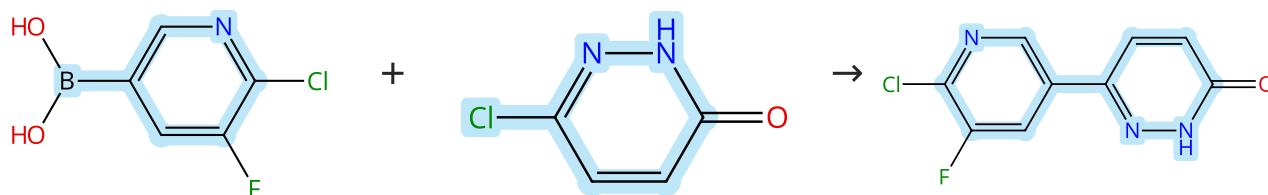
Structure-based design and evaluation of reversible KRAS G13D inhibitors

By: Nilewski, Christian; et al

ACS Medicinal Chemistry Letters (2024), 15(1), 21-28.

Scheme 433 (1 Reaction)

Steps: 1 Yield: 50%



Suppliers (74)

Suppliers (96)

31-614-CAS-39487070

Steps: 1 Yield: 50%

Discovery of Novel Potent and Fast BTK PROTACs for the Treatment of Osteoclasts-Related Inflammatory Diseases

1.1 **Reagents:** Potassium carbonate
Catalysts: Dichloro[1,1'-bis(diphenylphosphino)ferrocene] palladium(II) dichloromethane adduct
Solvents: 1,4-Dioxane; 3 h, rt

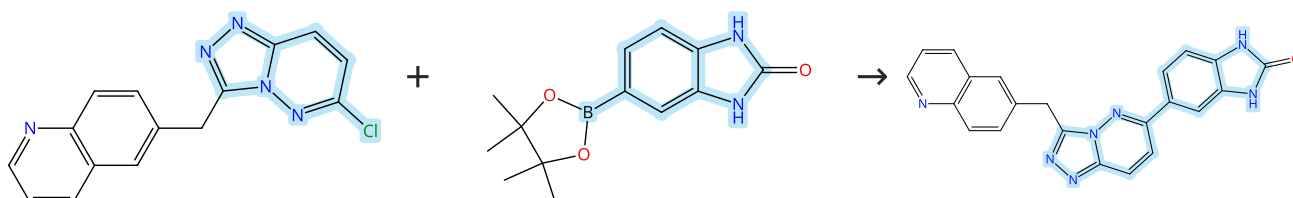
By: Huang, Junli; et al

Journal of Medicinal Chemistry (2024), 67(4), 2438-2465.

Experimental Protocols

Scheme 434 (1 Reaction)

Steps: 1 Yield: 50%



Suppliers (4)

Suppliers (72)

31-614-CAS-38906002

Steps: 1 Yield: 50%

Discovery of potent and selective c-Met inhibitors for MET-amplified hepatocellular carcinoma treatment

1.1 **Reagents:** Potassium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: 1,4-Dioxane, Water; 8 h, 90 °C

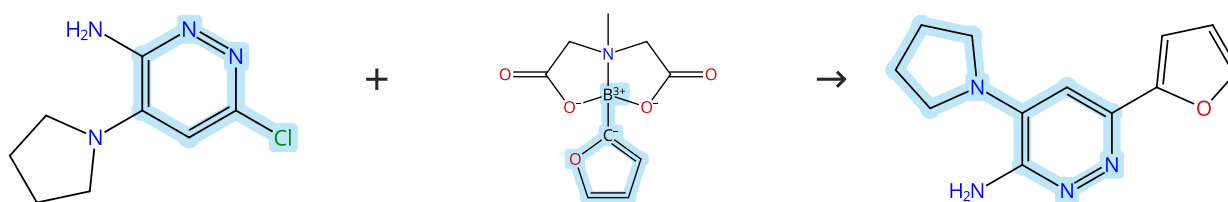
By: Min, Wenjian; et al

European Journal of Medicinal Chemistry (2024), 264, 116025.

Experimental Protocols

Scheme 435 (1 Reaction)

Steps: 1 Yield: 50%



Suppliers (2)

Suppliers (34)

31-614-CAS-37985267

Steps: 1 Yield: 50%

GNE-235: A Lead Compound Selective for the Second Bromodomain of PBRM1

1.1 **Reagents:** Lithium hydroxide, Tripotassium phosphate
Catalysts: Palladium diacetate, 2-Dicyclohexylphosphino-2',6'-dimethoxybiphenyl
Solvents: 1,4-Dioxane, Water; 18 h, rt → 60 °C

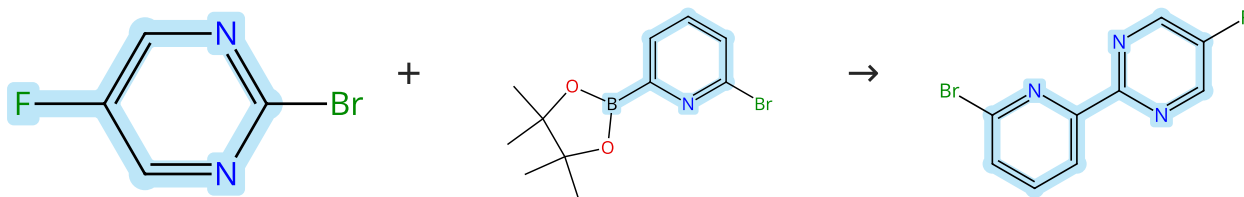
By: Cochran, Andrea G.; et al

Journal of Medicinal Chemistry (2023), 66(18), 13116-13134.

Experimental Protocols

Scheme 436 (1 Reaction)

Steps: 1 Yield: 50%



Suppliers (68)

Suppliers (79)

31-614-CAS-40870520

Steps: 1 Yield: 50%

Advanced Design, Synthesis, and Evaluation of Highly Selective Wee1 Inhibitors: Enhancing Pharmacokinetics and Antitumor Efficacy

By: Wang, Yong; et al

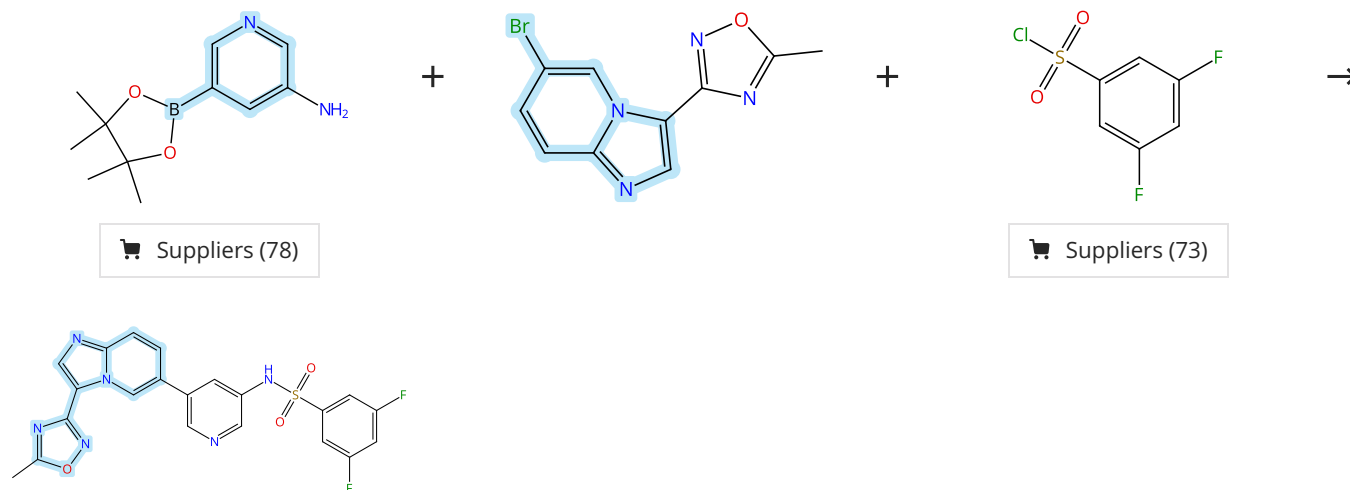
Journal of Medicinal Chemistry (2024), 67(12), 9927-9949.

1.1 **Reagents:** Cesium carbonate
Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium
Solvents: 1,4-Dioxane, Water; 15 h, 70 °C

Experimental Protocols

Scheme 437 (1 Reaction)

Steps: 1 Yield: 50%



Suppliers (78)

Suppliers (73)

Supplier (1)

31-179-CAS-14841716

Steps: 1 Yield: 50%

Design and Synthesis of Imidazopyridine Analogues as Inhibitors of Phosphoinositide 3-Kinase Signaling and Angiogenesis

By: Kim, Okseon; et al

Journal of Medicinal Chemistry (2011), 54(7), 2455-2466.

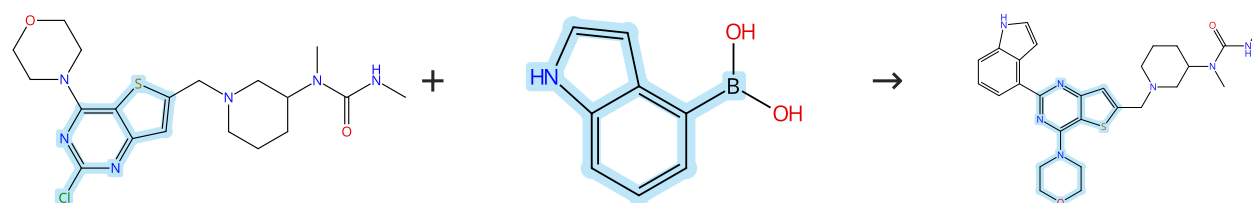
1.1 **Reagents:** Potassium carbonate
Catalysts: Dichloro[1,1'-bis(diphenylphosphino)ferrocene]palladium(II) dichloromethane adduct
Solvents: 1,4-Dioxane, Water; 12 h, 100 °C; 100 °C → rt

1.2 **Reagents:** Pyridine
Solvents: Dichloromethane; 24 h, rt

Experimental Protocols

Scheme 438 (1 Reaction)

Steps: 1 Yield: 50%



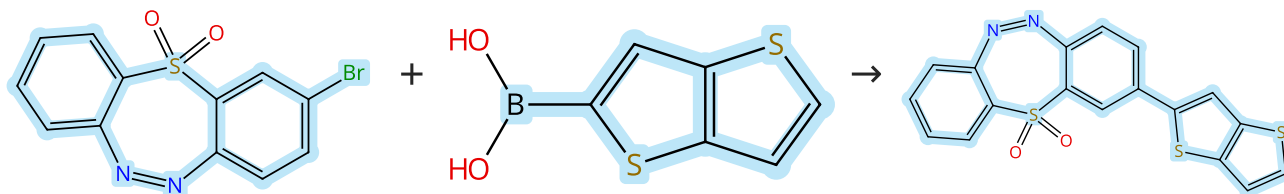
Suppliers (79)

Supplier (1)

31-179-CAS-22386575	Steps: 1 Yield: 50%	Free-Wilson Analysis of Comprehensive Data on Phosphoinositide-3-kinase (PI3K) Inhibitors Reveals Importance of N-Methylation for PI3Kδ Activity
1.1 Reagents: Tripotassium phosphate Catalysts: [Bis(bicyclo[2.2.1]hept-2-yl)phosphine]chloro[2-[(dimethylamino-κM)methyl]ferrocenyl-κC]palladium Solvents: Ethanol, Water; 0.5 h, rt → 140 °C		By: Barnes, Lydia; et al Journal of Medicinal Chemistry (2019), 62(22), 10402-10422.
Experimental Protocols		

Scheme 439 (1 Reaction)

Steps: 1 Yield: 50%

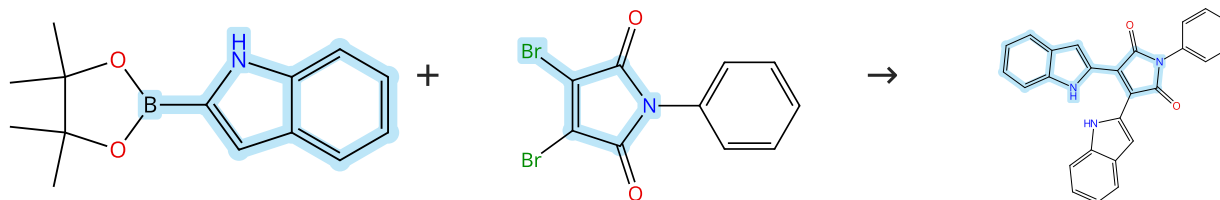


Suppliers (77)

31-614-CAS-44678477	Steps: 1 Yield: 50%	Light-intersecting Photoclick Reactions for Bioorthogonal Labeling on Single Cells: Dibenzo[b,f][1,4,5]thiadiazepine-11, 11-dioxide as a Photoswitchable Reporter
1.1 Reagents: Potassium carbonate, Boric acid (H ₃ BO ₃) Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: 1,4-Dioxane, Water; 15 h, rt → 80 °C		By: Liu, Zhihao; et al Angewandte Chemie, International Edition (2025), 64(18), e202501936.
Experimental Protocols		

Scheme 440 (1 Reaction)

Steps: 1 Yield: 50%



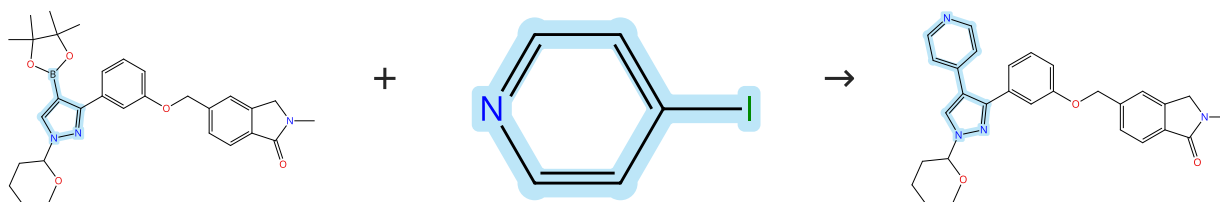
Suppliers (58)

Suppliers (10)

31-614-CAS-36018193	Steps: 1 Yield: 50%	Synthesis and Unexpected Optical Properties of Ionic Phosphorus Heterocycles with P-Regulated Noncovalent Interactions
1.1 Reagents: Tripotassium phosphate Catalysts: Tris(dibenzylideneacetone)dipalladium, Tri- <i>tert</i> -butylphosphonium tetrafluoroborate Solvents: Toluene, Water; overnight, 40 °C		By: Yang, Zi; et al Journal of Organic Chemistry (2023), 88(5), 2792-2800.
Experimental Protocols		

Scheme 441 (1 Reaction)

Steps: 1 Yield: 50%

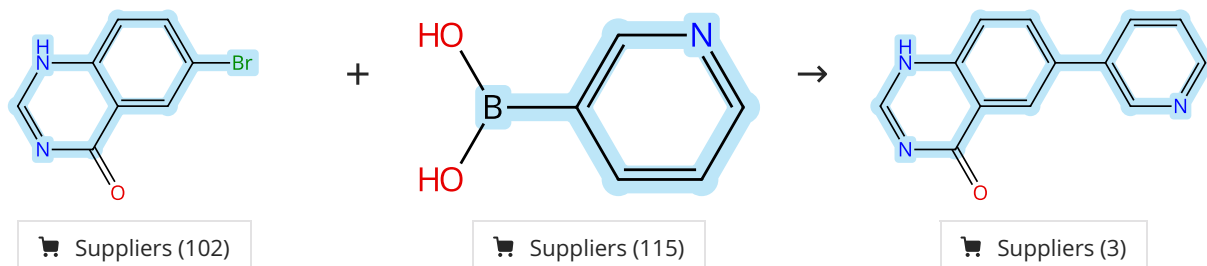


Suppliers (81)

31-614-CAS-40922279	Steps: 1 Yield: 50%	Discovery and Optimization of Potent, Efficacious and Selective Inhibitors Targeting EGFR Exon20 Insertion Mutations
1.1 Reagents: Cesium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: 1,4-Dioxane, Water; 3 h, 90 °C		By: Thomson, Clare; et al
Experimental Protocols		Journal of Medicinal Chemistry (2024), 67(11), 8988-9027.

Scheme 442 (1 Reaction)

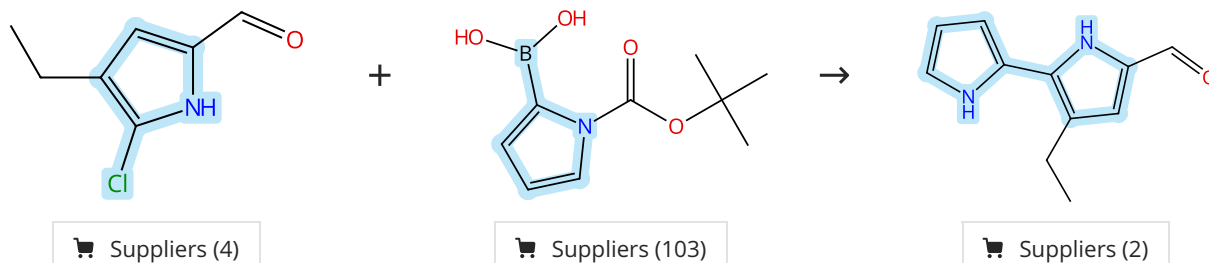
Steps: 1 Yield: 50%



31-179-CAS-9719311	Steps: 1 Yield: 50%	Potent and selective small molecule inhibitors of specific isoforms of Cdc2-like kinases (Clk) and dual specificity tyrosine-phosphorylation-regulated kinases (Dyrk)
1.1 Reagents: Sodium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: 1,2-Dimethoxyethane, Water; 1 h, 150 °C		By: Rosenthal, Andrew S.; et al
		Bioorganic & Medicinal Chemistry Letters (2011), 21(10), 3152-3158.

Scheme 443 (1 Reaction)

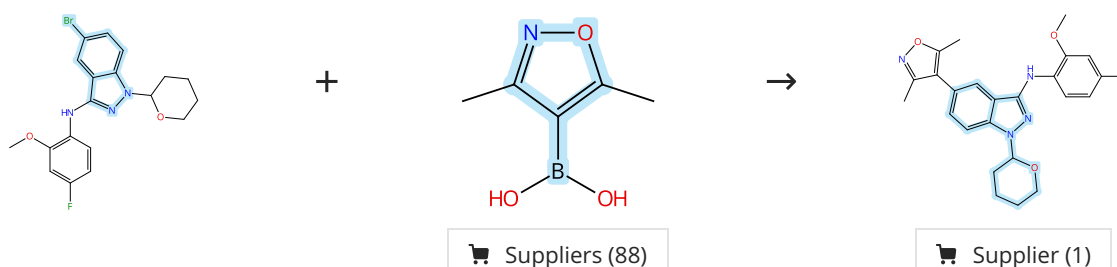
Steps: 1 Yield: 50%



31-179-CAS-12620091	Steps: 1 Yield: 50%	Synthesis of 2,2'-bipyrrole-5-carboxaldehydes and their application in the synthesis of B- ring functionalized prodiginines and tambjamines
1.1 Reagents: Sodium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Hexane, 1,4-Dioxane, Water; 3 h, 100 °C		By: Kancharla, Papireddy; et al
1.2 Reagents: Hydrochloric acid Solvents: Water; pH 7		Tetrahedron (2013), 69(39), 8375-8385.
1.3 Reagents: Lithium hydroxide Solvents: Methanol, Tetrahydrofuran; rt; 30 min, rt		
Experimental Protocols		

Scheme 444 (1 Reaction)

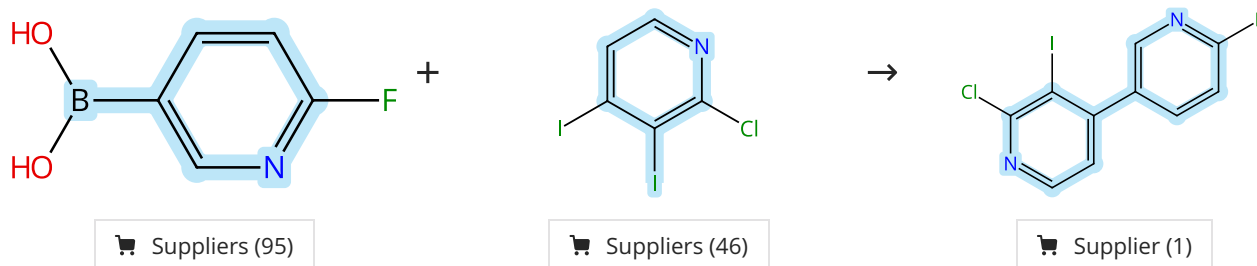
Steps: 1 Yield: 50%



31-179-CAS-22579913	Steps: 1 Yield: 50%	Design, synthesis and biological evaluation of 3, 5-dimethylisoxazole and pyridone derivatives as BRD4 inhibitors
1.1 Reagents: Sodium carbonate Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: 1,4-Dioxane, Water; 6 - 8 h, 90 °C; rt		By: Rong, Juan; et al Bioorganic & Medicinal Chemistry Letters (2019), 29(19), 126577.
Experimental Protocols		

Scheme 445 (1 Reaction)

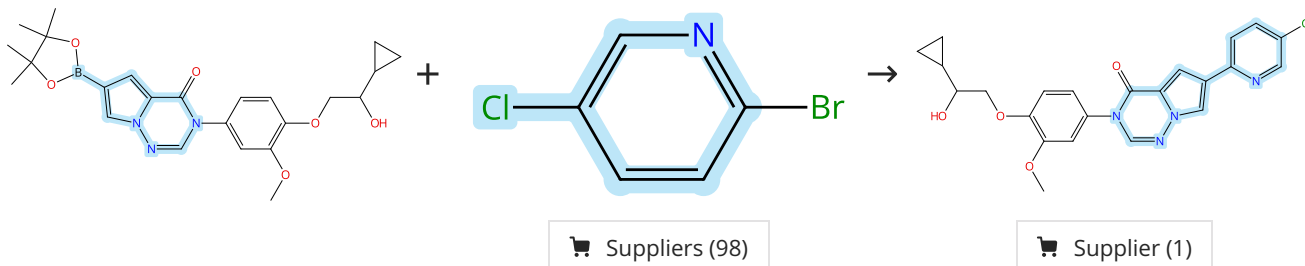
Steps: 1 Yield: 50%



31-179-CAS-9782041	Steps: 1 Yield: 50%	Iterative and regioselective cross-couplings of 2-chloro-3,4-diiodopyridine leading to 2,3,4-triheteroarylpyridines
1.1 Reagents: Sodium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: 1,4-Dioxane, Water; 24 h, reflux		By: Daykin, Laura M.; et al Tetrahedron (2010), 66(3), 668-675.
Experimental Protocols		

Scheme 446 (1 Reaction)

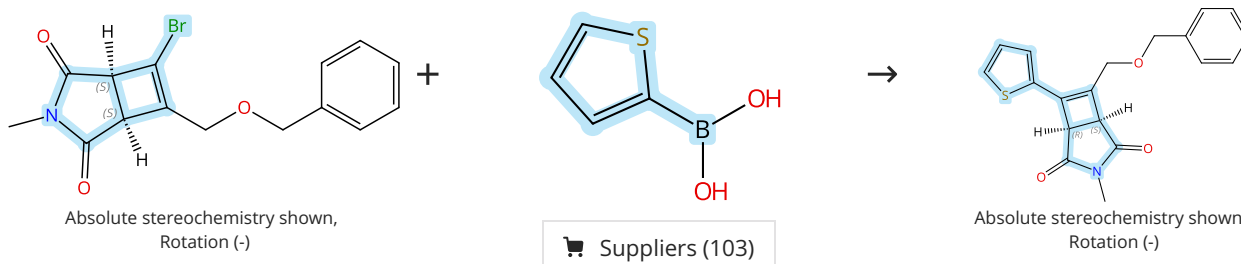
Steps: 1 Yield: 50%



31-179-CAS-11227505	Steps: 1 Yield: 50%	Non-basic azolotriazinone MCHR1 antagonists for the treatment of obesity: An empirical brain-exposures-driven candidate selection for in vivo efficacy studies
1.1 Reagents: Potassium carbonate Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: Dimethylformamide; 60 min, 100 °C		By: Devasthale, Pratik; et al Bioorganic & Medicinal Chemistry Letters (2015), 25(20), 4412-4418.

Scheme 447 (1 Reaction)

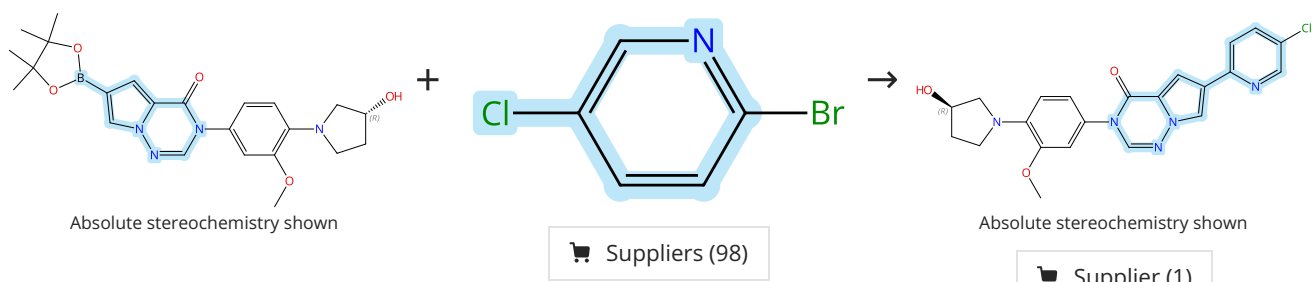
Steps: 1 Yield: 50%



31-614-CAS-38332818 Steps: 1 Yield: 50% 1.1 Reagents: Tripotassium phosphate Catalysts: Palladium diacetate, [2',6'-Bis(1-methylethoxy)[1,1'-biphenyl]-2-yl]dicyclohexylphosphine Solvents: Tetrahydrofuran, Water; 5 min, rt; rt → 70 °C; 18 h, 70 °C; 70 °C → rt 1.2 Reagents: Sodium bicarbonate Solvents: Water; rt Experimental Protocols	Divergent Synthesis of 1,2,3,4-Tetrasubstituted Cyclobutenes from a Common Scaffold: Enantioselective Desymmetrization by Dual-Catalyzed Photoredox Cross-Coupling By: Konowalchuk, Dawson J.; et al Angewandte Chemie, International Edition (2023), 62(49), e202313503.
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Scheme 448 (1 Reaction)

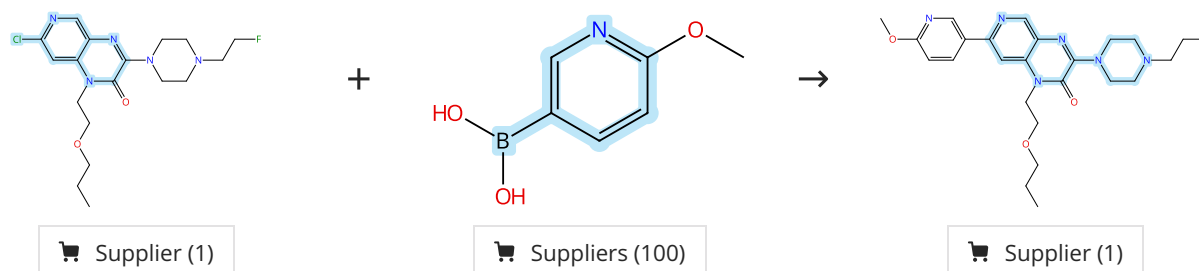
Steps: 1 Yield: 50%



31-179-CAS-2092357 Steps: 1 Yield: 50% 1.1 Reagents: Potassium carbonate Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: Dimethylformamide; 60 min, 100 °C	Non-basic azolotriazinone MCHR1 antagonists for the treatment of obesity: An empirical brain-exposures-driven candidate selection for in vivo efficacy studies By: Devasthale, Pratik; et al Bioorganic & Medicinal Chemistry Letters (2015), 25(20), 4412-4418.
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Scheme 449 (1 Reaction)

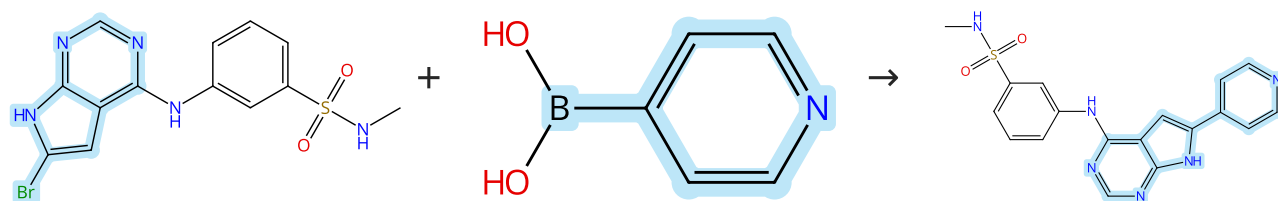
Steps: 1 Yield: 50%



31-179-CAS-16704059 Steps: 1 Yield: 50% 1.1 Reagents: Sodium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Ethanol, 1,4-Dioxane; 16 h, reflux Experimental Protocols	Carbon-11 and Fluorine-18 Radiolabeled Pyridopyrazinone Derivatives for Positron Emission Tomography (PET) Imaging of Phosphodiesterase-5 (PDE5) By: Chekol, Rufael; et al Journal of Medicinal Chemistry (2017), 60(1), 486-496.
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Scheme 450 (1 Reaction)

Steps: 1 Yield: 50%



Suppliers (2)

Suppliers (111)

Suppliers (2)

31-179-CAS-6590101

Steps: 1 Yield: 50%

Identification of Purines and 7-Deazapurines as Potent and Selective Type I Inhibitors of Troponin I-Interacting Kinase (TN NI3K)

By: Lawhorn, Brian G.; et al

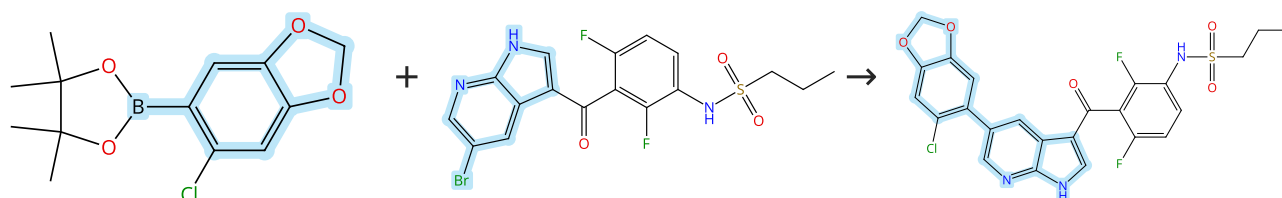
Journal of Medicinal Chemistry (2015), 58(18), 7431-7448.

1.1 **Reagents:** Sodium carbonate
Catalysts: Dichloro[1,1'-bis(diphenylphosphino)ferrocene]
 palladium(II) dichloromethane adduct
Solvents: 1,4-Dioxane; 15 min, 160 °C

Experimental Protocols

Scheme 451 (1 Reaction)

Steps: 1 Yield: 50%



Suppliers (13)

Suppliers (51)

Supplier (1)

31-614-CAS-24101333

Steps: 1 Yield: 50%

From off-to on-target: New BRAF-inhibitor-template-derived compounds selectively targeting mitogen activated protein kinase kinase 4 (MKK4)

By: Kloevekorn, Philip; et al

European Journal of Medicinal Chemistry (2021), 210, 112963.

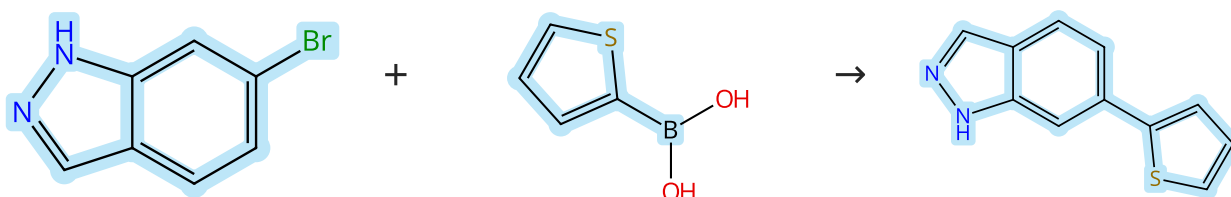
1.1 **Reagents:** Potassium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: 1,4-Dioxane, Water; 30 min, rt → 120 °C

1.2 **Reagents:** Ammonium chloride
Solvents: Ethyl acetate, Water; neutralized

Experimental Protocols

Scheme 452 (1 Reaction)

Steps: 1 Yield: 50%



Suppliers (89)

Suppliers (103)

Suppliers (45)

31-614-CAS-34399644

Steps: 1 Yield: 50%

Unified Access to Pyrimidines and Quinazolines Enabled by N-N Cleaving Carbon Atom Insertion

By: Hyland, Ethan E.; et al

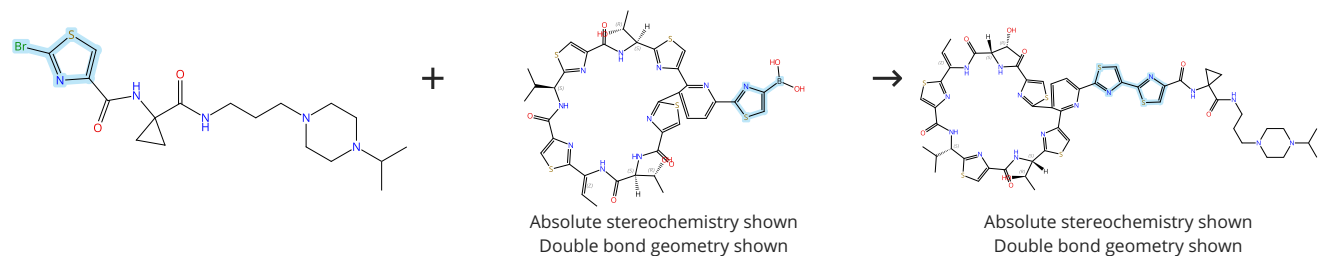
Journal of the American Chemical Society (2022), 144(42), 19258-19264.

1.1 **Reagents:** Sodium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: 1,4-Dioxane, Water; 2 h, 100 °C

Experimental Protocols

Scheme 453 (1 Reaction)

Steps: 1 Yield: 50%



31-614-CAS-37991971

Steps: 1 Yield: 50%

Identification of Micrococцин P2-Derivatives as Antibiotic Candidates against Two Gram-Positive Pathogens

By: Kim, Dahyun; et al

Journal of Medicinal Chemistry (2023), 66(20), 14263-14277.

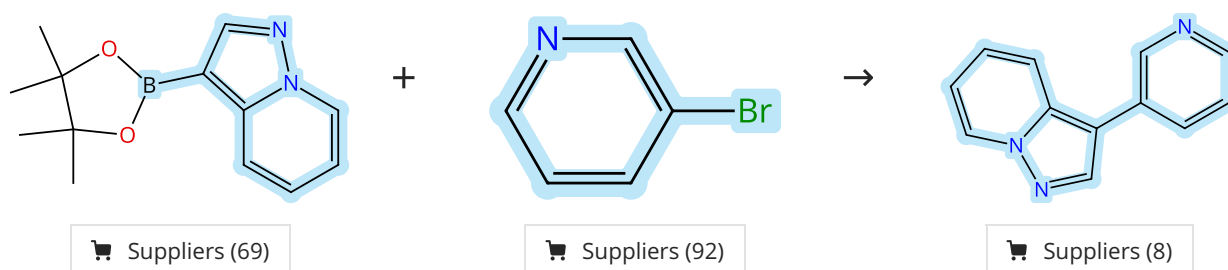
- 1.1 Reagents: Potassium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Tetrahydrofuran, Water; 12 h, 35 °C

- 1.2 Reagents: Water

Experimental Protocols

Scheme 454 (1 Reaction)

Steps: 1 Yield: 50%



31-179-CAS-6087066

Steps: 1 Yield: 50%

Optimized scale up of 3-pyrimidinylpyrazolo[1,5-a]pyridine via Suzuki coupling; a general method of accessing a range of 3-(hetero)arylpyrazolo[1,5-a]pyridines

By: Bethel, Paul A.; et al

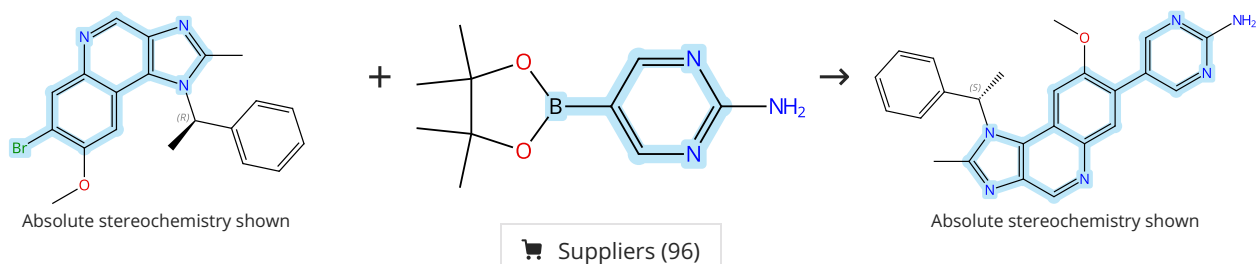
Tetrahedron (2012), 68(27-28), 5434-5444.

- 1.1 Reagents: Sodium carbonate
Catalysts: (*SP*-4-1)-Bis[4-[bis(1,1-dimethylethyl)phosphino-κ^P]-*N,N*-dimethylbenzenamine]dichloropalladium
Solvents: 1,2-Dimethoxyethane; 4 h, 80 °C

Experimental Protocols

Scheme 455 (1 Reaction)

Steps: 1 Yield: 50%



31-614-CAS-38841459

Steps: 1 Yield: 50%

Discovery of (R)-4-(8-methoxy-2-methyl-1-(1-phenylethyl)-1H-imidazo[4,5-c]quinnolin-7-yl)-3,5-dimethylisoxazole as a potent and selective BET inhibitor for treatment of acute myeloid leukemia (AML) guided by FEP calculation

By: Yu, Su; et al

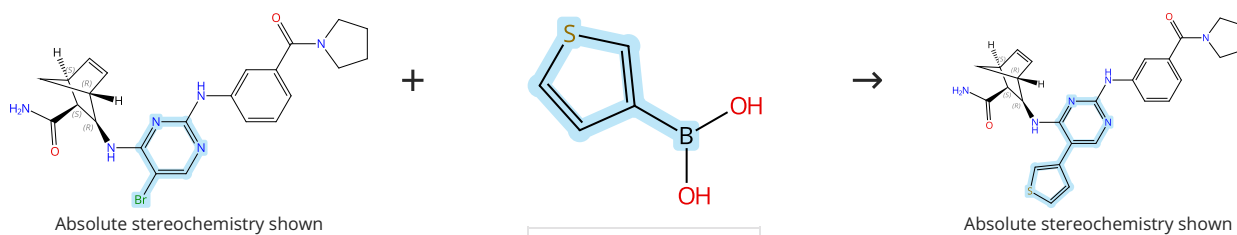
European Journal of Medicinal Chemistry (2024), 263, 115924.

- 1.1 Reagents: Potassium acetate
Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium
Solvents: 1,4-Dioxane, Water; overnight, 100 °C

Experimental Protocols

Scheme 456 (1 Reaction)

Steps: 1 Yield: 50%



Suppliers (101)

31-614-CAS-32044047

Steps: 1 Yield: 50%

Discovery of 5-Aryl-2,4-diaminopyrimidine Compounds as Potent and Selective IRAK4 Inhibitors

By: Chen, Yan; et al

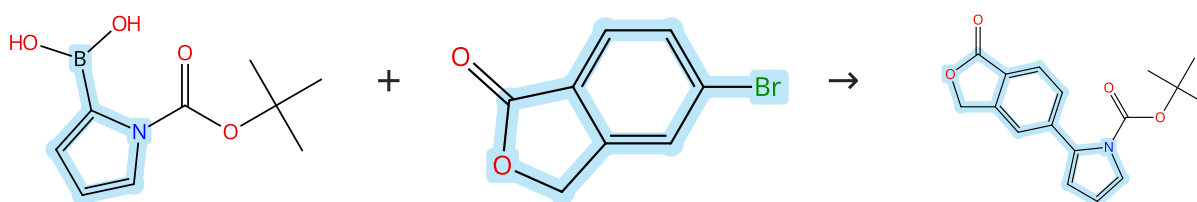
ACS Medicinal Chemistry Letters (2022), 13(4), 714-719.

1.1 Reagents: Sodium carbonate
 Catalysts: Tetrakis(triphenylphosphine)palladium
 Solvents: Ethanol, 1,2-Dimethoxyethane, Water; 30 min, 150 °C

Experimental Protocols

Scheme 457 (1 Reaction)

Steps: 1 Yield: 50%



Suppliers (103)

Suppliers (100)

Supplier (1)

31-179-CAS-7487992

Steps: 1 Yield: 50%

Inhibition of the pore-forming protein perforin by a series of aryl-substituted isobenzofuran-1(3H)-ones

By: Spicer, Julie A.; et al

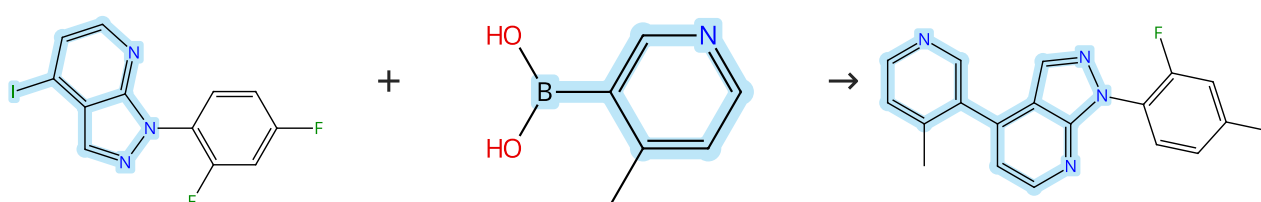
Bioorganic & Medicinal Chemistry (2012), 20(3), 1319-1336.

1.1 Reagents: Sodium carbonate
 Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium
 Solvents: Ethanol, Toluene, Water; 2 h, reflux

Experimental Protocols

Scheme 458 (1 Reaction)

Steps: 1 Yield: 50%



Suppliers (6)

Suppliers (84)

Supplier (1)

31-614-CAS-34555161

Steps: 1 Yield: 50%

The discovery of BMS-737 as a potent, CYP17 lyase-selective inhibitor for the treatment of castration-resistant prostate cancer

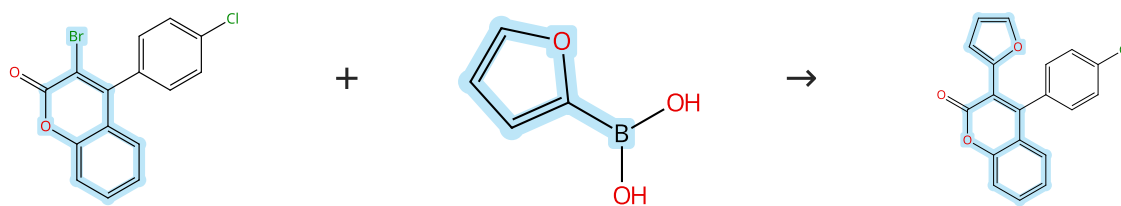
By: Padmakar Darne, Chetan; et al

Bioorganic & Medicinal Chemistry Letters (2022), 75, 128951.

1.1 Reagents: Potassium carbonate
 Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium
 Solvents: N-Methyl-2-pyrrolidone, Water; 4 h, 105 °C

Scheme 459 (1 Reaction)

Steps: 1 Yield: 50%



Suppliers (93)

31-179-CAS-19352719

Steps: 1 Yield: 50%

Synthesis of Polycyclic Heteroaromatic Coumarins via Photoinduced Dehydrogenative Annulation of 4-Phenyl-3-heteroaryl coumarins

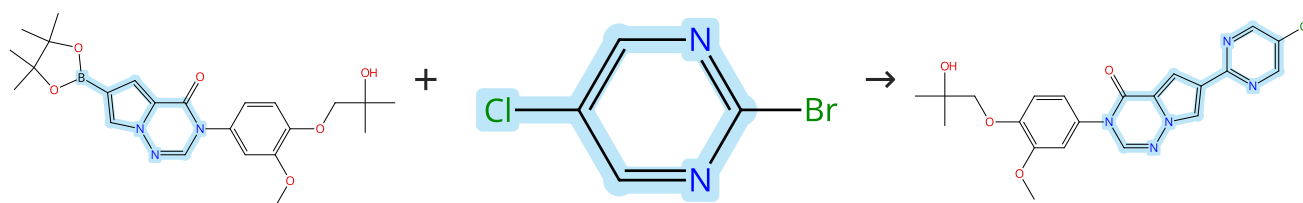
By: Shi, Juan; et al

Journal of Organic Chemistry (2018), 83(22), 13940-13948.

1.1 **Reagents:** Cesium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: 1,4-Dioxane; 90 °C

Scheme 460 (1 Reaction)

Steps: 1 Yield: 50%



Suppliers (65)

Supplier (1)

31-179-CAS-10611733

Steps: 1 Yield: 50%

Non-basic azolotriazinone MCHR1 antagonists for the treatment of obesity: An empirical brain-exposures-driven candidate selection for in vivo efficacy studies

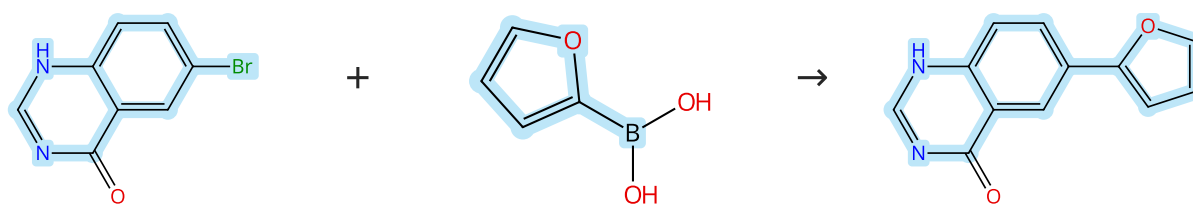
By: Devasthale, Pratik; et al

Bioorganic & Medicinal Chemistry Letters (2015), 25(20), 4412-4418.

1.1 **Reagents:** Potassium carbonate
Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium
Solvents: Dimethylformamide; 60 min, 100 °C

Scheme 461 (1 Reaction)

Steps: 1 Yield: 50%



Suppliers (102)

Suppliers (93)

Suppliers (3)

31-179-CAS-3053997

Steps: 1 Yield: 50%

Potent and selective small molecule inhibitors of specific isoforms of Cdc2-like kinases (Clk) and dual specificity tyrosine-phosphorylation-regulated kinases (Dyrk)

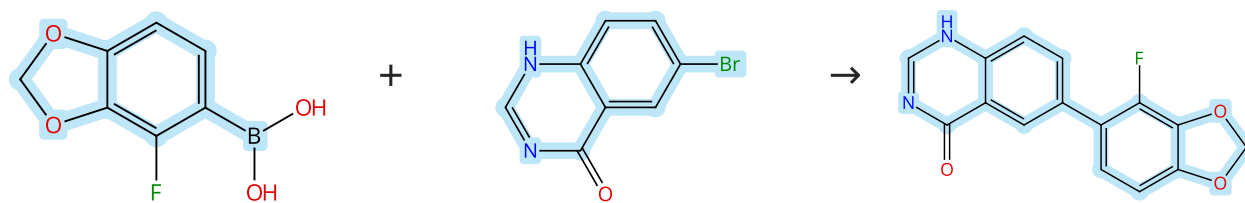
By: Rosenthal, Andrew S.; et al

Bioorganic & Medicinal Chemistry Letters (2011), 21(10), 3152-3158.

1.1 **Reagents:** Sodium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: 1,2-Dimethoxyethane, Water; 1 h, 150 °C

Scheme 462 (1 Reaction)

Steps: 1 Yield: 50%



Suppliers (49)

Suppliers (102)

31-179-CAS-13974304

Steps: 1 Yield: 50%

Potent and selective small molecule inhibitors of specific isoforms of Cdc2-like kinases (Clk) and dual specificity tyrosine-phosphorylation-regulated kinases (Dyrk)

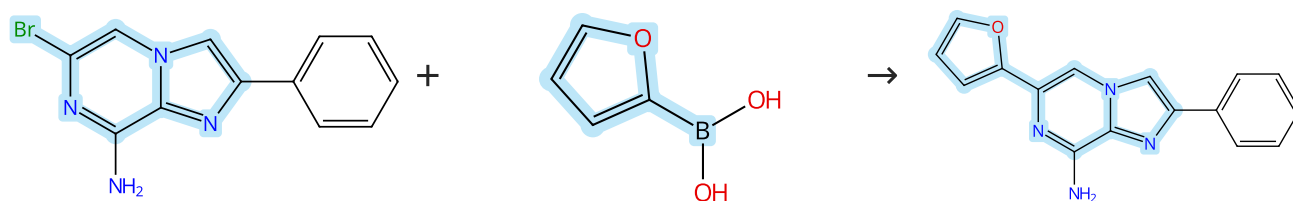
By: Rosenthal, Andrew S.; et al

Bioorganic & Medicinal Chemistry Letters (2011), 21(10), 3152-3158.

1.1 Reagents: Sodium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: 1,2-Dimethoxyethane, Water; 1 h, 150 °C

Scheme 463 (1 Reaction)

Steps: 1 Yield: 50%



Suppliers (13)

Suppliers (93)

Supplier (1)

31-179-CAS-16367118

Steps: 1 Yield: 50%

Imidazo[1,2-a]pyrazin-8-amine core for the design of new adenosine receptor antagonists: Structural exploration to target the A₃ and A_{2A} subtypes

By: Poli, Daniela; et al

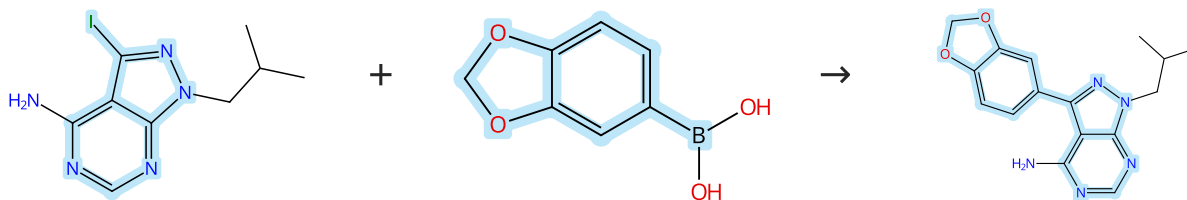
European Journal of Medicinal Chemistry (2017), 125, 611-628.

1.1 Reagents: Sodium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: 1,2-Dimethoxyethane, Water; 85 °C

Experimental Protocols

Scheme 464 (1 Reaction)

Steps: 1 Yield: 50%



Suppliers (2)

Suppliers (99)

31-179-CAS-9592698

Steps: 1 Yield: 50%

Microwave-assisted synthesis of N1- and C3- substituted pyrazolo[3,4-d]pyrimidine libraries

By: Todorovic, Nick; et al

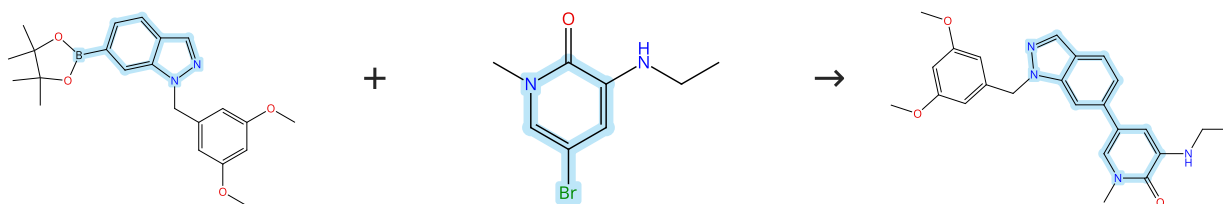
Tetrahedron Letters (2011), 52(44), 5761-5763.

1.1 Reagents: Cesium carbonate, 1,3,5,7-Tetramethyl-8-phenyl-2,4,6-trioxa-8-phosphatricyclo[3.3.1.1^{3,7}]decane
Catalysts: Tris(dibenzylideneacetone)dipalladium
Solvents: Toluene; 30 min, 70 °C

Experimental Protocols

Scheme 465 (1 Reaction)

Steps: 1 Yield: 50%



Supplier (1)

31-179-CAS-22579924

Steps: 1 Yield: 50%

Design, synthesis and biological evaluation of 3, 5-dimethylsoxazole and pyridone derivatives as BRD4 inhibitors

By: Rong, Juan; et al

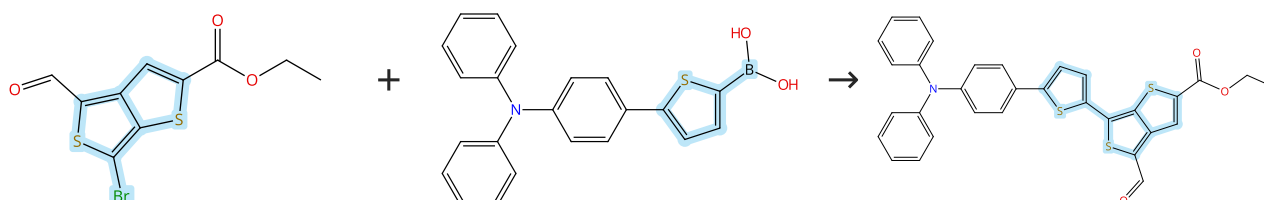
Bioorganic & Medicinal Chemistry Letters (2019), 29(19), 126577.

1.1 Reagents: Sodium carbonate
Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium
Solvents: 1,4-Dioxane, Water; 6 - 8 h, 95 °C; rt

Experimental Protocols

Scheme 466 (1 Reaction)

Steps: 1 Yield: 50%



31-179-CAS-7831608

Steps: 1 Yield: 50%

Thieno[3,4-b]Thiophene-Based Organic Dyes for Dye-Sensitized Solar Cells

By: Chen, Yung-Chung; et al

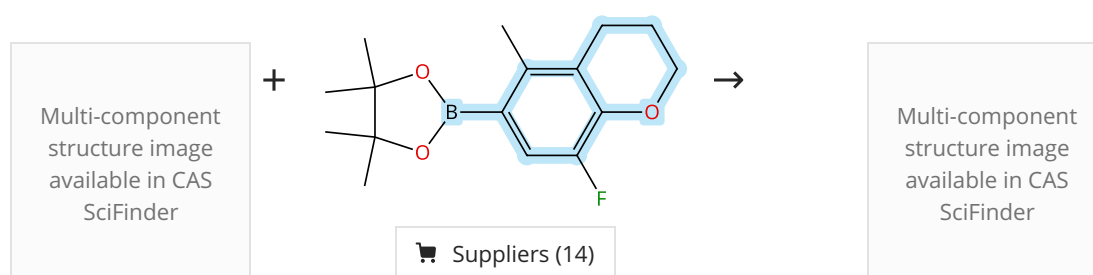
Chemistry - A European Journal (2012), 18(17), 5430-5437, S5430/1-S5430/20.

1.1 Reagents: Sodium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Ethanol, Toluene, Water; 18 h, reflux

Experimental Protocols

Scheme 467 (1 Reaction)

Steps: 1 Yield: 50%



Suppliers (14)

31-614-CAS-27641825

Steps: 1 Yield: 50%

Discovery and Optimization of Novel Pyrazolopyrimidines as Potent and Orally Bioavailable Allosteric HIV-1 Integrase Inhibitors

By: Li, Guo; et al

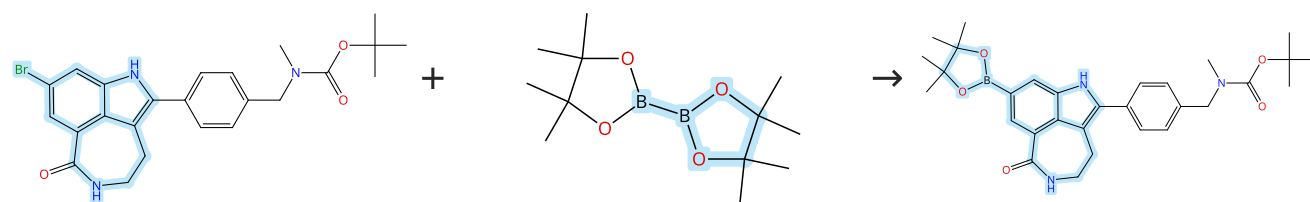
Journal of Medicinal Chemistry (2020), 63(5), 2620-2637.

1.1 Reagents: Tripotassium phosphate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Dimethylformamide, Water; 15 min, 130 °C

Experimental Protocols

Scheme 468 (1 Reaction)

Steps: 1 Yield: 50%



Suppliers (131)

31-614-CAS-39150495

Steps: 1 Yield: 50%

Synthesis of a Pinacol Boronate Precursor for [¹⁸F]Rucaparib Radiosynthesis

By: Seo, Ju-Ahn; et al

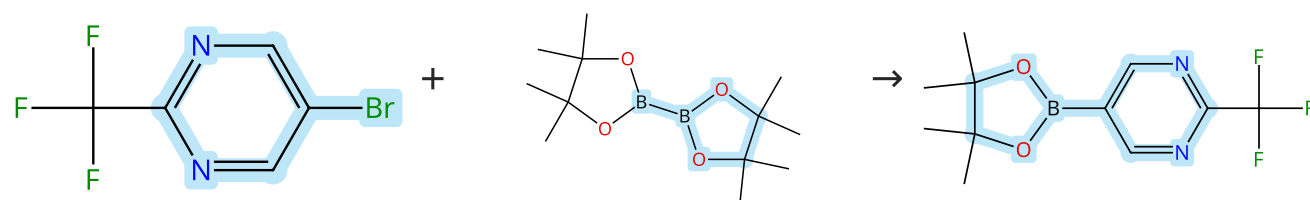
Asian Journal of Organic Chemistry (2024), 13(1), e202300501.

1.1 **Reagents:** Potassium acetate
Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium
Solvents: Dimethylformamide; 18 h, 80 °C

Experimental Protocols

Scheme 469 (1 Reaction)

Steps: 1 Yield: 50%



Suppliers (76)

Suppliers (131)

Suppliers (57)

31-142-CAS-21041419

Steps: 1 Yield: 50%

Discovery of a Potent (4R,5S)-4-Fluoro-5-methylproline Sulfonamide Transient Receptor Potential Ankyrin 1 Antagonist and Its Methylene Phosphate Prodrug Guided by Molecular Modeling

By: Chen, Huifen; et al

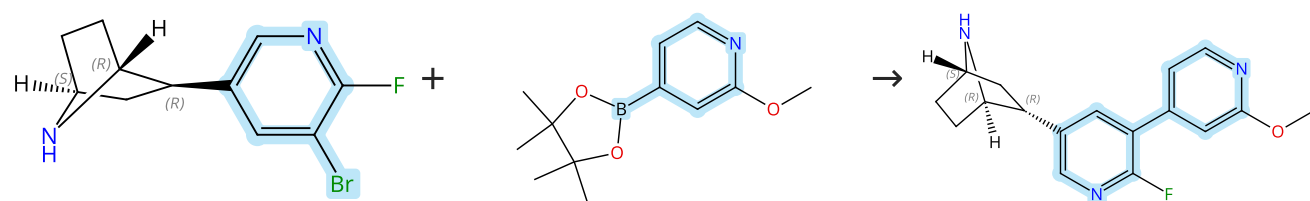
Journal of Medicinal Chemistry (2018), 61(8), 3641-3659.

1.1 **Reagents:** Potassium acetate
Catalysts: Palladium diacetate
Solvents: Dimethylformamide; 3 h, 80 °C

Experimental Protocols

Scheme 470 (1 Reaction)

Steps: 1 Yield: 50%



Relative stereochemistry shown

Relative stereochemistry shown

Suppliers (2)

Suppliers (83)

31-179-CAS-10903344

Steps: 1 Yield: 50%

Synthesis, Nicotinic Acetylcholine Receptor Binding, and Antinociceptive Properties of 2'-Fluoro-3'-(substituted pyridinyl)-7-deschloroepibatidine Analogues

By: Ondachi, Pauline W.; et al

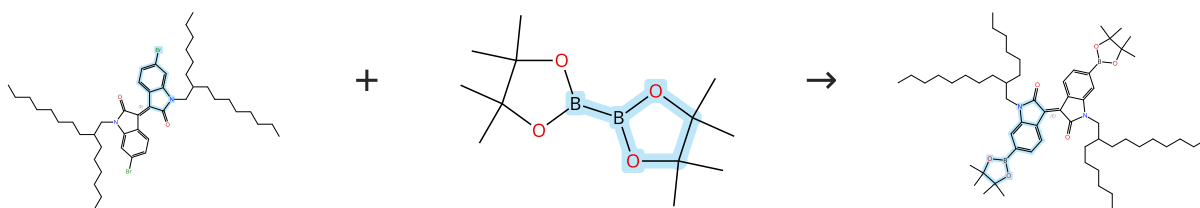
Journal of Medicinal Chemistry (2014), 57(3), 836-848.

1.1 **Reagents:** Potassium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: 1,4-Dioxane, Water; 18 h, 110 °C

Experimental Protocols

Scheme 471 (1 Reaction)

Steps: 1 Yield: 50%



Double bond geometry shown

Suppliers (131)

Double bond geometry shown

Suppliers (2)

31-614-CAS-28939906

Steps: 1 Yield: 50%

Improved photovoltaic performance of D-A-D-type small molecules with isoindigo and pyrene units by inserting different π -conjugated bridge

By: Yu, Delong; et al

Tetrahedron (2016), 72(30), 4543-4549.

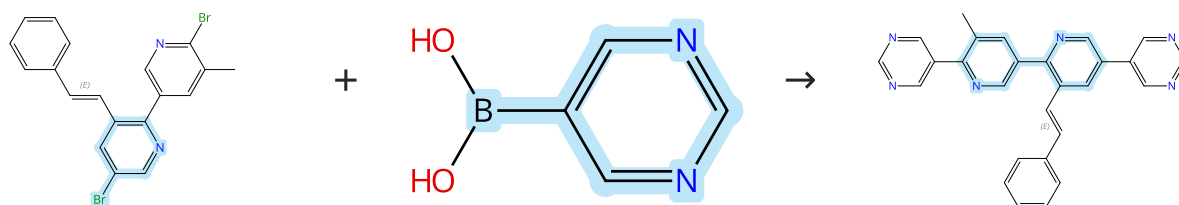
1.1 **Reagents:** Potassium acetate
Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium
Solvents: 1,4-Dioxane; 24 h, rt $\rightarrow$ reflux

1.2 **Solvents:** Water

Experimental Protocols

Scheme 472 (1 Reaction)

Steps: 1 Yield: 50%



Double bond geometry shown

Suppliers (113)

Double bond geometry shown

31-179-CAS-6003328

Steps: 1 Yield: 50%

First Evidence That Oligopyridines, α -Helix Foldamers, Inhibit Mcl-1 and Sensitize Ovarian Carcinoma Cells to Bcl-x_L-Targeting Strategies

By: Gloaguen, Celine; et al

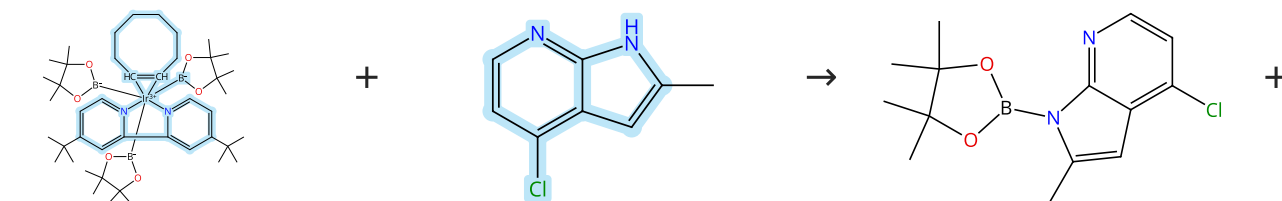
Journal of Medicinal Chemistry (2015), 58(4), 1644-1668.

1.1 **Reagents:** Sodium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: 1,4-Dioxane; 24 h, reflux

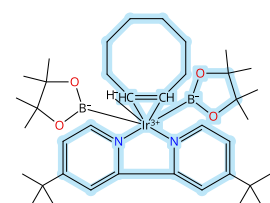
Experimental Protocols

Scheme 473 (1 Reaction)

Steps: 1 Yield: 100%



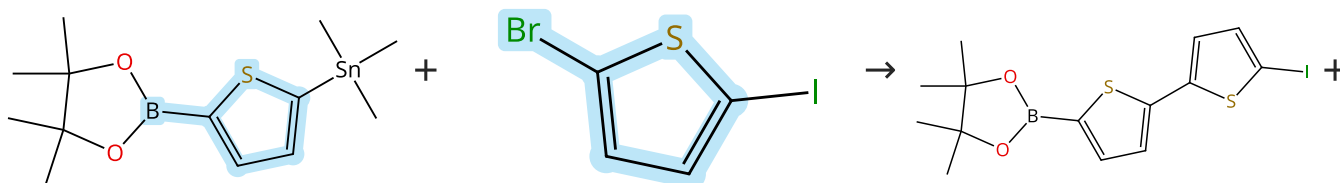
Suppliers (59)



31-614-CAS-29836002 Steps: 1 Yield: 100%	Iridium-Catalyzed C-H Borylation of Heteroarenes: Scope, Regioselectivity, Application to Late-Stage Functionalization, and Mechanism By: Larsen, Matthew A.; et al Journal of the American Chemical Society (2014), 136(11), 4287-4299.
1.1 Solvents: THF- <i>d</i> ₈ Experimental Protocols	

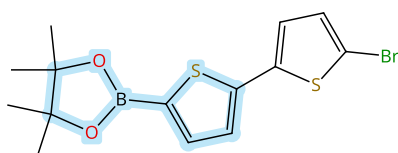
Scheme 474 (1 Reaction)

Steps: 1 Yield: 90%



Suppliers (3)

Suppliers (54)

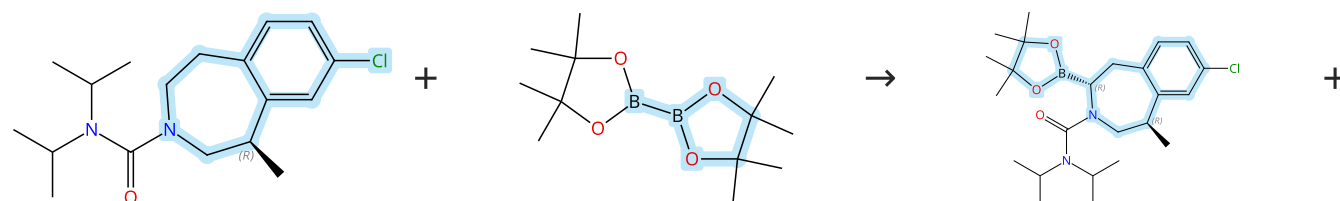


Suppliers (31)

31-130-CAS-1648718 Steps: 1 Yield: 90%	Dual Selectivity: Electrophile and Nucleophile Selective Cross-Coupling Reactions on a Single Aromatic Substrate By: Heinrich, Annika C. J.; et al Organic Letters (2013), 15(18), 4666-4669.
1.1 Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Dimethylformamide; 15 h, 70 °C Experimental Protocols	

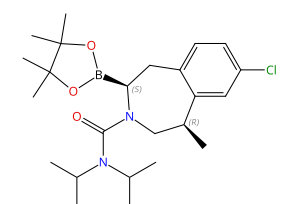
Scheme 475 (1 Reaction)

Steps: 1 Yield: 89%


 Absolute stereochemistry shown,
 Rotation (+)

Suppliers (131)

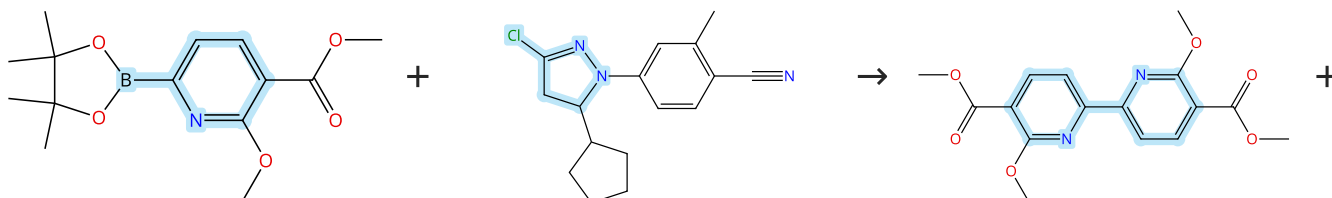
Absolute stereochemistry shown


 Absolute stereochemistry shown,
 Rotation (+)

31-614-CAS-24150831 Steps: 1 Yield: 89% 1.1 Catalysts: Di- μ -chlorobis[(1,2,5,6- η)-1,5-cyclooctadiene]diiridium, Pyridine, 2-[(4 <i>R</i> ,5 <i>R</i>)-3-(2-cyclohexylphenyl)-2-(dimethylphenylsilyl)-4,5-diphenyl-1,3,2-diazaborolidin-1-yl]-5-[1,1':3',1''-terphenyl]-2'-yl- Solvents: Hexane; 24 - 48 h, 70 °C	Iridium-Catalyzed Enantioselective α-C(sp³)-H Borylation of Azacycles By: Chen, Lili; et al Journal of the American Chemical Society (2020), 142(28), 12062-12068.
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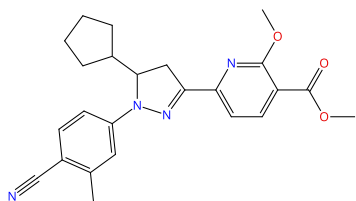
Scheme 476 (1 Reaction)

Steps: 1 Yield: 85%



Suppliers (51)

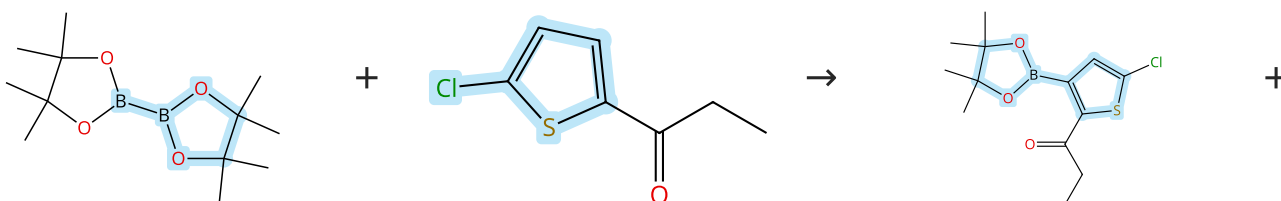
Supplier (1)



31-179-CAS-11575760 Steps: 1 Yield: 85% 1.1 Reagents: Potassium carbonate Catalysts: Tris(dibenzylideneacetone)dipalladium, X-Phos Solvents: Tetrahydrofuran, Water; 4 h, 60 - 65 °C Experimental Protocols	Synthesis and cross coupling of a highly substituted 2-pyridylboronate: application to the large scale synthesis of a mineralocorticoid antagonist By: Boos, Charles; et al Tetrahedron Letters (2011), 52(52), 7025-7029.
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Scheme 477 (1 Reaction)

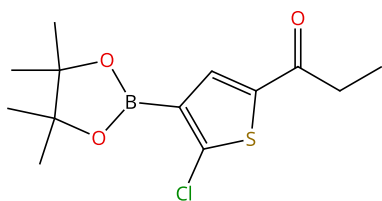
Steps: 1 Yield: 81%



Suppliers (131)

Suppliers (41)

Supplier (1)

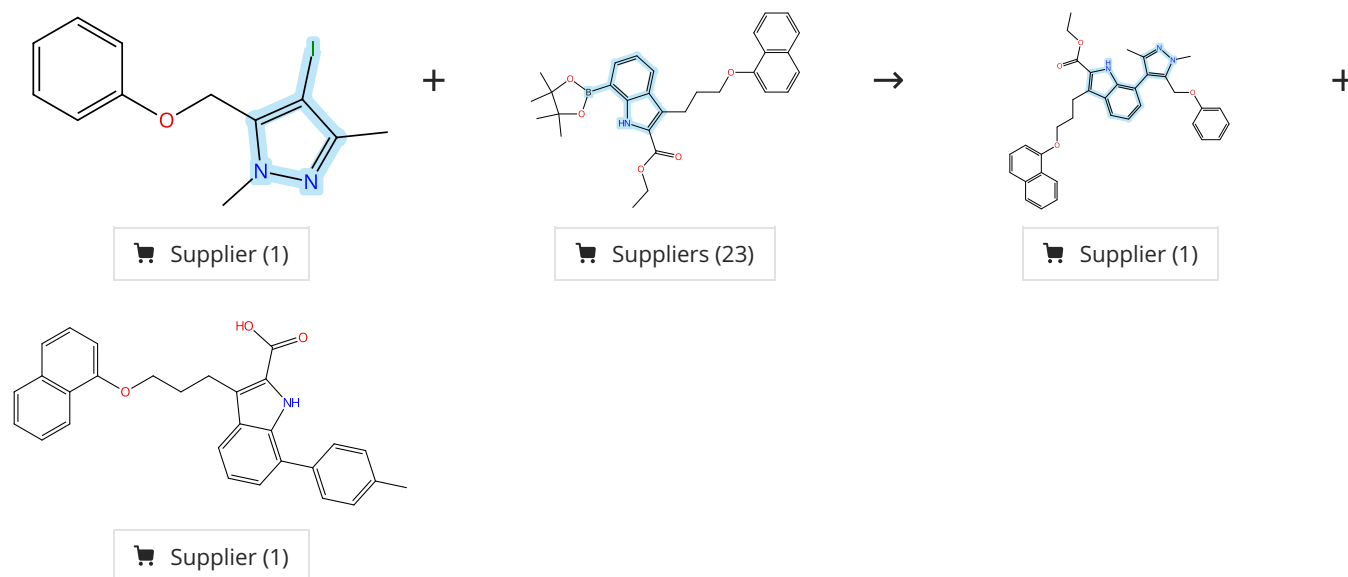


Suppliers (4)

31-142-CAS-4808736 Steps: 1 Yield: 81% 1.1 Catalysts: Bis[(1,2,5,6- η)-1,5-cyclooctadiene]di- μ -methoxydiiridium, 4,4'-Bis(1,1-dimethylethyl)-2,2'-bipyridine Solvents: Octane; 15 min, rt 1.2 16 h, rt Experimental Protocols	Catalyst-controlled regiodivergent C-H borylation of multifunctionalized heteroarenes by using iridium complexes By: Sasaki, Ikuo; et al Chemistry - A European Journal (2015), 21(25), 9236-9241.
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Scheme 478 (1 Reaction)

Steps: 1 Yield: 78%



31-179-CAS-493719

Steps: 1 Yield: 78%

Structure-Guided Design of a Series of MCL-1 Inhibitors with High Affinity and Selectivity

By: Bruncko, Milan; et al

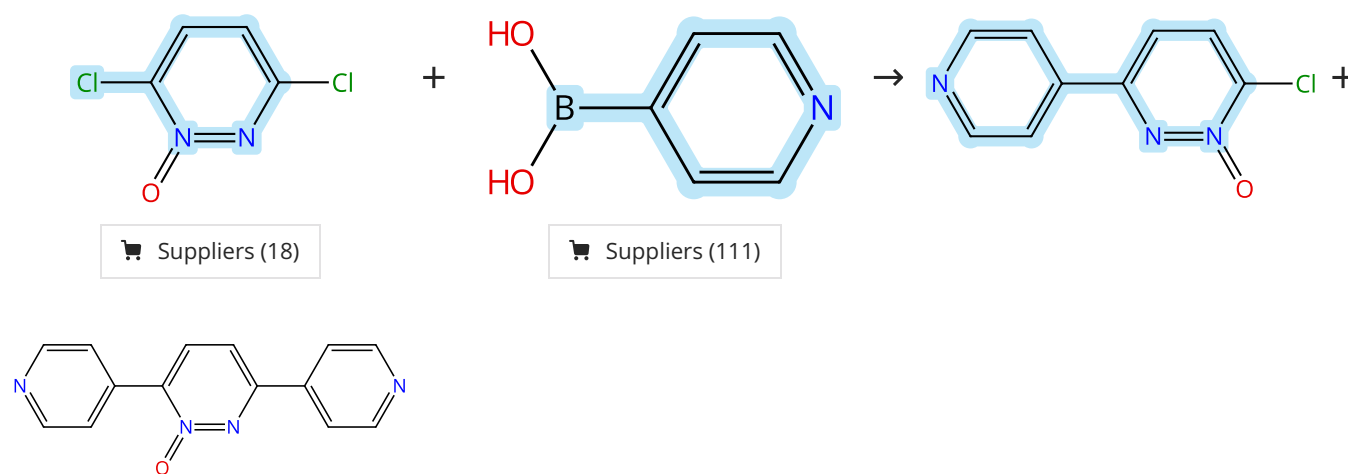
Journal of Medicinal Chemistry (2015), 58(5), 2180-2194.

1.1 **Reagents:** Potassium carbonate
Catalysts: Dichlorobis(triphenylphosphine)palladium
Solvents: Dimethylformamide, Water; 4 h, rt → 100 °C

Experimental Protocols

Scheme 479 (1 Reaction)

Steps: 1 Yield: 73%



31-179-CAS-16200158

Steps: 1 Yield: 73%

Synthesis of Nitrogen Heterocycles via Photochemical Ring Opening of Pyridazine N-Oxides

By: Portillo, Maribel; et al

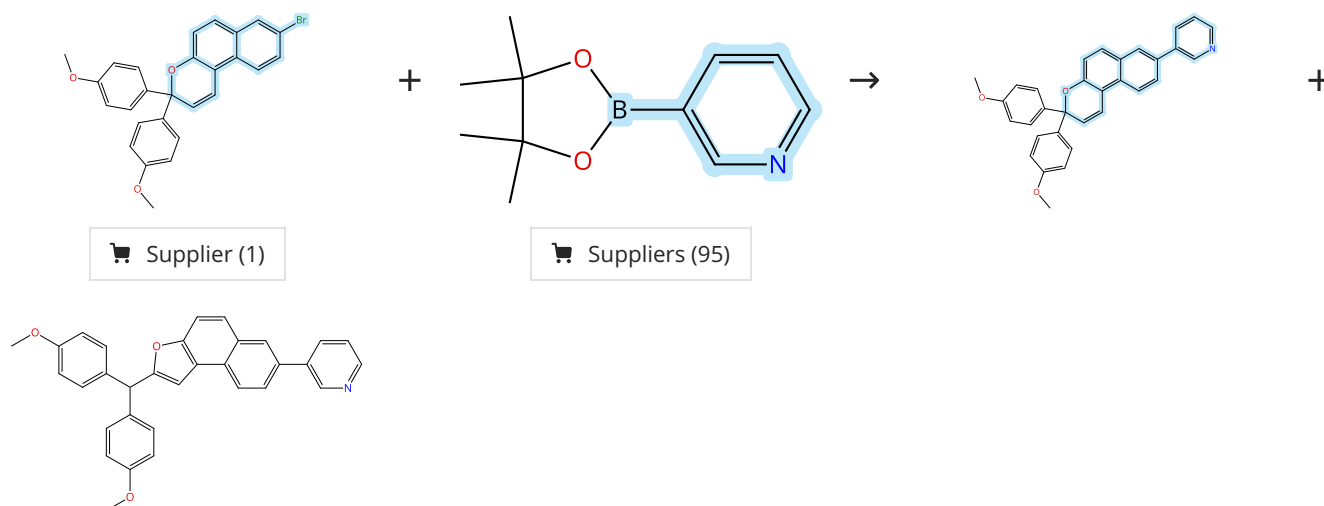
Organic Letters (2016), 18(19), 5142-5145.

1.1 **Reagents:** Tripotassium phosphate
Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium
Solvents: 1,4-Dioxane, Water; 1 min, rt; rt → 95 °C; 95 °C

Experimental Protocols

Scheme 480 (1 Reaction)

Steps: 1 Yield: 70%



31-614-CAS-28210586

Steps: 1 Yield: 70%

Synthesis and Photochromism of Novel Pyridyl-Substituted Naphthopyrans

By: de Azevedo, Orlando D. C. C.; et al

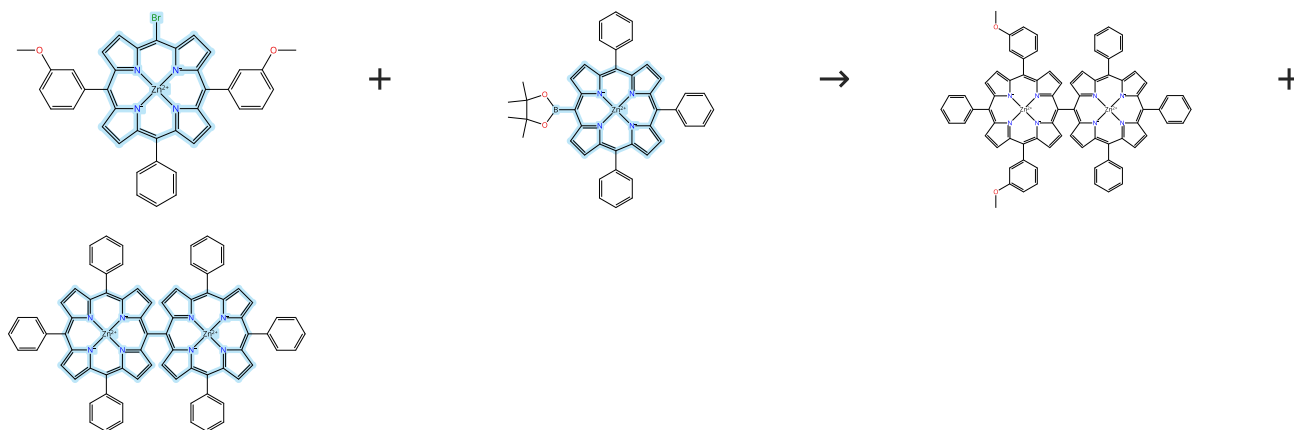
Journal of Organic Chemistry (2020), 85(16), 10772-10796.

1.1 **Reagents:** Potassium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Ethanol, Toluene; 16 h, reflux

Experimental Protocols

Scheme 481 (1 Reaction)

Steps: 1 Yield: 69%



31-226-CAS-12879673

Steps: 1 Yield: 69%

Synthesis and Functionalization of Triply Fused Porphyrin Dimers

By: Ryan, Aoife A.; et al

European Journal of Organic Chemistry (2013), 2013(18), 3700-3711.

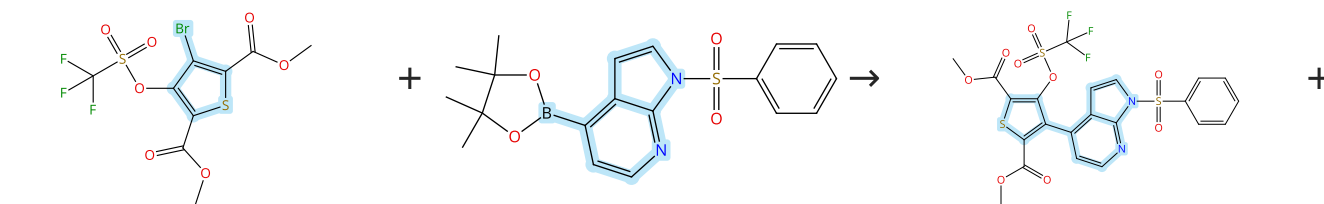
1.1 **Reagents:** Cesium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Dimethylformamide, Toluene; 18 - 24 h, 80 °C

1.2 **Reagents:** Water

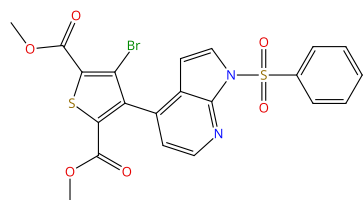
Experimental Protocols

Scheme 482 (2 Reactions)

Steps: 1 Yield: 69%



Suppliers (59)



31-179-CAS-14589121

Steps: 1 Yield: 69%

Unusual ligand-dependent chemoselective Suzuki-Miyaura cross-coupling reactions of 3-bromo-4-trifloyl-thiophenes

1.1 **Reagents:** Potassium carbonate
Catalysts: Bis(tri-*tert*-butylphosphine)palladium
Solvents: 1,2-Dimethoxyethane, Water; rt → 70 °C; 70 °C

By: Wang, Jizhou; et al

Tetrahedron Letters (2011), 52(48), 6346-6348.

Experimental Protocols

31-179-CAS-12456895

Steps: 1 Yield: 69%

Unusual ligand-dependent chemoselective Suzuki-Miyaura cross-coupling reactions of 3-bromo-4-trifloyl-thiophenes

1.1 **Reagents:** Potassium carbonate
Catalysts: Bis(tri-*tert*-butylphosphine)palladium
Solvents: 1,4-Dioxane, Water; rt → 70 °C; 70 °C

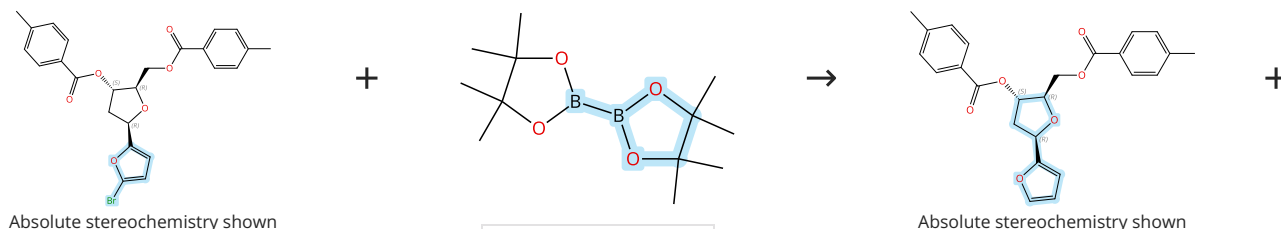
By: Wang, Jizhou; et al

Tetrahedron Letters (2011), 52(48), 6346-6348.

Experimental Protocols

Scheme 483 (1 Reaction)

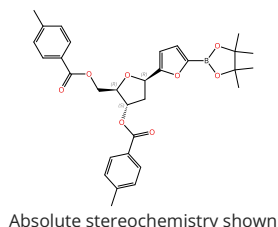
Steps: 1 Yield: 68%



Absolute stereochemistry shown

Absolute stereochemistry shown

Suppliers (131)



Absolute stereochemistry shown

31-113-CAS-7063270

Steps: 1 Yield: 68%

Modular synthesis of 5-substituted furan-2-yl C-2'-deoxyribonucleosides and biaryl covalent base-pair analogues

1.1 **Reagents:** Potassium acetate
Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium
Solvents: 1,4-Dioxane; 40 min, 105 °C

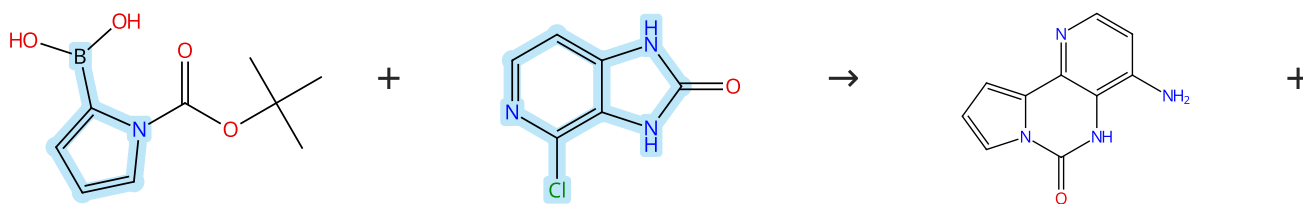
By: Birta, Jan; et al

European Journal of Organic Chemistry (2010), (28), 5432-5443, S5432/1-S5432/29.

Experimental Protocols

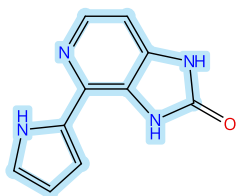
Scheme 484 (1 Reaction)

Steps: 1 Yield: 67%



Suppliers (103)

Suppliers (45)



31-614-CAS-31810109

Steps: 1 Yield: 67%

Aza-BODIPY Route to Ageladine A

1.1 Reagents: Cesium carbonate

By: Tolle, Christian; et al

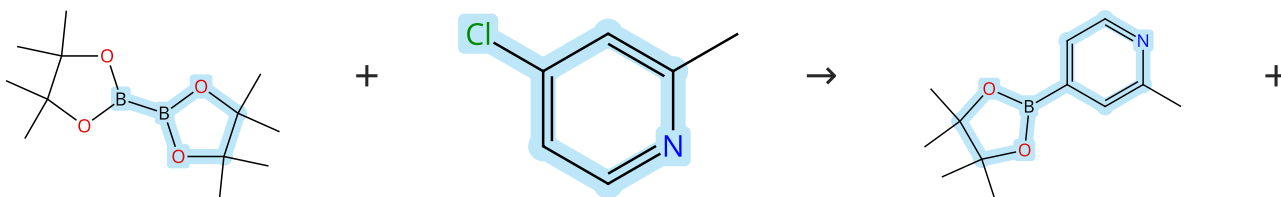
Catalysts: (*SP*-4-3)-[2'-(Amino-κ*M*)[1,1'-biphenyl]-2-yl-κ*C*][dicyclohexyl[2',4',6'-tris(1-methylethyl)[1,1'-biphenyl]-2-yl]phosphine](methanesulfonato-κ*O*)palladium

Organic Letters (2022), 24(6), 1308-1312.

Solvents: 1,4-Dioxane, Water; 6 h, 110 °C

Scheme 485 (1 Reaction)

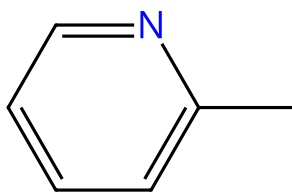
Steps: 1 Yield: 64%



Suppliers (131)

Suppliers (88)

Suppliers (91)



Suppliers (77)

31-113-CAS-22759718

Steps: 1 Yield: 64%

Visible-Light-Induced Ni-Catalyzed Radical Borylation of Chloroarenes

1.1 Reagents: Potassium methoxide

By: Tian, Ya-Ming; et al

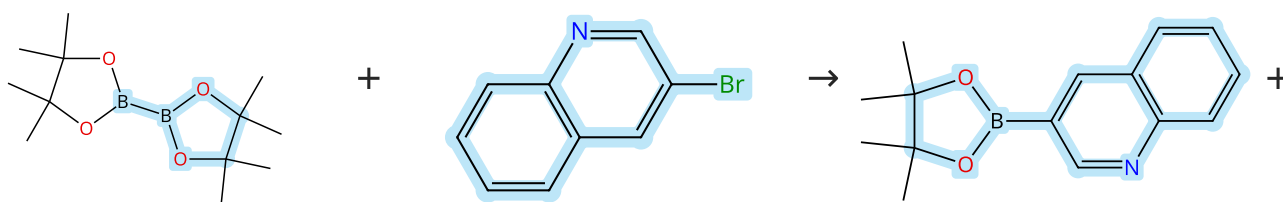
Catalysts: Bis[1,3-dihydro-1,3-bis(2,4,6-trimethylphenyl)-2*H*-imidazol-2-ylidene]nickel

Journal of the American Chemical Society (2020), 142(42), 18231-18242.

Solvents: Hexane; 6 h, rt

Scheme 486 (1 Reaction)

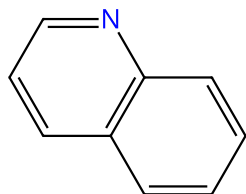
Steps: 1 Yield: 64%



Suppliers (131)

Suppliers (101)

Suppliers (85)



Suppliers (120)

31-113-CAS-2459584

Steps: 1 Yield: 64%

1.1 **Catalysts:** Zinc bromide, 1,3-Dihydro-1,3-bis(2,4,6-trimethylphenyl)-2*H*-imidazol-2-ylidene

Solvents: *tert*-Butyl methyl ether; 30 min, rt

1.2 **Reagents:** Potassium methoxide; 30 min, rt

1.3 12 h, rt

Experimental Protocols

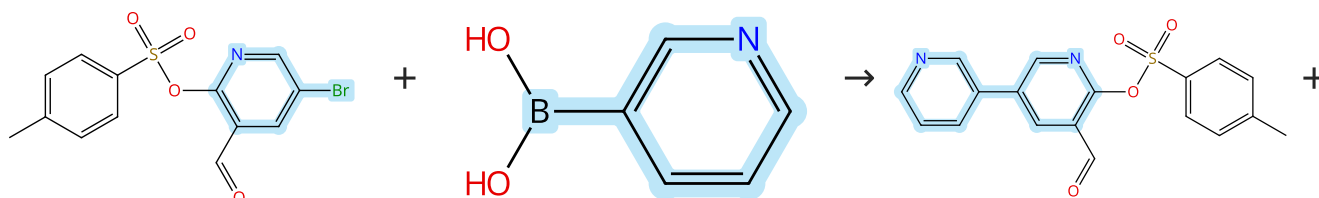
Efficient Synthesis of Aryl Boronates via Zinc-Catalyzed Cross-Coupling of Alkoxy Diboron Reagents with Aryl Halides at Room Temperature

By: Bose, Shubhankar Kumar; et al

Organic Letters (2014), 16(17), 4562-4565.

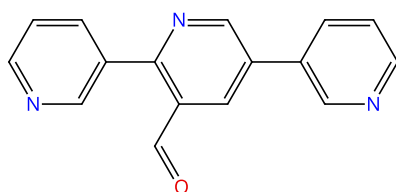
Scheme 487 (1 Reaction)

Steps: 1 Yield: 60%



Suppliers (2)

Suppliers (115)



Suppliers (6)

31-179-CAS-22252118

Steps: 1 Yield: 60%

1.1 **Reagents:** Potassium carbonate
Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium

Solvents: 1,4-Dioxane, Water; 1 h, 100 °C

Experimental Protocols

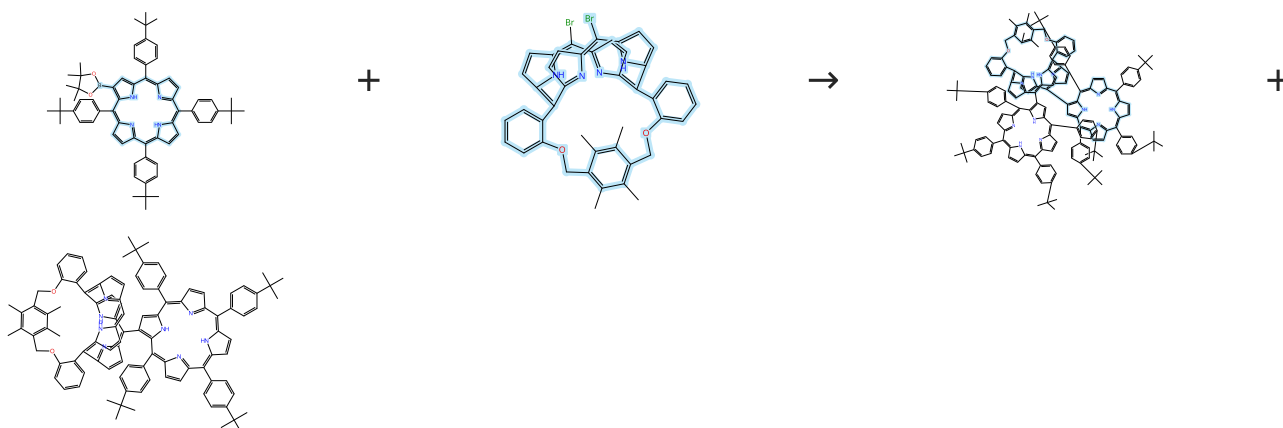
Br vs. TsO Chemoselective Suzuki-Miyaura Cross-Coupling Reaction on Nicotinaldehyde Moiety for the Preparation of 2,3,5-Trisubstituted Pyridines

By: Hedou, Damien; et al

European Journal of Organic Chemistry (2020), 2020(24), 3640-3649.

Scheme 488 (1 Reaction)

Steps: 1 Yield: 60%



31-179-CAS-5639517

Steps: 1 Yield: 60%

Axial, Helical, and Planar Chirality in Directly Linked Basket-Handle Porphyrin Arrays

By: Gehrold, Andreas C.; et al

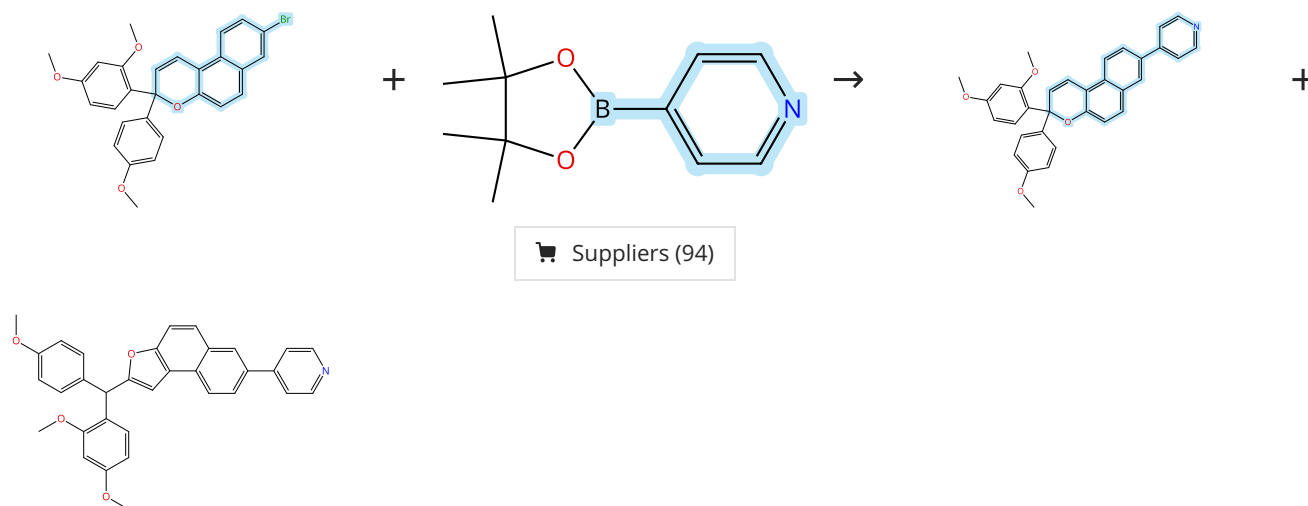
Journal of Organic Chemistry (2016), 81(3), 1075-1088.

1.1 **Reagents:** Barium hydroxide
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Toluene, Water; 20 min, rt; rt → 110 °C; 15 h, 110 °C

Experimental Protocols

Scheme 489 (1 Reaction)

Steps: 1 Yield: 60%



31-179-CAS-22623299

Steps: 1 Yield: 60%

Synthesis and Photochromism of Novel Pyridyl-Substituted Naphthopyrans

By: de Azevedo, Orlando D. C. C.; et al

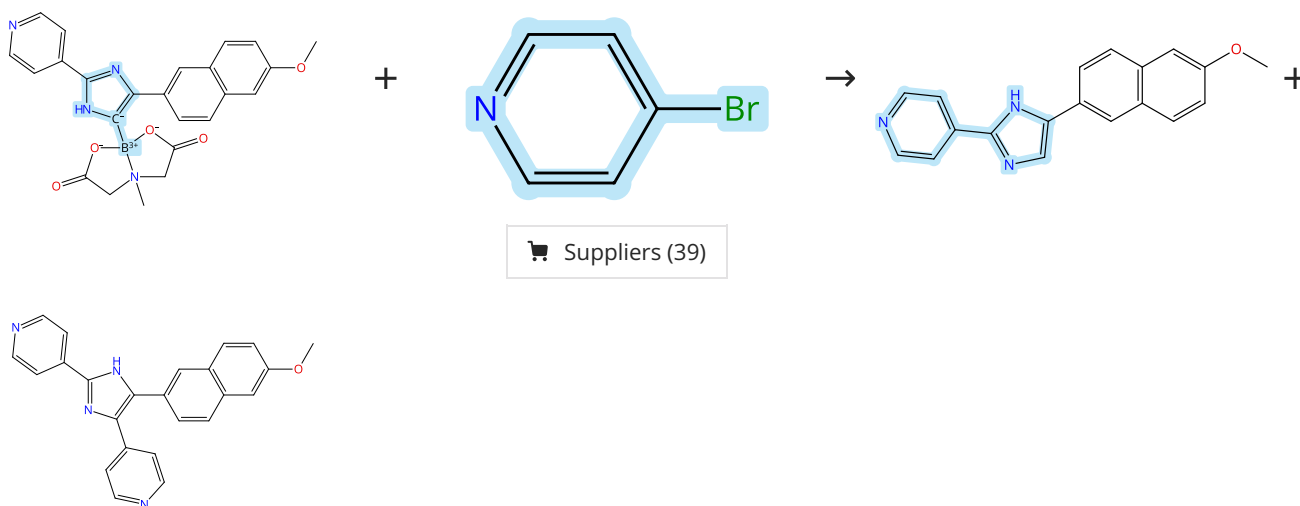
Journal of Organic Chemistry (2020), 85(16), 10772-10796.

1.1 **Reagents:** Potassium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Ethanol, Toluene; 15 h, reflux

Experimental Protocols

Scheme 490 (1 Reaction)

Steps: 1 Yield: 59%



31-177-CAS-17082839

Steps: 1 Yield: 59%

Oxalyl Boronates Enable Modular Synthesis of Bioactive Imidazoles

- 1.1 **Reagents:** Sodium carbonate
Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium
Solvents: Dimethylformamide, Water; 1 h, 70 °C

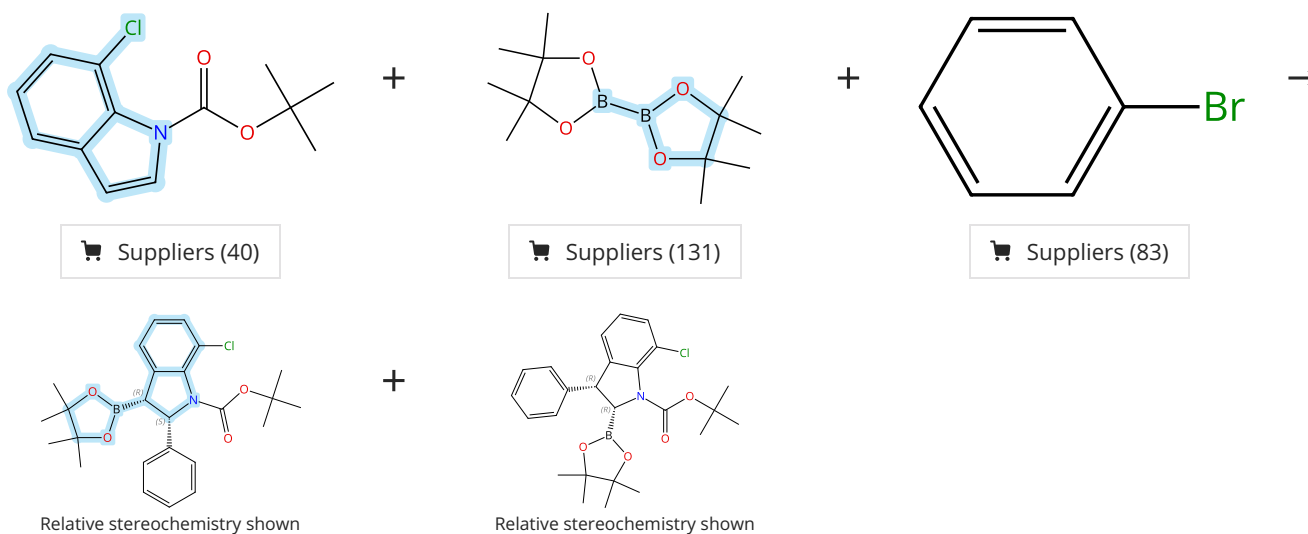
By: Lee, C. Frank; et al

Angewandte Chemie, International Edition (2017), 56(22), 6264-6267.

Experimental Protocols

Scheme 491 (1 Reaction)

Steps: 1 Yield: 58%



31-614-CAS-24363500

Steps: 1 Yield: 58%

Nickel-Catalyzed Dearomative Arylboration of Indoles: Regioselective Synthesis of C2- and C3-Borylated Indolines

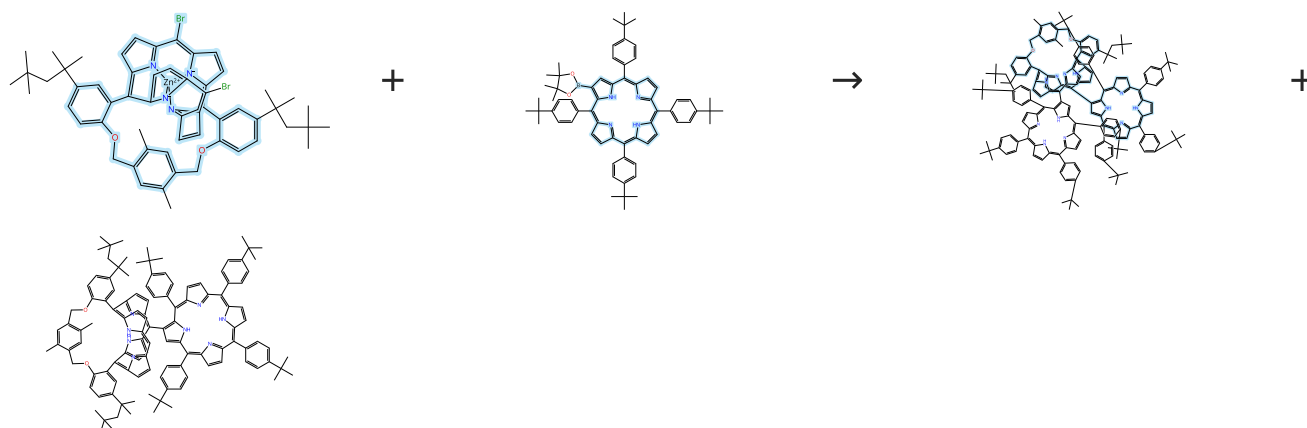
- 1.1 **Reagents:** Sodium *tert*-butoxide
Solvents: Tetrahydrofuran; 25 °C → 0 °C
 1.2 **Catalysts:** Dichloro[1,2-di(methoxy-κO)ethane]nickel
Solvents: Dimethylacetamide; 24 h, 2 - 10 °C
 1.3 **Reagents:** Water

By: Trammel, Grace L.; et al

Journal of the American Chemical Society (2021), 143(40), 16502-16511.

Scheme 492 (1 Reaction)

Steps: 1 Yield: 58%



31-614-CAS-28106000

Steps: 1 Yield: 58%

Axial, Helical, and Planar Chirality in Directly Linked Basket-Handle Porphyrin Arrays

By: Gehrold, Andreas C.; et al

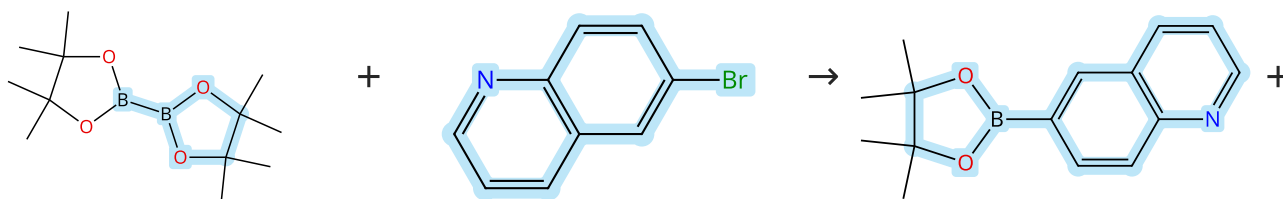
Journal of Organic Chemistry (2016), 81(3), 1075-1088.

1.1 **Reagents:** Barium hydroxide
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Toluene, Water; 20 min, rt; rt → 110 °C; 15 h, 110 °C

Experimental Protocols

Scheme 493 (1 Reaction)

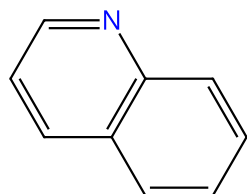
Steps: 1 Yield: 58%



Suppliers (131)

Suppliers (104)

Suppliers (74)



Suppliers (120)

31-113-CAS-8575043

Steps: 1 Yield: 58%

Efficient Synthesis of Aryl Boronates via Zinc-Catalyzed Cross-Coupling of Alkoxy Diboron Reagents with Aryl Halides at Room Temperature

By: Bose, Shubhankar Kumar; et al

Organic Letters (2014), 16(17), 4562-4565.

1.1 **Catalysts:** Zinc bromide, 1,3-Dihydro-1,3-bis(2,4,6-trimethylphenyl)-2*H*-imidazol-2-ylidene
Solvents: *tert*-Butyl methyl ether; 30 min, rt

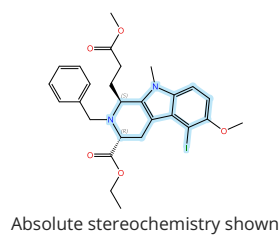
1.2 **Reagents:** Potassium methoxide; 30 min, rt

1.3 12 h, rt

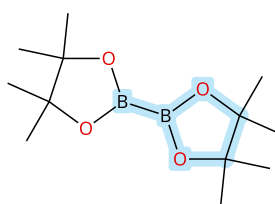
Experimental Protocols

Scheme 494 (1 Reaction)

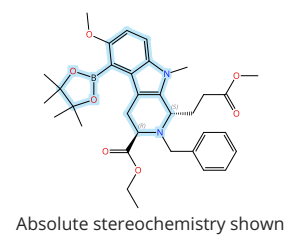
Steps: 1 Yield: 55%



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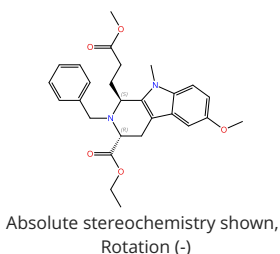


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Suppliers (131)



31-113-CAS-11155237

Steps: 1 Yield: 55%

1.1 **Reagents:** Potassium acetate
Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium
Solvents: Dimethyl sulfoxide; rt → 100 °C; 10 h, 100 °C

Experimental Protocols

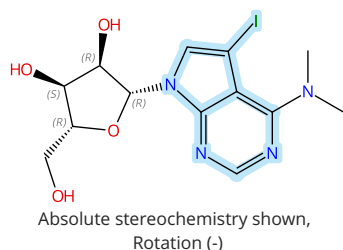
Stereospecific Approach to the Synthesis of Ring-A Oxygenated Sarpagine Indole Alkaloids. Total Synthesis of the Dimeric Indole Alkaloid P-(+)-Dispegatine and Six Other Monomeric Indole Alkaloids

By: Edwankar, Chitra R.; et al

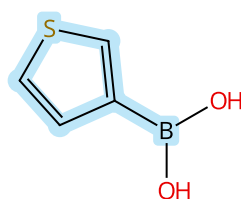
Journal of Organic Chemistry (2013), 78(13), 6471-6487.

Scheme 495 (1 Reaction)

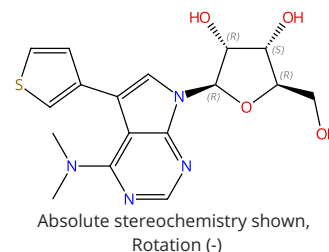
Steps: 1 Yield: 55%



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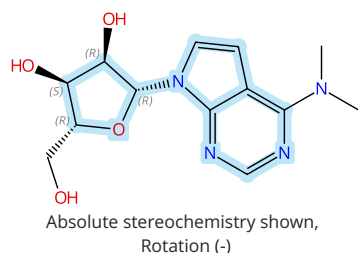


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Supplier (1)

Suppliers (101)

Supplier (1)



Suppliers (3)

31-113-CAS-4715238

Steps: 1 Yield: 55%

1.1 **Reagents:** Sodium carbonate
Catalysts: Palladium diacetate, Benzenesulfonic acid, 3,3',3''-phosphinidynetris-, sodium salt (1:3)
Solvents: Acetonitrile, Water; 3 h, 100 °C; cooled

1.2 **Reagents:** Hydrochloric acid
Solvents: Water; neutralized, cooled

Experimental Protocols

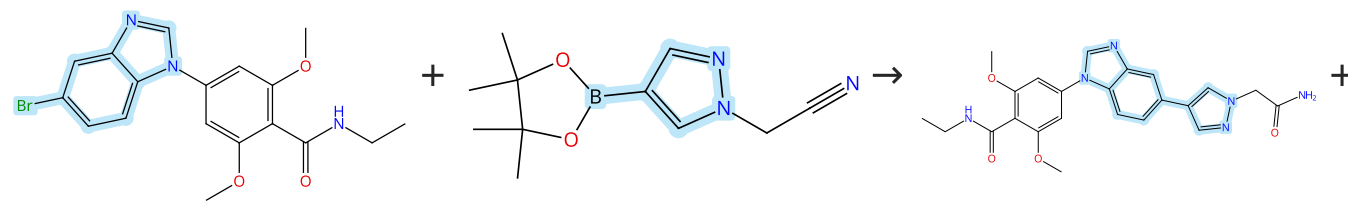
Synthesis, Cytostatic, Antimicrobial, and Anti-HCV Activity of 6-Substituted 7-(Het)aryl-7-deazapurine Ribonucleosides

By: Naus, Petr; et al

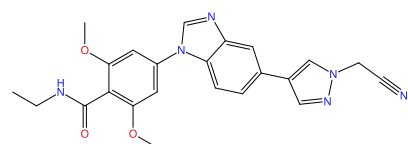
Journal of Medicinal Chemistry (2014), 57(3), 1097-1110.

Scheme 496 (1 Reaction)

Steps: 1 Yield: 55%



Suppliers (62)



31-614-CAS-38699943

Steps: 1 Yield: 55%

Optimization of selectivity and pharmacokinetic properties of salt-inducible kinase (SIK) inhibitors that led to the discovery of the pan-SIK inhibitor GLPG3312

By: Temal-Laib, Taoues; et al

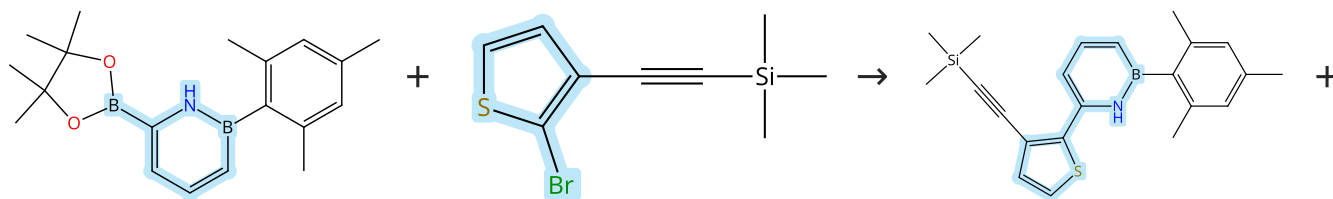
Journal of Medicinal Chemistry (2024), 67(1), 380-401.

1.1 **Reagents:** Cesium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: 1,4-Dioxane, Water; 0.25 - 16 h, 90 - 110 °C

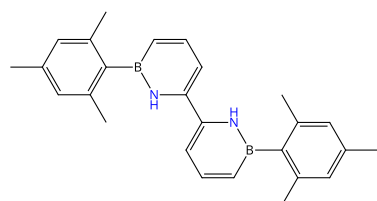
Experimental Protocols

Scheme 497 (1 Reaction)

Steps: 1 Yield: 53%



Suppliers (4)



31-614-CAS-42237233

Steps: 1 Yield: 53%

From a 1,2-azaborinine to large BN-PAHs via electrophilic cyclization: synthesis, characterization and promising optical properties

By: Appiarius, Yannik; et al

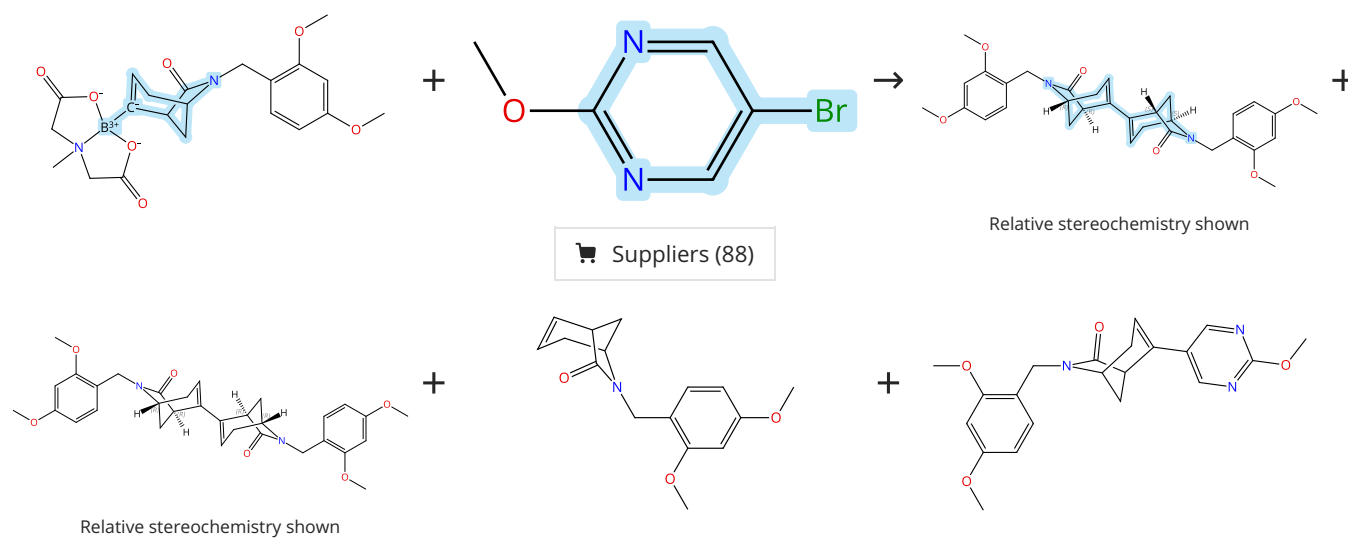
Organic Chemistry Frontiers (2021), 8(1), 10-17.

1.1 **Reagents:** Tripotassium phosphate
Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium
Solvents: *tert*-Butyl methyl ether, Water; 6 h, 80 °C

Experimental Protocols

Scheme 498 (1 Reaction)

Steps: 1 Yield: 63%



31-177-CAS-23376241

Steps: 1 Yield: 63%

Synthesis and functionalization of a bifunctional normorphan 3D building block for medicinal chemistry

1.1 **Reagents:** Dipotassium phosphate
Catalysts: Palladium diacetate, 2-Dicyclohexylphosphino-2',6'-dimethoxybiphenyl
Solvents: 1,4-Dioxane; 18 h, 60 °C

By: Gomez-Angel, Andres R.; et al
 Tetrahedron (2021), 83, 131961.